

EIGHTH EDITION

8

FRANK H. NETTER, MD

NETTER ATLAS *of* HUMAN ANATOMY

Classic Regional Approach



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1600 John F. Kennedy Blvd.
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Philadelphia, PA 19103-2899

**ATLAS OF HUMAN ANATOMY: CLASSIC REGIONAL APPROACH,
EIGHTH EDITION**

Standard Edition: 978-0-323-68042-4
Professional Edition: 978-0-323-79373-5
International Edition: 978-0-323-79374-2

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International Standard Book Number: 978-0-323-68042-4

Publisher: Elyse O'Grady
Senior Content Strategist: Marybeth Thiel
Publishing Services Manager: Catherine Jackson
Senior Project Manager/Specialist: Carrie Stetz
Book Design: Renee Duenow

Printed in China

9 8 7 6 5 4 3 2 1



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PREFACE

The illustrations comprising the *Netter Atlas of Human Anatomy* were painted by physician-artists, Frank H. Netter, MD, and Carlos Machado, MD. Dr. Netter was a surgeon and Dr. Machado is a cardiologist. Their clinical insights and perspectives have informed their approaches to these works of art. The collective expertise of the anatomists, educators, and clinicians guiding the selection, arrangement, labeling, and creation of the illustrations ensures the accuracy, relevancy, and educational power of this outstanding collection.

You have a copy of the **regionally organized** 8th edition with English language terminology. This is the traditional organization and presentation that has been used since the first edition. Also available is a Latin terminology option that is also regionally organized, as well as an option with English terminology organized by body system. In all cases, the same beautiful and instructive Art Plates and Table information are included.

New to this Edition

New Art

More than 20 new illustrations have been added and over 30 art modifications have been made throughout this edition. Highlights include new views of the temporal and infratemporal fossa, pelvic fascia, nasal cavity and paranasal sinuses, plus multiple new perspectives of the heart, a cross-section of the foot, enhanced surface anatomy plates, and overviews of many body systems. In these pages you will find the most robust illustrated coverage to date for modern clinical anatomy courses.

Terminology and Label Updates

This 8th edition incorporates terms of the *Terminologia Anatomica* (2nd edition), as published by the Federative International Programme on Anatomical Terminology in 2019 (<https://fipat.library.dal.ca/ta2>) and adopted by the International Federation of Associations of Anatomy in 2020. A fully searchable database of the updated *Terminologia Anatomica* can be accessed at <https://ta2viewer.openanatomy.org>. Common clinical eponyms and former terminologies are selectively included, parenthetically, for clarity. In addition, a strong effort has been made to reduce label text on the page while maximizing label information through the use of abbreviations (muscle/s = m/mm.; artery/ies = a./aa.; vein/s = v./vv.; and nerve/s = n./nn.) and focusing on the labels most relevant to the subject of each Plate.

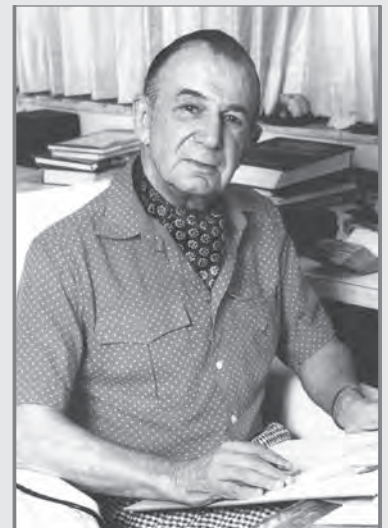
Nerve Tables

The muscle tables and clinical tables of previous editions have been so positively received that new tables have been added to cover four major nerve groups: cranial nerves and the nerves of the cervical, brachial, and lumbosacral plexuses.

Future of the Netter Anatomy Atlas

As the Netter Atlas continues to evolve to meet the needs of students, educators, and clinicians, we welcome suggestions! Please use the following form to provide your feedback:

<https://tinyurl.com/NetterAtlas8>



To my dear wife, Vera

PREFACE TO THE FIRST EDITION

I have often said that my career as a medical artist for almost 50 years has been a sort of “command performance” in the sense that it has grown in response to the desires and requests of the medical profession. Over these many years, I have produced almost 4,000 illustrations, mostly for *The CIBA (now Netter) Collection of Medical Illustrations* but also for *Clinical Symposia*. These pictures have been concerned with the varied subdivisions of medical knowledge such as gross anatomy, histology, embryology, physiology, pathology, diagnostic modalities, surgical and therapeutic techniques, and clinical manifestations of a multitude of diseases. As the years went by, however, there were more and more requests from physicians and students for me to produce an atlas purely of gross anatomy. Thus, this atlas has come about, not through any inspiration on my part but rather, like most of my previous works, as a fulfillment of the desires of the medical profession.

It involved going back over all the illustrations I had made over so many years, selecting those pertinent to gross anatomy, classifying them and organizing them by system and region, adapting them to page size and space, and arranging them in logical sequence. Anatomy of course does not change, but our understanding of anatomy and its clinical significance does change, as do anatomical terminology and nomenclature. This therefore required much updating of many of the older pictures and

even revision of a number of them in order to make them more pertinent to today’s ever-expanding scope of medical and surgical practice. In addition, I found that there were gaps in the portrayal of medical knowledge as pictorialized in the illustrations I had previously done, and this necessitated my making a number of new pictures that are included in this volume.

In creating an atlas such as this, it is important to achieve a happy medium between complexity and simplification. If the pictures are too complex, they may be difficult and confusing to read; if oversimplified, they may not be adequately definitive or may even be misleading. I have therefore striven for a middle course of realism without the clutter of confusing minutiae. I hope that the students and members of the medical and allied professions will find the illustrations readily understandable, yet instructive and useful.

At one point, the publisher and I thought it might be nice to include a foreword by a truly outstanding and renowned anatomist, but there are so many in that category that we could not make a choice. We did think of men like Vesalius, Leonardo da Vinci, William Hunter, and Henry Gray, who of course are unfortunately unavailable, but I do wonder what their comments might have been about this atlas.

Frank H. Netter, MD
(1906–1991)

FRANK H. NETTER, MD

Frank H. Netter was born in New York City in 1906. He studied art at the Art Students League and the National Academy of Design before entering medical school at New York University, where he received his Doctor of Medicine degree in 1931. During his student years, Dr. Netter’s notebook sketches attracted the attention of the medical faculty and other physicians, allowing him to augment his income by illustrating articles and textbooks. He continued illustrating as a sideline after establishing a surgical practice in 1933, but he ultimately opted to give up his practice in favor of a full-time commitment to art. After service in the United States Army during World War II, Dr. Netter began his long collaboration with the CIBA Pharmaceutical Company (now Novartis Pharmaceuticals). This 45-year partnership resulted in the production of the extraordinary collection of medical art so familiar to physicians and other medical professionals worldwide.

Icon Learning Systems acquired the Netter Collection in July 2000 and continued to update Dr. Netter’s original paintings and to add newly commissioned paintings by artists trained in the style of Dr. Netter. In 2005, Elsevier Inc. purchased the Netter Collection and all publications from Icon Learning Systems. There are now over

50 publications featuring the art of Dr. Netter available through Elsevier Inc.

Dr. Netter’s works are among the finest examples of the use of illustration in the teaching of medical concepts. The 13-book *Netter Collection of Medical Illustrations*, which includes the greater part of the more than 20,000 paintings created by Dr. Netter, became and remains one of the most famous medical works ever published. *The Netter Atlas of Human Anatomy*, first published in 1989, presents the anatomic paintings from the Netter Collection. Now translated into 16 languages, it is the anatomy atlas of choice among medical and health professions students the world over.

The Netter illustrations are appreciated not only for their aesthetic qualities, but, more importantly, for their intellectual content. As Dr. Netter wrote in 1949 “clarification of a subject is the aim and goal of illustration. No matter how beautifully painted, how delicately and subtly rendered a subject may be, it is of little value as a *medical illustration* if it does not serve to make clear some medical point.” Dr. Netter’s planning, conception, point of view, and approach are what inform his paintings and what make them so intellectually valuable.

Frank H. Netter, MD, physician and artist, died in 1991.

ABOUT THE EDITORS

Carlos A.G. Machado, MD was chosen by Novartis to be Dr. Netter's successor. He continues to be the main artist who contributes to the Netter collection of medical illustrations.

Self-taught in medical illustration, cardiologist Carlos Machado has contributed meticulous updates to some of Dr. Netter's original plates and has created many paintings of his own in the style of Netter as an extension of the Netter collection. Dr. Machado's photorealistic expertise and his keen insight into the physician/patient relationship inform his vivid and unforgettable visual style. His dedication to researching each topic and subject he paints places him among the premier medical illustrators at work today.

Learn more about his background and see more of his art at: <https://netterimages.com/artist-carlos-a-g-machado.html>

Paul E. Neumann, MD was clinically trained in anatomical pathology and neuropathology. Most of his research publications have been in mouse neurogenetics and molecular human genetics. In the past several years, he has concentrated on the anatomical sciences, and has frequently written about anatomical terminology and anatomical ontology in the journal *Clinical Anatomy*. As an officer of the Federative International Programme for Anatomical Terminology (FIPAT), he participated in the production of *Terminologia Anatomica* (2nd edition), *Terminologia Embryologica* (2nd edition), and *Terminologia Neuroanatomica*. In addition to serving as the lead Latin editor of the 8th edition of Netter's Atlas, he was a contributor to the 33rd edition of *Dorland's Illustrated Medical Dictionary*.

R. Shane Tubbs, MS, PA-C, PhD is a native of Birmingham, Alabama and a clinical anatomist. His research interests are centered around clinical/surgical problems that are identified and solved with anatomical studies. This investigative paradigm in anatomy as resulted in over 1,700 peer reviewed publications. Dr. Tubbs' laboratory has made novel discoveries in human anatomy including a new nerve to the skin of the lower eyelid, a new space of the face, a new venous sinus over the spinal cord, new connections between the parts of the sciatic nerve, new ligaments of the neck, a previously undescribed cutaneous branch of the inferior gluteal nerve, and an etiology for postoperative C5 nerve palsies. Moreover, many anatomical feasibility studies from Dr. Tubbs' laboratory have gone on to be used by surgeons from around the world and have thus resulted in new surgical/clinical procedures such as treating hydrocephalus by shunting cerebrospinal fluid into various bones, restoration of upper limb function in paralyzed patients with neurotization procedures using the contralateral spinal accessory nerve, and harvesting of clavicle for anterior cervical discectomy and fusion procedures in patients with cervical instability or degenerative spine disease.

Dr. Tubbs sits on the editorial board of over 15 anatomical journals and has reviewed for over 150 scientific journals. He has been a visiting professor to major institutions in the United States and worldwide. Dr. Tubbs has authored over 40 books and over 75 book chapters. His published books by Elsevier include *Gray's Anatomy Review*, *Gray's Clinical Photographic Dissector of the Human Body*, *Netter's Introduction to Clinical Procedures*, and *Nerves and Nerve Injuries* volumes I and II. He is an editor for the 41st and 42nd editions of the over 150-year-old *Gray's Anatomy*, the 5th through 8th editions of *Netter's Atlas of Anatomy*, and is the editor-in-chief of the journal *Clinical Anatomy*. He is the Chair of the Federative International Programme on Anatomical Terminologies (FIPAT).

Jennifer K. Brueckner-Collins, PhD is a proud Kentucky native. She pursued her undergraduate and graduate training at the University of Kentucky. During her second year of graduate school there, she realized that her professional calling was not basic science research in skeletal muscle biology, but was instead helping medical students master the anatomical sciences. She discovered this during a required teaching assistantship in medical histology, where working with students at the 10-headed microscope changed her career path.

The next semester of graduate school, she assisted in teaching dissection-based gross anatomy, although she had taken anatomy when the lab component was prosection based. After teaching in the first lab, she knew that she needed to learn anatomy more thoroughly through dissection on her own, so she dissected one to two labs ahead of the students that semester; that was when she really learned anatomy and was inspired to teach this discipline as a profession. All of this occurred in the early 1990s, when pursuing a teaching career was frowned upon by many; it was thought that you only pursued this track if you were unsuccessful in research. She taught anatomy part-time during the rest of her graduate training, on her own time, to gain requisite experience to ultimately secure a faculty position.

Dr. Brueckner-Collins spent 10 years at the University of Kentucky as a full-time faculty member teaching dissection-based gross anatomy to medical, dental, and allied health students. Then, after meeting the love of her life, she moved to the University of Louisville and has taught medical and dental students there for more than a decade. Over 20 years of teaching full time at two medical schools in the state, her teaching efforts have been recognized through receipt of the highest teaching honor at each medical school in the state, the Provost's Teaching Award at University of Kentucky, and the Distinguished Teaching Professorship at University of Louisville.

Martha Johnson Gdowski, PhD earned her BS in Biology cum laude from Gannon University in 1990, followed by a PhD in Anatomy from the Pennsylvania State University College of Medicine in 1995. She completed postdoctoral

fellowships at the Cleveland Clinic and Northwestern University School of Medicine prior to accepting a faculty position in the Department of Neuroscience at the University of Rochester School of Medicine and Dentistry in 2001. Previous research interests include the development of an adult model of hydrocephalus, sensorimotor integration in the basal ganglia, and sensorimotor integration in normal and pathological aging.

Her passion throughout her career has been in her service as an educator. Her teaching has encompassed a variety of learning formats, including didactic lecture, laboratory, journal club, and problem-based learning. She has taught for four academic institutions in different capacities (The Pennsylvania State University School of Medicine, Northwestern University School of Medicine, Ithaca College, and The University of Rochester School of Medicine and Dentistry). She has taught in the following curricula: Undergraduate and Graduate Neuroscience, Graduate Neuroanatomy, Graduate Human Anatomy and Physiology for Physical Therapists, Undergraduate Medical Human Anatomy and Histology, and Undergraduate and Graduate Human Anatomy. These experiences have provided an opportunity to instruct students that vary in age, life experience, race, ethnicity, and economic background, revealing how diversity in student populations enriches learning environments in ways that benefit everyone. She has been honored to be the recipient of numerous awards for her teaching and mentoring of students during their undergraduate medical education. Martha enjoys gardening, hiking, and swimming with her husband, Greg Gdowski, PhD, and their dogs, Sophie and Ivy.

Virginia T. Lyons, PhD is an Associate Professor of Medical Education and the Associate Dean for Preclinical Education at the Geisel School of Medicine at Dartmouth. She received her BS in Biology from Rochester Institute of Technology and her PhD in Cell Biology and Anatomy from the University of North Carolina at Chapel Hill. Dr. Lyons has devoted her career to education in the anatomical sciences, teaching gross anatomy, histology, embryology, and neuroanatomy to medical students and other health professions students. She has led courses and curricula in human gross anatomy and embryology for more than 20 years and is a strong advocate for incorporating engaged pedagogies into preclinical medical education. Dr. Lyons has been recognized with numerous awards for teaching and mentoring students and was elected to the Dartmouth chapter of the Alpha Omega Alpha Honor Medical Society. She is the author of *Netter's Essential Systems-Based Anatomy* and co-author of the Human Anatomy Learning Modules website accessed by students worldwide. Dr. Lyons also serves as the Discipline Editor for Anatomy on the Aquifer Sciences Curriculum Editorial Board, working to integrate anatomical concepts into virtual patient cases that are used in multiple settings including clerkships and residency training.

Peter J. Ward, PhD grew up in Casper, Wyoming, graduating from Kelly Walsh High School and then attending Carnegie Mellon University in Pittsburgh, Pennsylvania.

He began graduate school at Purdue University, where he first encountered gross anatomy, histology, embryology, and neuroanatomy. Having found a course of study that engrossed him, he helped teach those courses in the veterinary and medical programs at Purdue. Dr. Ward completed a PhD program in anatomy education and, in 2005, he joined the faculty at the West Virginia School of Osteopathic Medicine (WVSOM) in Lewisburg, West Virginia. There he has taught gross anatomy, embryology, neuroscience, histology, and the history of medicine. Dr. Ward has received numerous teaching awards, including the WVSOM Golden Key Award, the Basmajian Award from the American Association of Anatomists, and has been a two-time finalist in the West Virginia Merit Foundation's Professor of the Year selection. Dr. Ward has also been director of the WVSOM plastination facility, coordinator of the anatomy graduate teaching assistants, chair of the curriculum committee, chair of the faculty council, creator and director of a clinical anatomy elective course, and host of many anatomy-centered events between WVSOM and two Japanese Colleges of Osteopathy. Dr. Ward has also served as council member and association secretary for the American Association of Clinical Anatomists. In conjunction with Bone Clones, Inc., Dr. Ward has produced tactile models that mimic the feel of anatomical structures when intact and when ruptured during the physical examination. He created the YouTube channel Clinical Anatomy Explained! and continues to pursue interesting ways to present the anatomical sciences to the public. Dr. Ward was the Senior Associate Editor for the three volumes of *The Netter Collection: The Digestive System*, 2nd Edition, a contributor to *Gray's Anatomy*, 42nd Edition, and is author of *Netter's Integrated Musculoskeletal System: Clinical Anatomy Explained*.

Brion Benninger, MD, MBChB, MSc currently teaches surgical, imaging, and dynamic anatomy to medical students and residents in several countries (United States, New Zealand, China, Japan, Korea, Taiwan, The Caribbean, Mexico). He develops, invents, and assesses ultrasound probes, medical equipment, simulations, and software while identifying dynamic anatomy. He enjoys mixing educational techniques integrating macro imaging and surgical anatomy. Dr. Benninger developed the teaching theory of anatomy deconstruction/reconstruction and was the first to combine ultrasound with Google Glass during physical examination, coining the term "triple feedback examination." An early user of ultrasound, he continues to develop eFAST teaching and training techniques, has developed and shares a patent on a novel ultrasound finger probe, and is currently developing a new revolutionary ultrasound probe for breast screening. He is a reviewer for several ultrasound, clinical anatomy, surgical, and radiology journals and edits and writes medical textbooks. His research interests integrate clinical anatomy with conventional and emerging technologies to improve training techniques in situ and simulation. Dr. Benninger pioneered and coined the term "dynamic anatomy,"

developed a technique to deliver novel contrast medium to humans, and was the first to reveal vessels and nerves not previously seen using CT and MRI imaging. He has mentored more than 200 students on over 350 research projects presented at national and international conferences and has received numerous awards for projects related to emergency procedures, ultrasound, sports medicine, clinical anatomy, medical simulation, reverse translational research, medical education, and technology. He is proud to have received medical teaching awards from several countries and institutions, including being the first recipient in more than 25 years to receive the Commendation Medal Award from the Commission of Osteopathic Accreditation for innovative clinical anatomy teaching that he designed and facilitated in Lebanon, Oregon. Dr. Benninger has received sports medicine accolades from Sir Roger Bannister regarding his medical invention on shoulder proprioception. He is also Executive Director of the Medical Anatomy Center and collaborates with colleagues globally from surgical and nonsurgical specialties. He is also an invited course speaker for surgical anatomy in New Zealand. Dr. Benninger collects medical history books, loves mountains and sports, and is an anonymous restaurant critic. British mentors directly responsible for his training include Prof. Peter Bell (surgery), Prof. Sir Alec Jeffreys (genetic fingerprinting), Profs. David deBono and Tony Gershlick (cardiology), Prof. Roger Greenhalgh (vascular surgery), Profs. Chris Colton, John Webb, and Angus Wallace (orthopaedics), Prof. Harold Ellis CBE (surgery and clinical anatomy), and Prof. Susan Standing (Guys Hospital/Kings College).

Todd M. Hoagland, PhD is Clinical Professor of Biomedical Sciences and Occupational Therapy at Marquette University in the College of Health Sciences. Previously he was Professor of Anatomy at the Medical College of Wisconsin (MCW). Prior to MCW, Dr. Hoagland was at

Boston University School of Medicine (BUSM) and he still holds an adjunct faculty position at Boston University Goldman School of Dental Medicine. Dr. Hoagland is a passionate teacher and is dedicated to helping students achieve their goals. He believes in being a strong steward of the anatomical sciences, which involves teaching it to students while contemporaneously developing resources to improve the transfer of knowledge and preparing the next generation to be even better teachers. While at BUSM, Dr. Hoagland was a leader for the Carnegie Initiative on the Doctorate in Neuroscience and helped develop the Vesalius Program (teacher training) for graduate students. The program ensures that graduate students learn about effective teaching, receive authentic experiences in the classroom, and understand how to share what they learn via scholarship.

Dr. Hoagland's dedication to health professions education has been richly rewarded by numerous teaching awards from the University of Notre Dame, BUSM, and MCW. Dr. Hoagland received the Award for Outstanding Ethical Leadership in 2009, was inducted into the Alpha Omega Alpha Honor Medical Society in 2010, received the American Association of Anatomists Basmajian Award in 2012, and was inducted into the Society of Teaching Scholars in 2012 and was their director from 2016–2020.

Dr. Hoagland's scholarly activity centers on (1) evaluating content and instructional/learning methodology in Clinical Human Anatomy and Neuroanatomy courses, especially as relevant to clinical practice, (2) translating basic anatomical science research findings into clinically meaningful information, and (3) evaluating professionalism in students to enhance their self-awareness and improve patient care outcomes. Dr. Hoagland is also consulting editor for *Netter's Atlas of Human Anatomy*, co-author for the digital anatomy textbook *AnatomyOne*, and lead author for *Clinical Human Anatomy Dissection Guide*.

ACKNOWLEDGMENTS

Carlos A. G. Machado, MD

With the completion of this 8th edition, I celebrate 27 years contributing to the Netter brand of educational products, 25 years of which have been dedicated to the update—seven editions—of this highly prestigious, from birth, *Atlas of Human Anatomy*. For these 25 years I have had the privilege and honor of working with some of the most knowledgeable anatomists, educators, and consulting editors—my treasured friends—from whom I have learned considerably.

For the last 16 years it has also been a great privilege to be part of the Elsevier team and be under the skillful coordination and orientation of Marybeth Thiel, Elsevier's Senior Content Development Specialist, and Elyse O'Grady, Executive Content Strategist. I thank both for their friendship, support, sensibility, and very dedicated work.

Once more I thank my wife Adriana and my daughter Beatriz for all their love and encouragement, and for patiently steering me back on track when I get lost in philosophical divagations about turning scientific research into artistic inspiration—and vice-versa!

It is impossible to put in words how thankful I am to my much-loved parents, Carlos and Neide, for their importance in my education and in the formation of my moral and ethical values.

I am eternally grateful to the body donors for their inestimable contribution to the correct understanding of human anatomy; to the students, teachers, health professionals, colleagues, educational institutions, and friends who have, anonymously or not, directly or indirectly, been an enormous source of motivation and invaluable scientific references, constructive comments, and relevant suggestions.

My last thanks, but far from being the least, go to my teachers Eugênio Cavalcante, Mário Fortes, and Paulo Carneiro, for their inspiring teachings on the practical application of the knowledge of anatomy.

Paul E. Neumann, MD

It has been a privilege to work on the English and Latin editions of *Netter's Atlas of Human Anatomy*. I thank the staff at Elsevier (especially Elyse O'Grady, Marybeth Thiel and Carrie Stetz), Dr. Carlos Machado, and the other editors for their efforts to produce a new, improved edition. I am also grateful to my wife, Sandra Powell, and my daughter, Eve, for their support of my academic work.

R. Shane Tubbs, MS, PA-C, PhD

I thank Elyse O'Grady and Marybeth Thiel for their dedication and hard work on this edition. As always, I thank my wife, Susan, and son, Isaiah, for their patience with me on such projects. Additionally, I thank Drs. George and Frank Salter who inspired and encouraged me along my path to anatomy.

Jennifer K. Brueckner-Collins, PhD

Reba McEntire once said "To succeed in life, you need three things: a wishbone, a backbone and a funny bone."

My work with the *Netter Atlas* and the people associated with it over the past 15 years has played an instrumental role in helping me develop and sustain these three metaphorical bones in my professional and personal life.

I am forever grateful to John Hansen, who believed in my ability to serve as an editor starting with the 4th edition.

I extend my sincere thanks to Marybeth Thiel and Elyse O'Grady for not only being the finest of colleagues, but part of my professional family as well. Thanks to you both for your professionalism, support, patience, and collegiality.

To Carlos Machado, you continue to amaze me and inspire me with your special gift of bringing anatomy to life through your art.

For this edition, I also count in my blessings, the ability to work closely with the talented team of educational leaders, including Martha Gdowski, Virginia Lyons, and Peter Ward. It is humbling to work with such brilliant and dedicated teachers as we collectively assembled the systems-based *Netter Atlas* concept.

Finally, I dedicate my work on this edition with unconditional and infinite love to Kurt, Lincoln, my Dad in Heaven, as well as my dog boys, Bingo and Biscuit.

Martha Johnson Gdowski, PhD

I am grateful for the honor to work with the team of editors that Elsevier has selected for the preparation of this 8th edition; they are exceptional in their knowledge, passion as educators, and collegiality. I especially would like to thank Elyse O'Grady and Marybeth Thiel, who have been outstanding in their expertise, patience, and guidance. I am grateful to John T. Hansen, PhD, for his guidance, mentorship, and friendship as a colleague at the University of Rochester and for giving me the opportunity to participate in this work. He continues to be an outstanding role model who has shaped my career as an anatomical sciences educator. Special thanks to Carlos Machado for his gift for making challenging anatomical dissections and difficult concepts accessible to students of anatomy through his artistry, research of the details, and thoughtful discussions. I am indebted to the selfless individuals who have gifted their bodies for anatomical study, the students of anatomy, and my colleagues at the University of Rochester, all of whom motivate me to work to be the best educator I can be. I am most grateful for my loving husband and best friend, Greg, who is my greatest source of support and inspiration.

Virginia T. Lyons, PhD

It has been a joy to work with members of the editorial team on the iconic *Atlas of Human Anatomy* by Frank Netter. I would like to thank Elyse O'Grady and Marybeth Thiel for their expert guidance and ability to nourish the creative process while also keeping us focused (otherwise we would have reveled in debating anatomy minutiae for hours!). I am amazed by the talent of Carlos Machado, who is able to transform our ideas into beautiful, detailed illustrations that simplify concepts for students. I appreciate the patience and support of my husband, Patrick, and my

children, Sean and Nora, who keep me sane when things get busy. Finally, I am grateful for the opportunity to teach and learn from the outstanding medical students at the Geisel School of Medicine at Dartmouth. I am fulfilled by their energy, curiosity, and love of learning.

Peter J. Ward, PhD

It is a thrill and honor to contribute to the 8th edition of *Netter's Atlas of Human Anatomy*. It still amazes me that I am helping to showcase the incomparable illustrations of Frank Netter and Carlos Machado. I hope that this atlas continues to bring these works of medical art to a new generation of students as they begin investigating the awesome enigma of the human body. Thanks to all the amazing contributors and to the hardworking team at Elsevier, especially Marybeth Thiel and Elyse O'Grady, for keeping all of us moving forward. Thank you especially to Todd Hoagland for recommending me to the team. I have immense gratitude to James Walker and Kevin Hannon, who introduced me to the world of anatomy. They both seamlessly combined high expectations for their students along with enthusiastic teaching that made the topic fascinating and rewarding. Great thanks to my parents, Robert and Lucinda Ward, for their lifelong support of my education and for the many formative museum trips to stare at dinosaur bones. Sarah, Archer, and Dashiell, you are all the reason I work hard and try to make the world a slightly better place. Your love and enthusiasm mean everything to me.

Brion Benninger, MD, MBChB, MSc

I thank all the healthcare institutions worldwide and the allopathic and osteopathic associations who have provided me the privilege to wake up each day and focus on how to improve our knowledge of teaching and healing the anatomy of the mind, body, and soul while nurturing humanism. I am grateful and fortunate to have my lovely wife, Alison, and thoughtful son, Jack, support my efforts during late nights and long weekends. Their laughs and experiences complete my life. I thank Elsevier, especially Marybeth Thiel, Elyse O'Grady, and Madelene Hyde for expecting the highest standards and providing guidance, enabling my fellow coeditors to work in a fluid diverse environment. Many thanks to Carlos Machado and Frank Netter: the world is proud. I thank clinicians who trained

me, especially my early gifted surgeon/anatomist/teacher mentors, Drs. Gerald Tressidor and Harold Ellis CBE (Cambridge & Guy's Hospital); Dr. S. Standring and Dr. M. England, who embody professionalism; Drs. P. Crone, E. Szeto, and J. Heatherington, for supporting innovative medical education; my past, current, and future students and patients; and clinical colleagues from all corners of the world who keep medicine and anatomy dynamic, fresh, and wanting. Special thanks to Drs. J.L. Horn, S. Echols, J. Anderson, and J. Underwood, friends, mentors and fellow visionaries who also see "outside the box," challenging the status quo. Heartfelt tribute to my late mentors, friends, and sister, Jim McDaniel, Bill Bryan, and Gail Hendricks, who represent what is good in teaching, caring, and healing. They made this world a wee bit better. Lastly, I thank my mother for her love of education and equality and my father for his inquisitive and creative mind.

Todd M. Hoagland, PhD

It is a privilege to teach clinical human anatomy, and I am eternally grateful to all the body donors and their families for enabling healthcare professionals to train in the dissection laboratory. It is my honor to work with occupational therapy and health professions students and colleagues at Marquette University. I am grateful to John Hansen and the professionals of the Elsevier team for the opportunity to be a steward of the incomparable *Netter Atlas*. Marybeth Thiel and Elyse O'Grady were especially helpful and a pleasure to work with. It was an honor to collaborate with the brilliant Carlos Machado and all the consulting editors. I thank Dave Bolender, Brian Bear, and Rebecca Lufler for being outstanding colleagues, and I thank all the graduate students I've worked with for helping me grow as a person; it is such a pleasure to see them flourish. I am deeply appreciative of Stan Hillman and Jack O'Malley for inspiring me with masterful teaching and rigorous expectations. I am indebted to Gary Kolesari and Richard Hoyt Jr for helping me become a competent clinical anatomist, and to Rob Bouchie for the intangibles and his camaraderie. I am most grateful to my brother, Bill, for his unwavering optimism and for always being there. I thank my mother, Liz, for her dedication and love, and for instilling a strong work ethic. I am humbled by my three awesome children, Ella, Caleb, and Gregory, for helping me redefine love, wonder, and joy. Olya, ty moye solntse!

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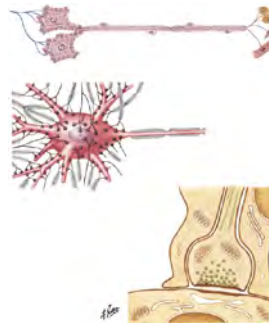
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BP 4 Features of a Typical Peripheral Nerve



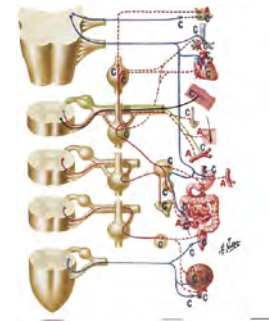
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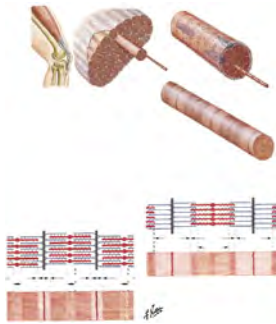
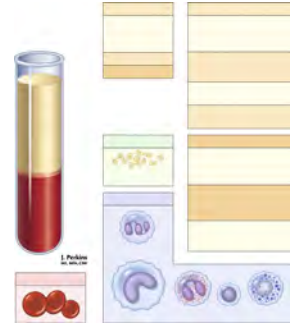


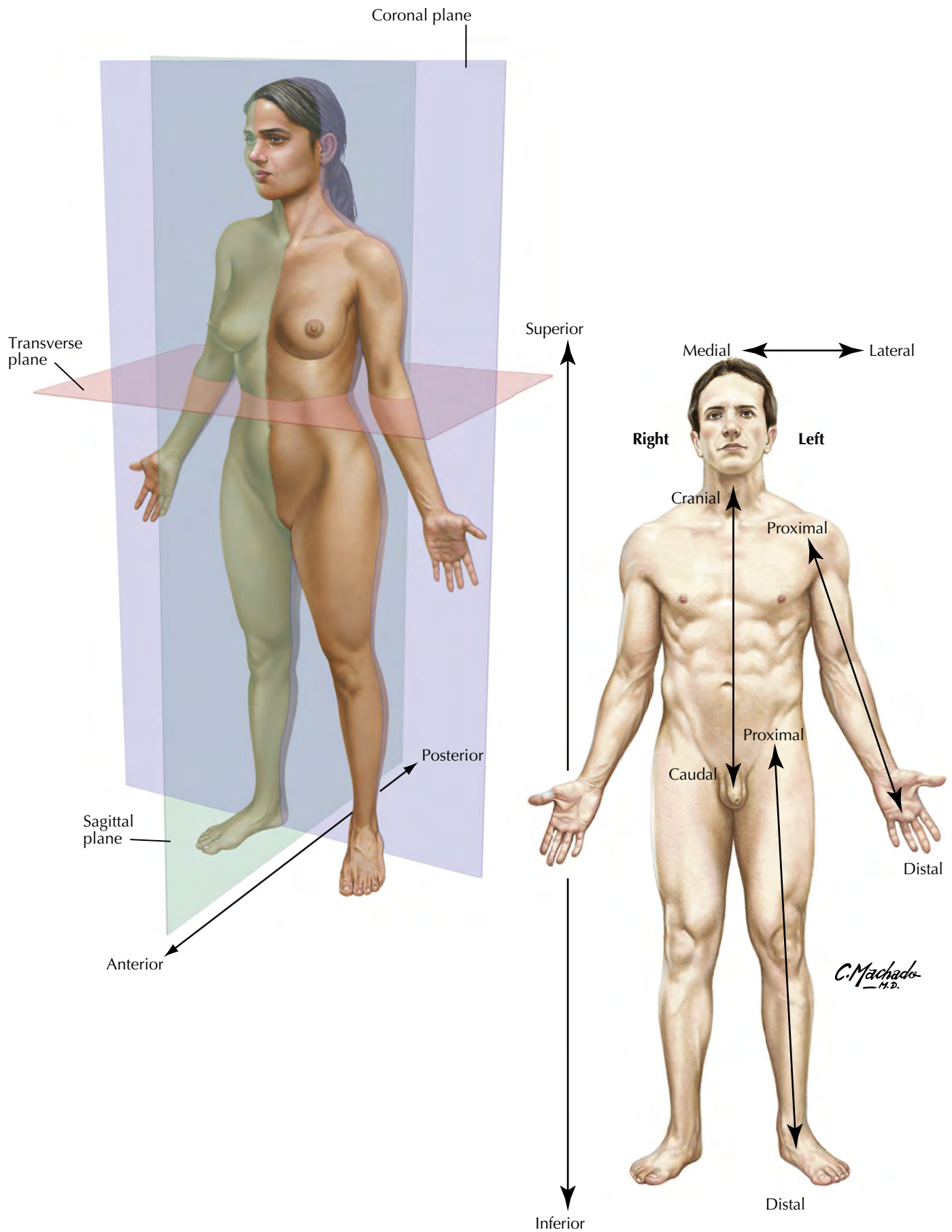
BP 7 Parasympathetic Nervous System: General Topography



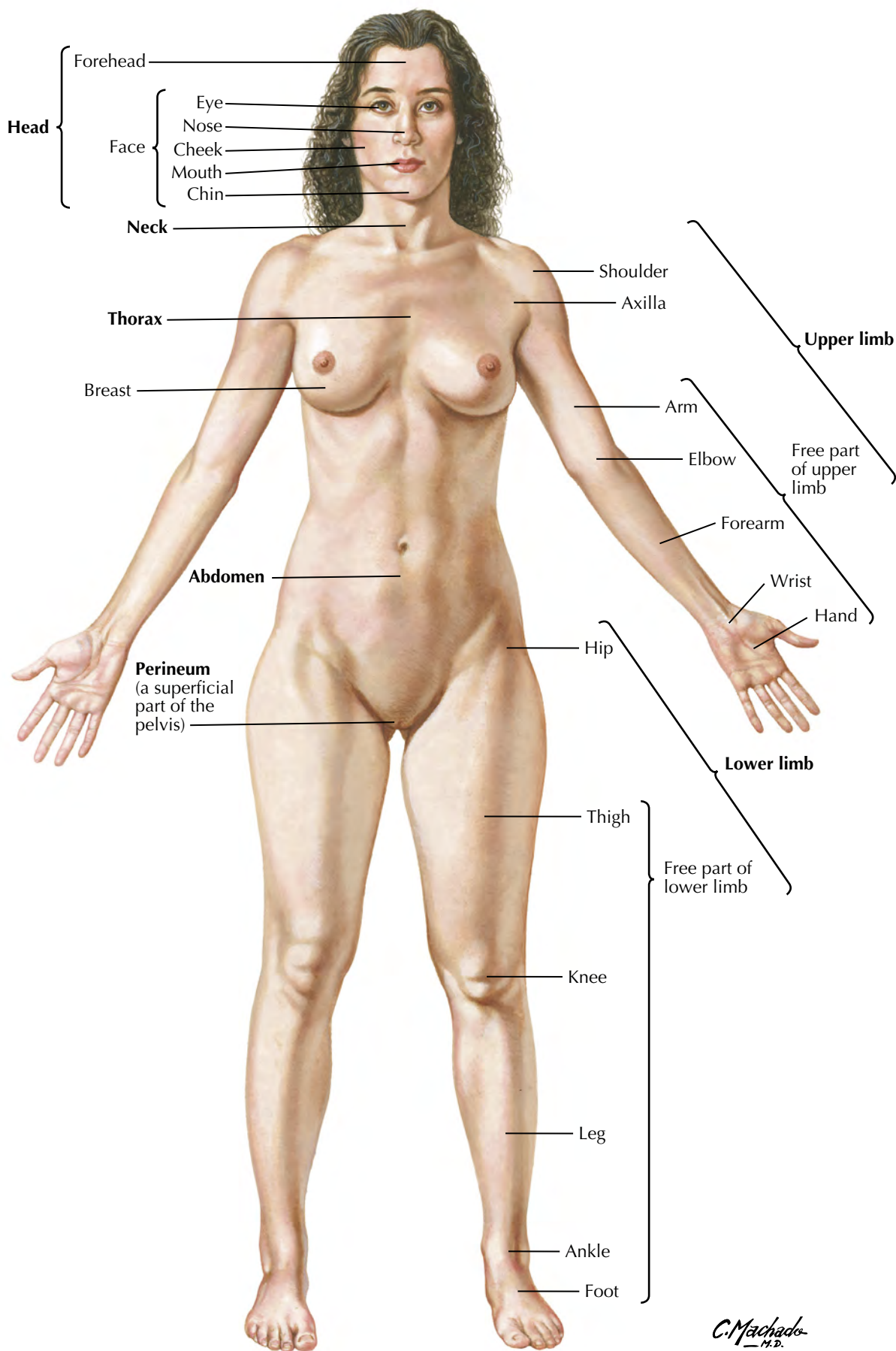
BP 8 Cholinergic and Adrenergic Synapses: Schema

ELECTRONIC BONUS PLATES—*cont'd*

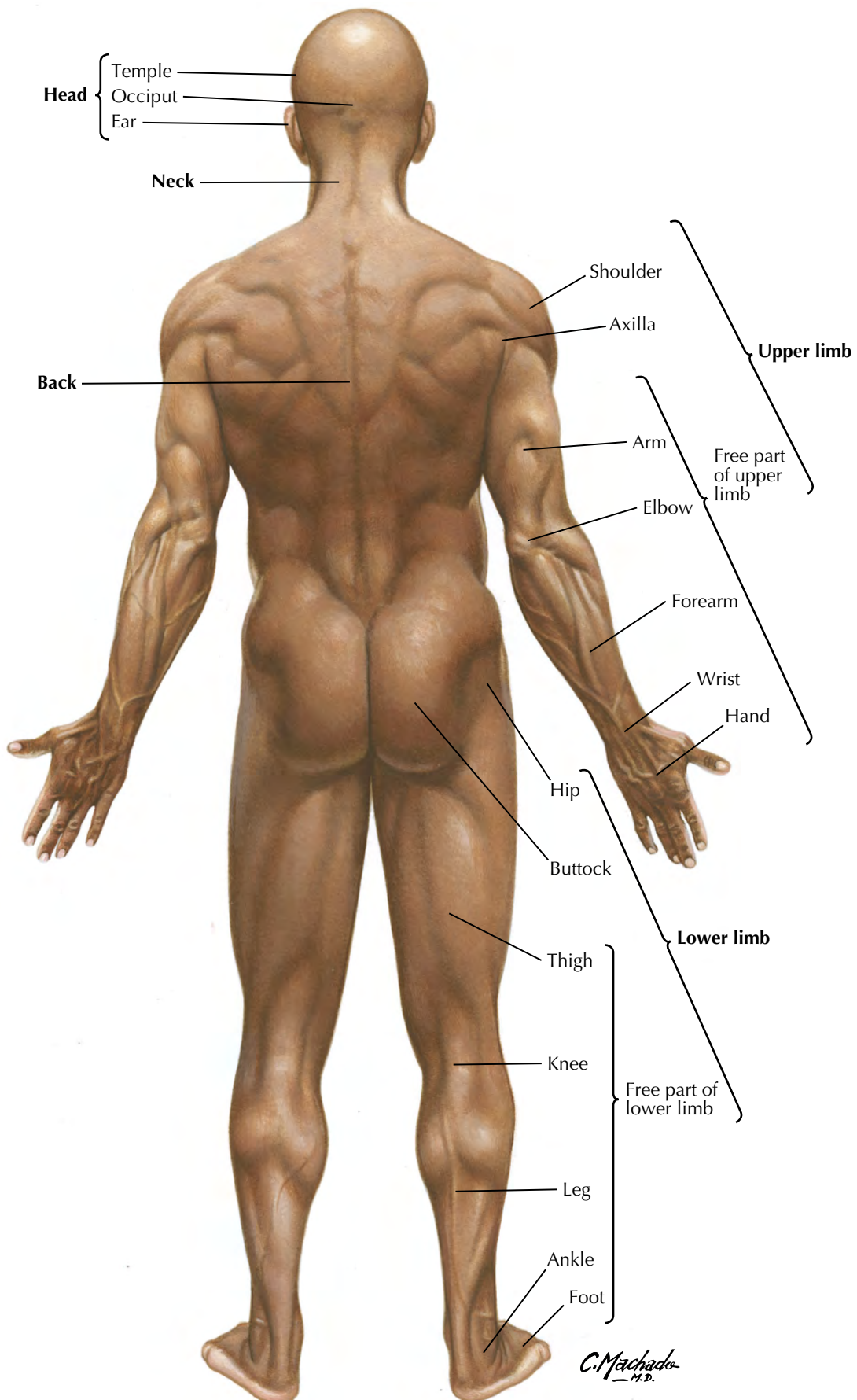
**BP 9** Architecture of Bone**BP 10** Muscle Structure**BP 11** Joints: Connective Tissues and Articular Cartilage**BP 12** Cardiovascular System: Composition of Blood**BP 13** Arterial Wall



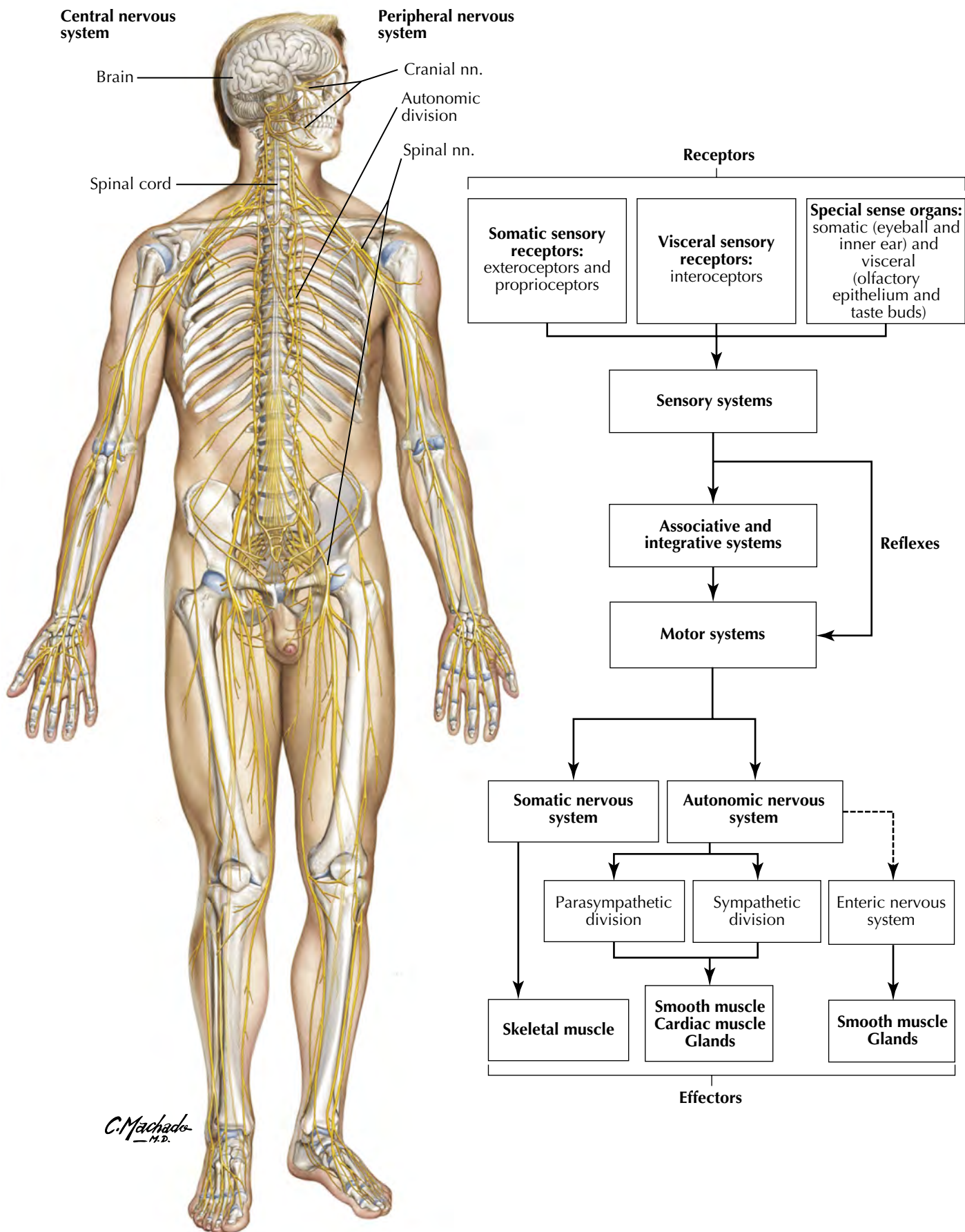
Body Parts: Anterior View of Female

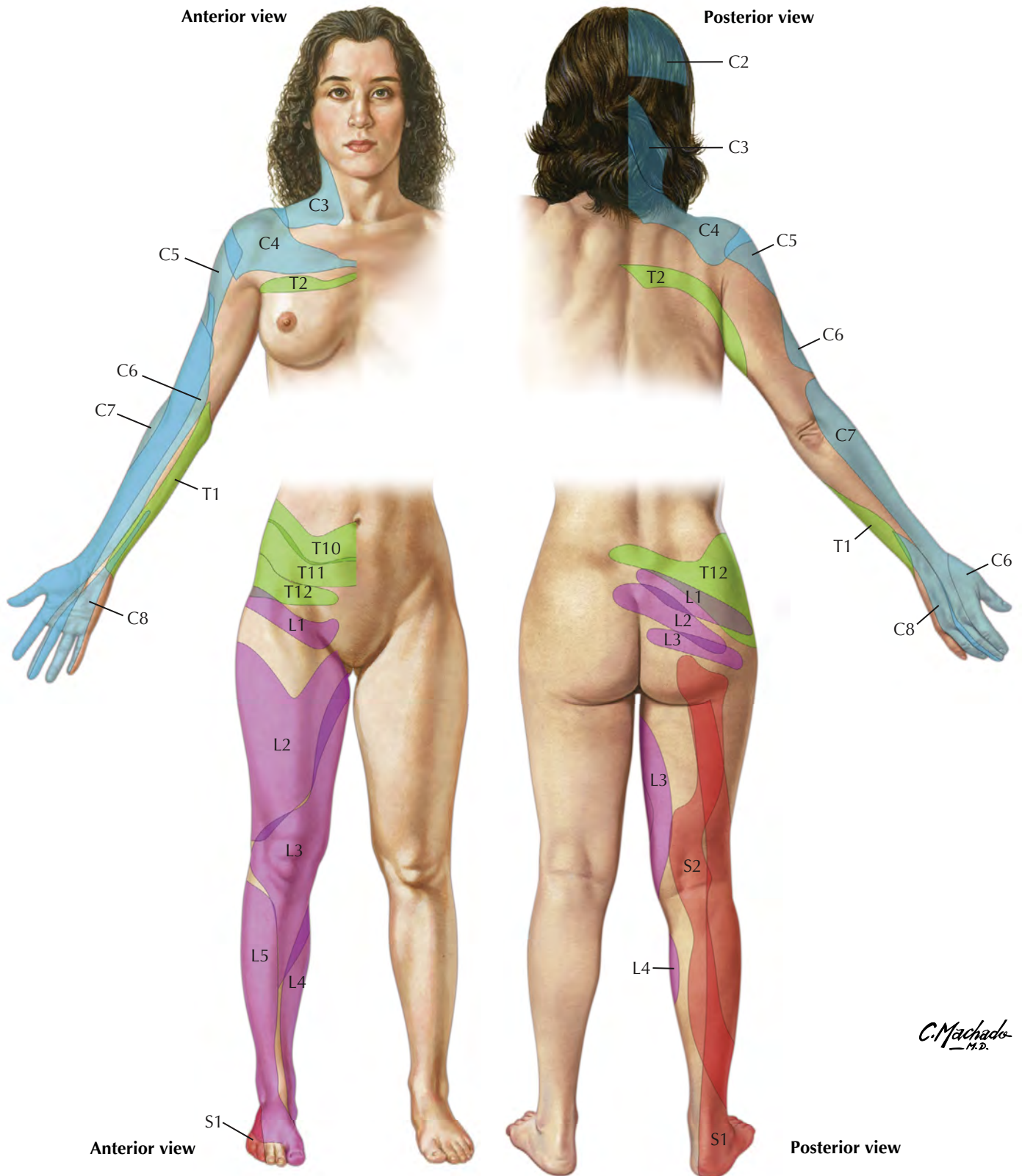


C. Machado
—M.D.—



Overview of Nervous System

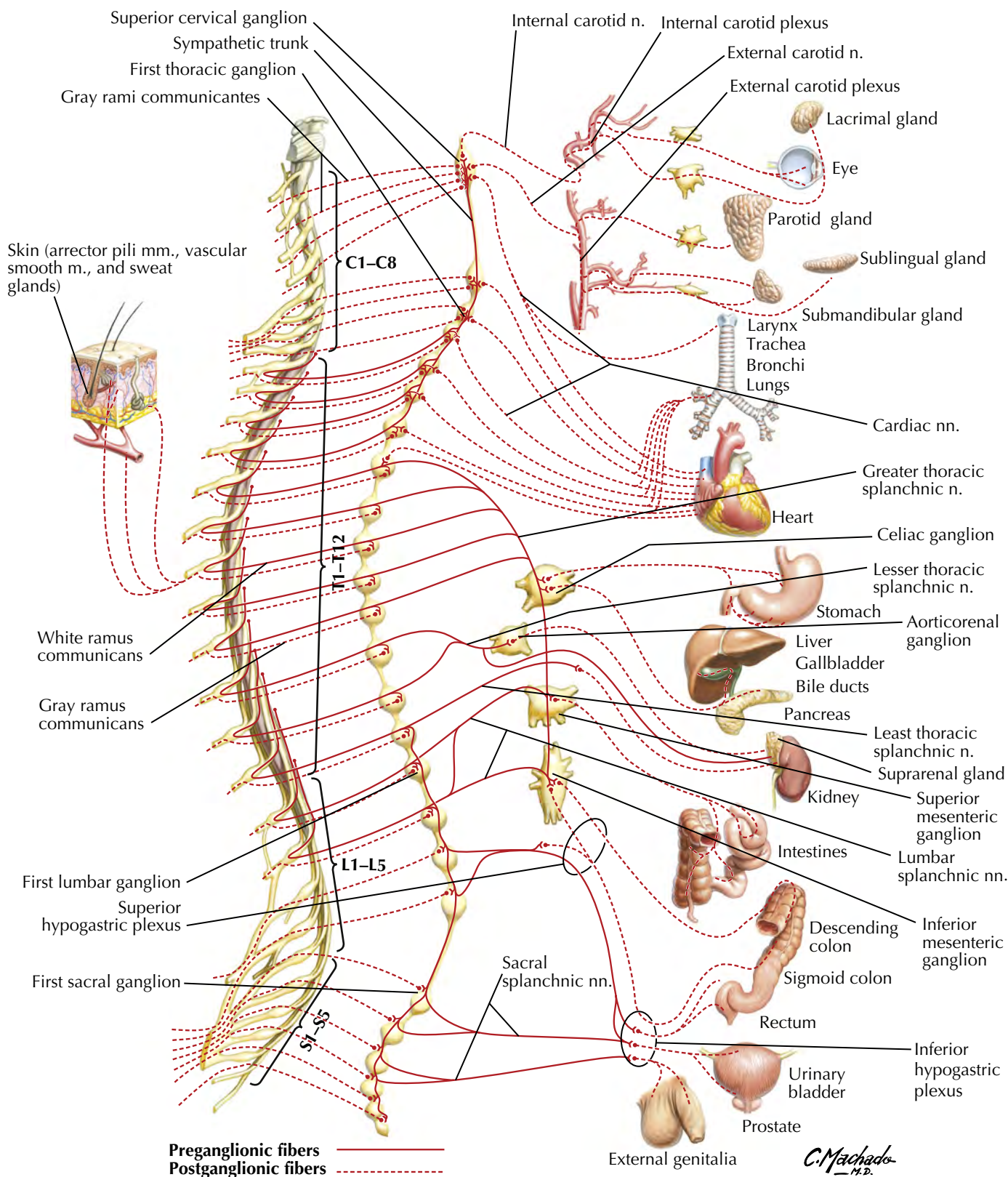


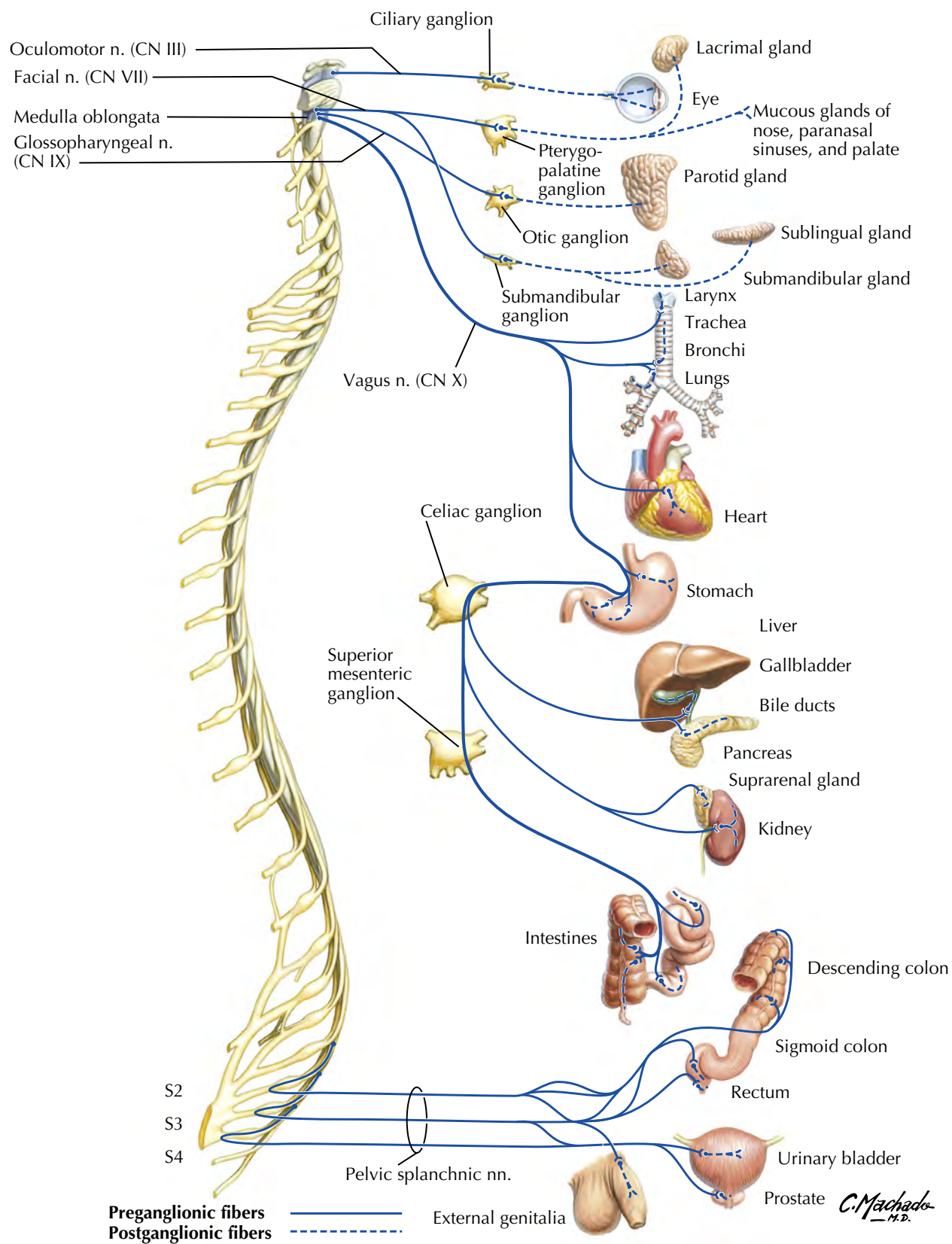


Schematic based on Lee MW, McPhee RW, Stringer MD. An evidence-based approach to human dermatomes. *Clin Anat.* 2008;21(5):363–373. doi: 10.1002/ca.20636. PMID: 18470936. Please note that these areas are not absolute and vary from person to person. S3, S4, S5, and Co supply the perineum but are not shown for reasons of clarity. Of note, the dermatomes are larger than illustrated as the figure is based on best evidence; gaps represent areas in which the data are inconclusive.

Sympathetic Nervous System: Schema

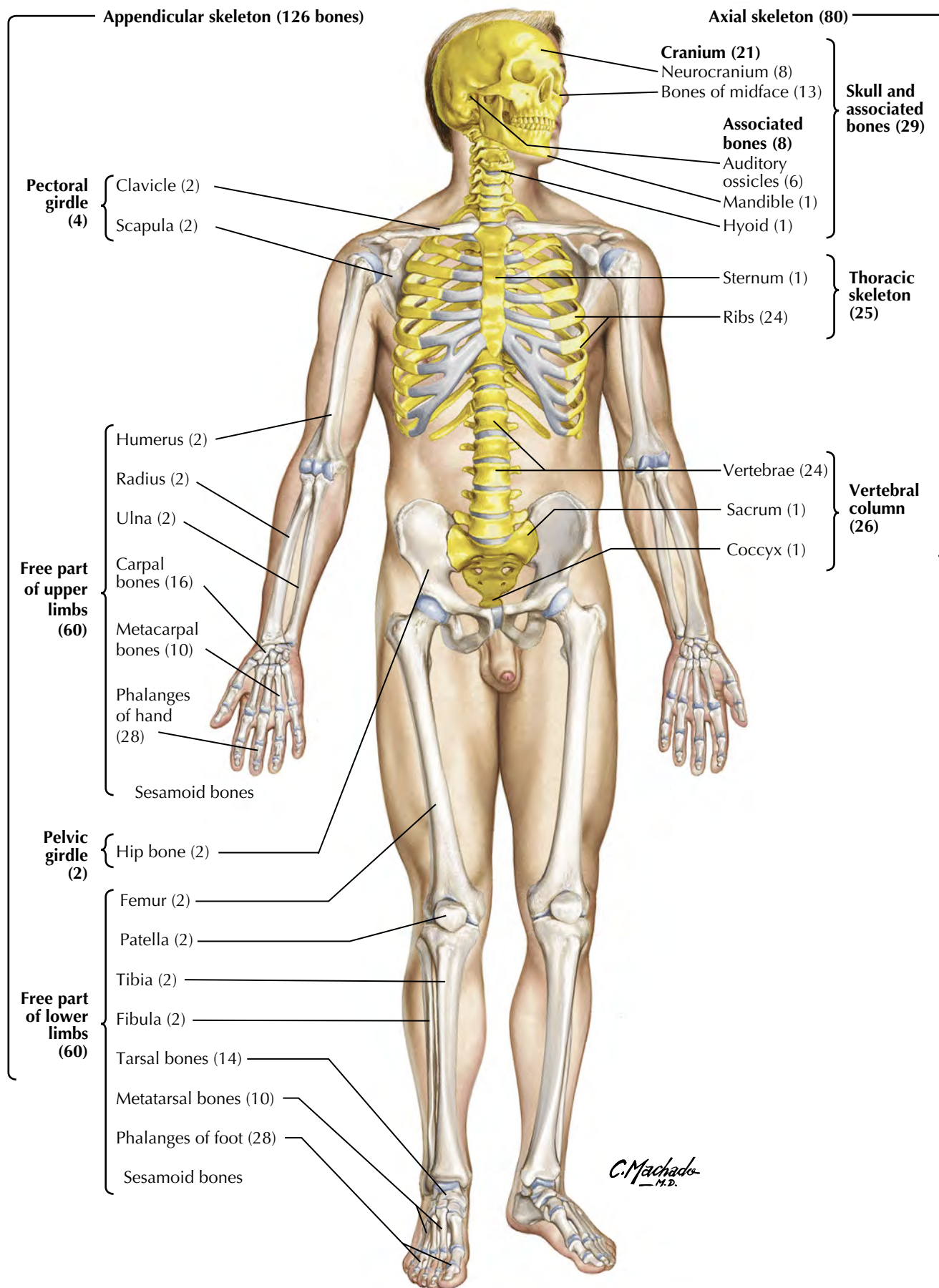
See also Plates 229, 249, 341, BP 6, BP 8, BP 52



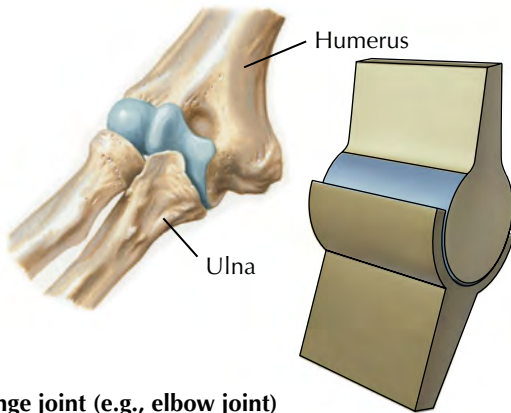
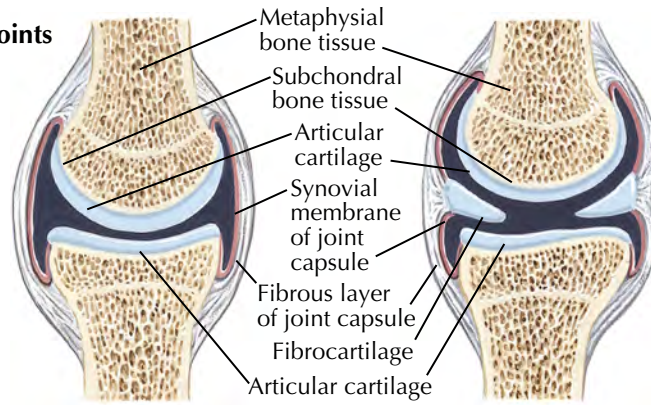


Overview of Skeletal System

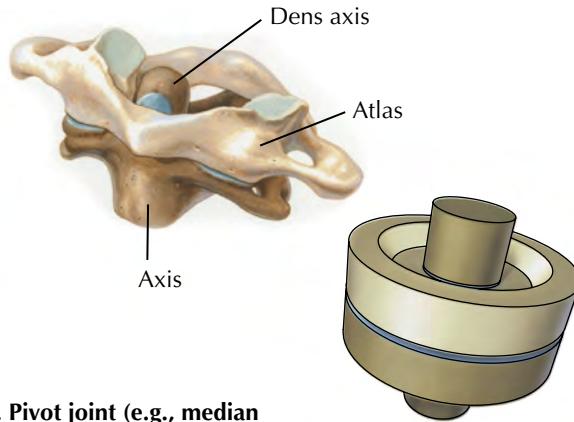
See also Plate BP 9



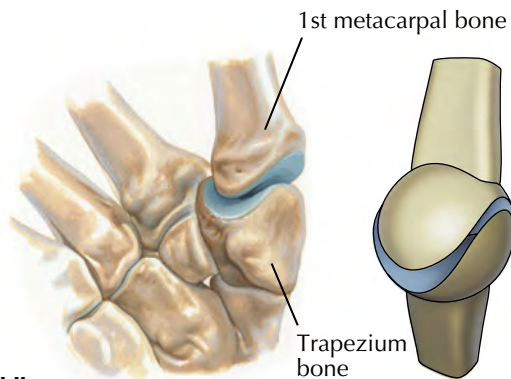
Structure of synovial joints



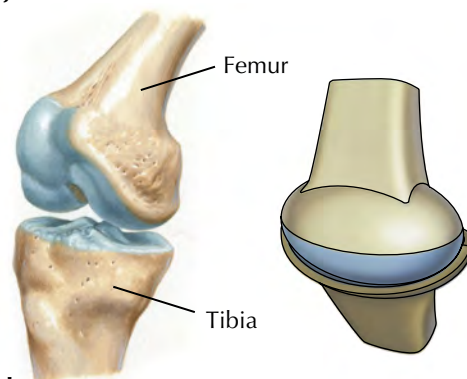
A. Hinge joint (e.g., elbow joint)



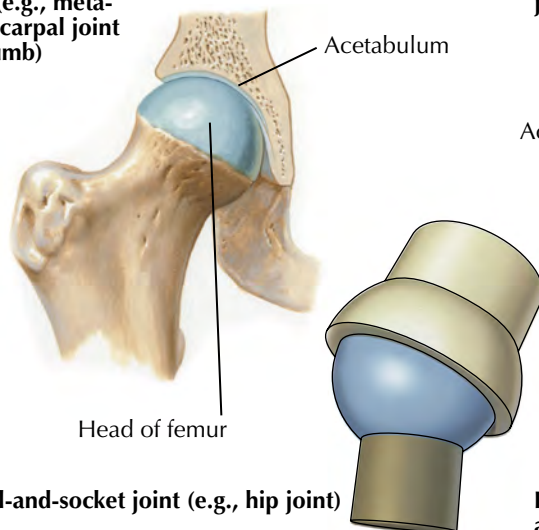
B. Pivot joint (e.g., median atlantoaxial joint)



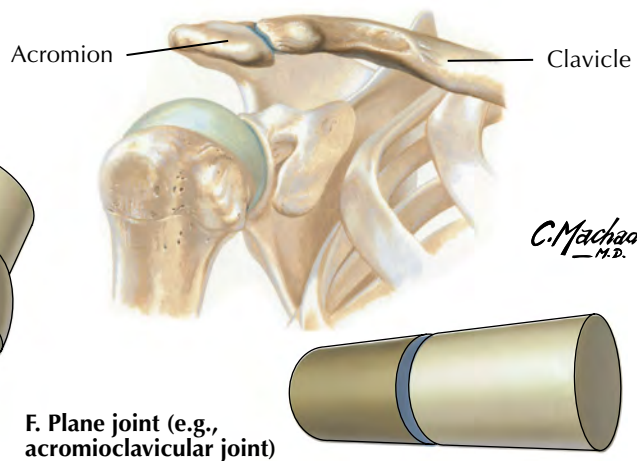
C. Saddle joint (e.g., metacarpophalangeal joint of thumb)



D. Condyloid joint (e.g., knee joint)

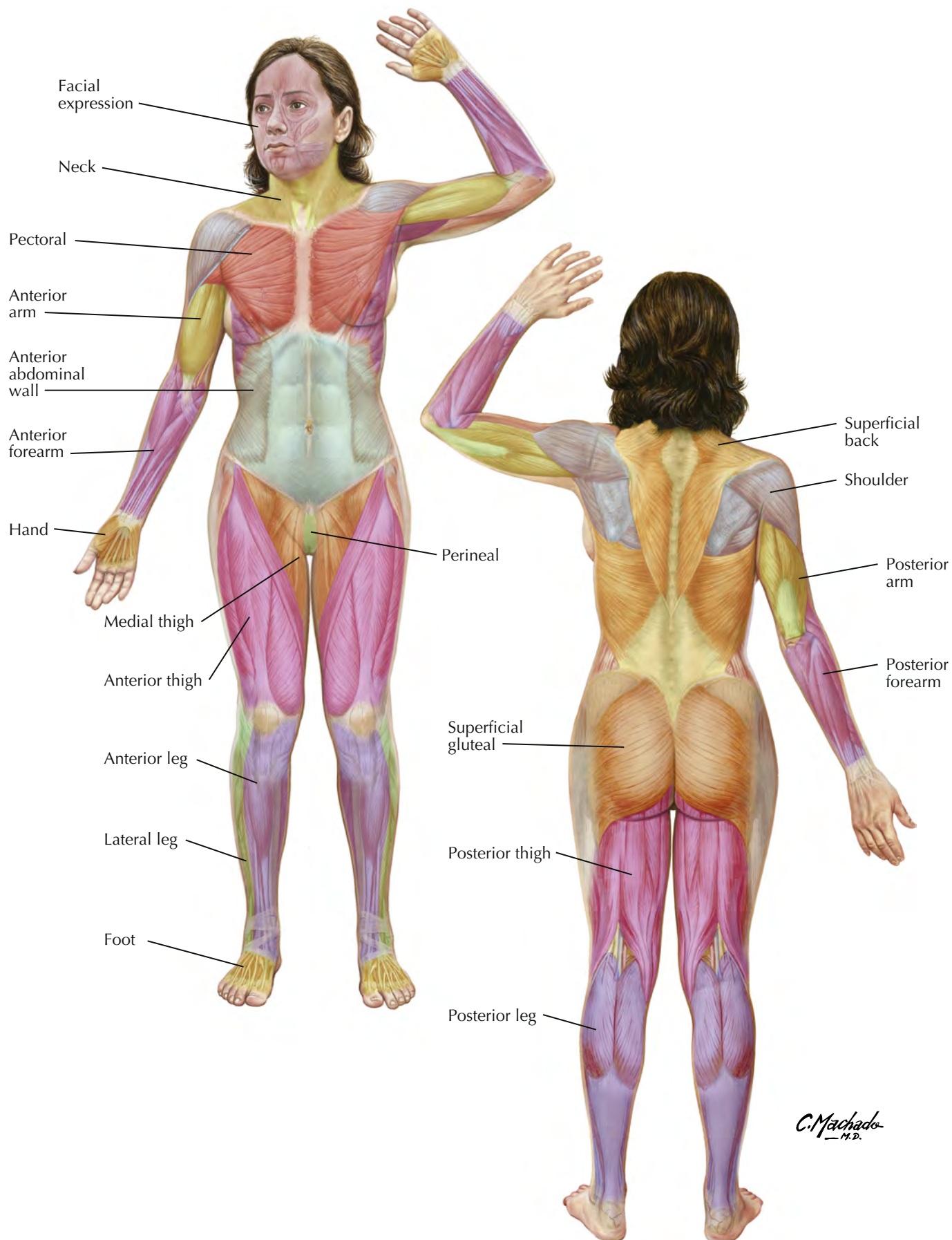


E. Ball-and-socket joint (e.g., hip joint)



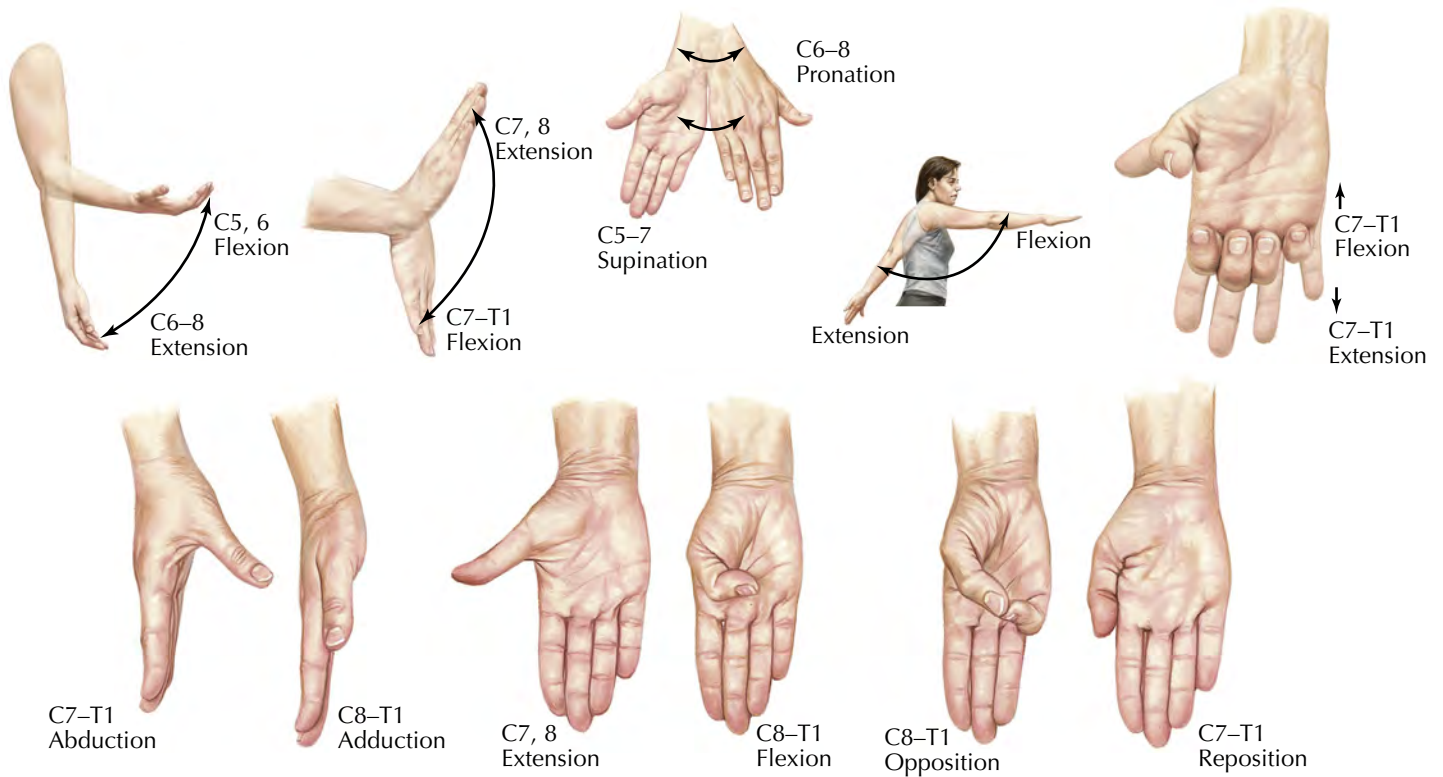
F. Plane joint (e.g., acromioclavicular joint)

C. Machado
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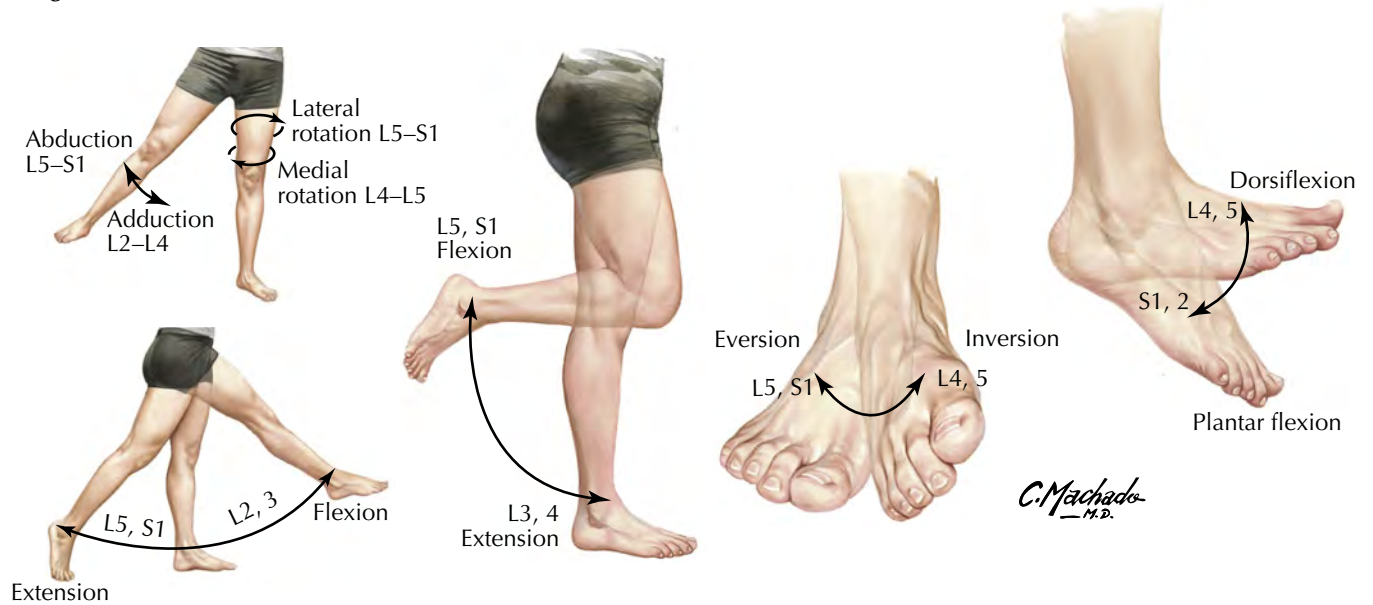


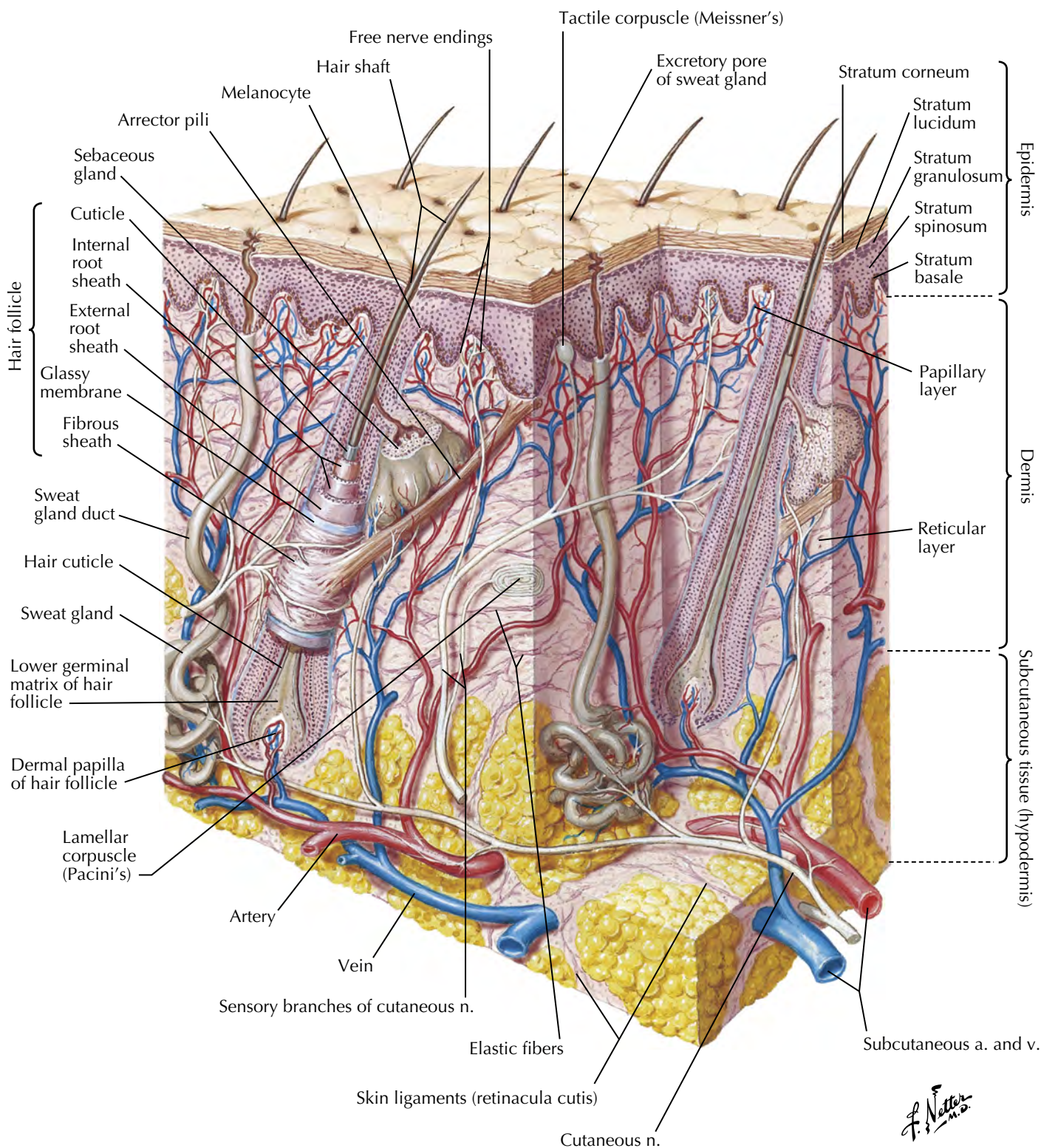
C. Machado
—M.D.

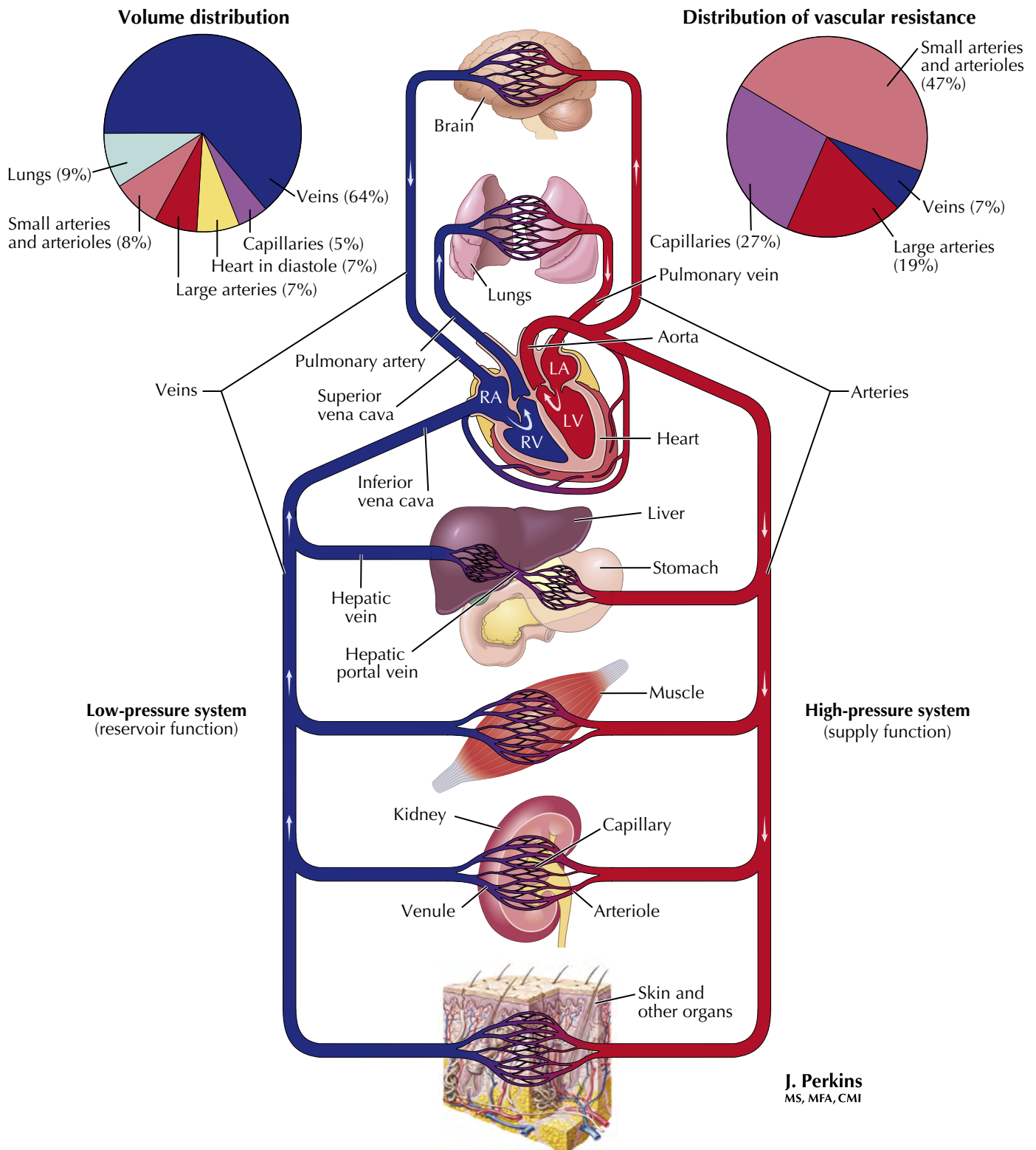
Segmental innervation of upper limb movements

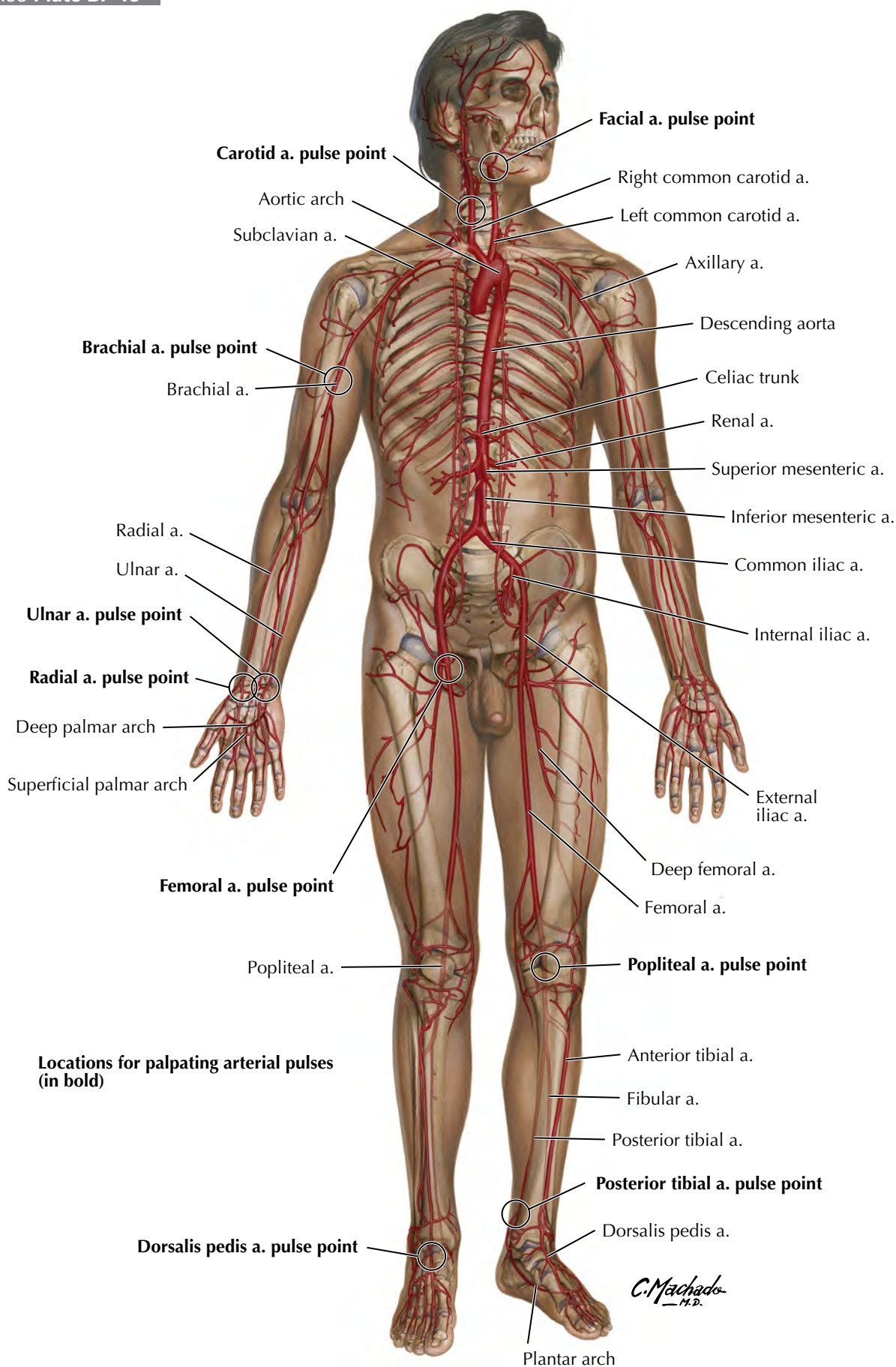


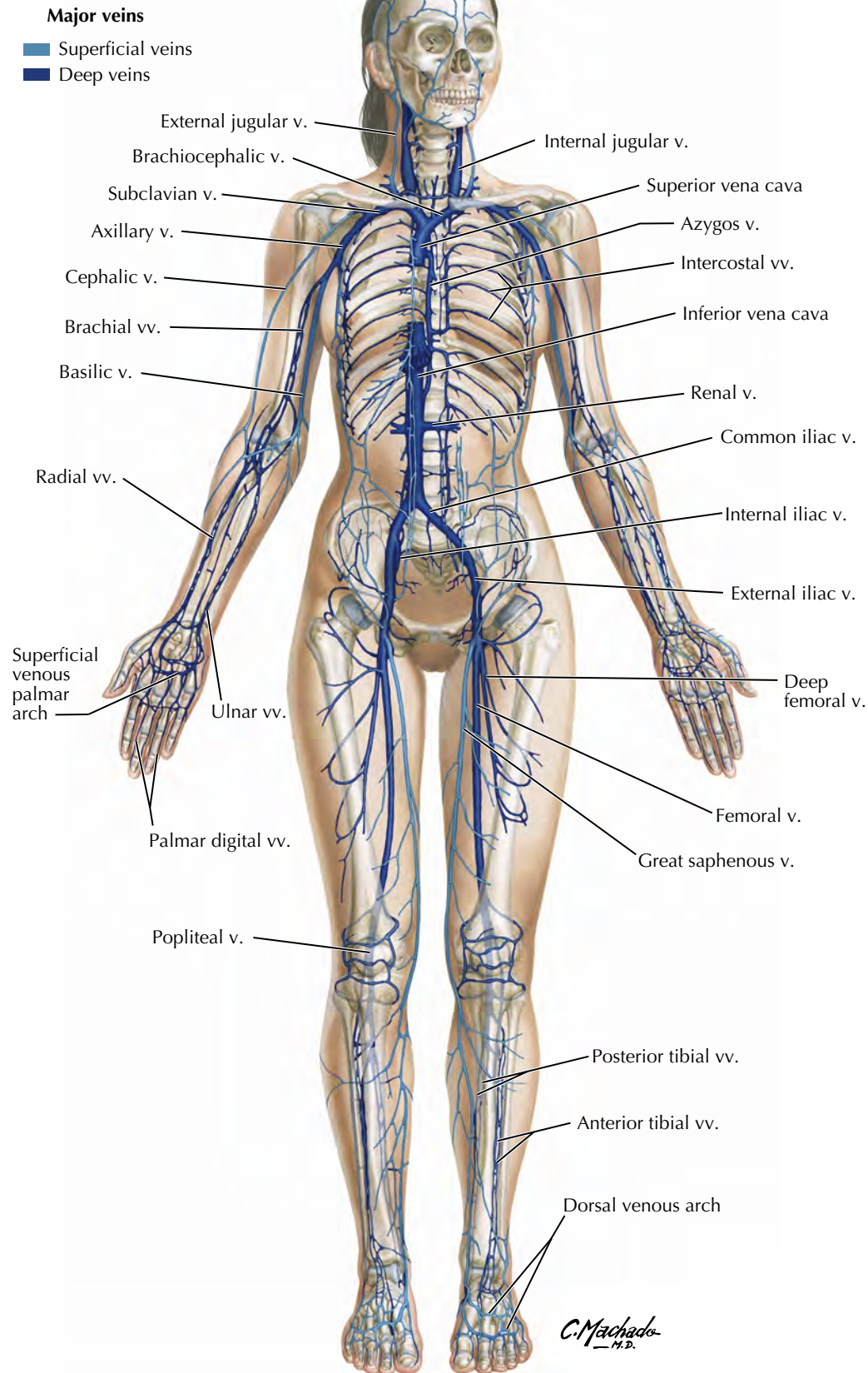
Segmental innervation of lower limb movements

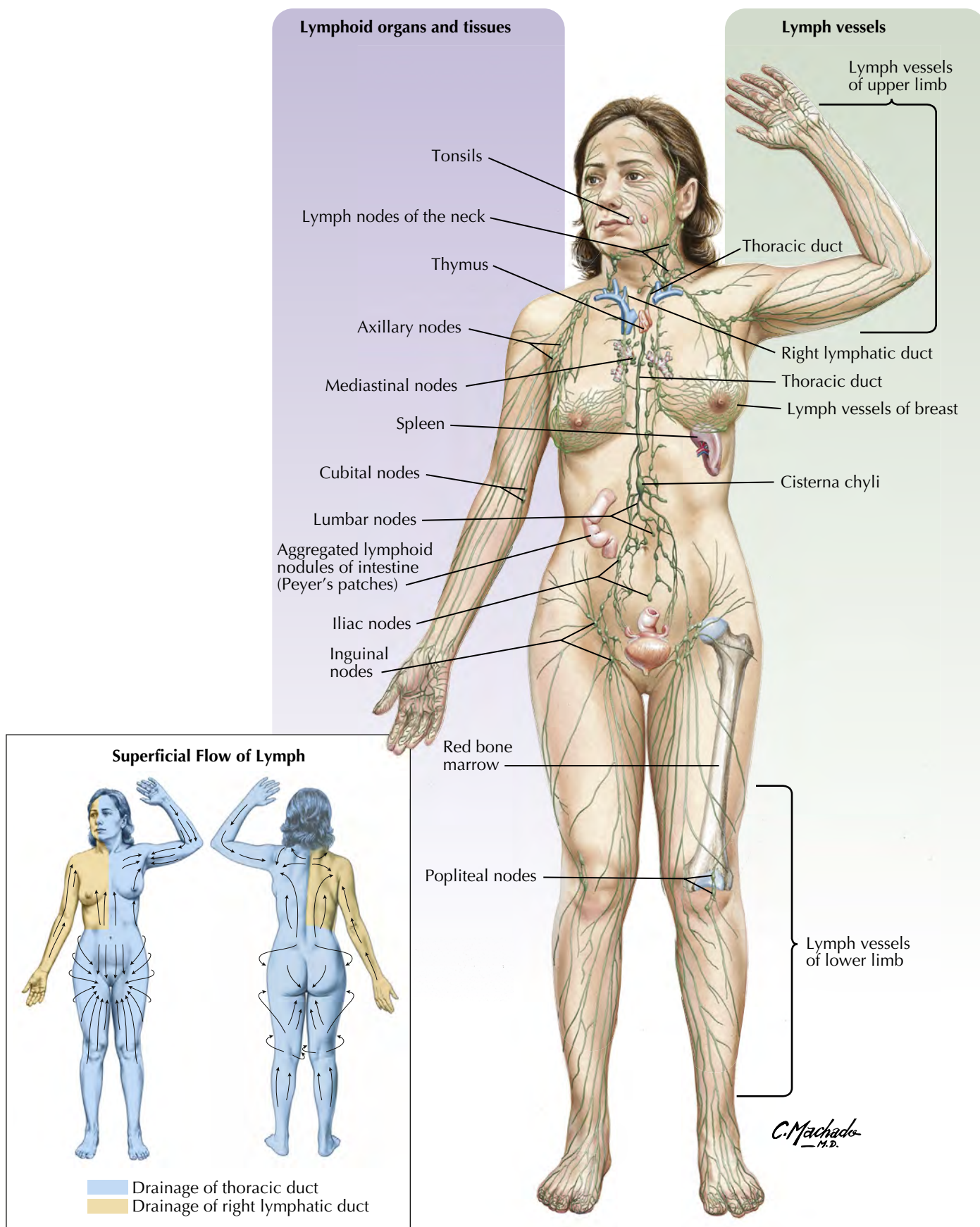


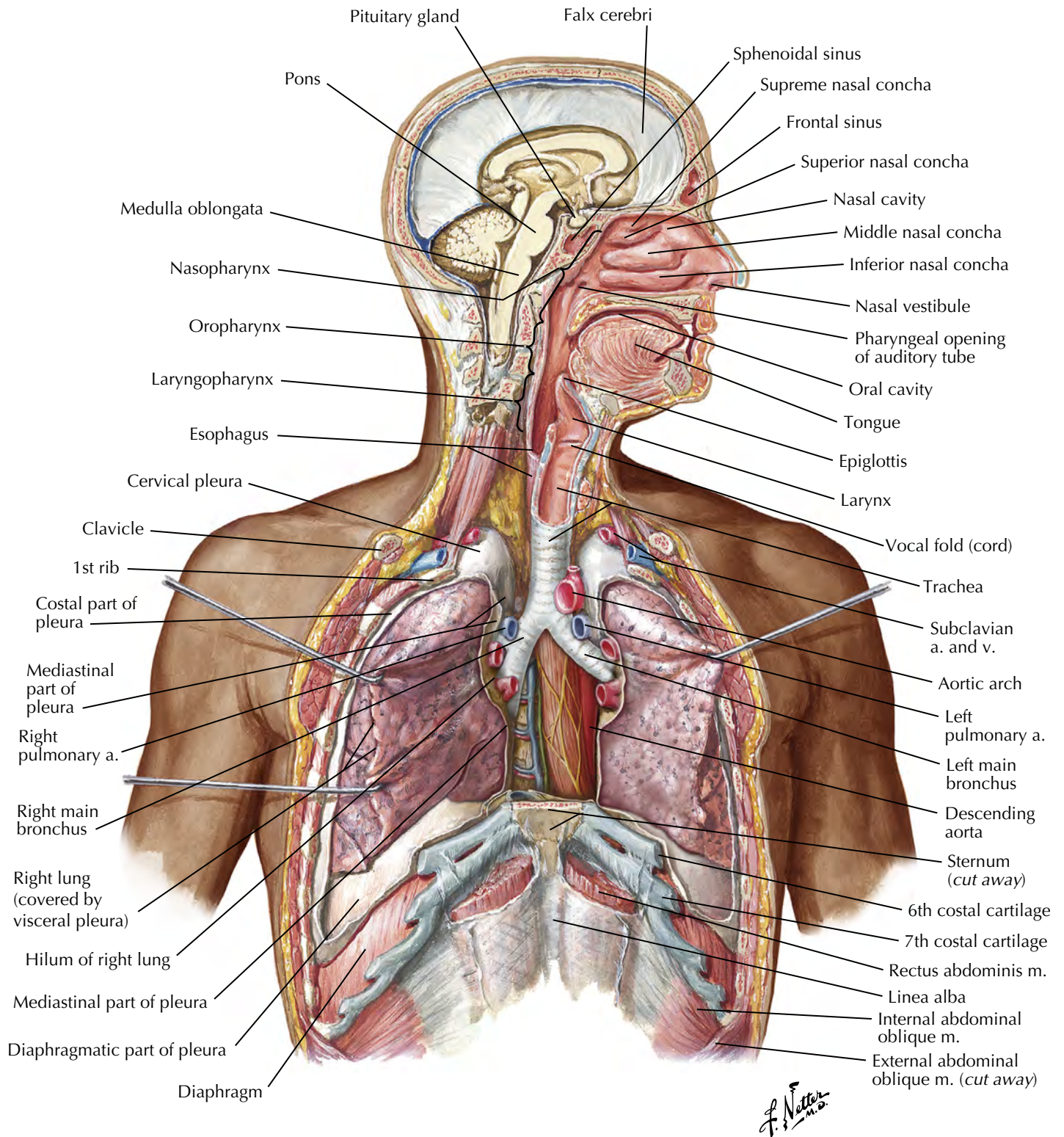




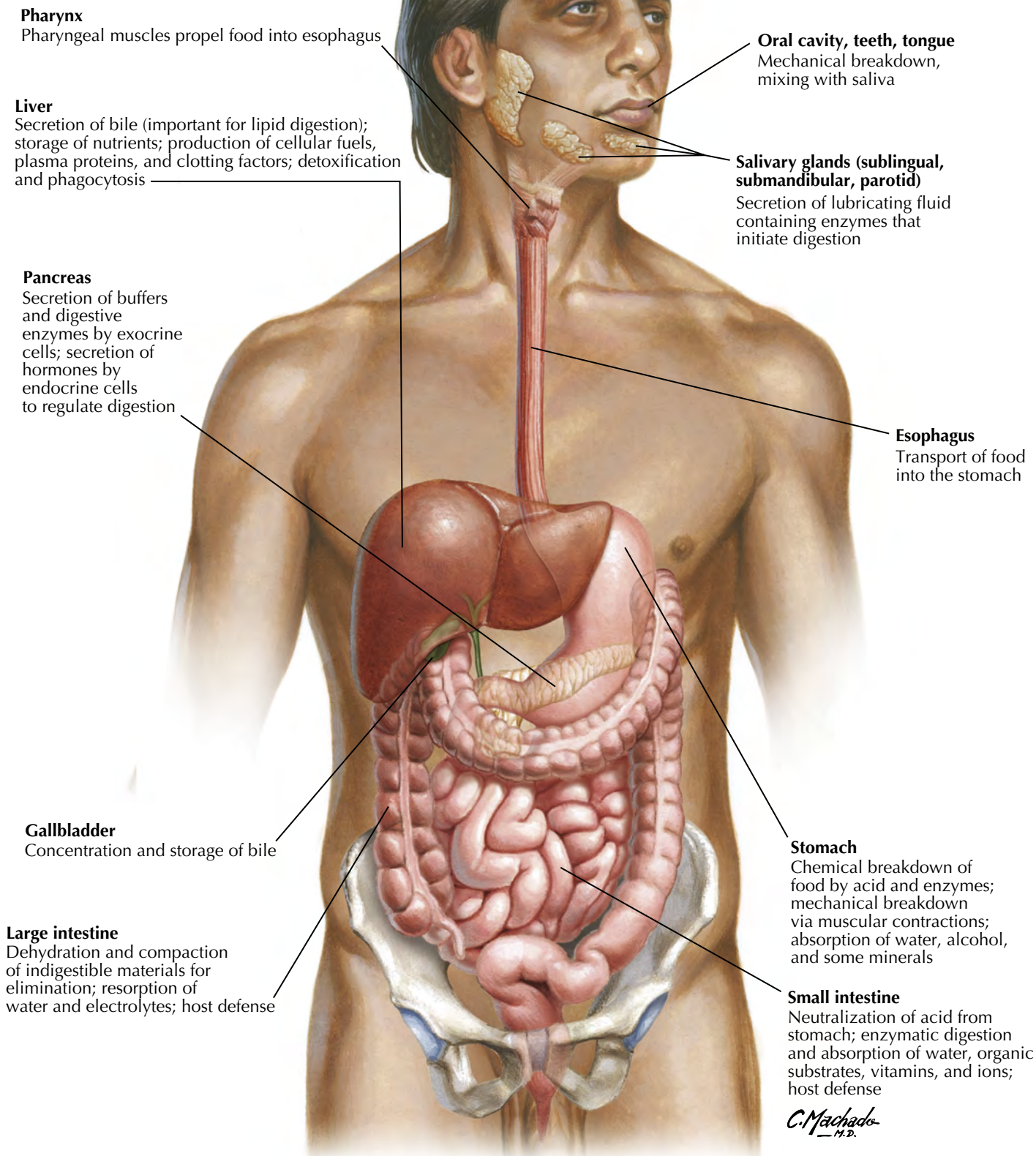


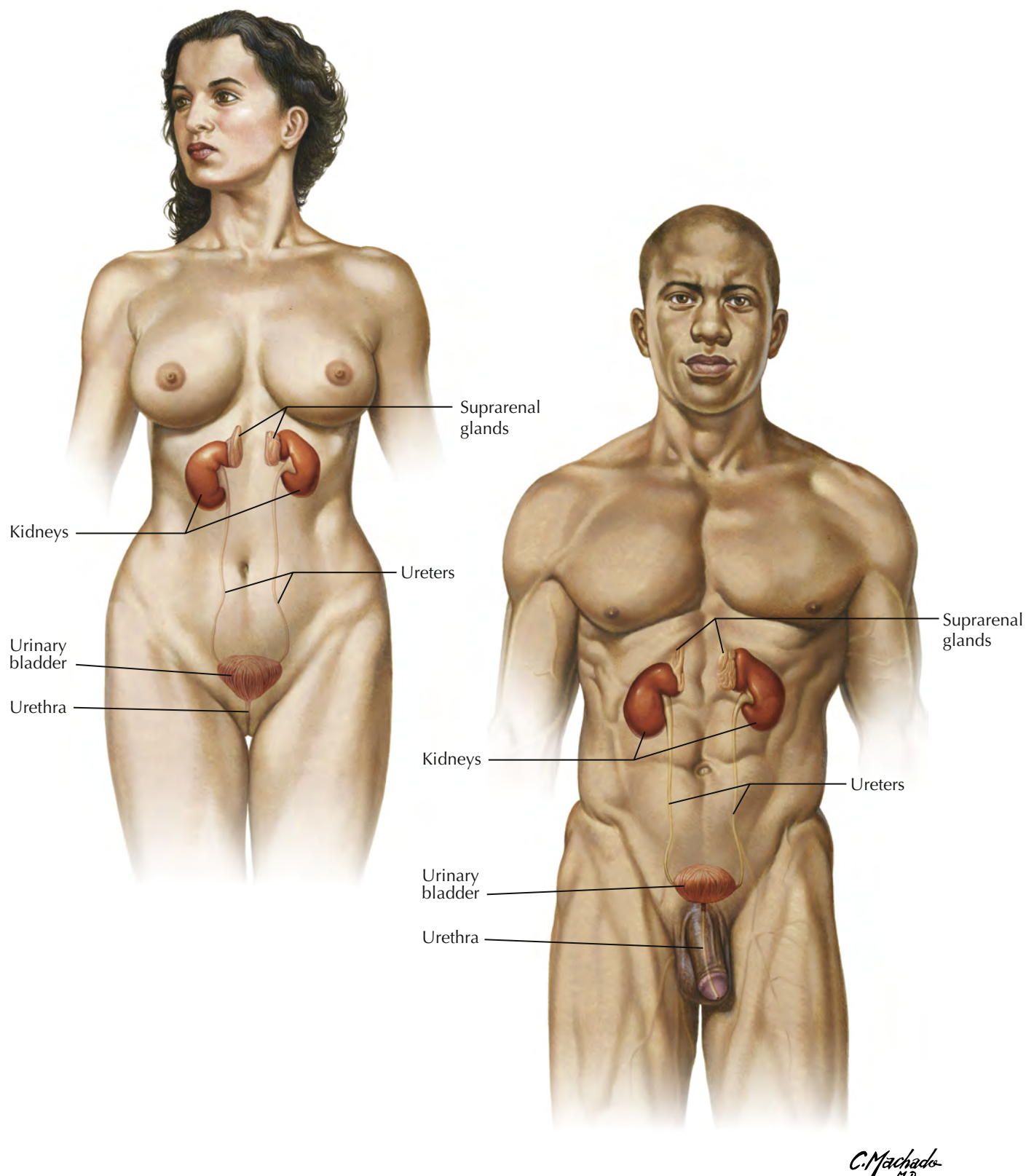


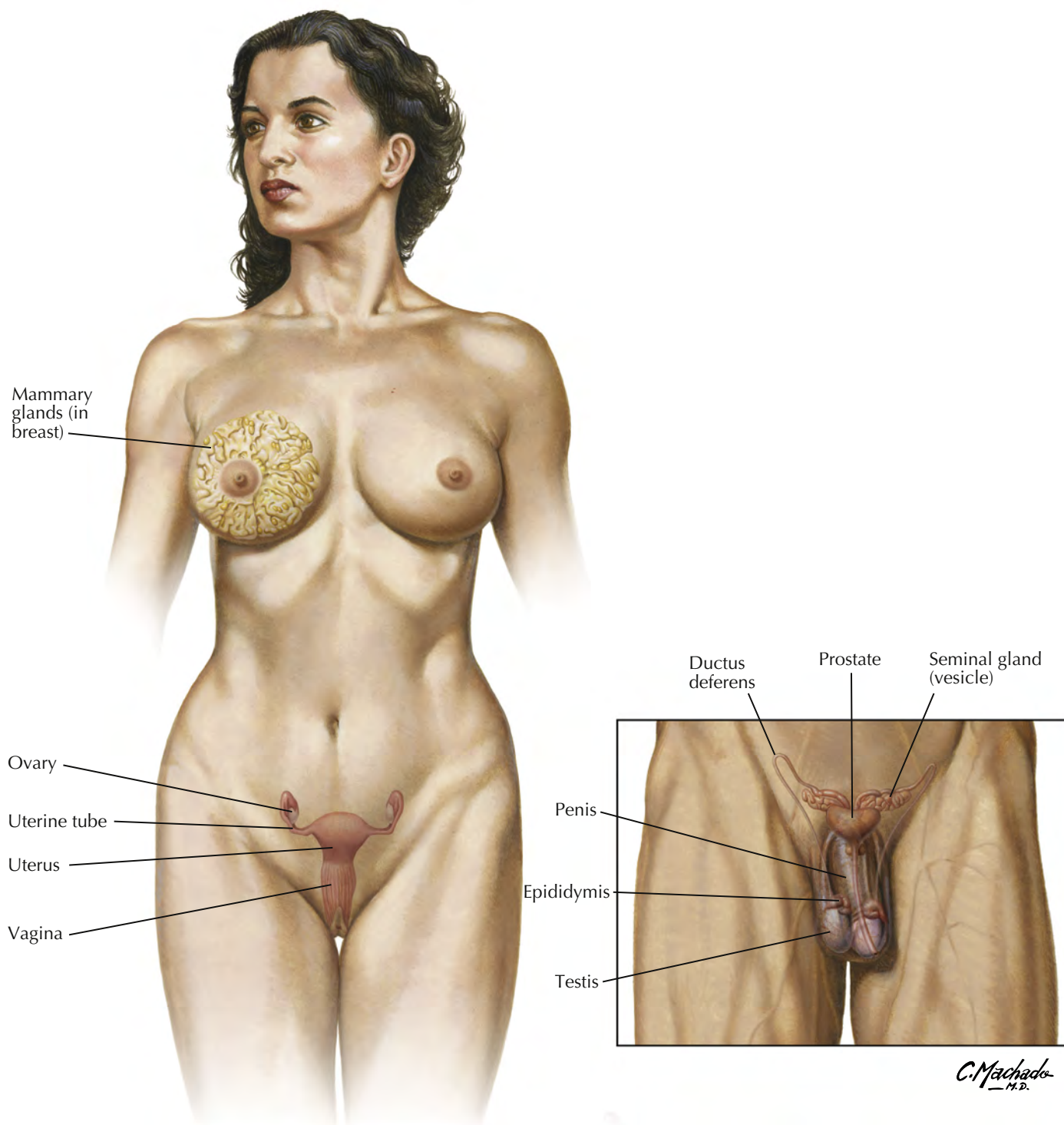


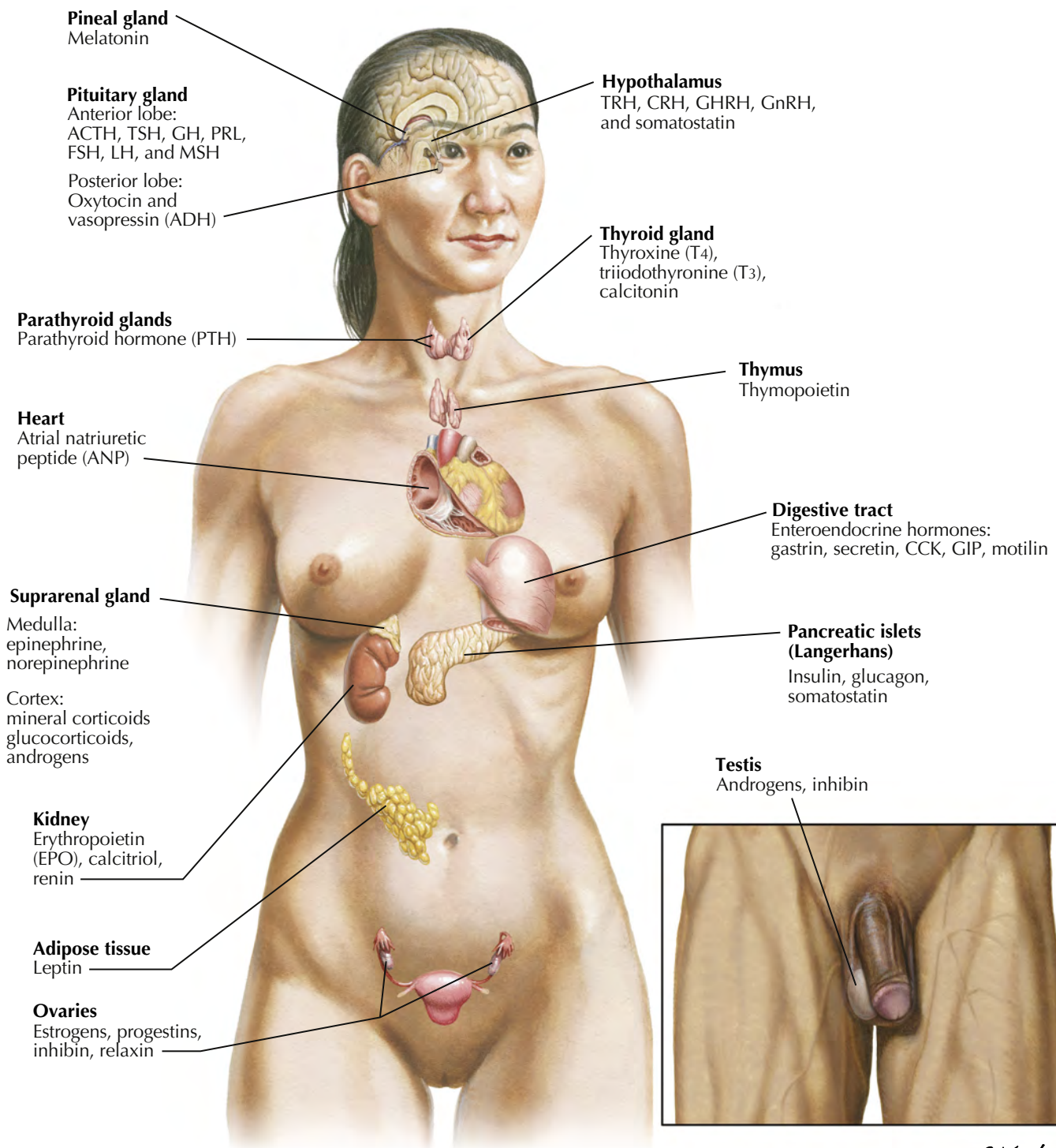


Overview of Digestive System

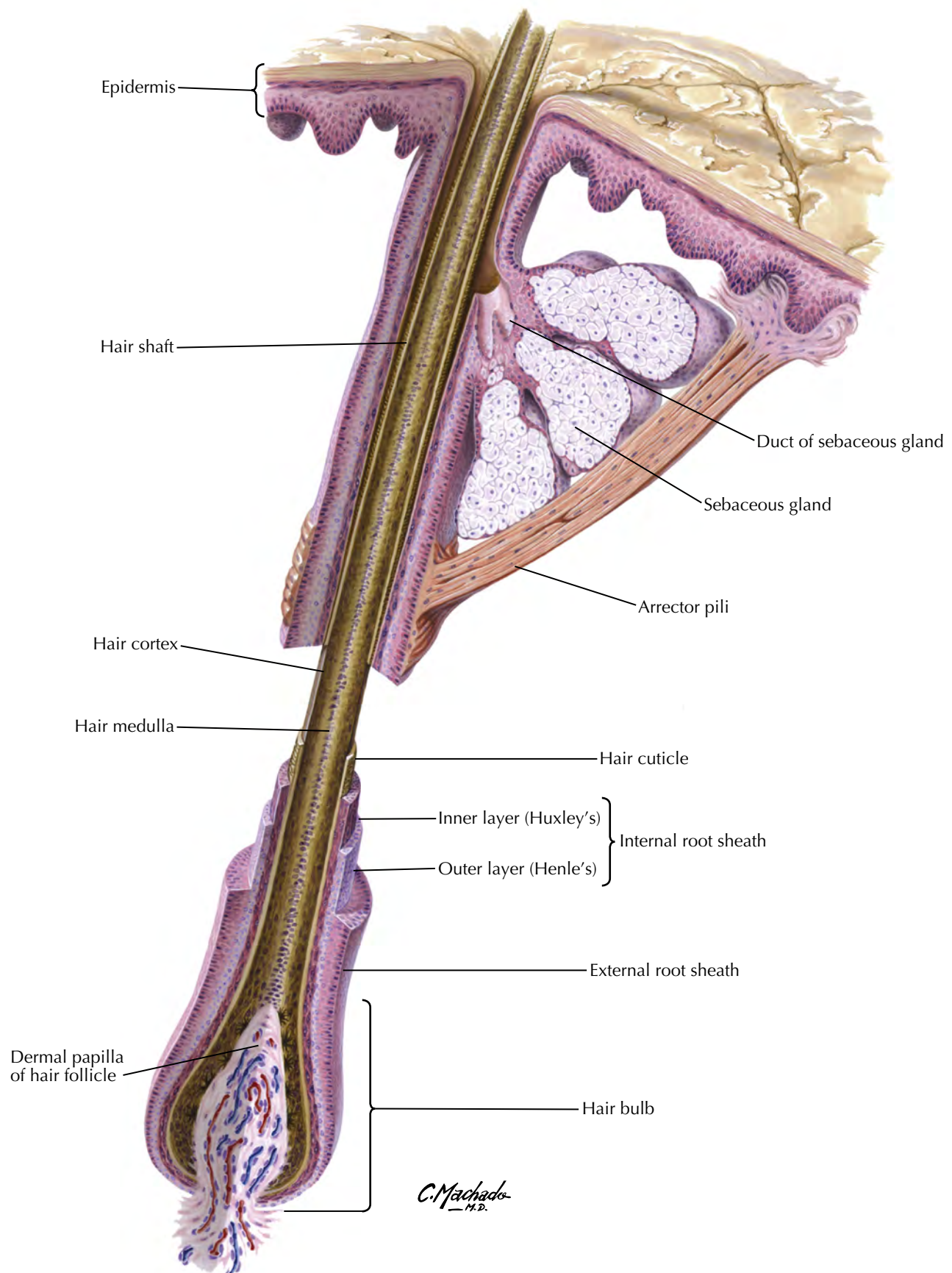


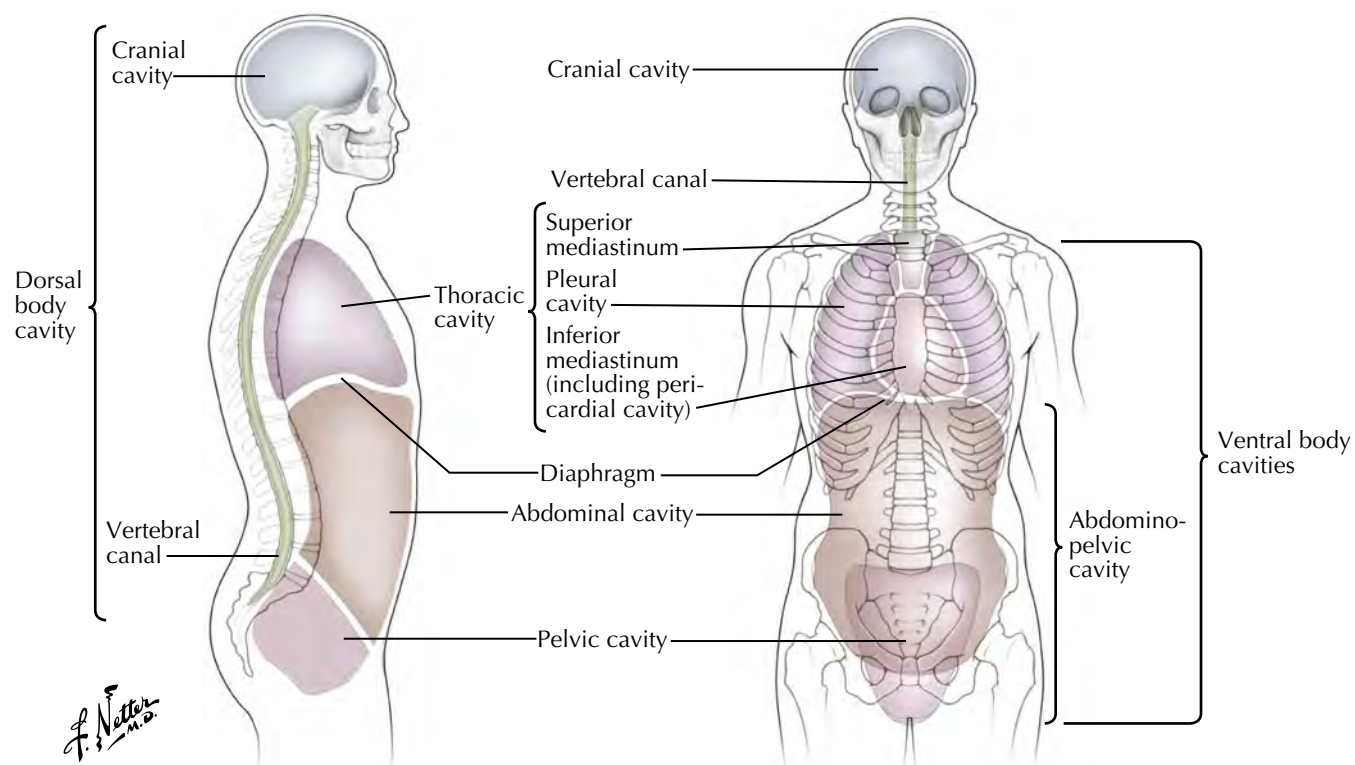






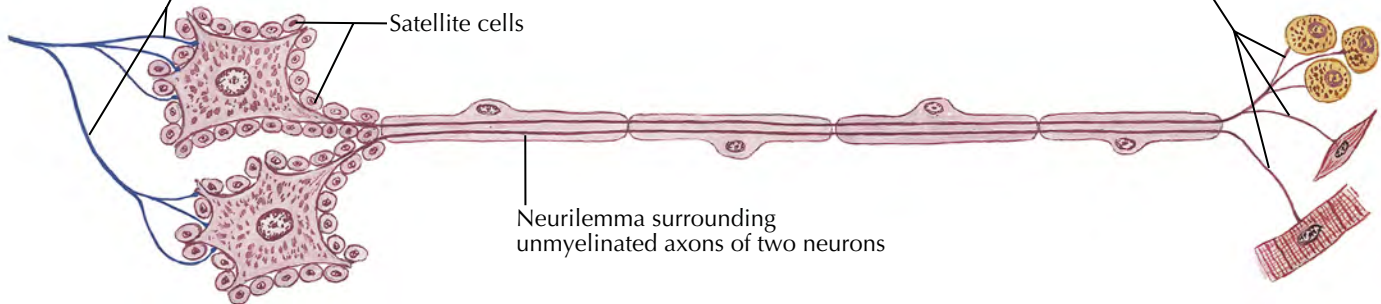
C. Machado
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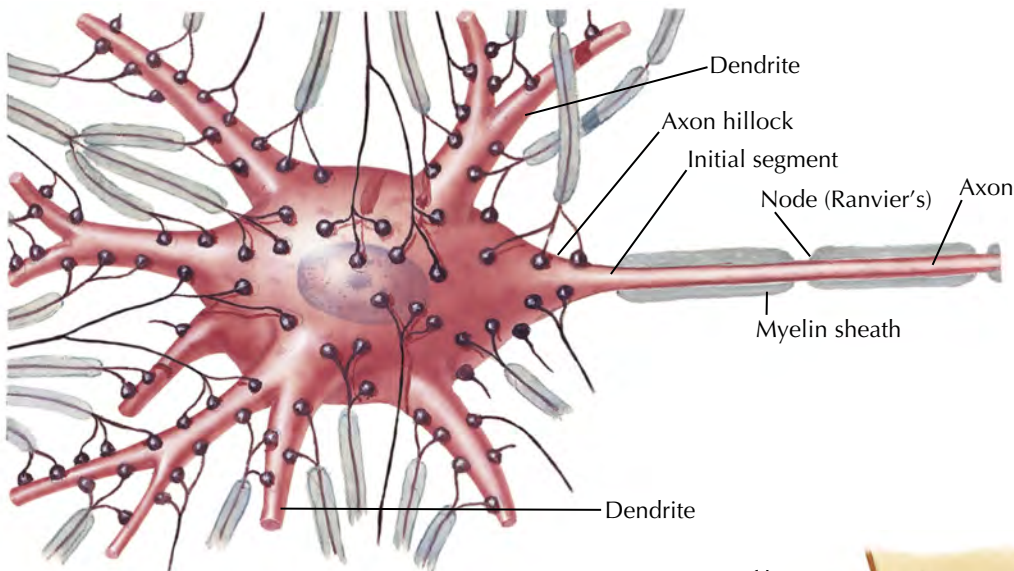


Endings of autonomic preganglionic fiber synapsing onto cell bodies of ganglionic neurons

Autonomic nerve endings on glandular epithelium, or smooth or cardiac muscle cells

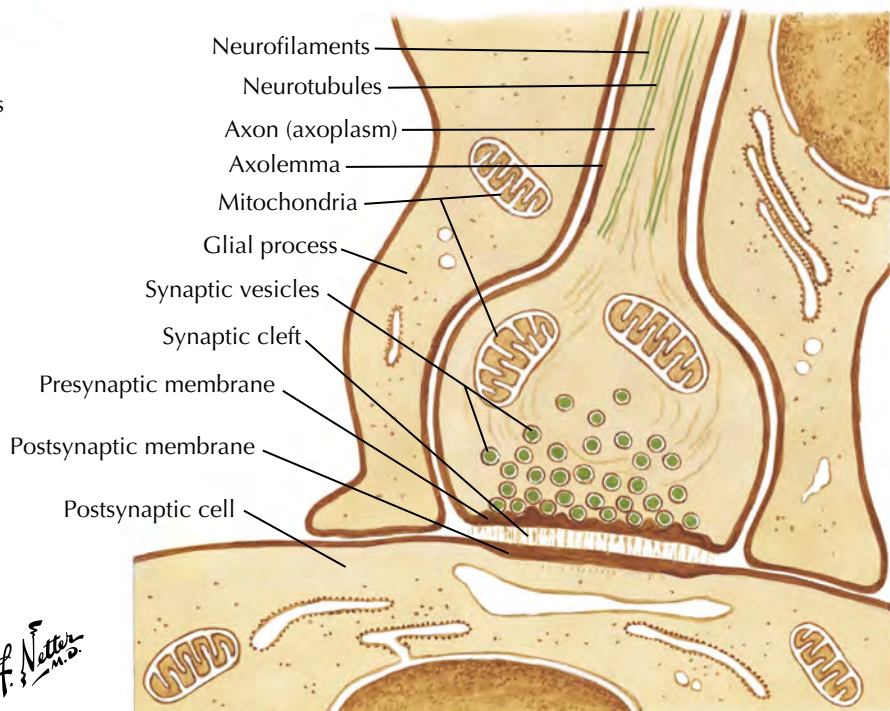


Two postganglionic autonomic neurons of a sympathetic or parasympathetic ganglion



Schematic of synapses on neuron

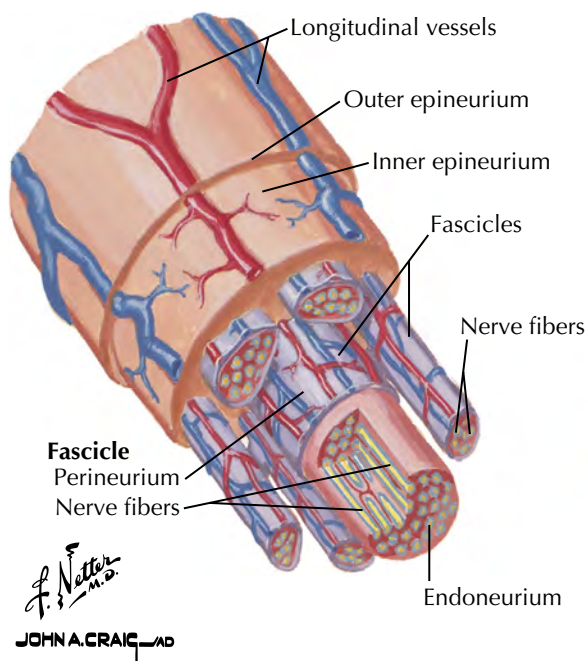
Numerous terminal boutons forming synapses on body and dendrites of a motor neuron



Enlarged section of bouton

Features of a Typical Peripheral Nerve

See also Plate BP 3



Cell body of an oligodendrocyte (Schwann cells play similar role in peripheral nervous system)

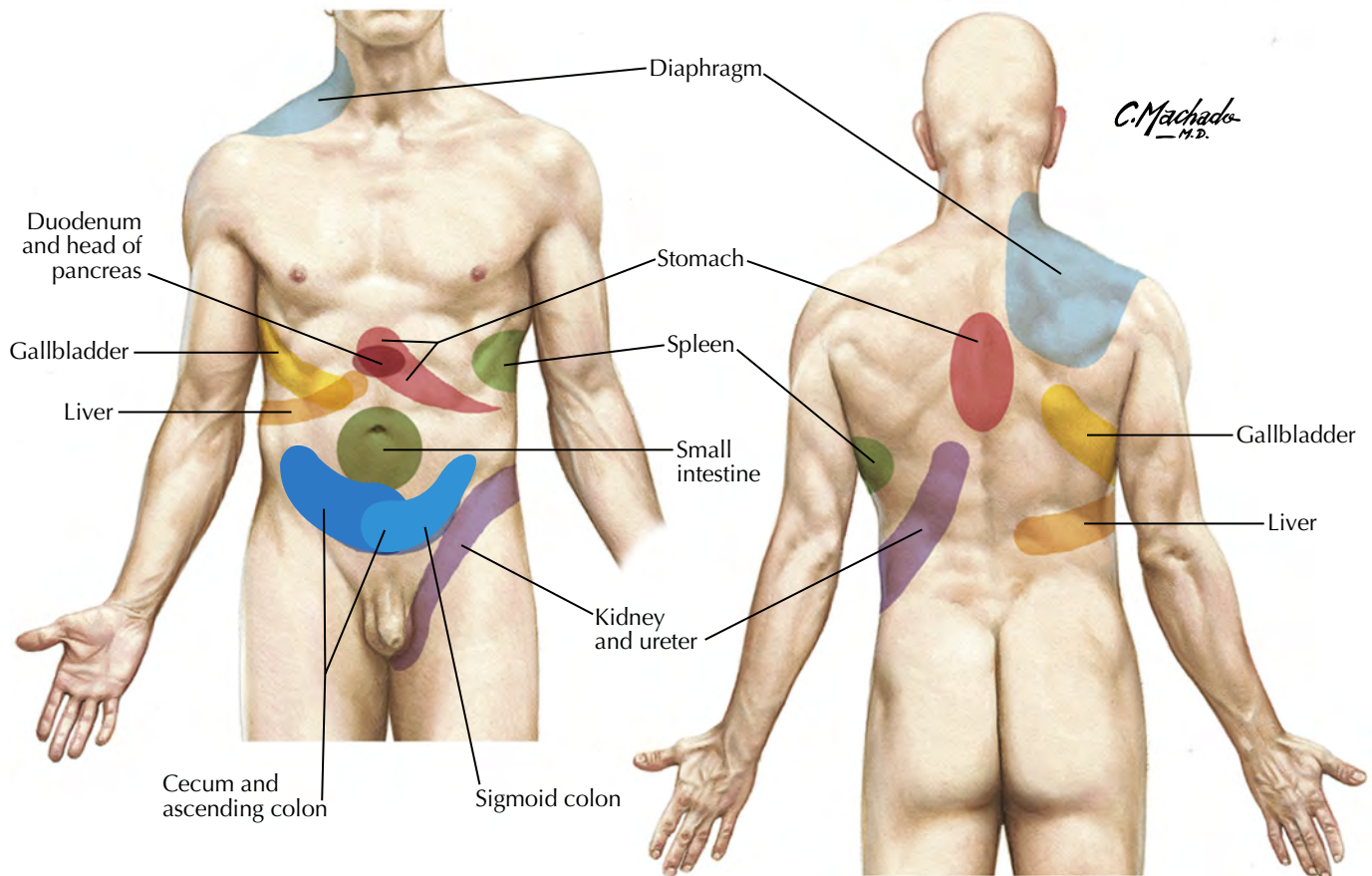
Cell membrane of myelinated axon (axolemma)

Mitochondrion in cytoplasm of axon

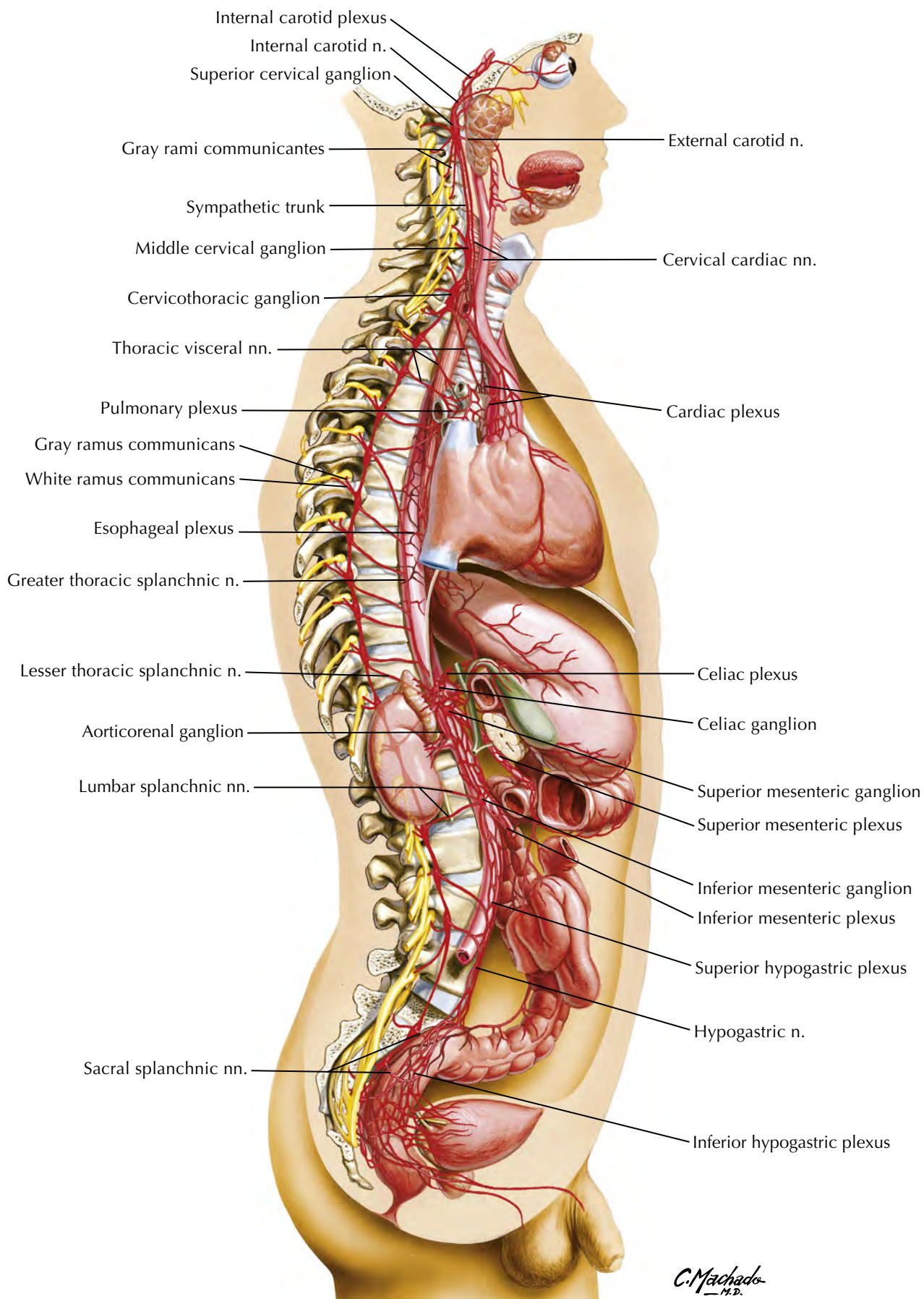
Node (Ranvier's)

Minute masses of cytoplasm trapped between fused layers of cell membrane of oligodendrocyte

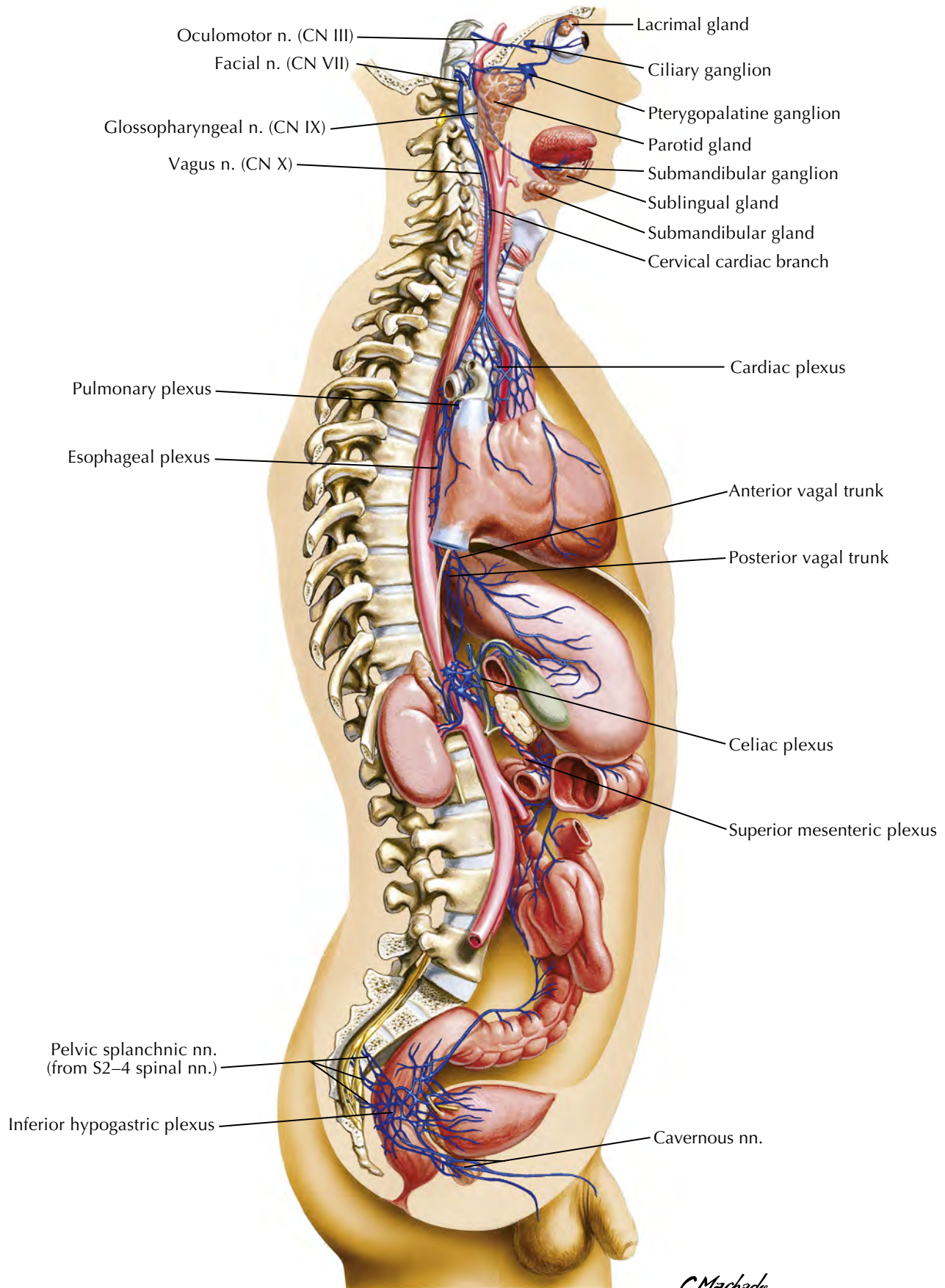
Fused layers of cell membrane of oligodendrocyte wrapped around axon of a myelinated neuron of central nervous system (the lipid of lipoprotein constituting fused cell membrane is myelin, which gives myelinated axon a white, glistening appearance)



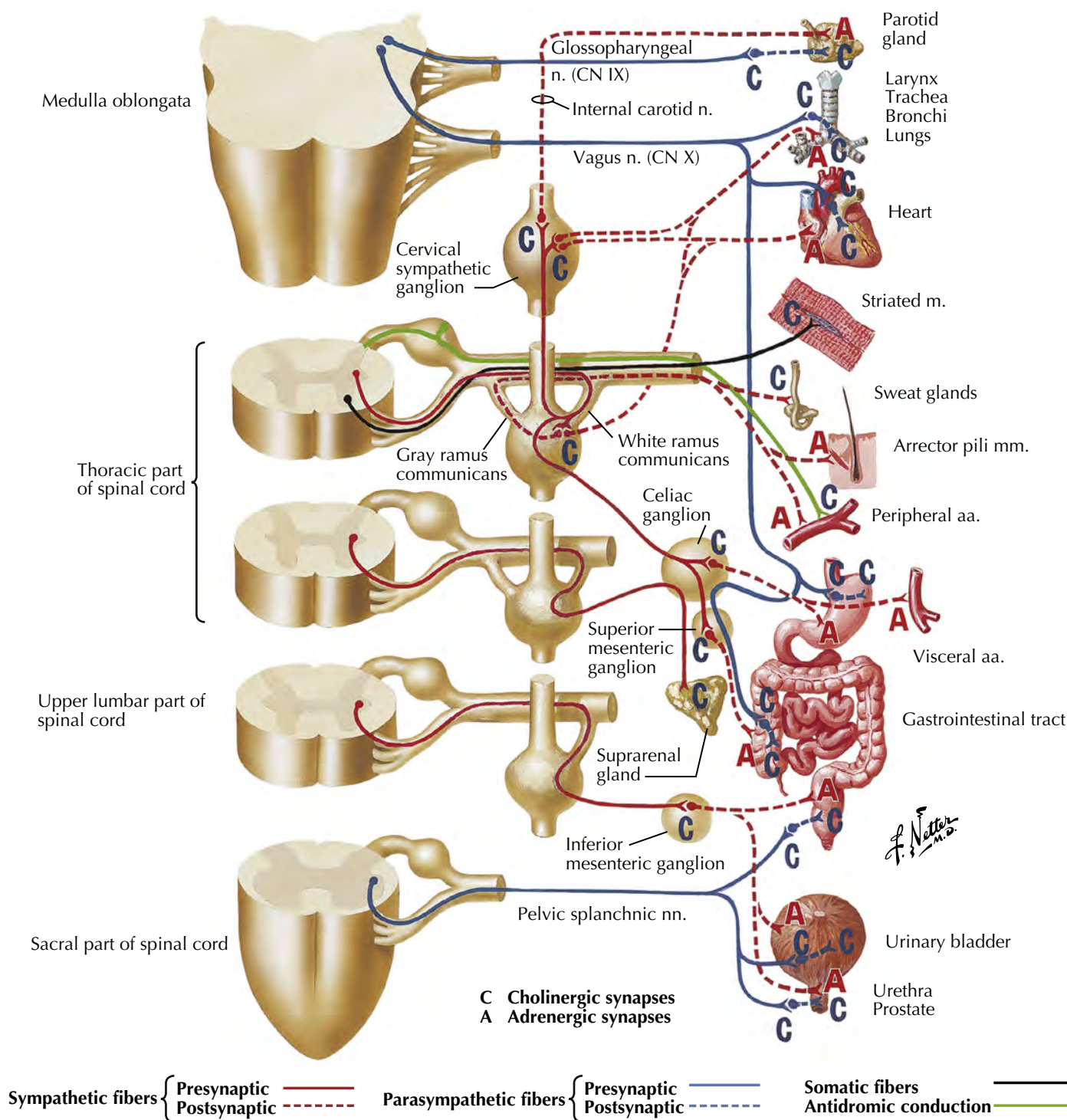
Sympathetic Nervous System: General Topography

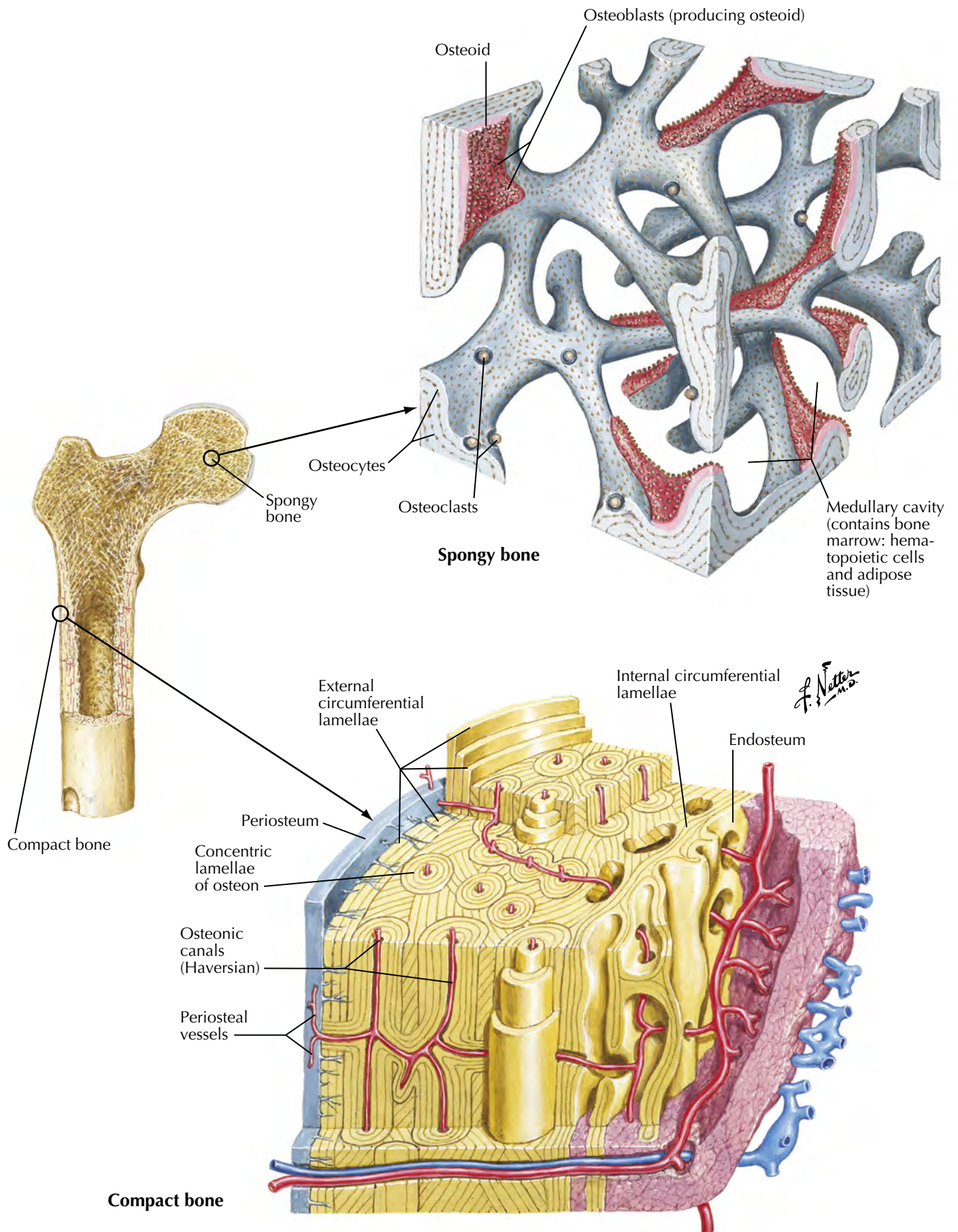


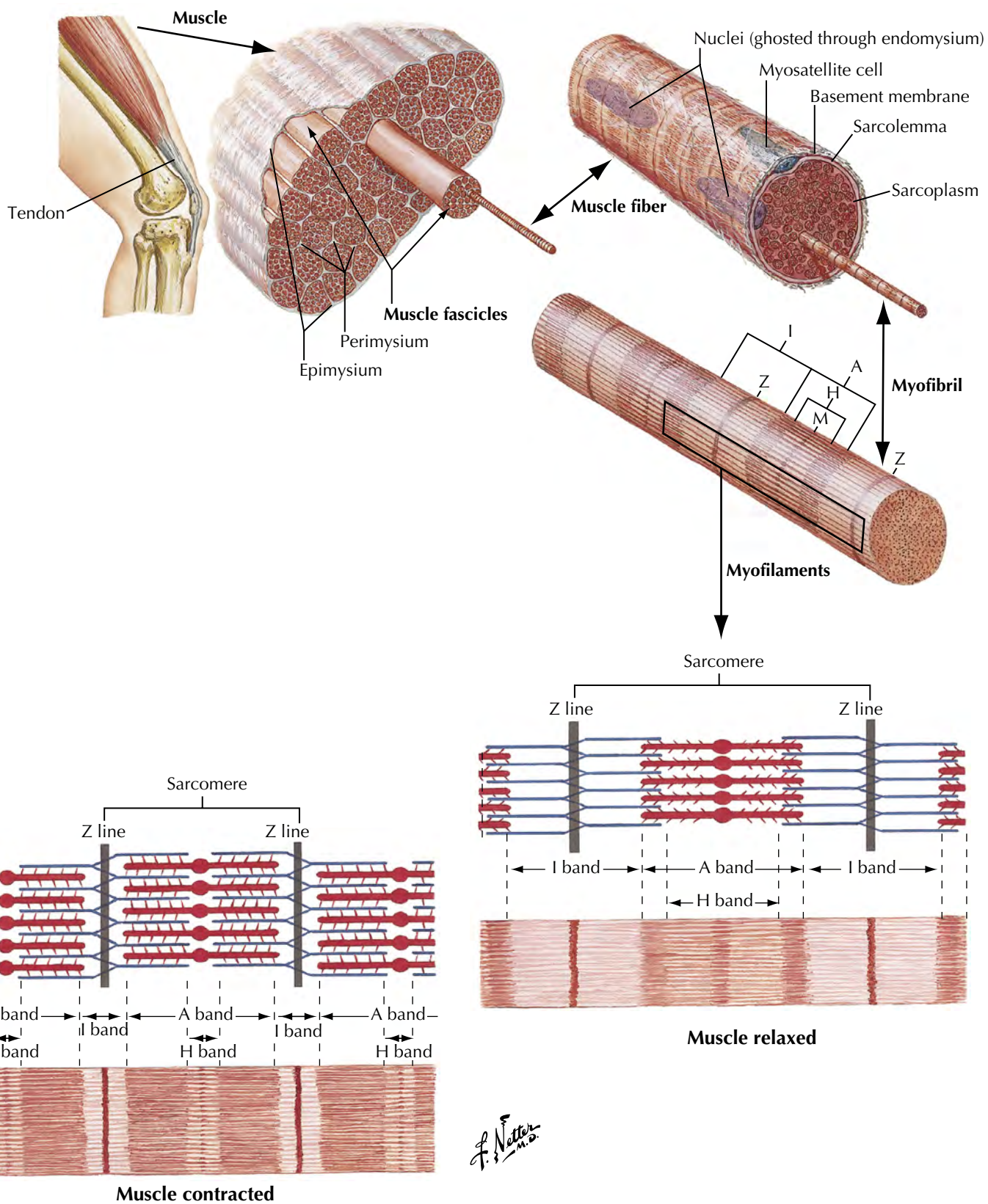
C. Machado
M.D.

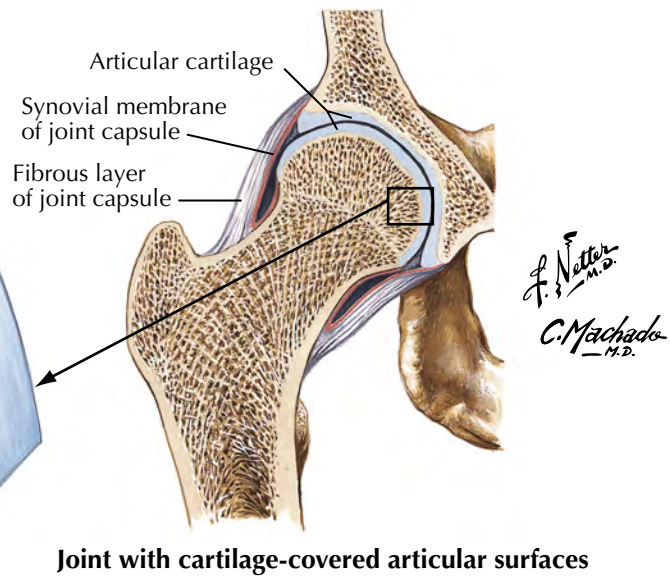
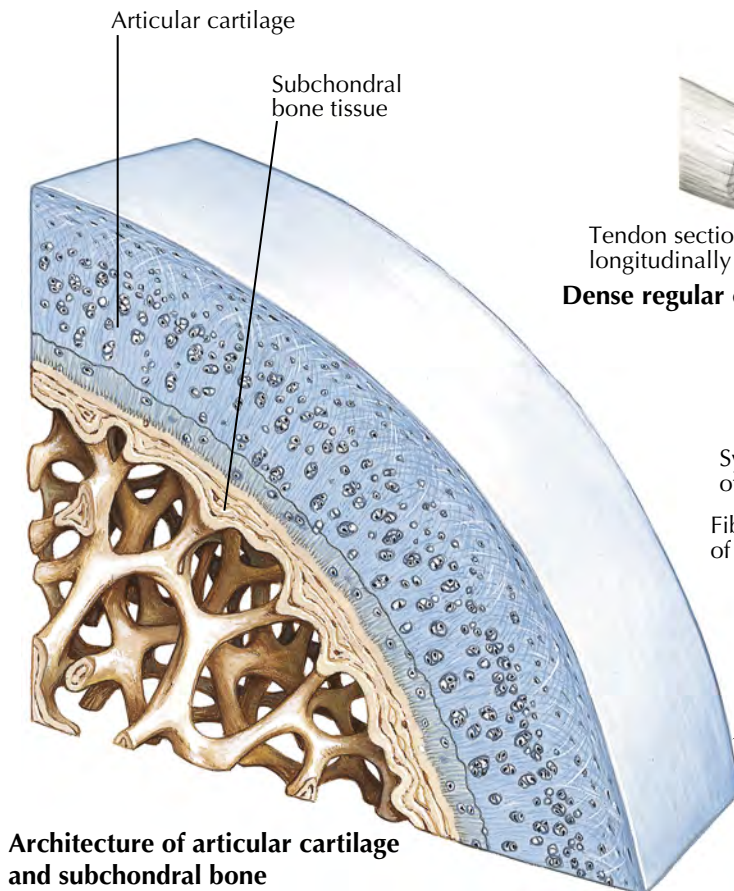
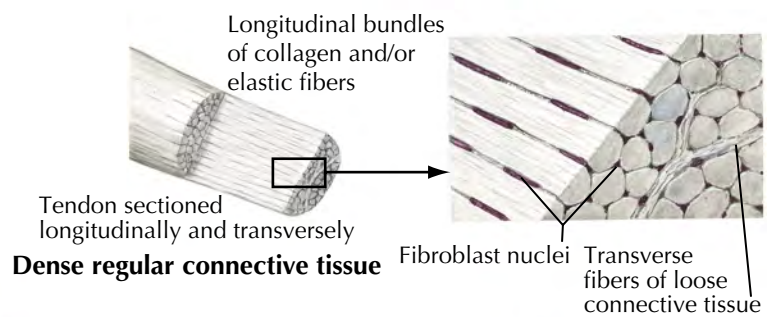
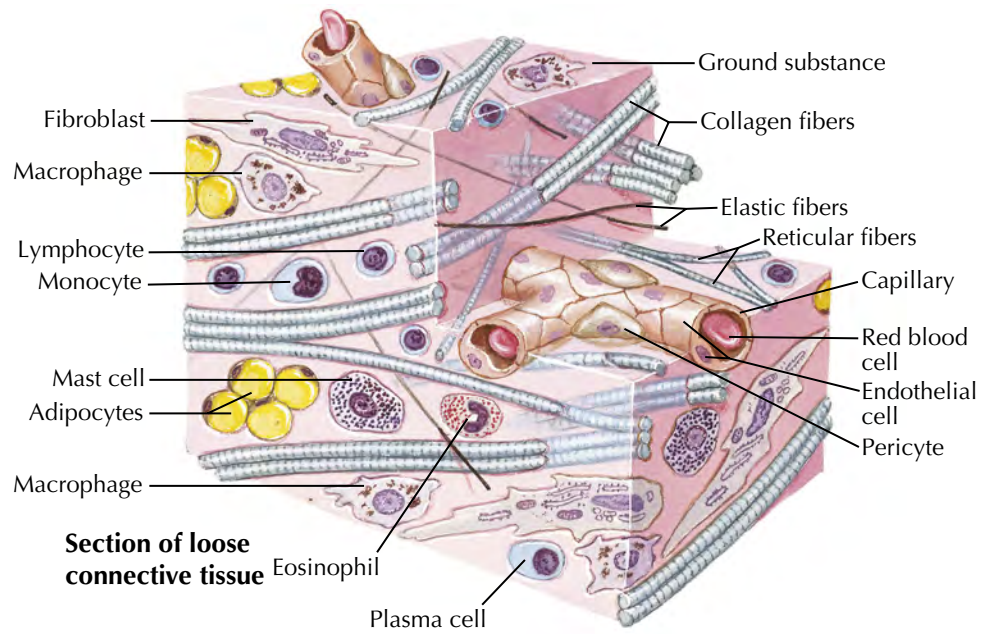


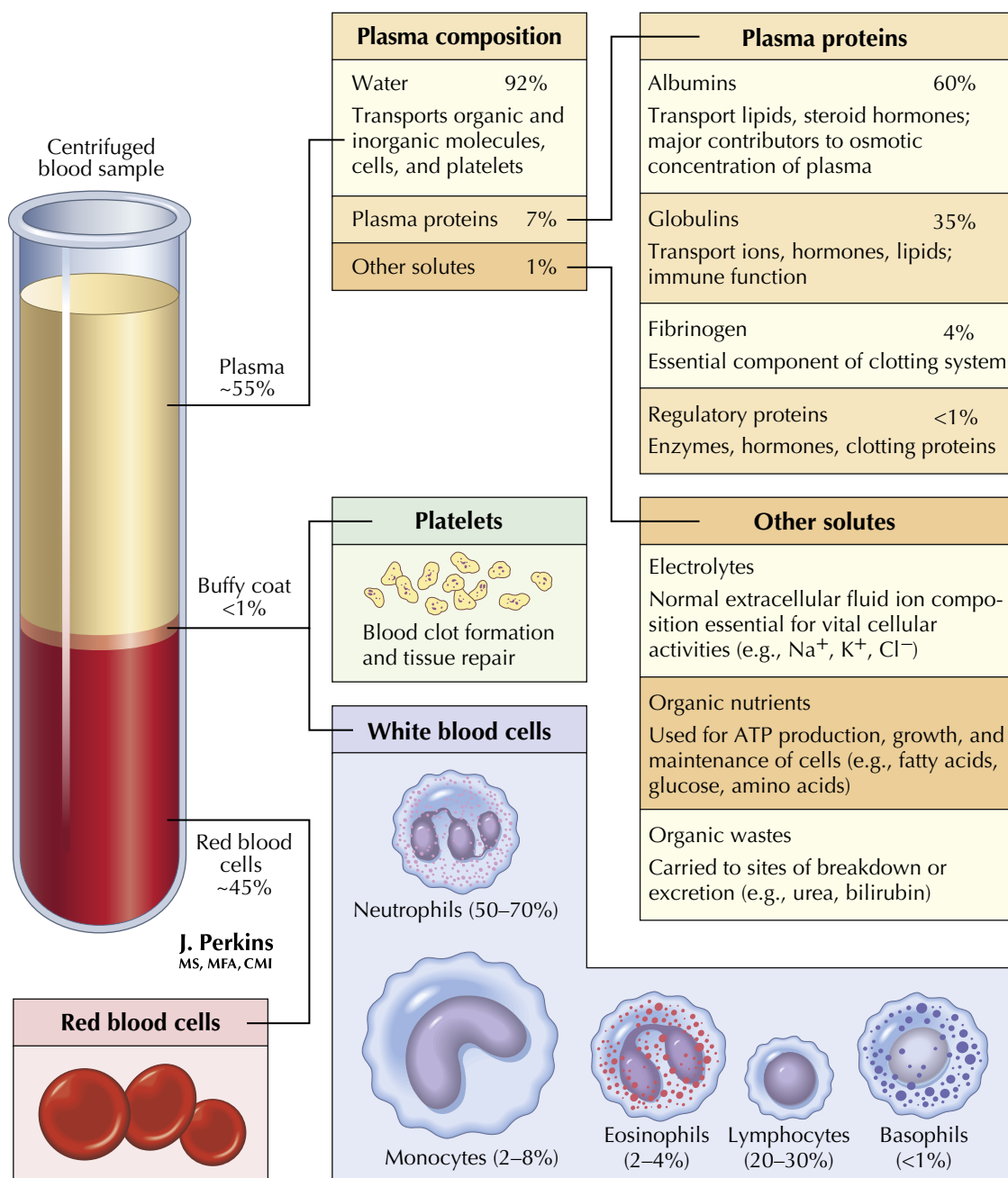
Cholinergic and Adrenergic Synapses: Schema

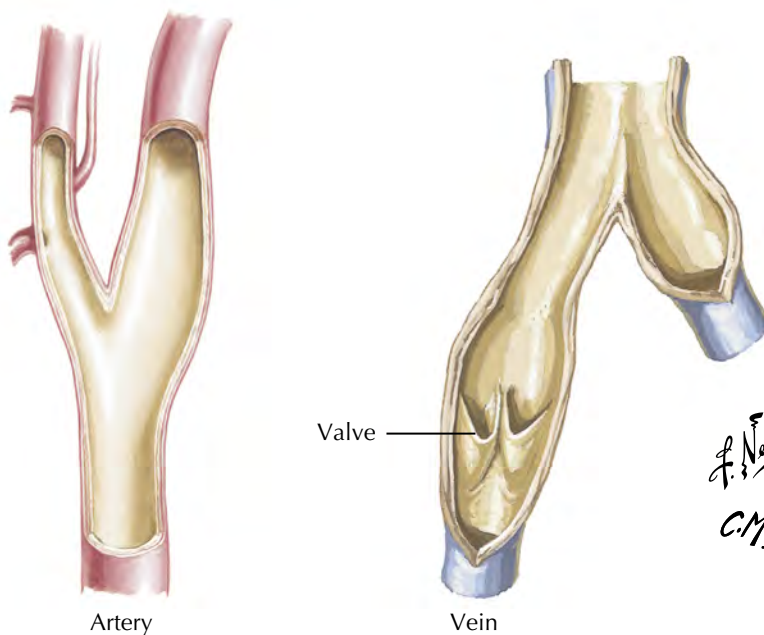
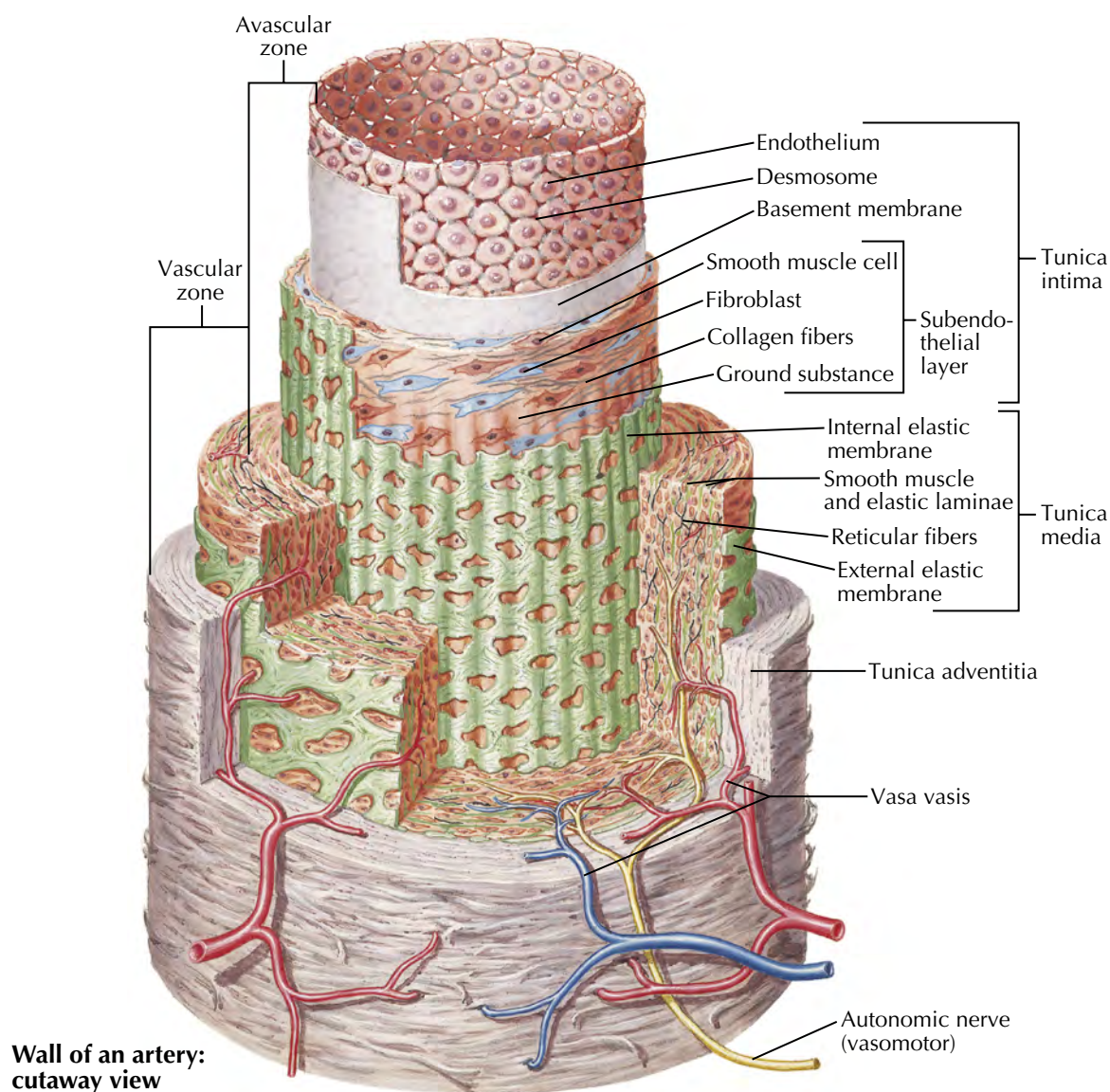












HEAD AND NECK 2

Surface Anatomy	22–24	Cranial and Cervical Nerves	143–162
Bones and Joints	25–47	Cerebral Vasculature	163–175
Neck	48–58	Regional Imaging	176–177
Nose	59–82	Structures with High Clinical Significance	Tables 2.1–2.4
Mouth	83–90	Cranial Nerves	Tables 2.5–2.8
Pharynx	91–102	Branches of Cervical Plexus	Table 2.9
Larynx and Endocrine Glands	103–109	Muscles	Tables 2.10–2.14
Eye	110–120	Electronic Bonus Plates	BP 14–BP 32
Ear	121–126		
Brain and Meninges	127–142		

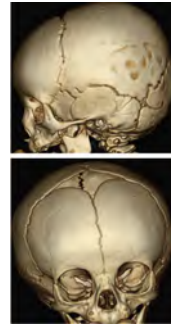
ELECTRONIC BONUS PLATES



BP 14 Somatosensory System: Trunk and Limbs



BP 15 Pyramidal System



BP 16 3D Skull Reconstruction CTs



BP 17 Degenerative Changes in Cervical Vertebrae



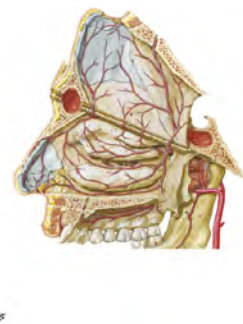
BP 18 Atlantooccipital Junction



BP 19 Muscles of Facial Expression: Anterior View



BP 20 Musculature of Face



BP 21 Arteries of Nasal Cavity: Bony Nasal Septum Turned Up

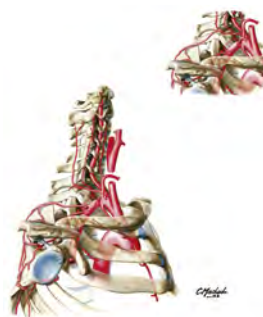
ELECTRONIC BONUS PLATES—*cont'd*



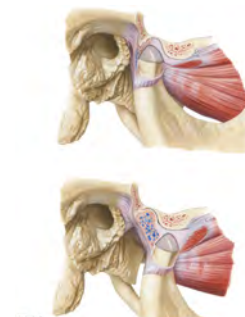
BP 22 Nose and Maxillary Sinus: Transverse Section



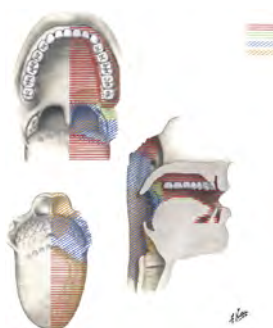
BP 23 Paranasal Sinuses



BP 24 Subclavian Artery



BP 25 Opening of the Mandible



BP 26 Afferent Innervation of Oral Cavity and Pharynx



BP 27 Fasciae of Orbit and Eyeball



BP 28 Tympanic Cavity: Medial and Lateral Views



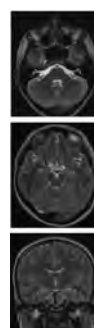
BP 29 Anatomy of the Pediatric Ear



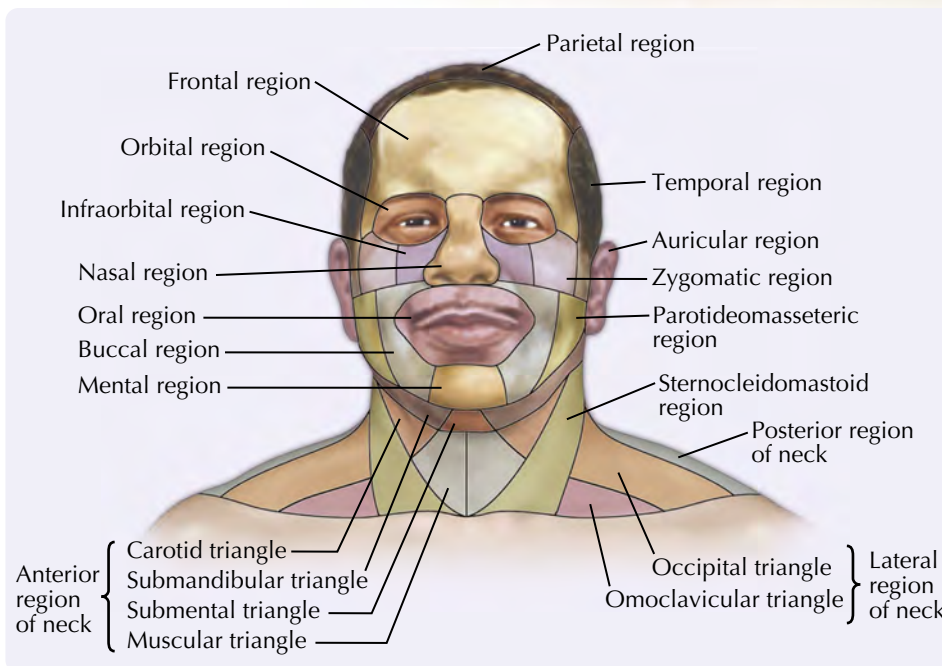
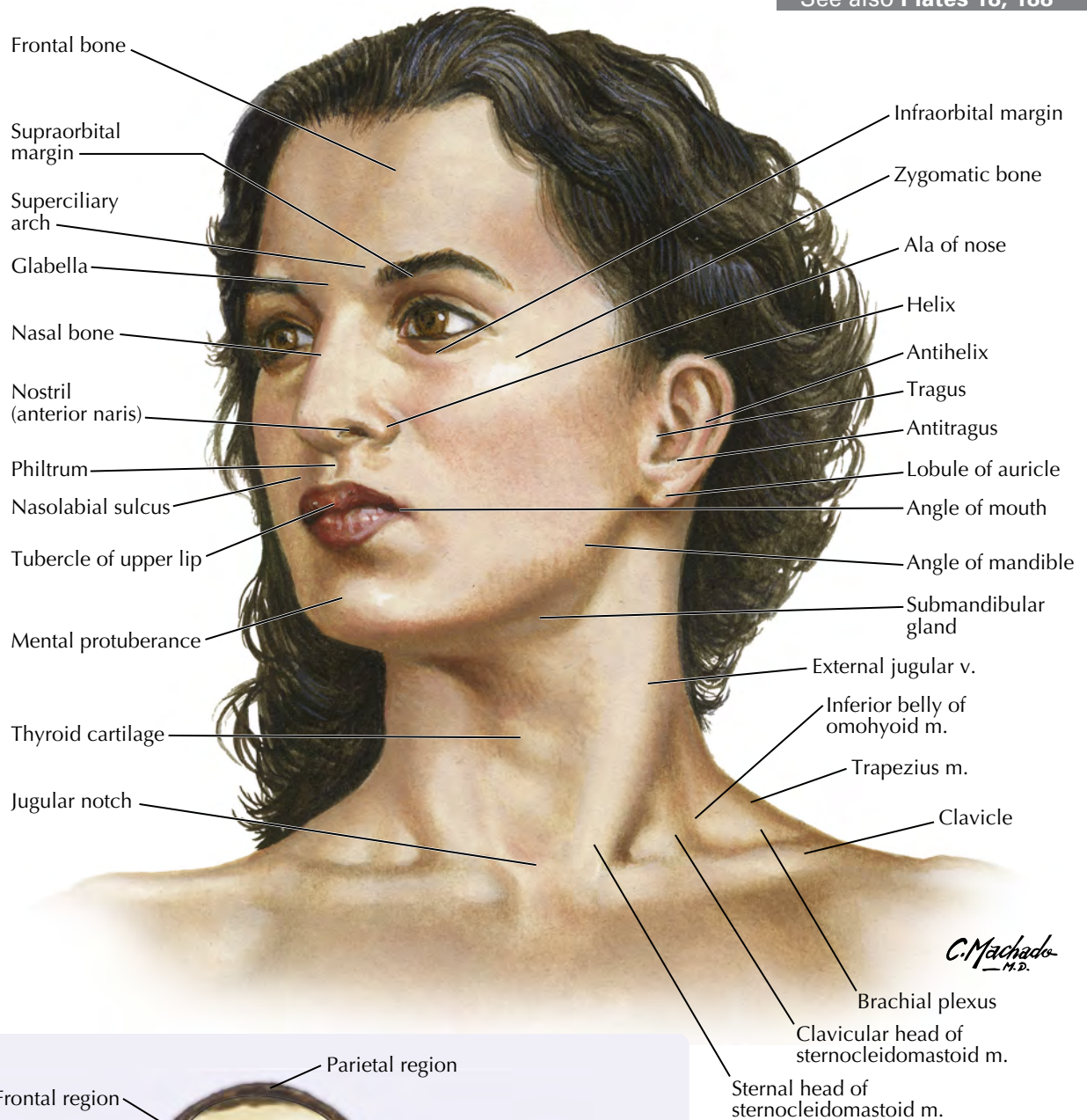
BP 30 Auditory Tube (Eustachian)



BP 31 Cranial Imaging (MRV and MRA)

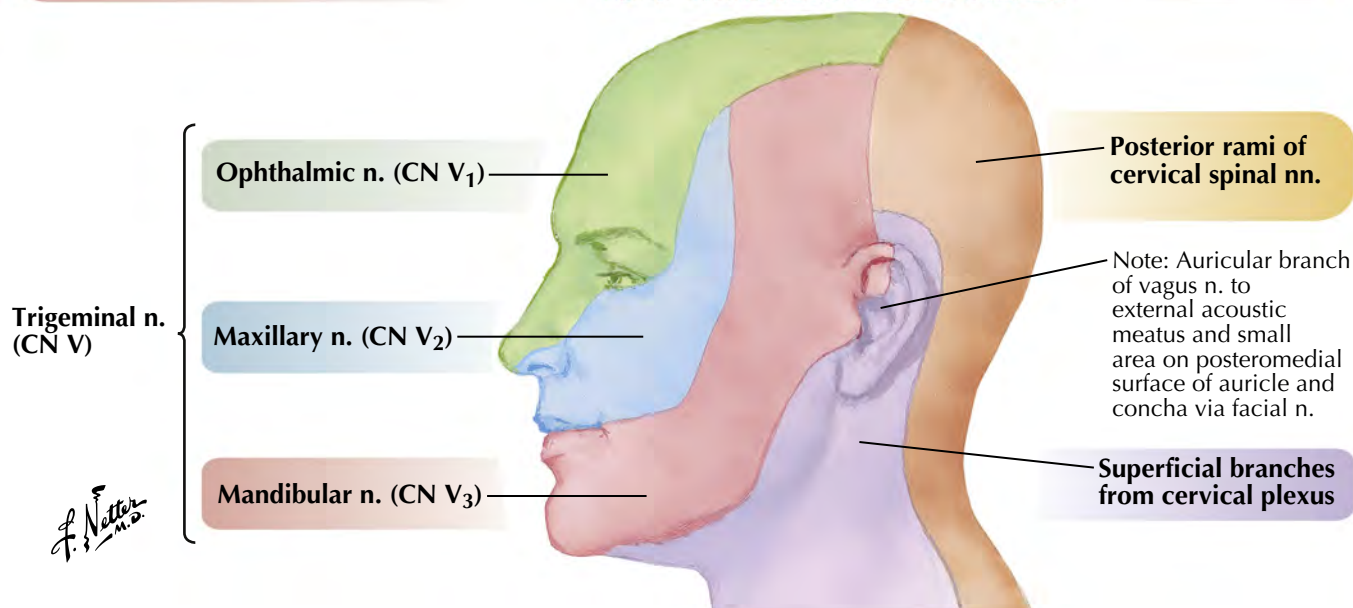
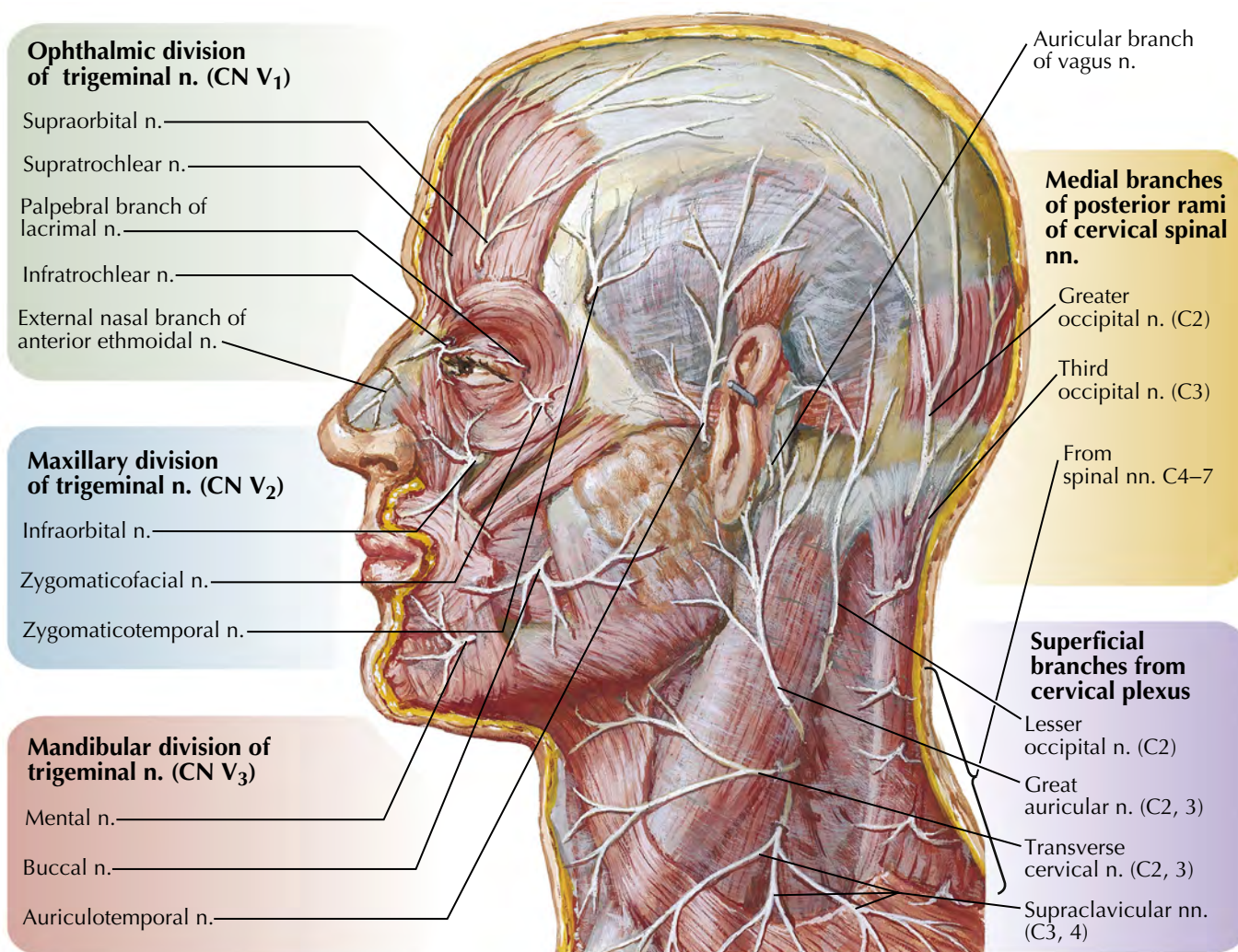


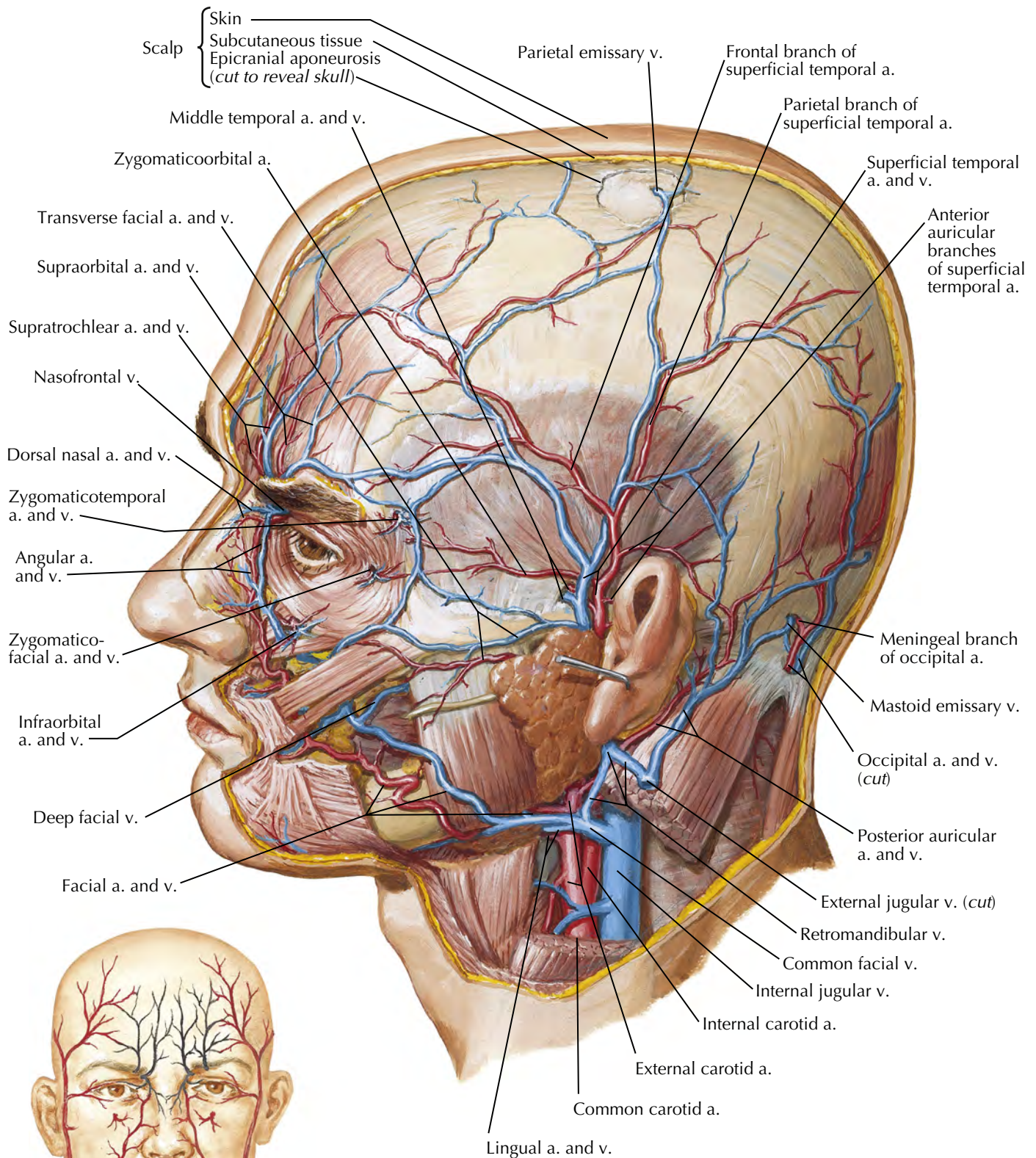
BP 32 Axial and Coronal MRIs of Brain



Cutaneous Nerves of Head and Neck

See also Plates 6, 59, 60, 79





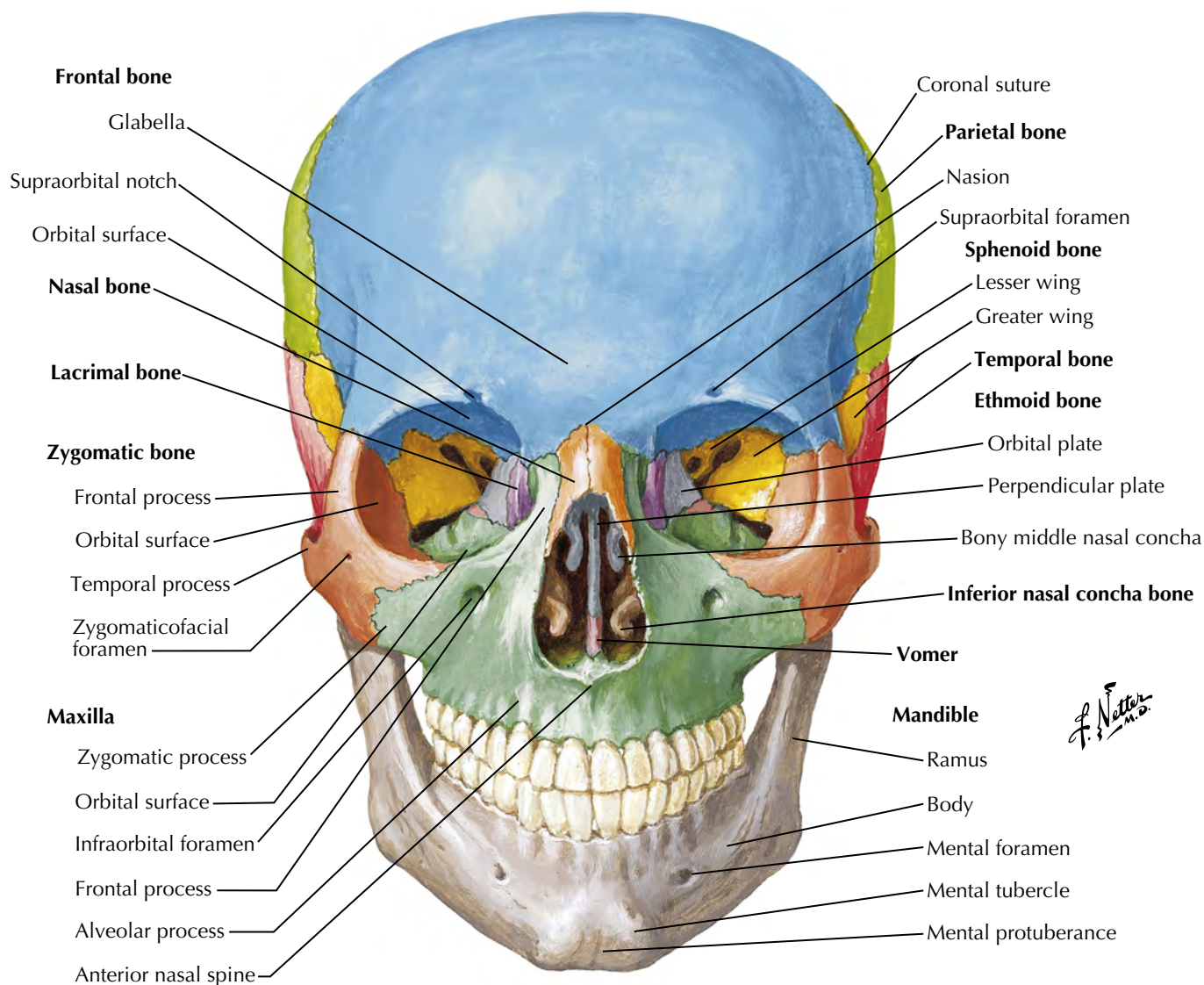
Sources of arterial supply of face

Black: from internal carotid artery (via ophthalmic artery)
Red: from external carotid artery

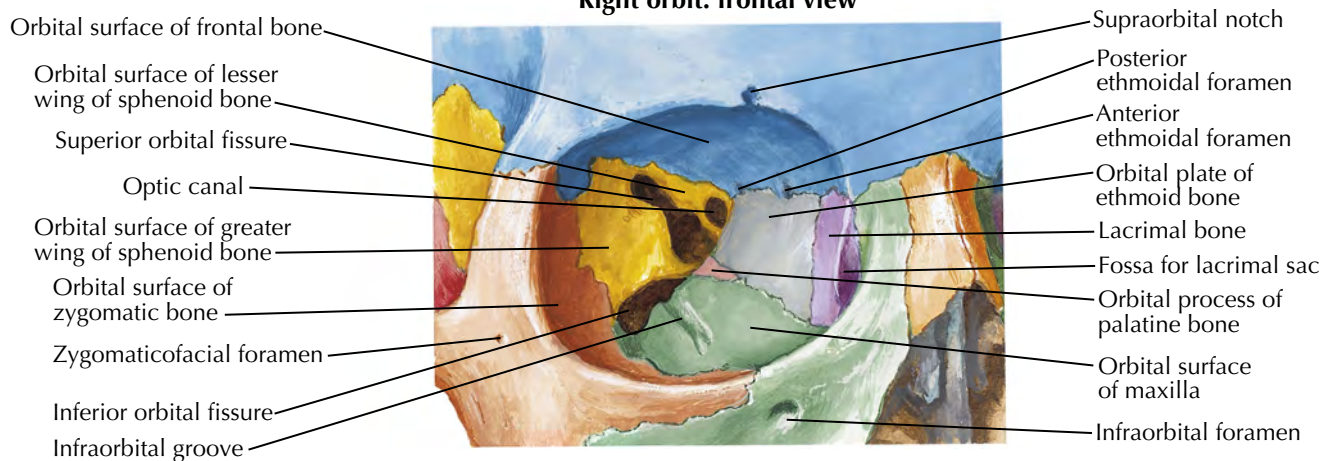
F. Netter M.D.

Skull: Anterior View

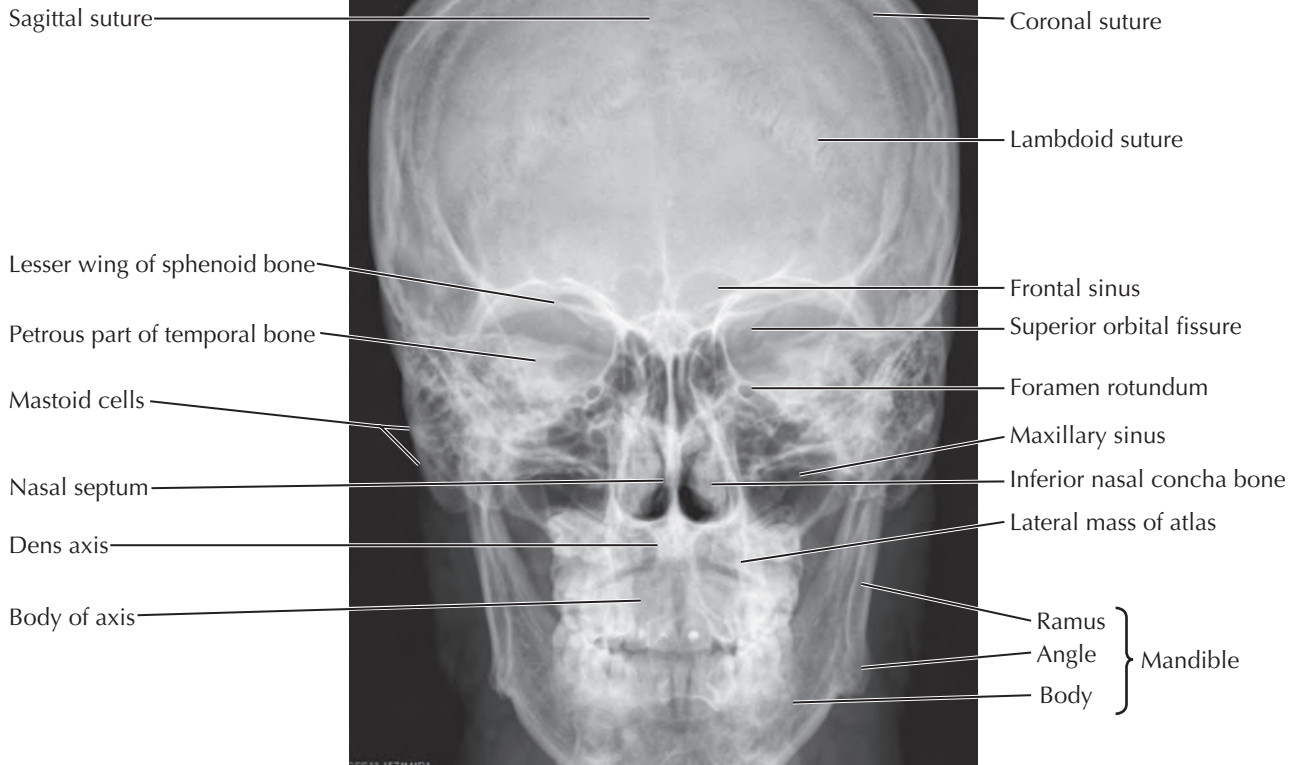
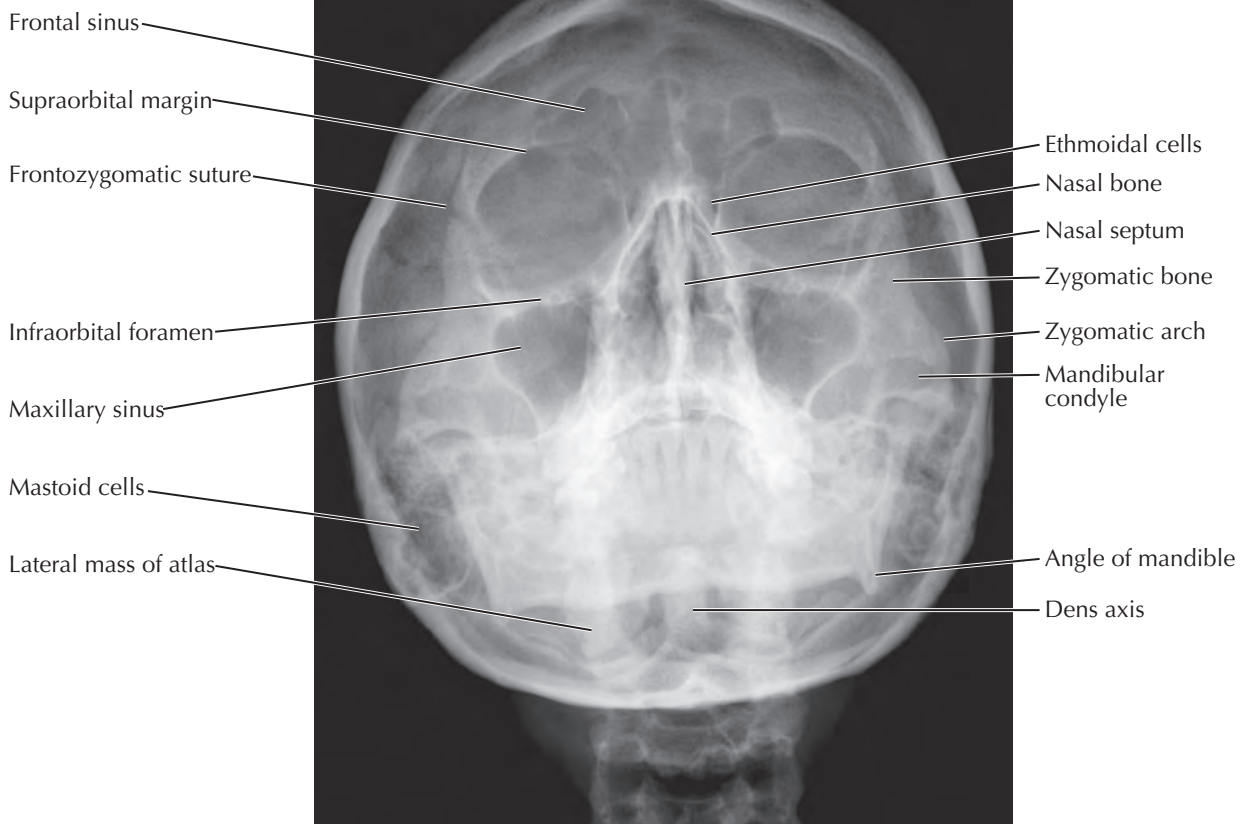
See also Plate 27



Right orbit: frontal view

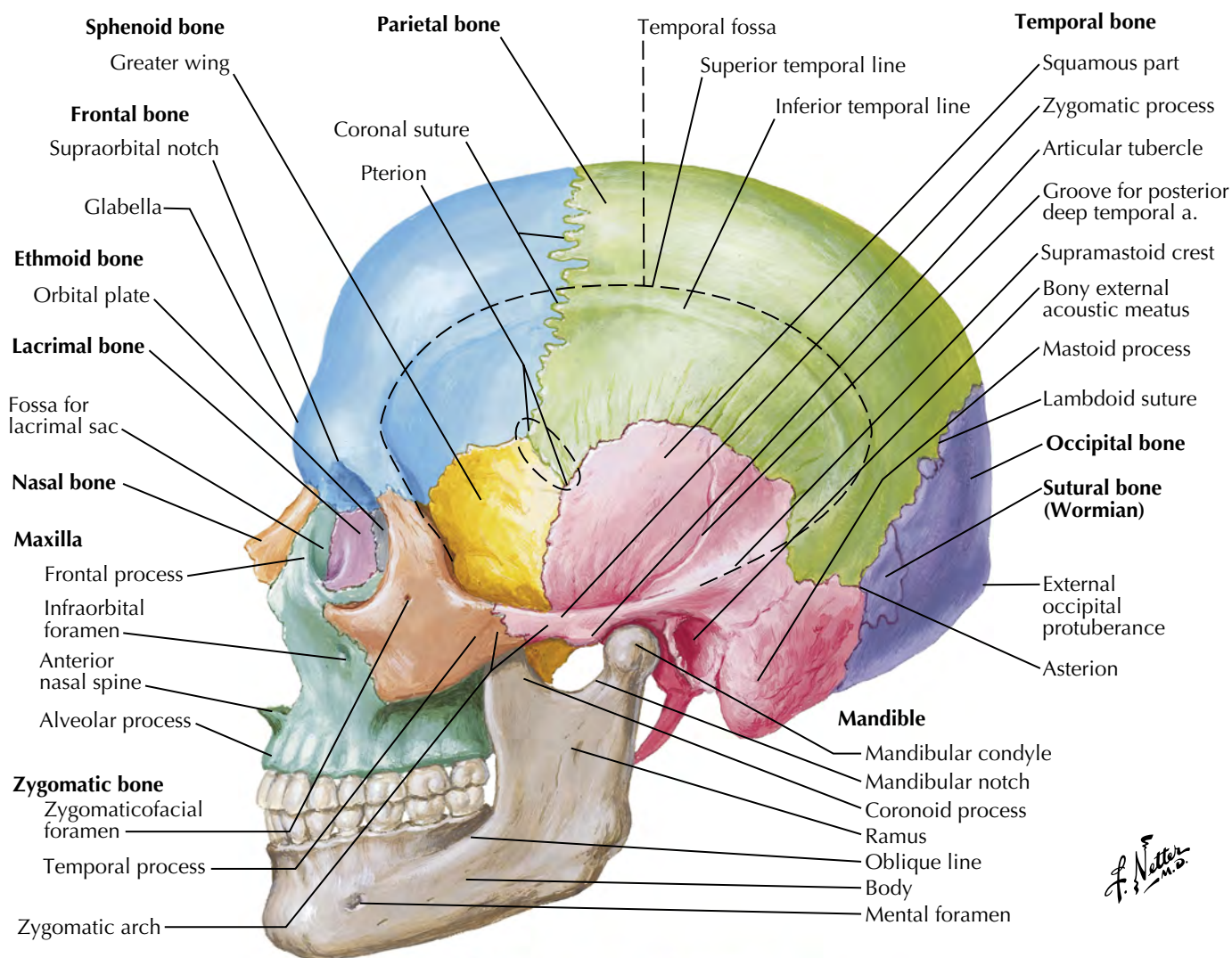


See also Plate 25

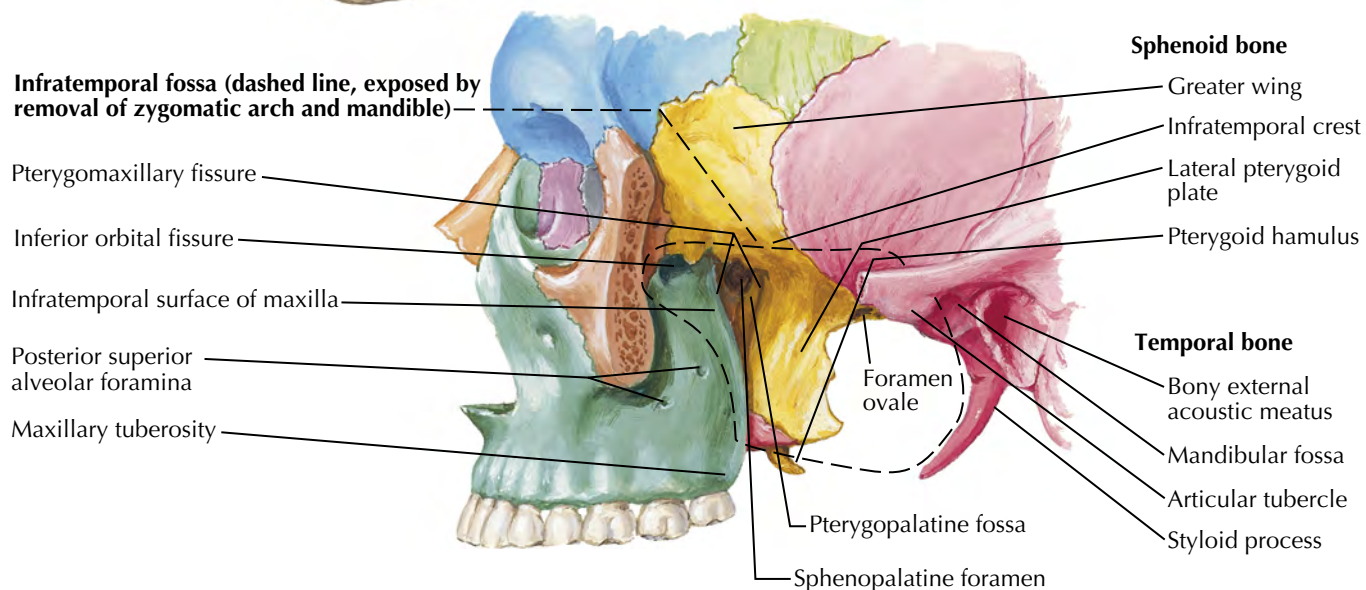
Posterior anterior view**Waters' view**

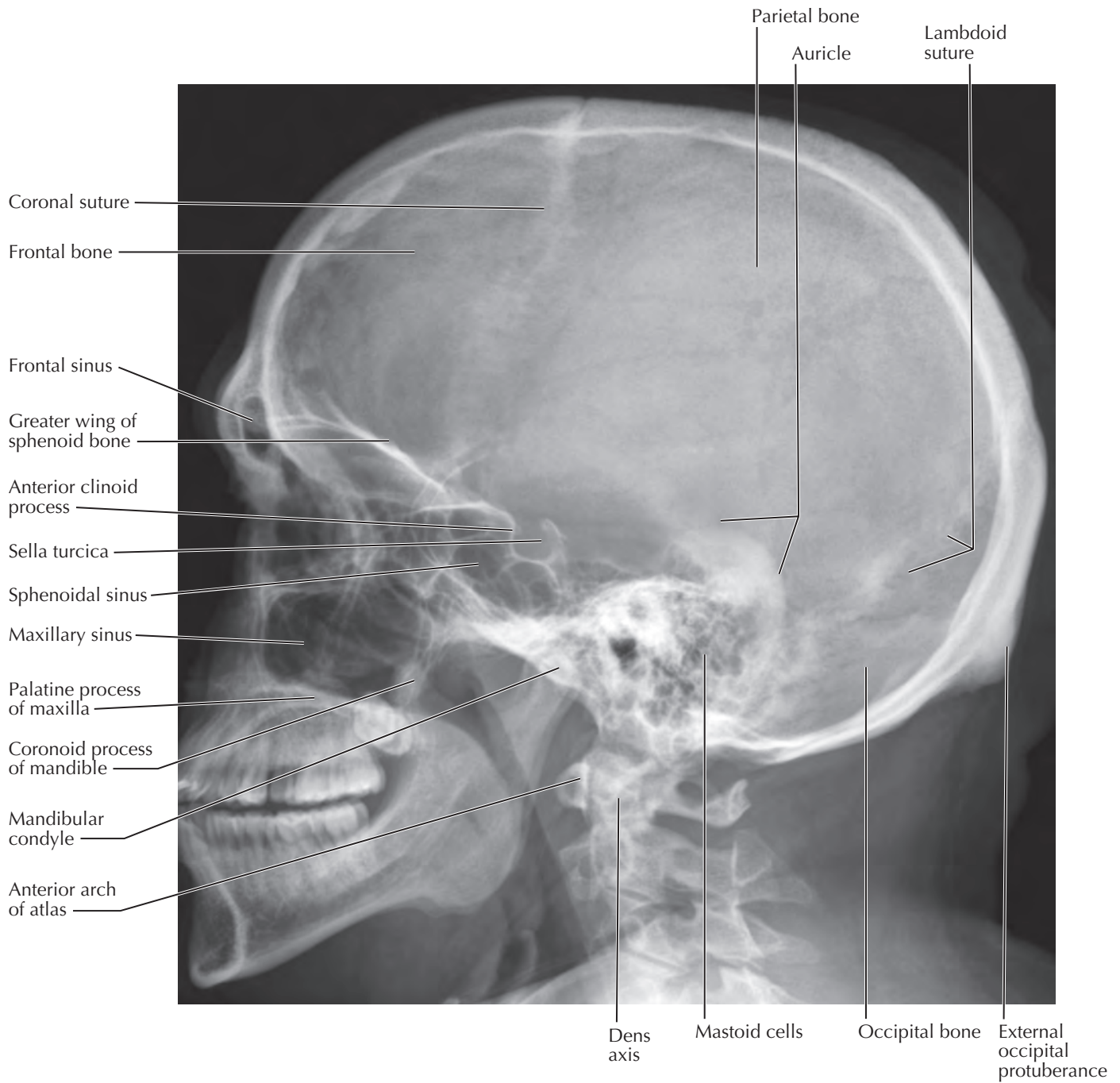
Skull: Lateral View

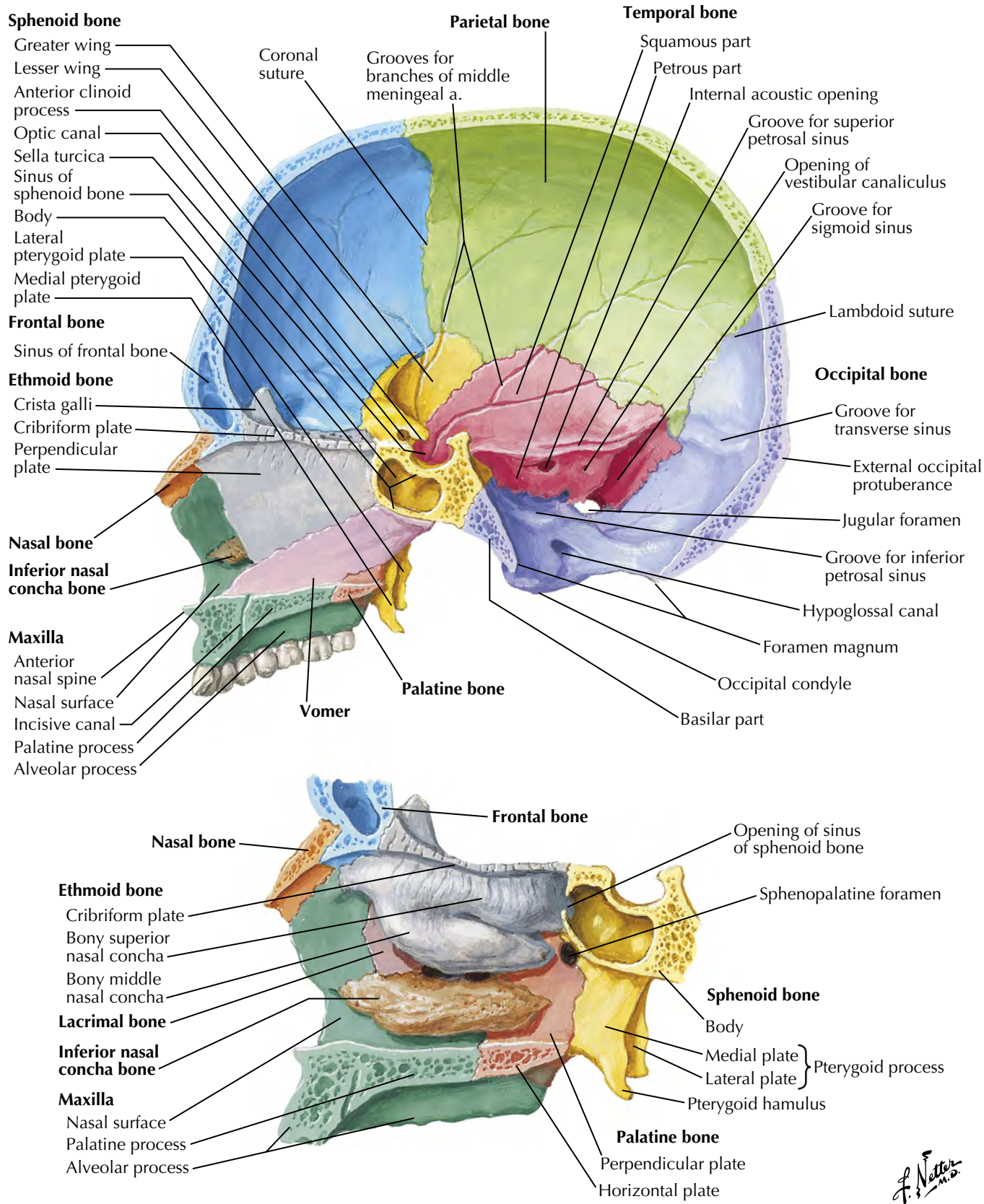
See also Plates 28, 29, 36

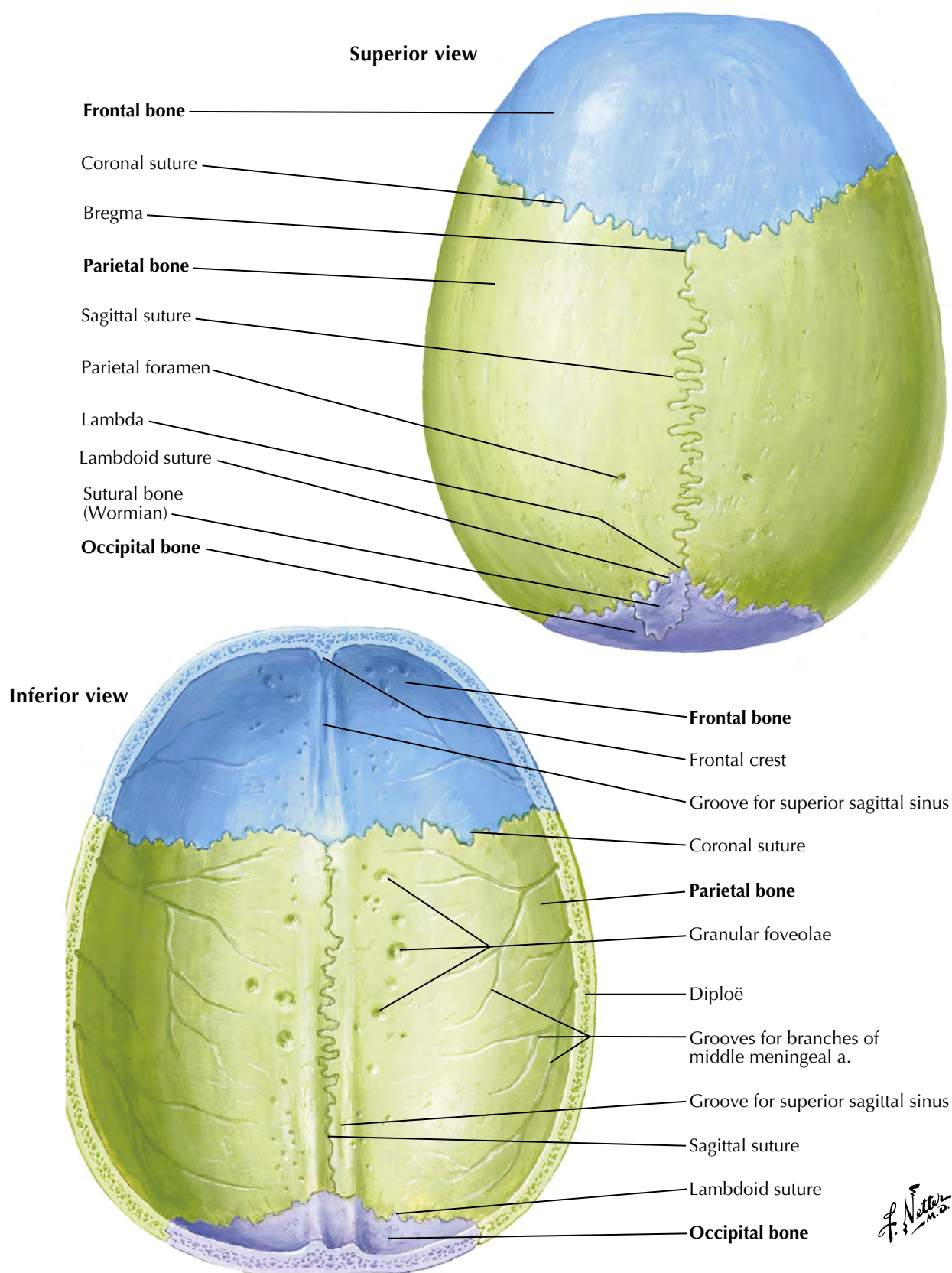


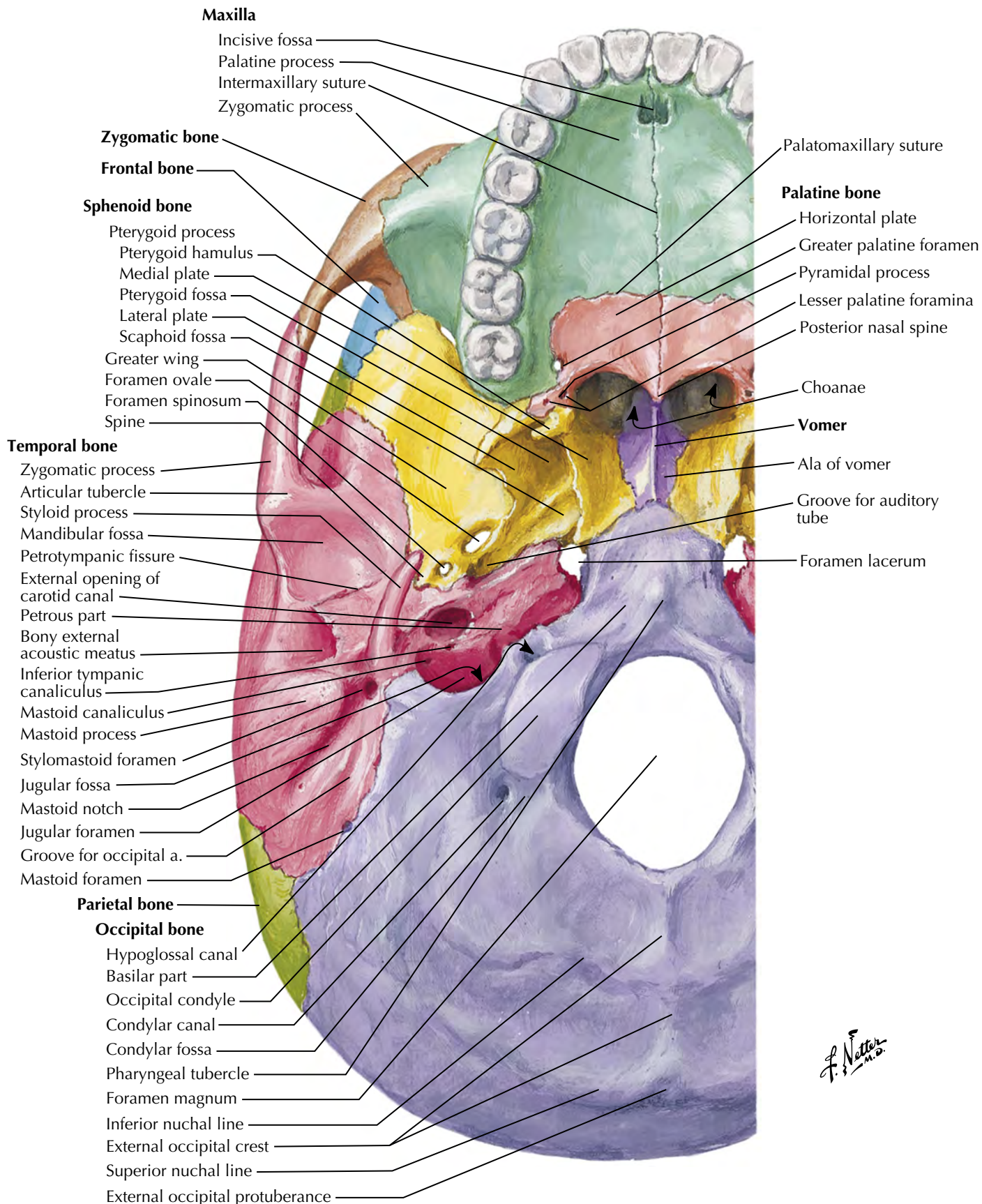
F. Netter M.D.



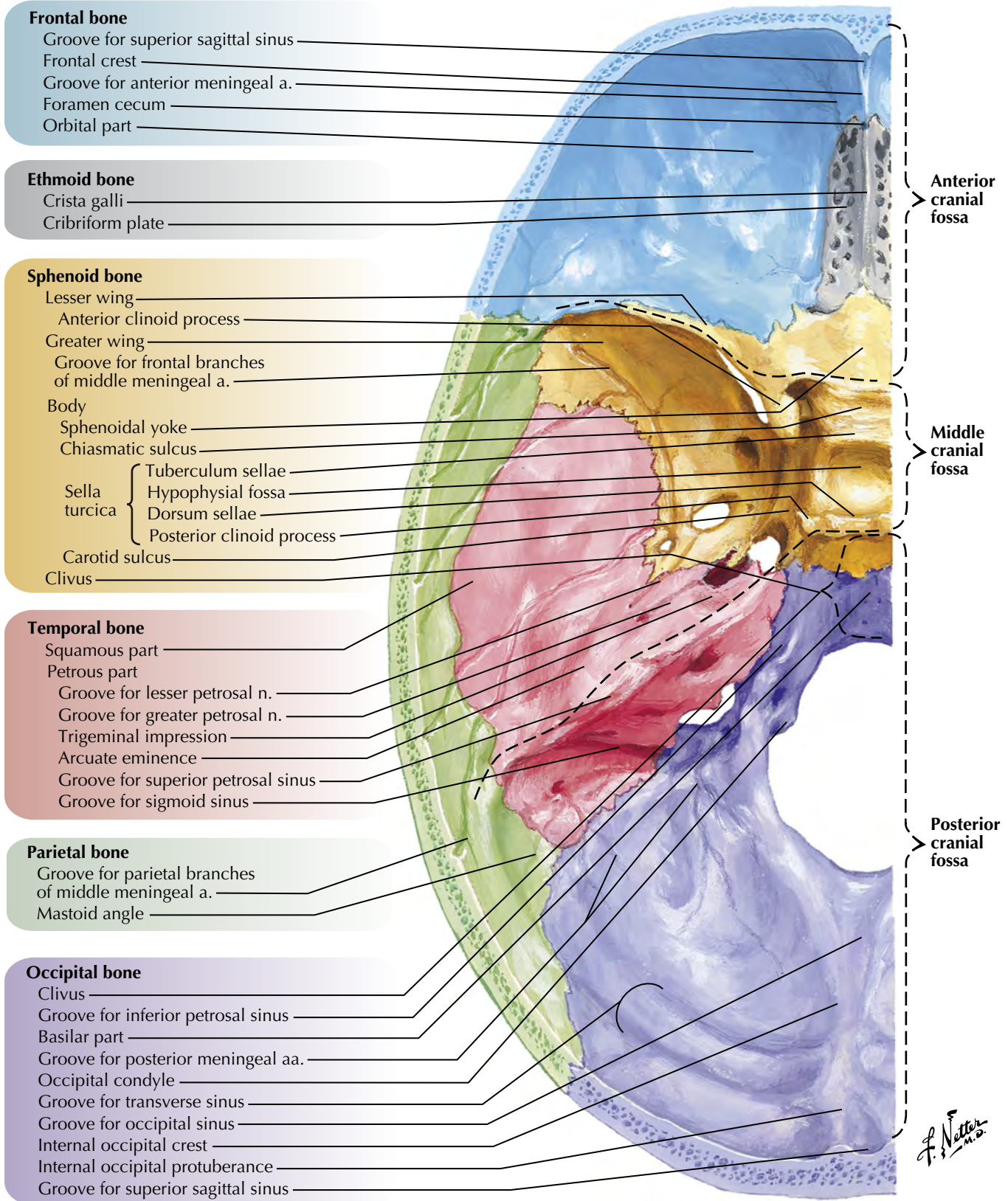






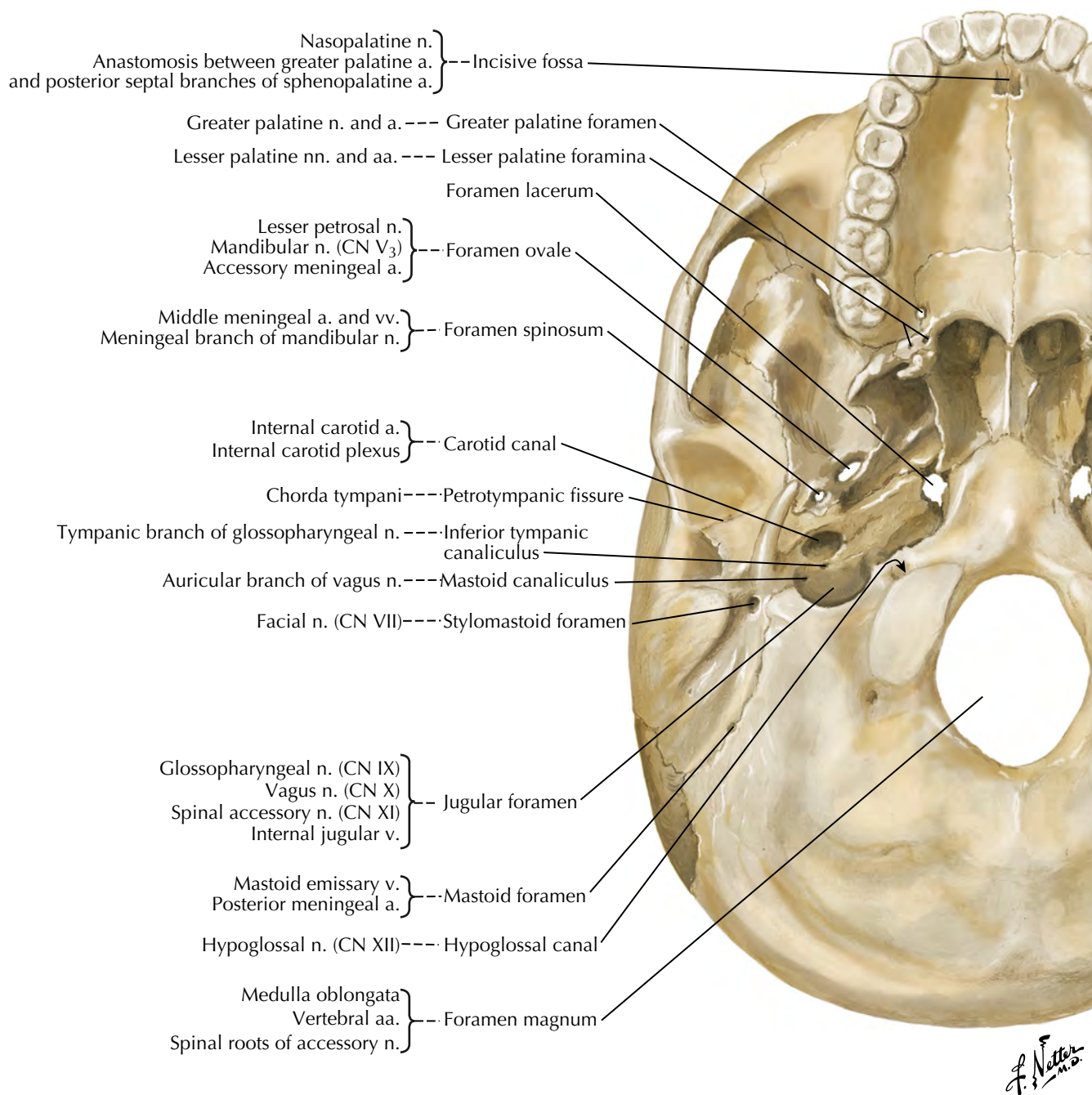


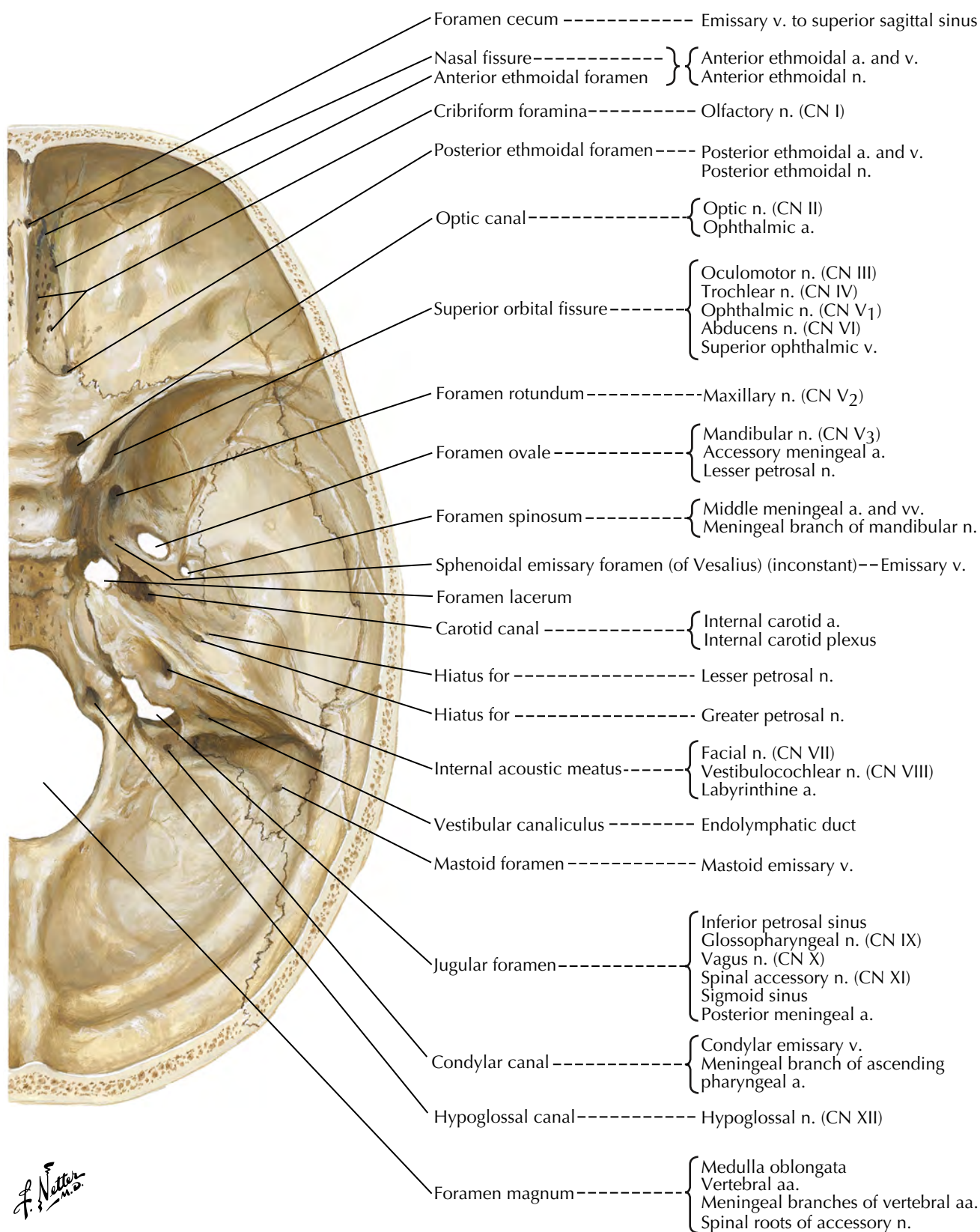
F. Netter M.D.

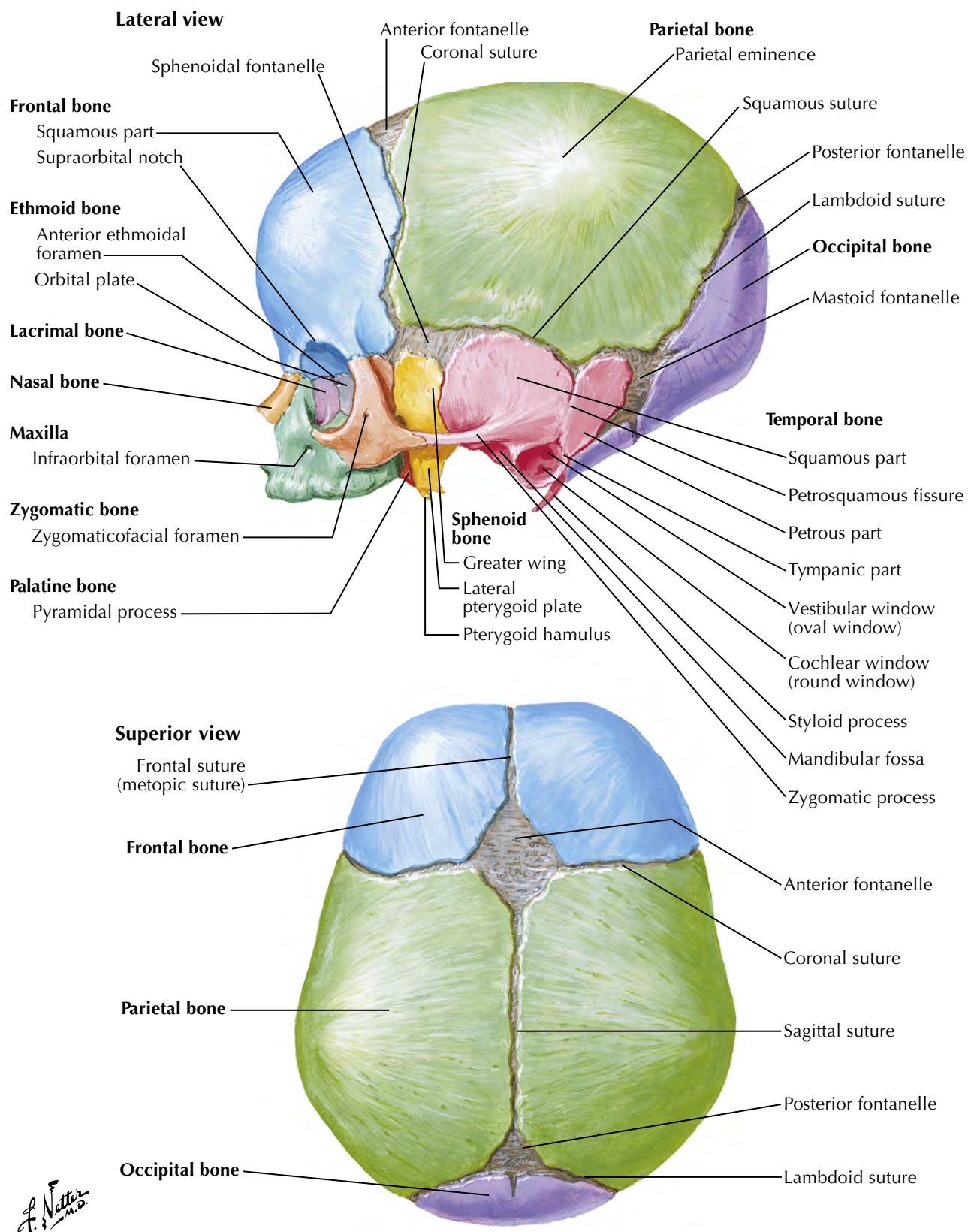


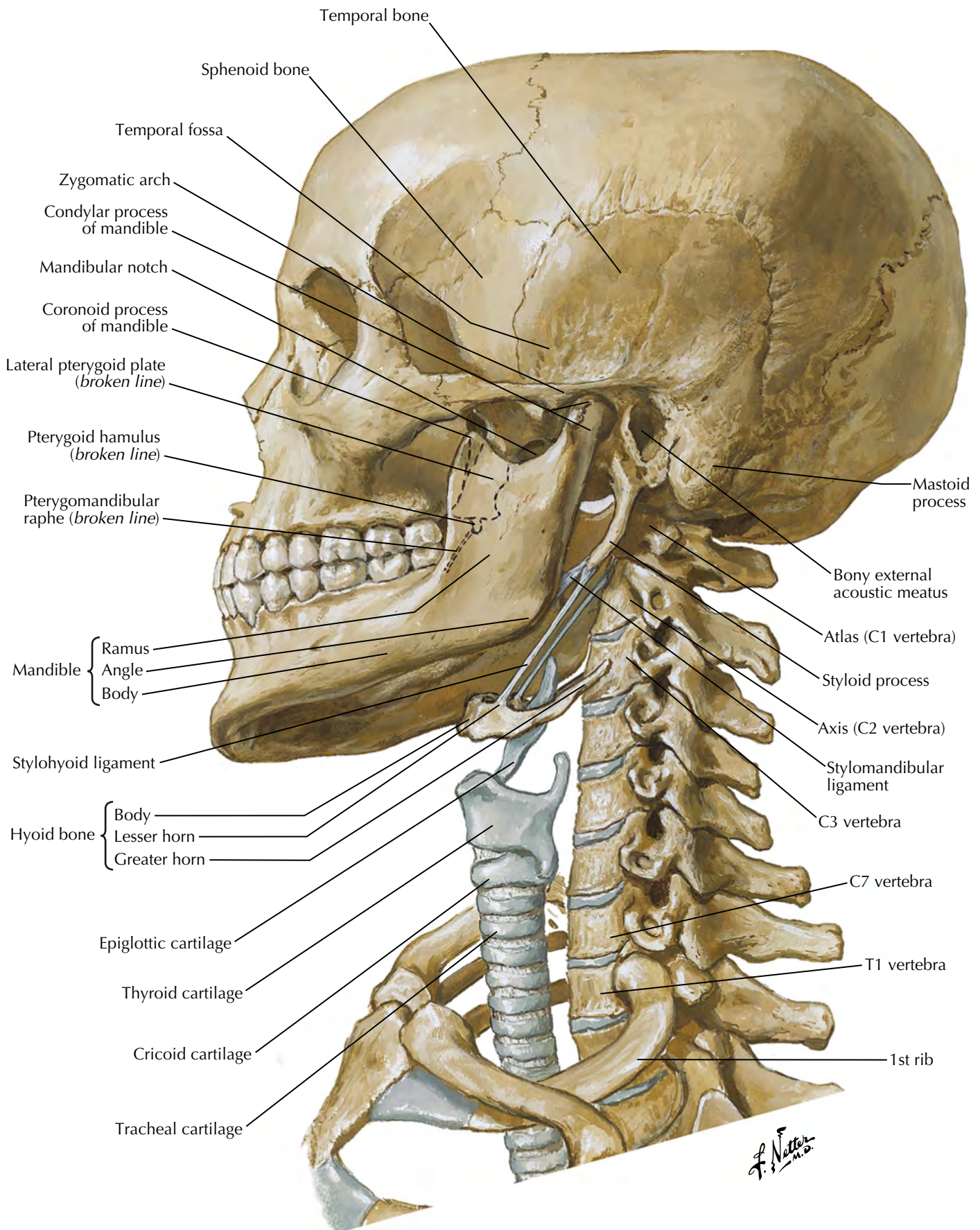
Foramina and Canals of Cranial Base: Inferior View

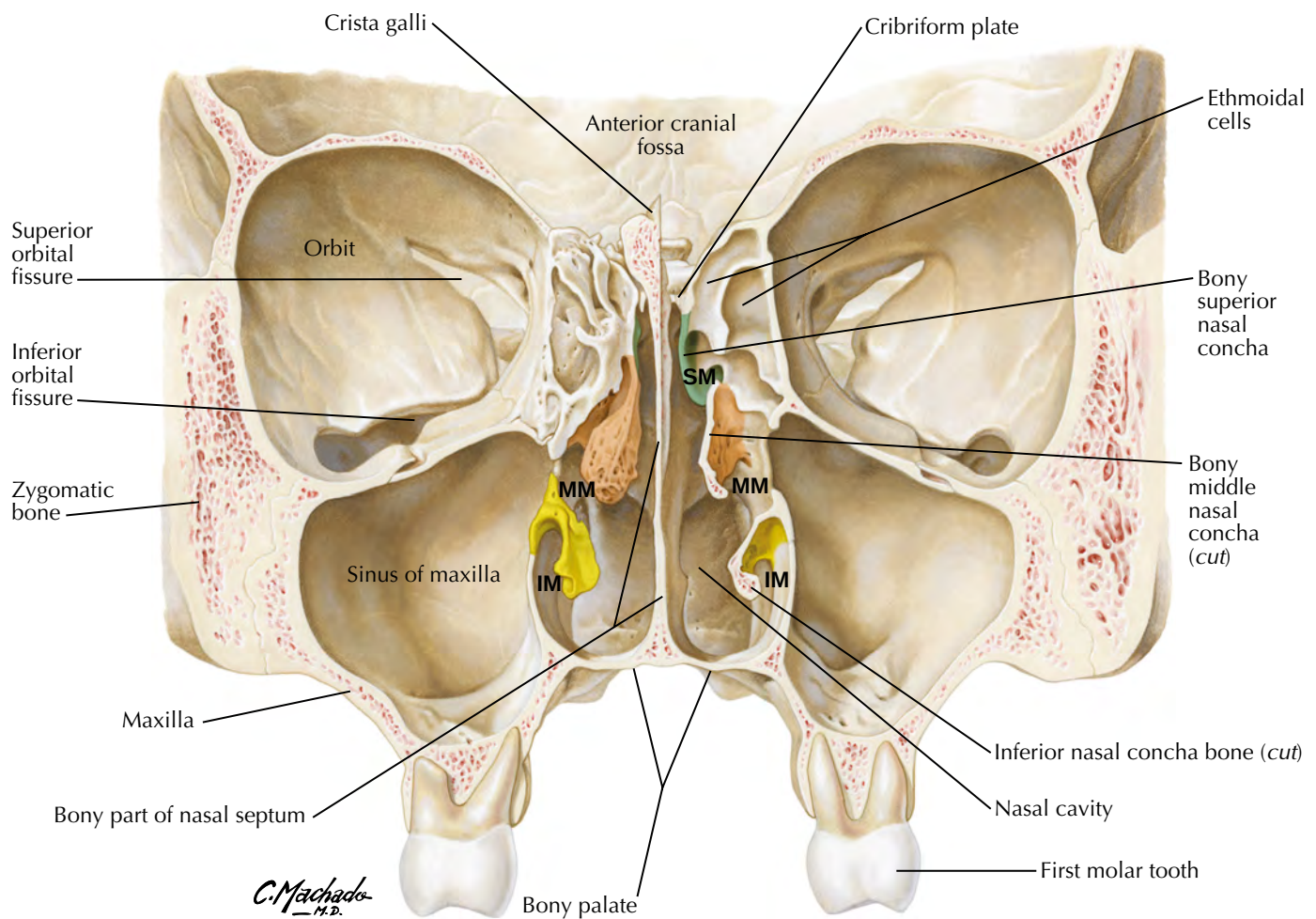
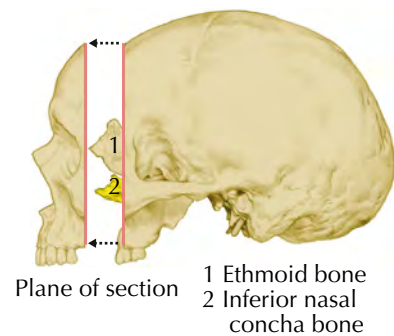
See also Plate 31







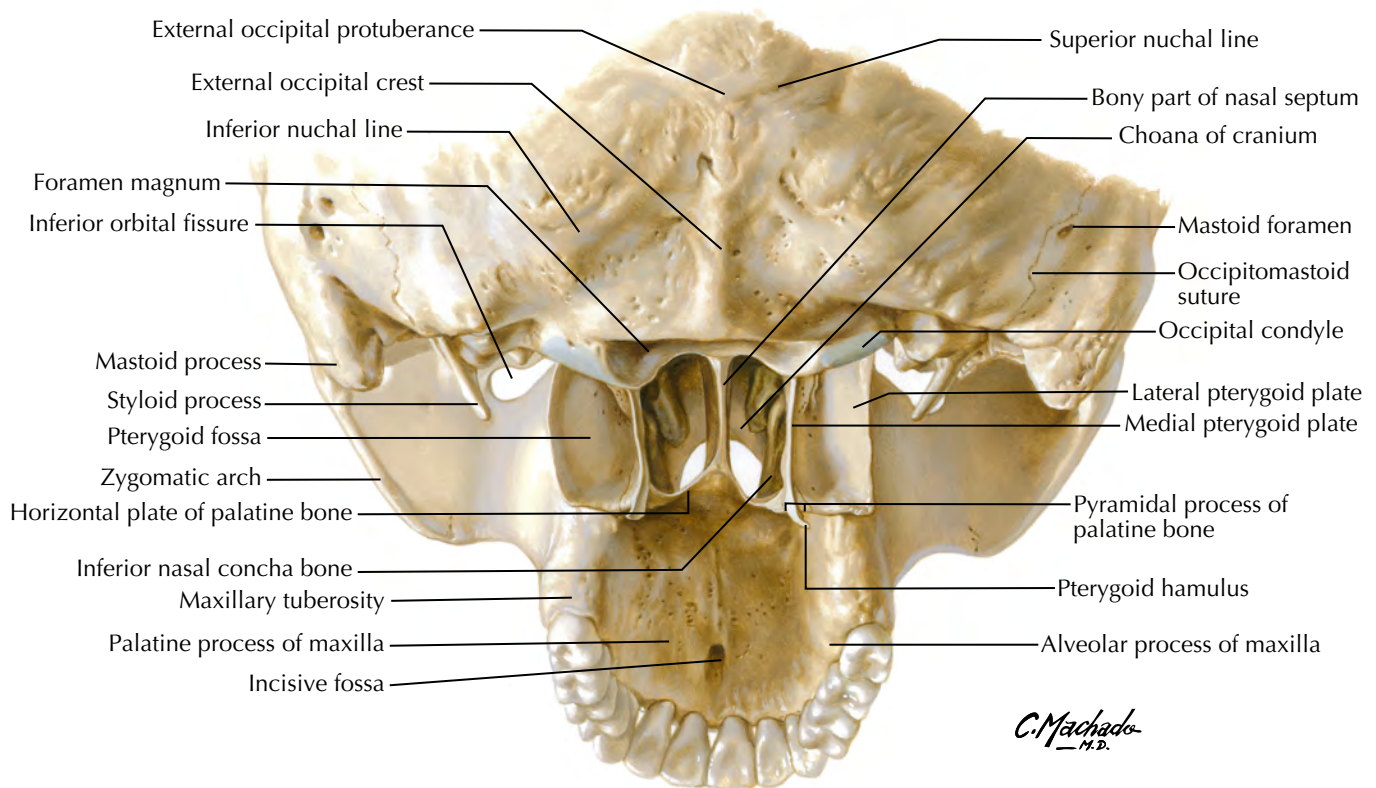




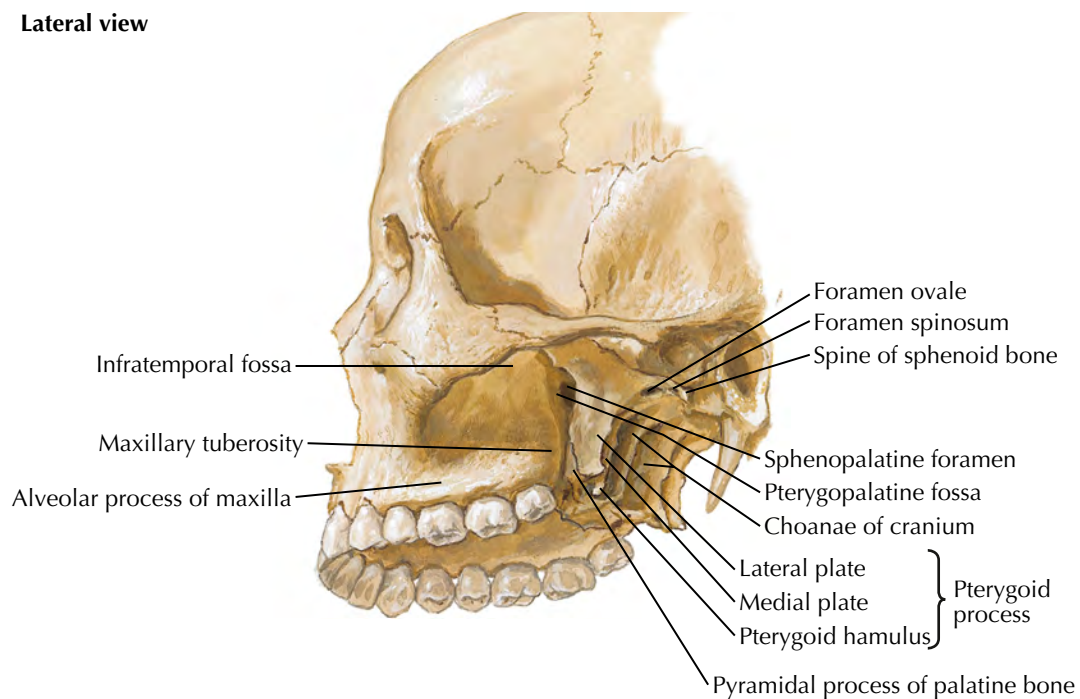
- Bony superior nasal concha
- Bony middle nasal concha
- Inferior nasal concha bone

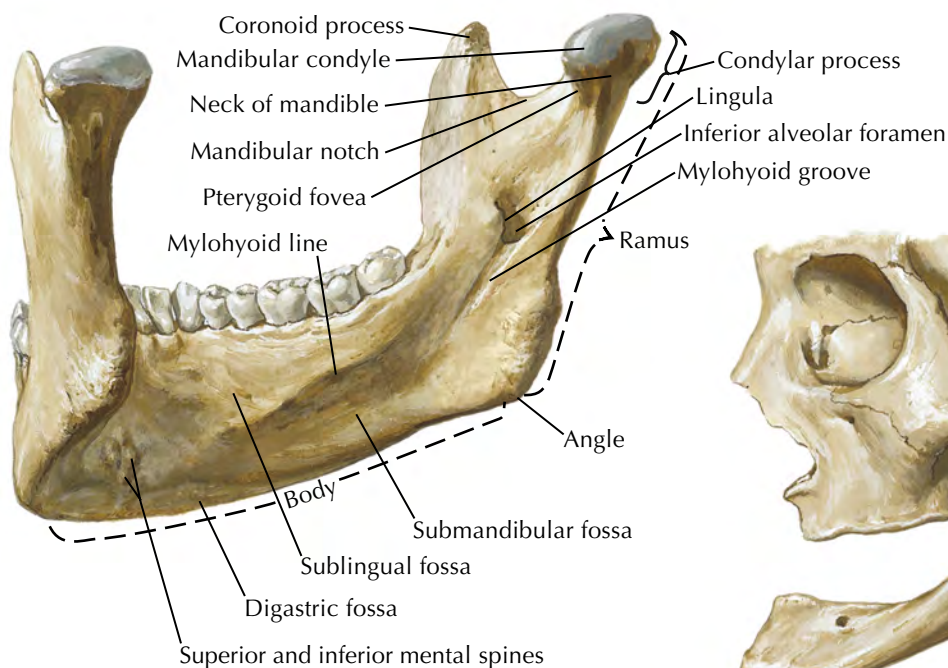
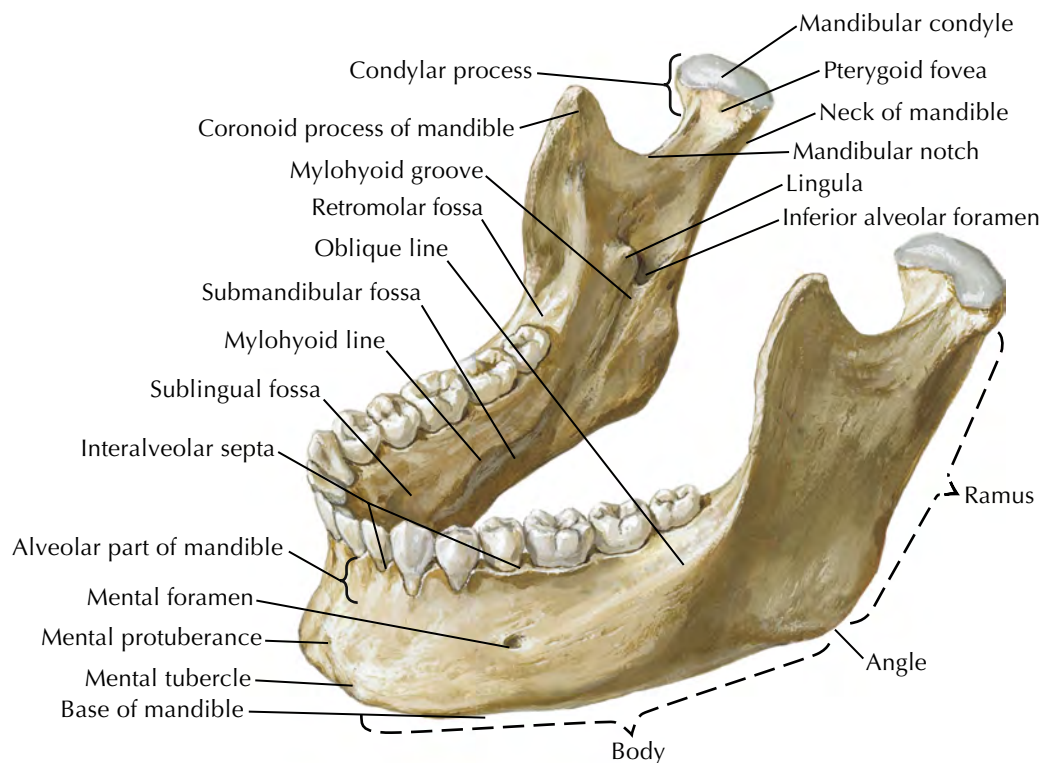
- SM** Superior nasal meatus
- MM** Middle nasal meatus
- IM** Inferior nasal meatus

Posterior view



Lateral view





Mandible of aged, edentulous person

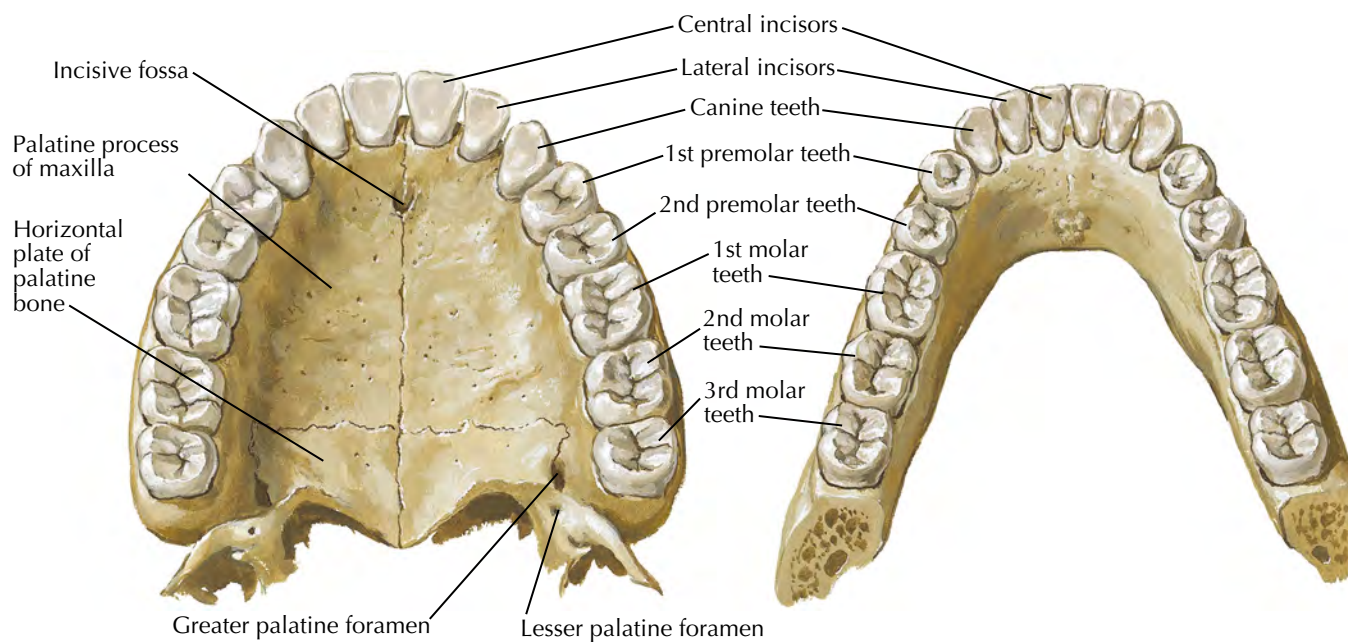
**Deciduous teeth
(usual age of
eruption)****Permanent
(usual age
of eruption)**

Central incisor
(8–10 months)
Lateral incisor
(8–10 months)
Canine tooth
(16–20 months)
1st molar tooth
(15–21 months)
2nd molar tooth
(20–24 months)

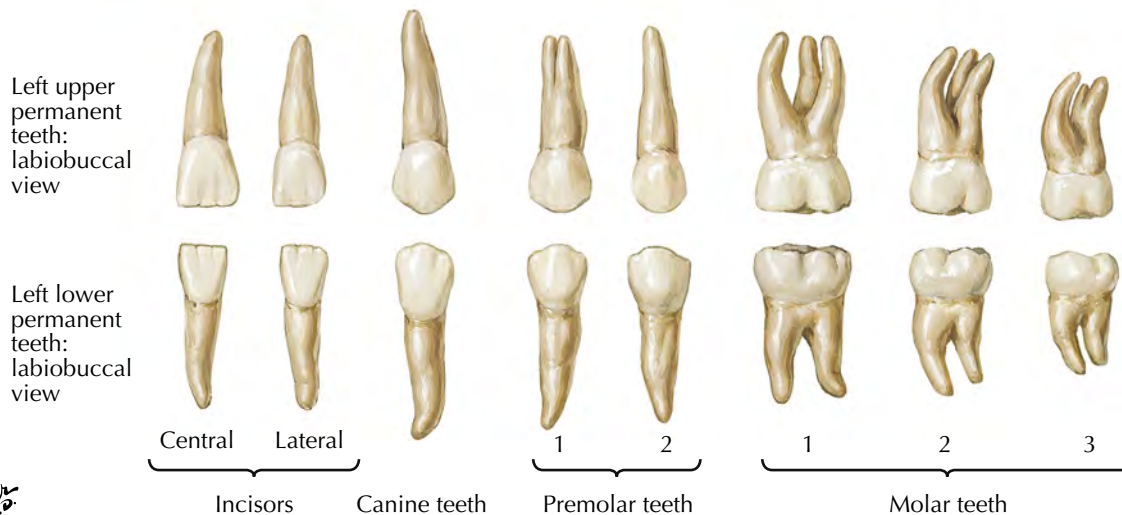
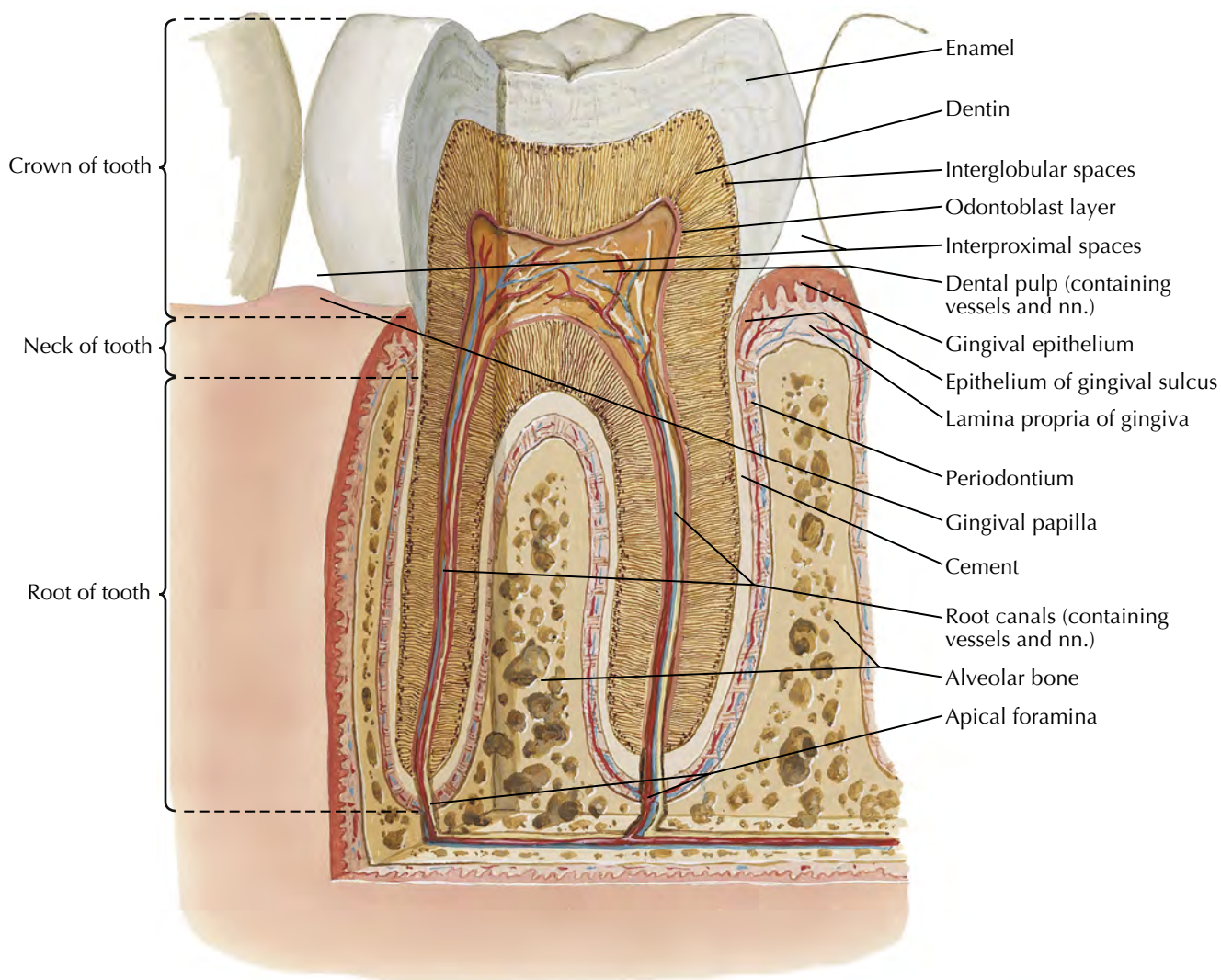
2nd molar tooth
(20–24 months)
1st molar tooth
(15–21 months)
Canine tooth
(16–20 months)
Lateral incisor
(15–21 months)
Central incisor
(6–9 months)

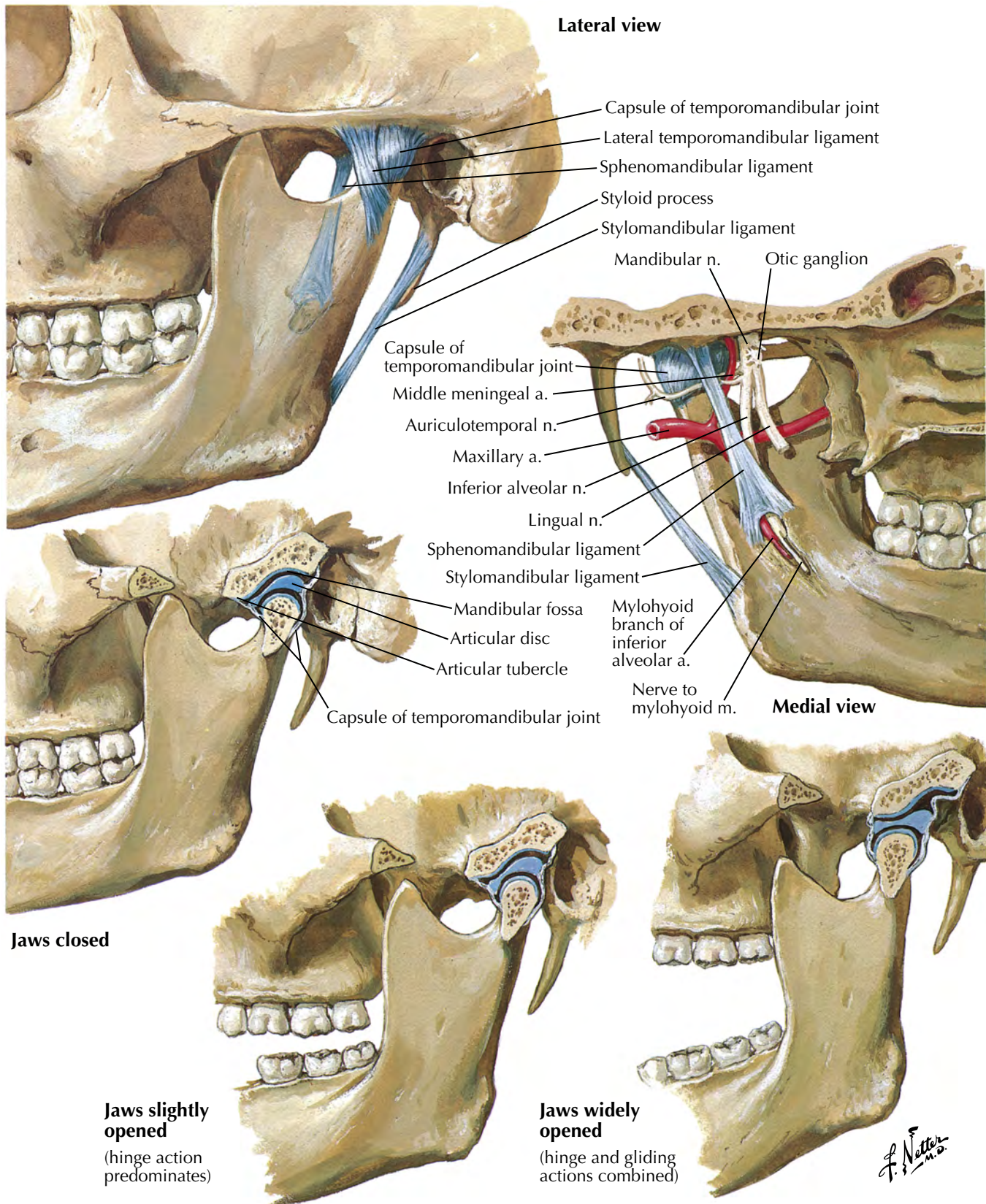
Central incisor (7th year)
Lateral incisor (8th year)
Canine tooth (11th–12th year)
1st premolar tooth (9th year)
2nd premolar tooth (10th year)
1st molar tooth (6th year)
2nd molar tooth (12th–13th year)
---3rd molar teeth (17th–25th year)

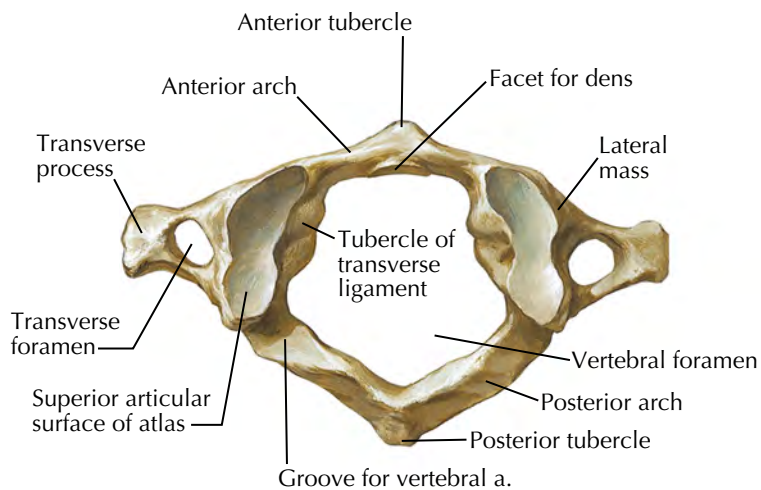
2nd molar tooth (12th–13th year)
1st molar tooth (6th year)
2nd premolar tooth (10th year)
1st premolar tooth (9th year)
Canine tooth (11th–12th year)
Lateral incisor (8th year)
Central incisor (7th year)

**Upper permanent teeth****Lower permanent teeth**

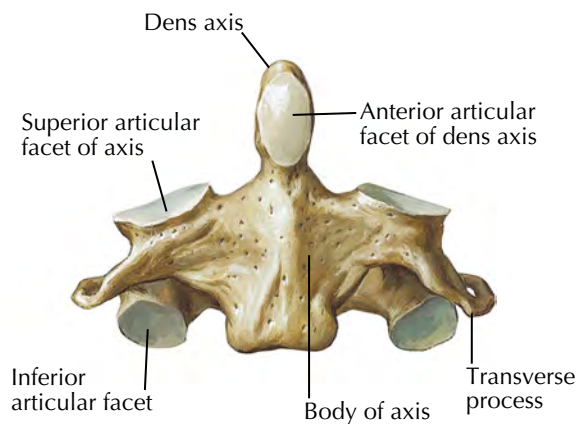
F. Netter M.D.



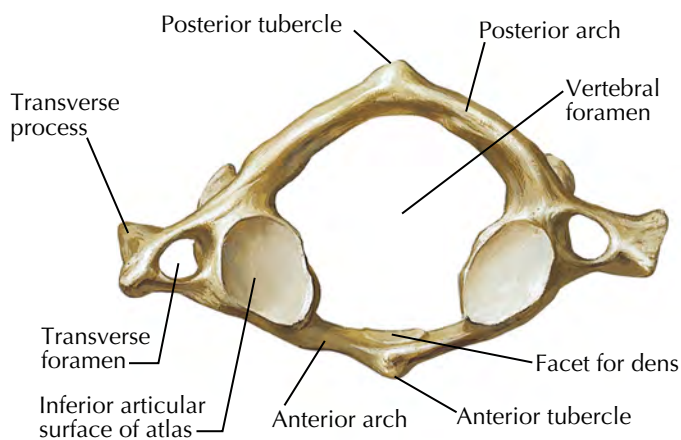




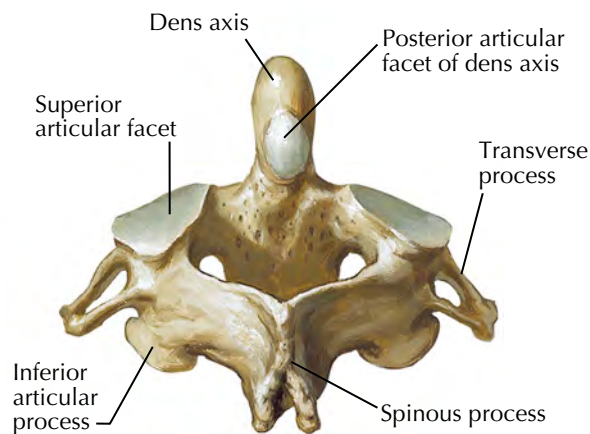
Atlas: superior view



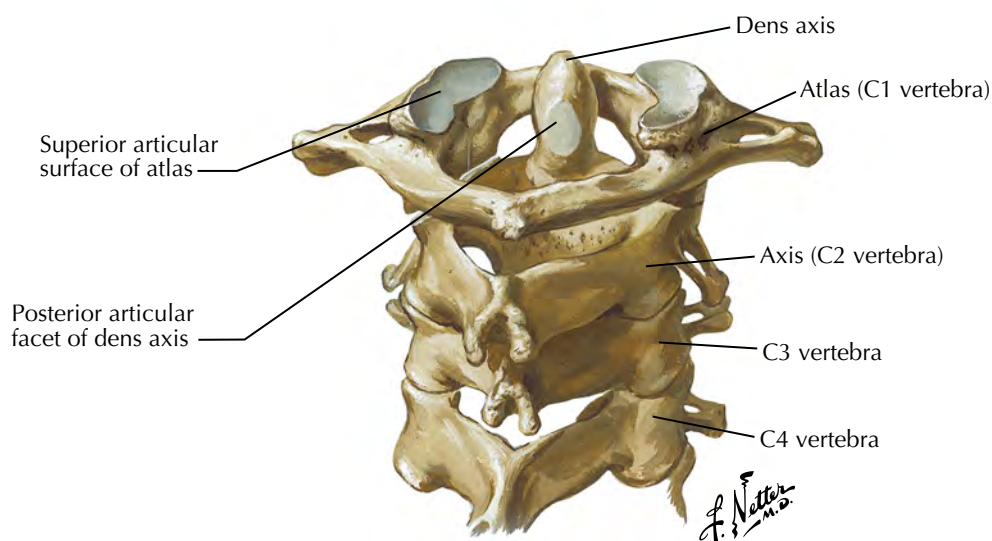
Axis: anterior view



Atlas: inferior view

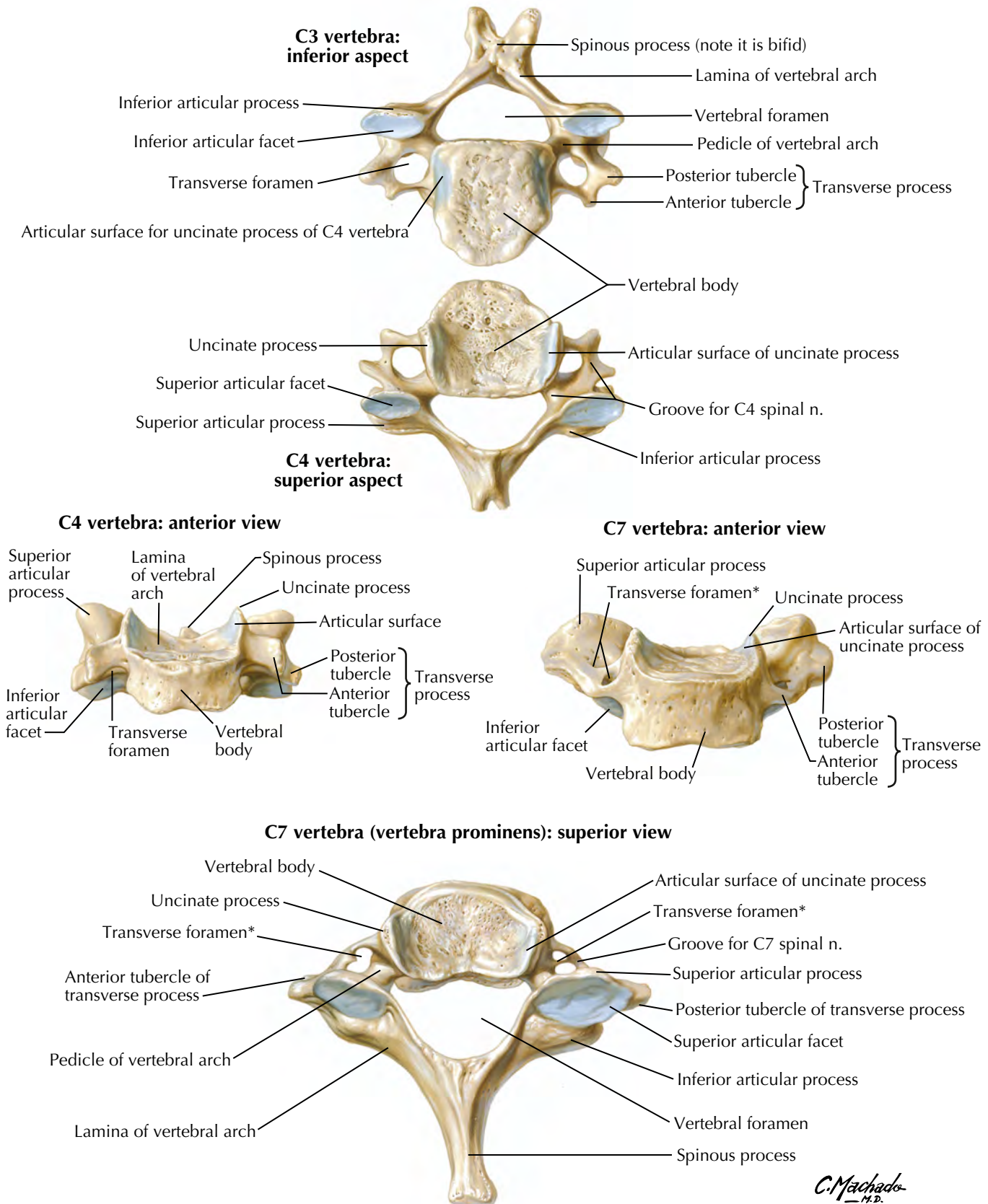


Axis: posterosuperior view



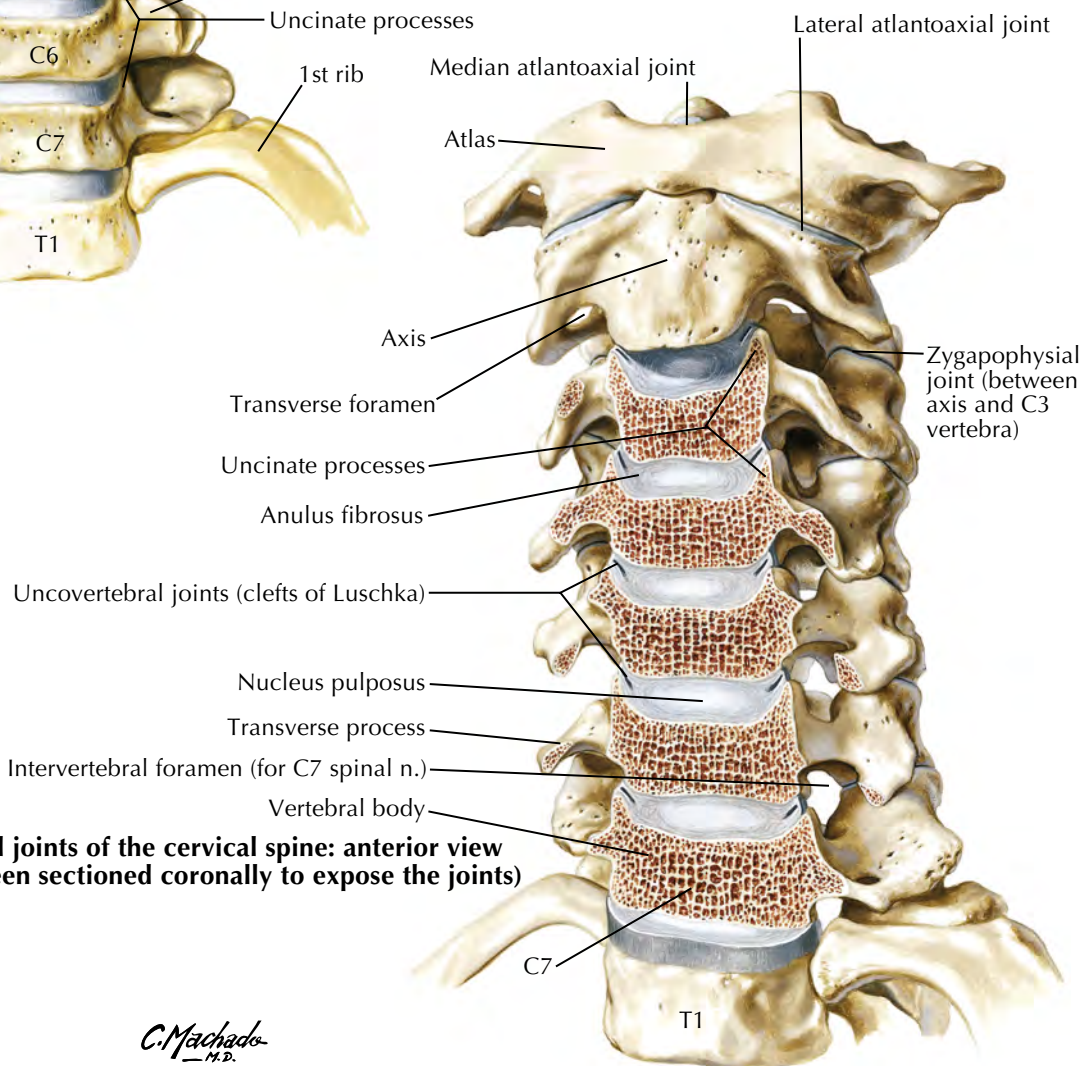
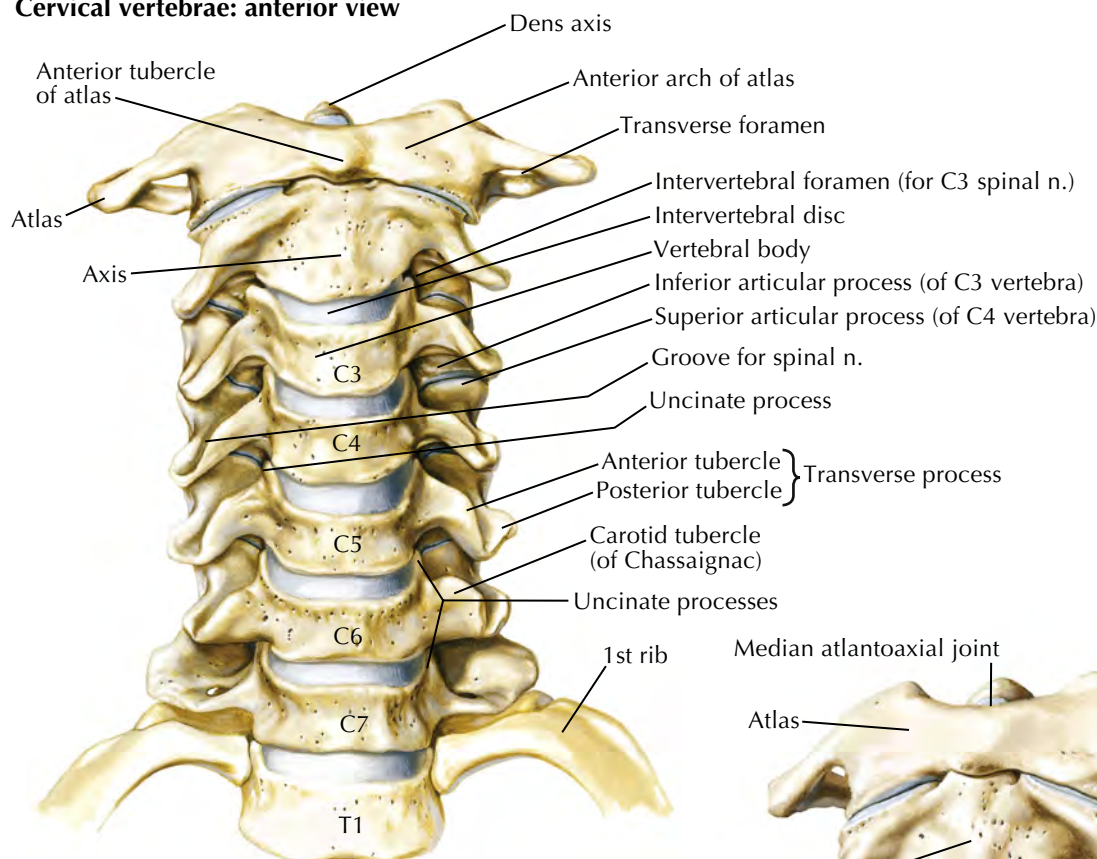
Upper cervical vertebrae: posterosuperior view

Inferior aspect of C3 vertebra and superior aspect of C4 vertebra showing the sites of the articular surfaces of the uncovertebral joints



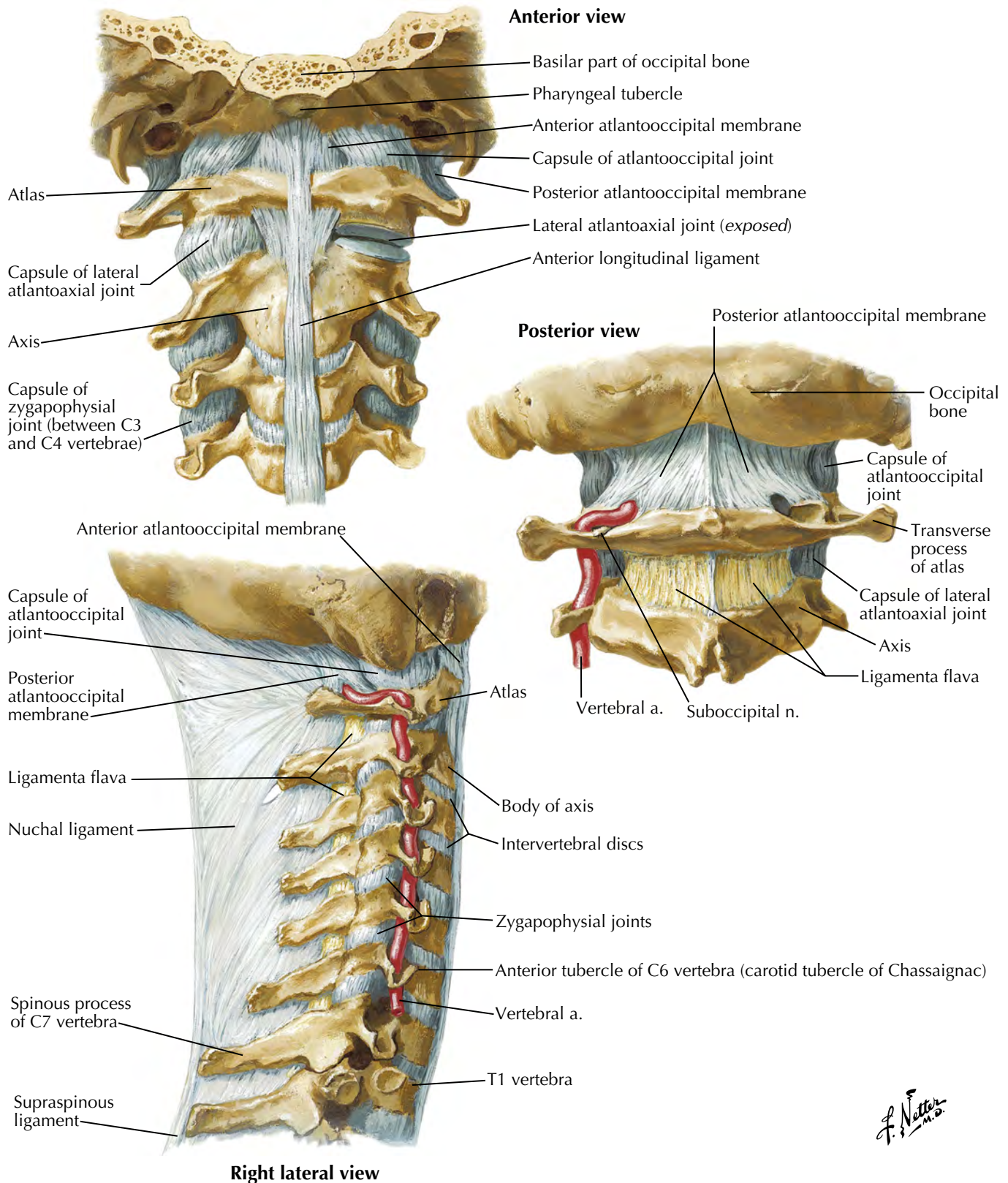
*The transverse foramina of C7 transmit vertebral veins, but usually not the vertebral artery, and are asymmetrical in these drawings. Note the right transverse foramen is septated.

Cervical vertebrae: anterior view

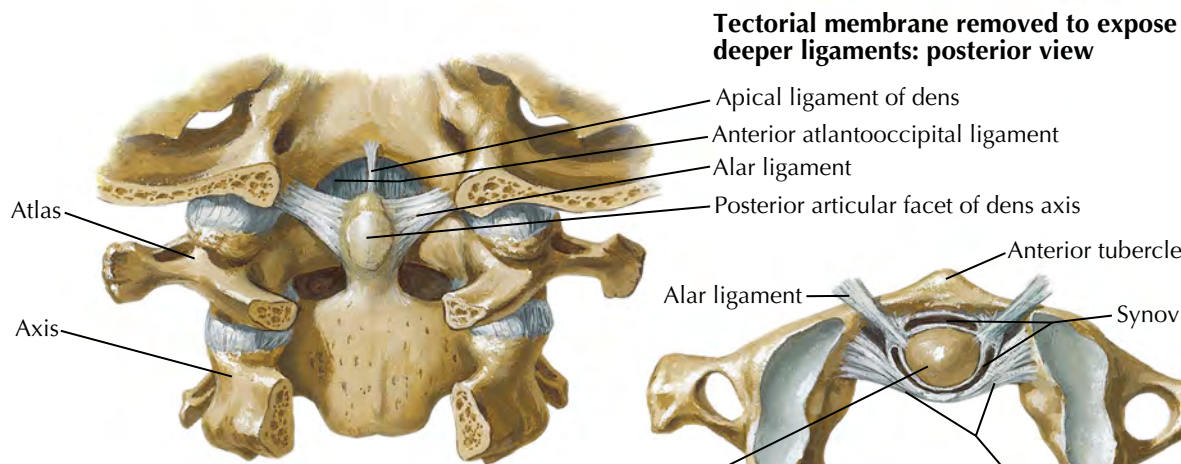
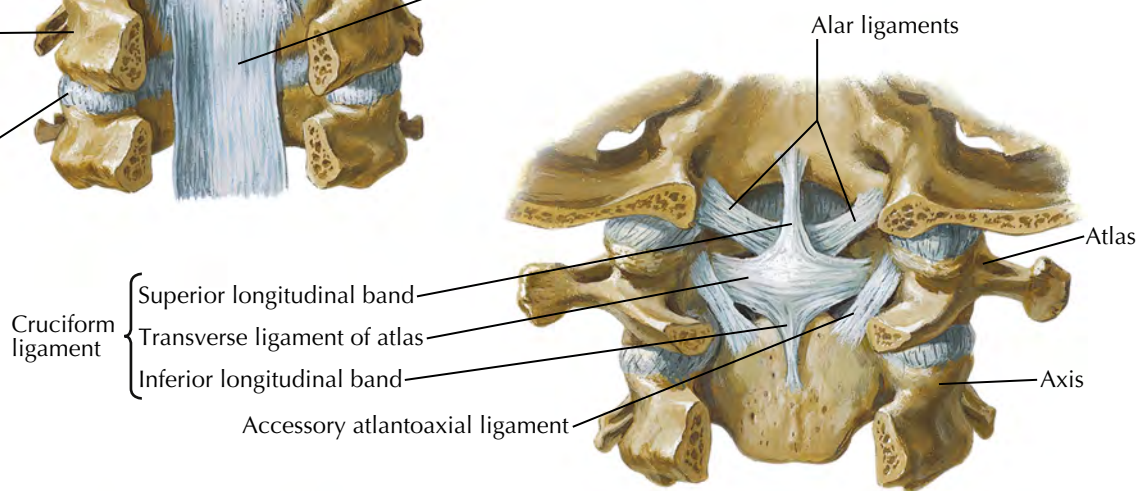
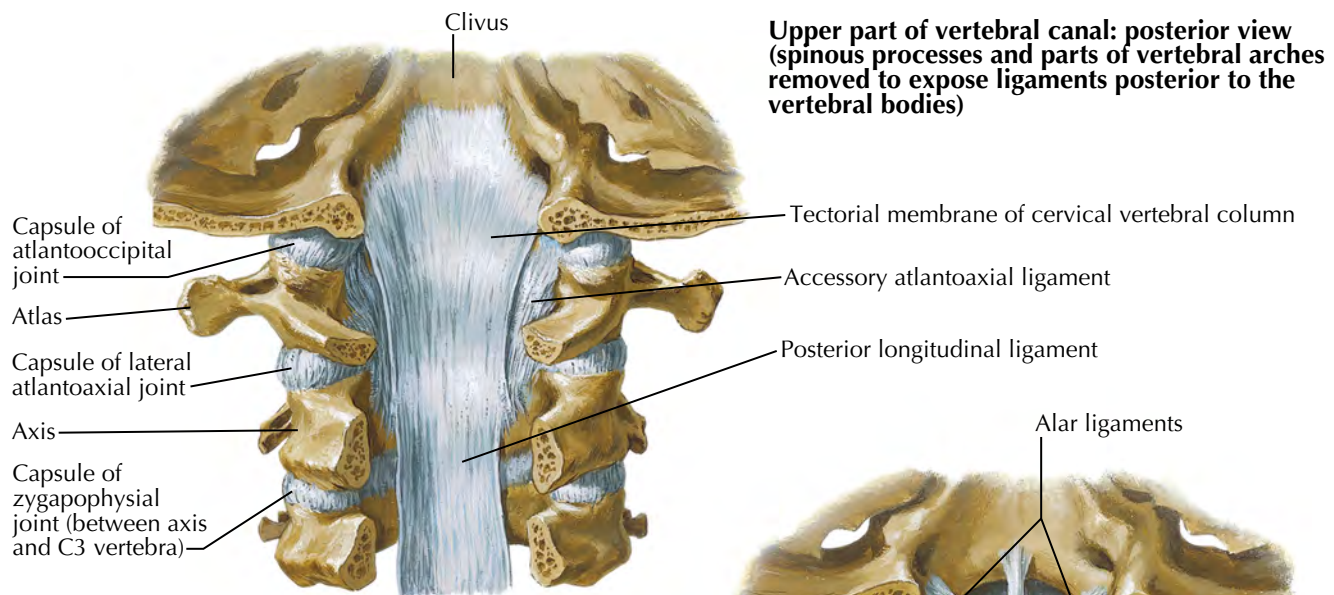


The uncovertebral joints of the cervical spine: anterior view
(C3 to C7 have been sectioned coronally to expose the joints)

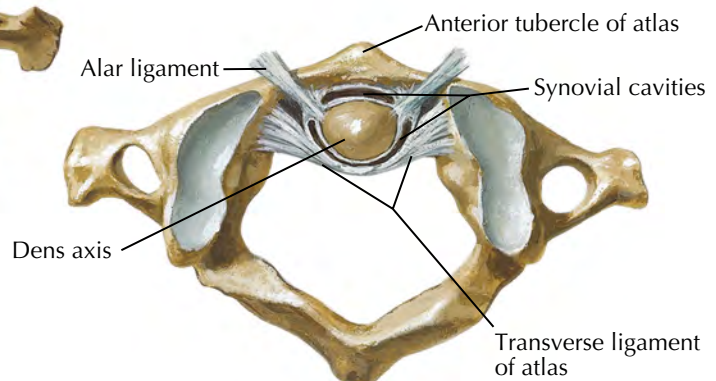
C. Machado
—M.D.—



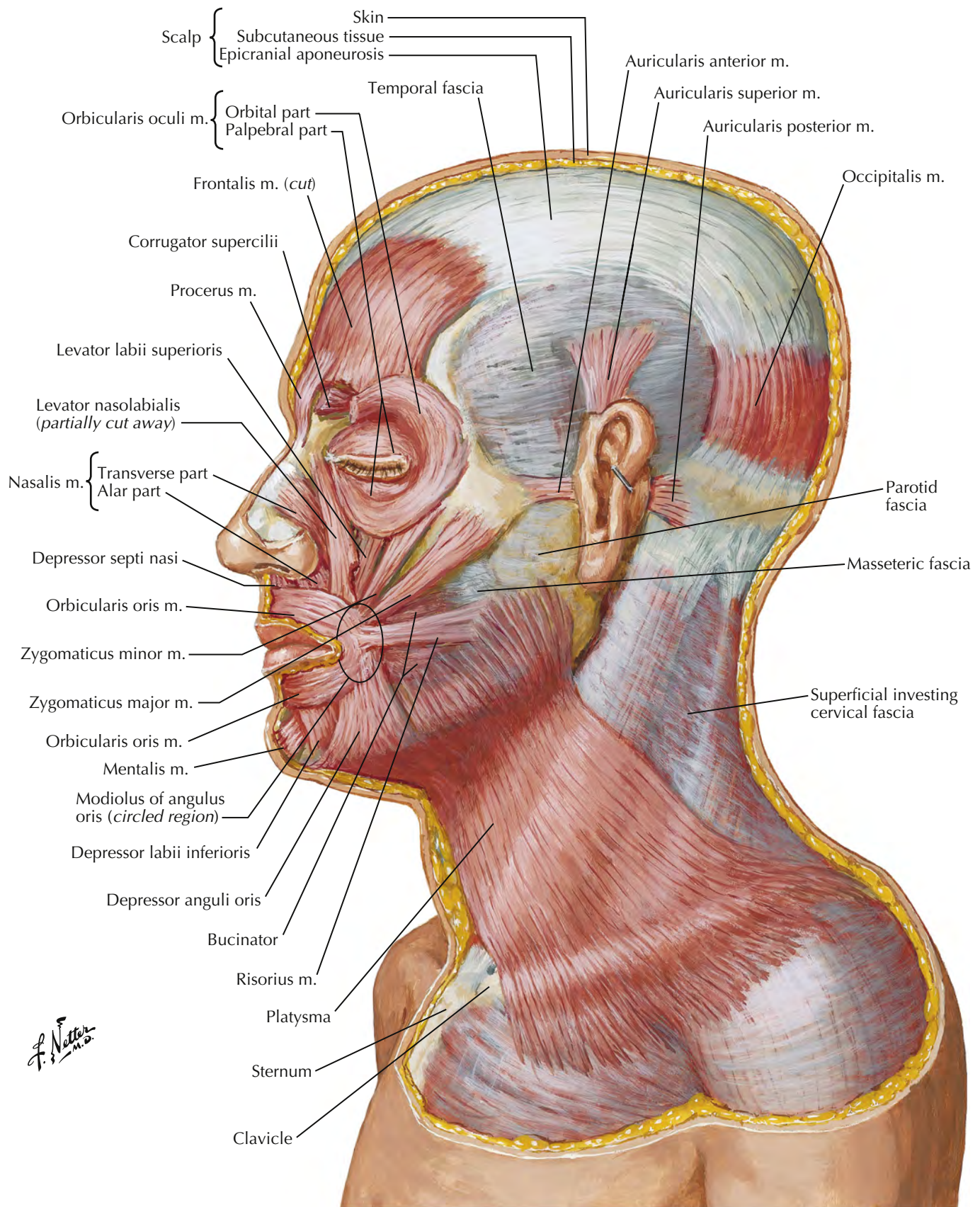
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Cruciform ligament removed to show deeper ligaments: posterior view

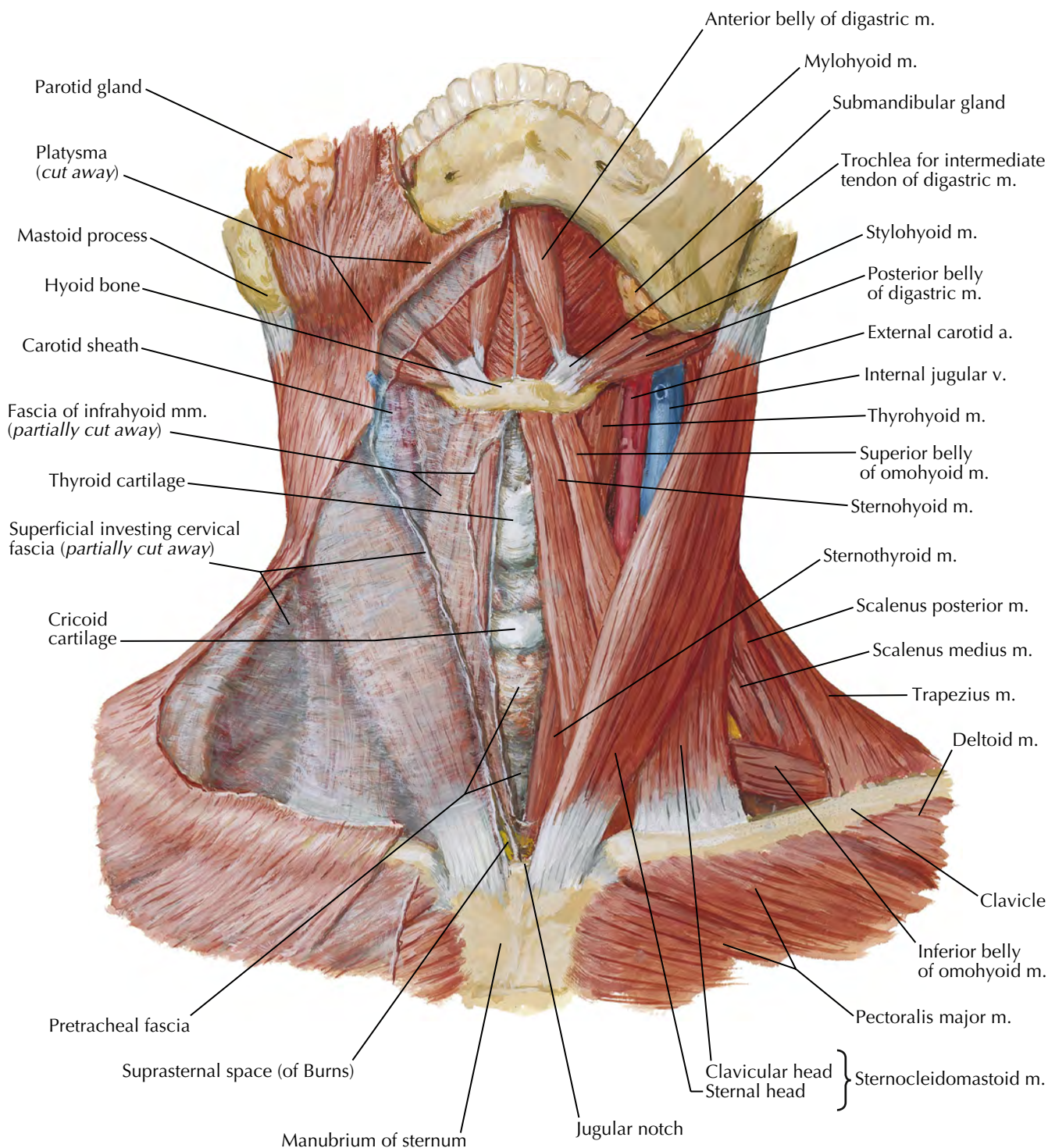


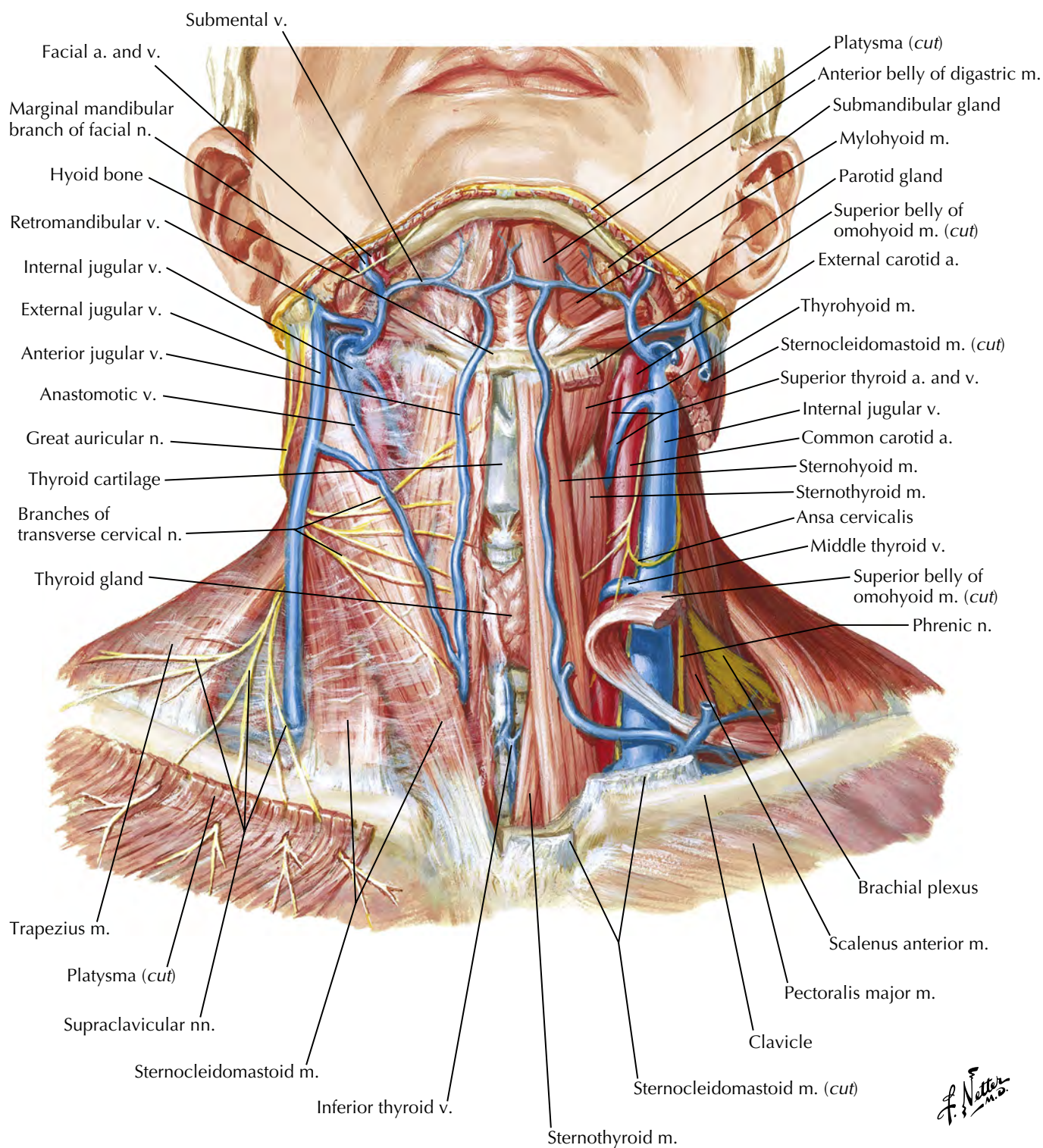
Median atlantoaxial joint: superior view



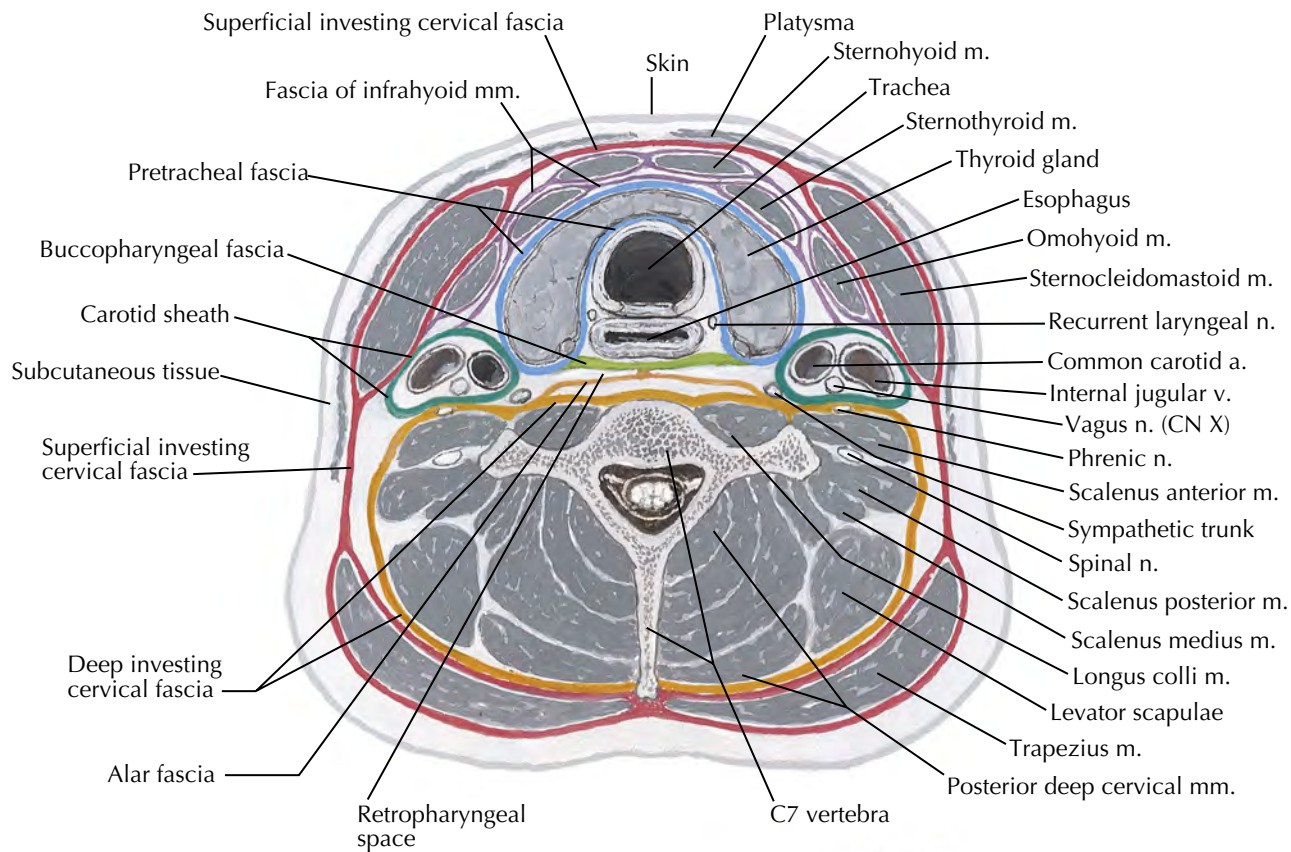
Muscles of Neck: Anterior View

See also Plates 37, 50, 51, 53

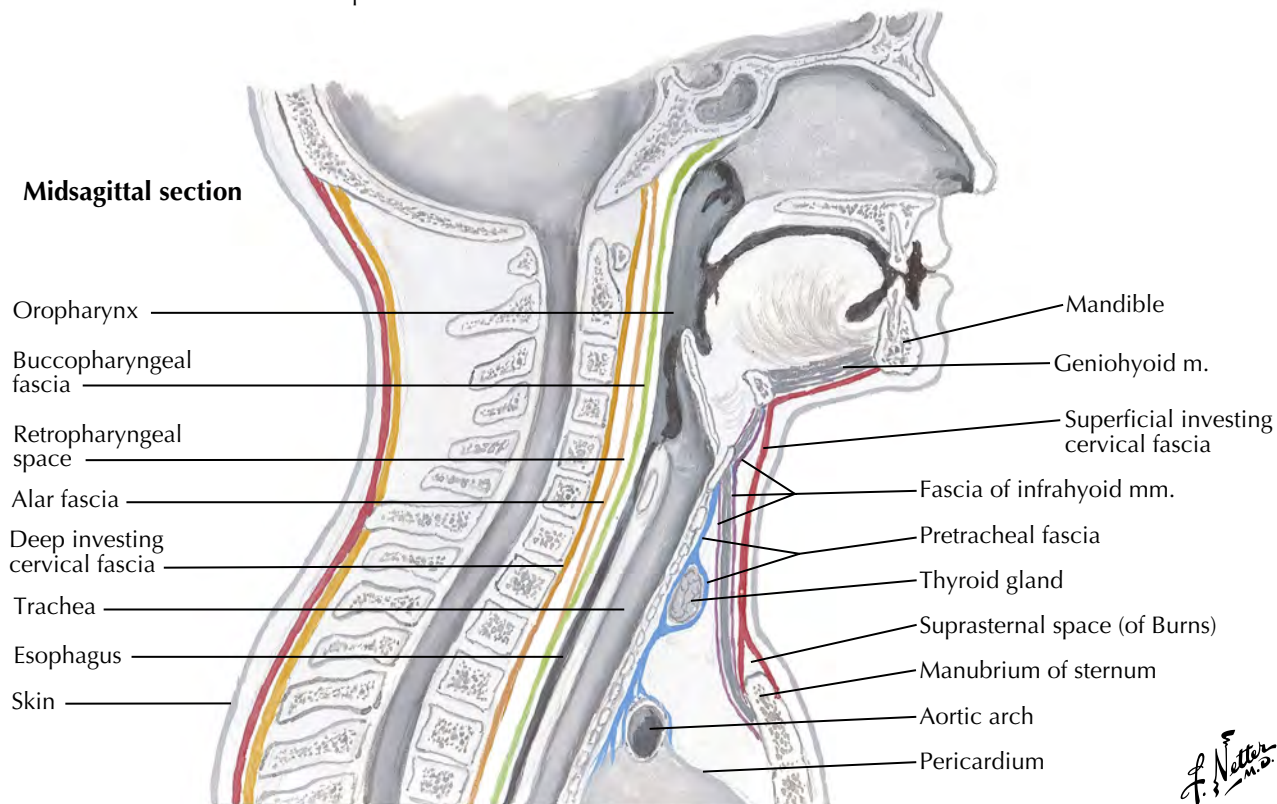


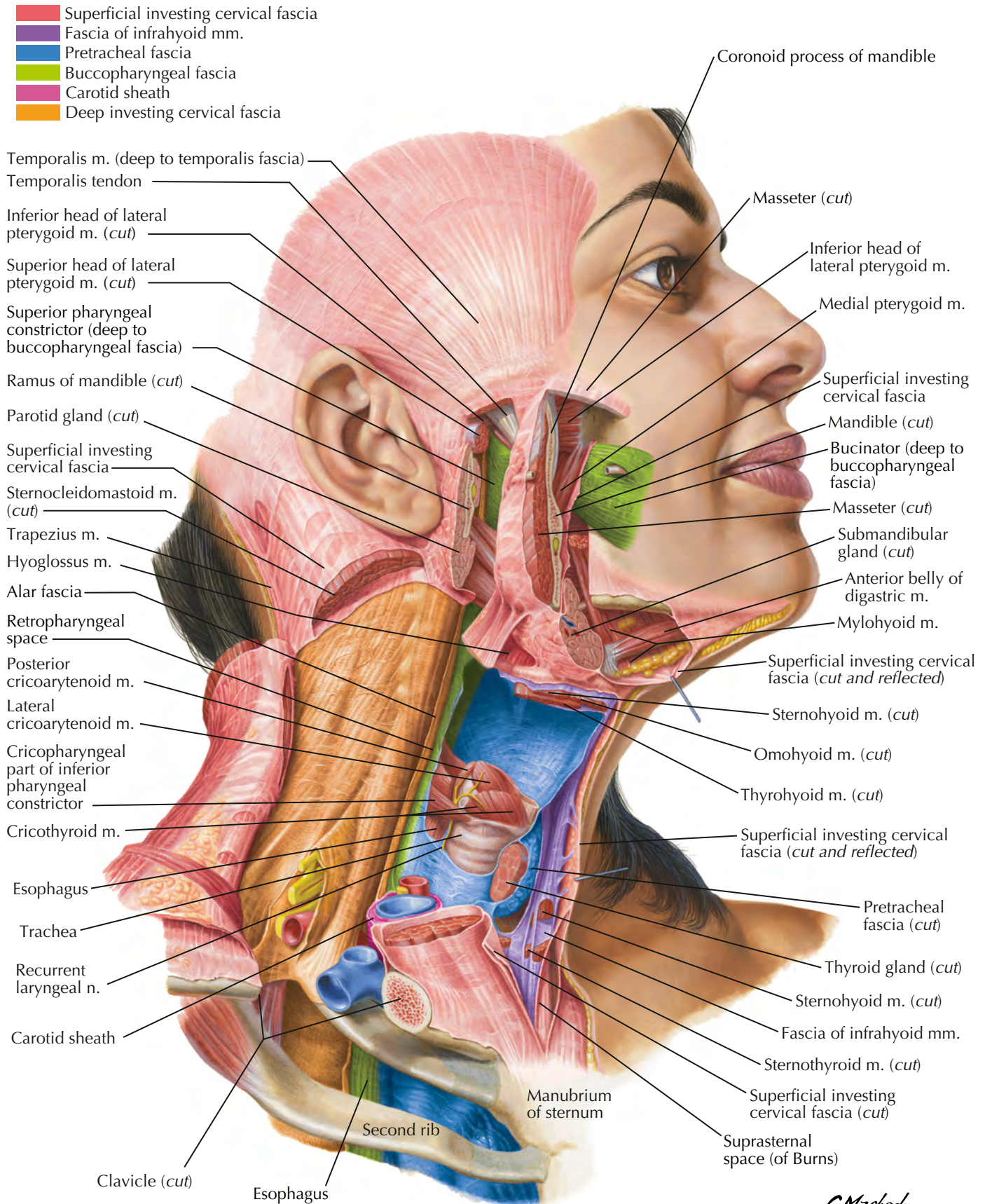


Cross section



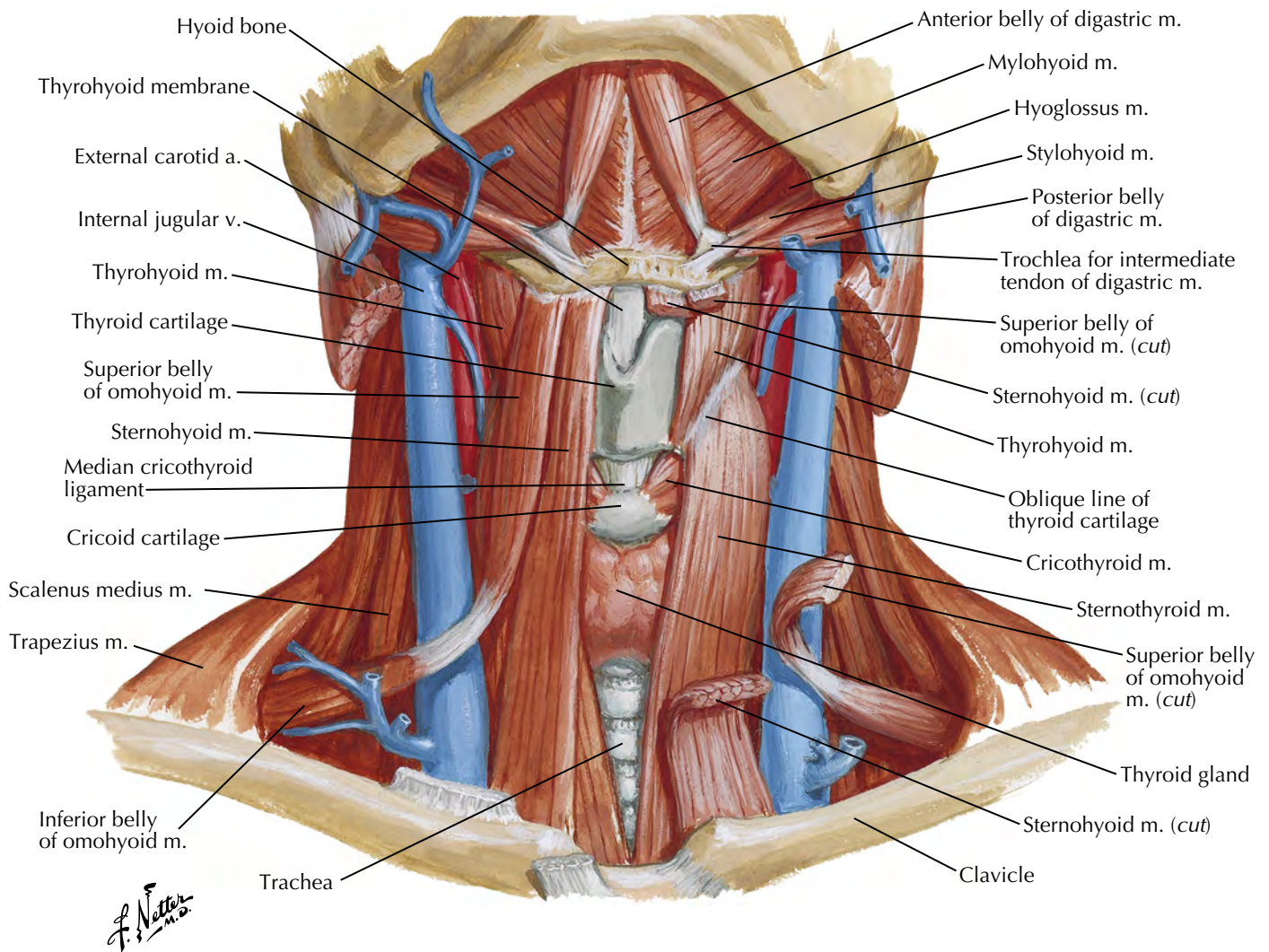
Midsagittal section

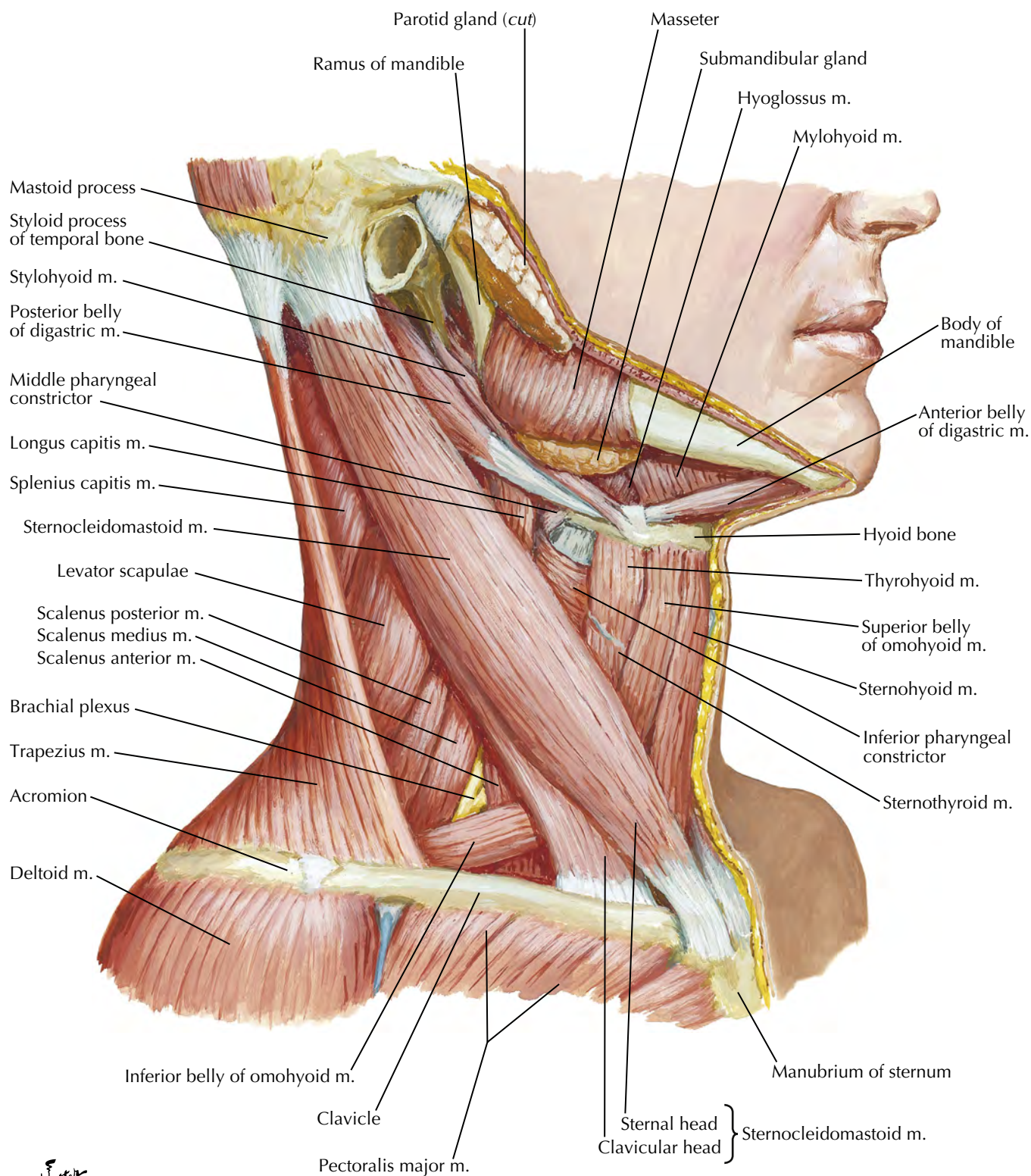


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—M.D.

Infrahyoid and Suprahyoid Muscles

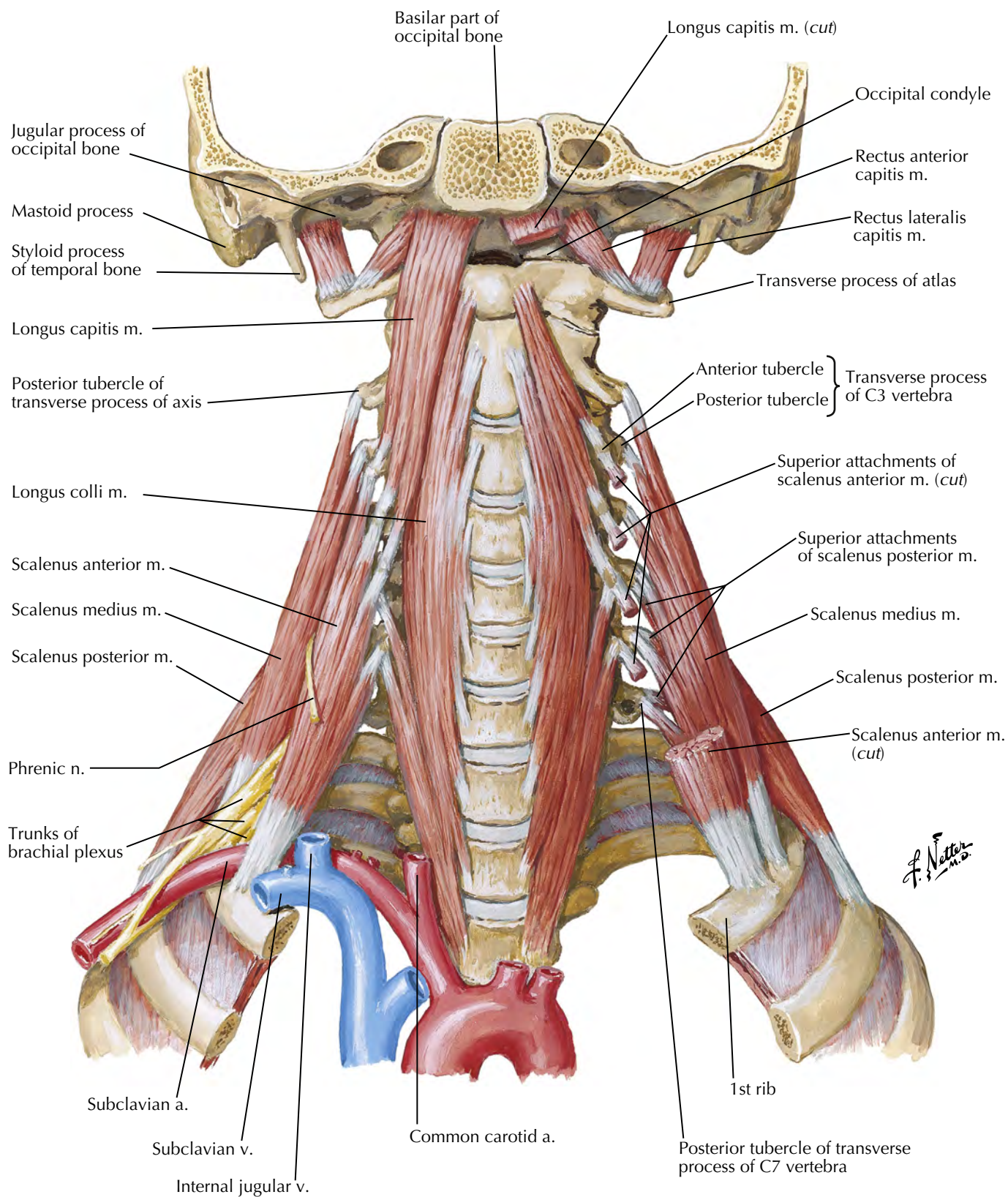
See also Plates 49, 54, 87

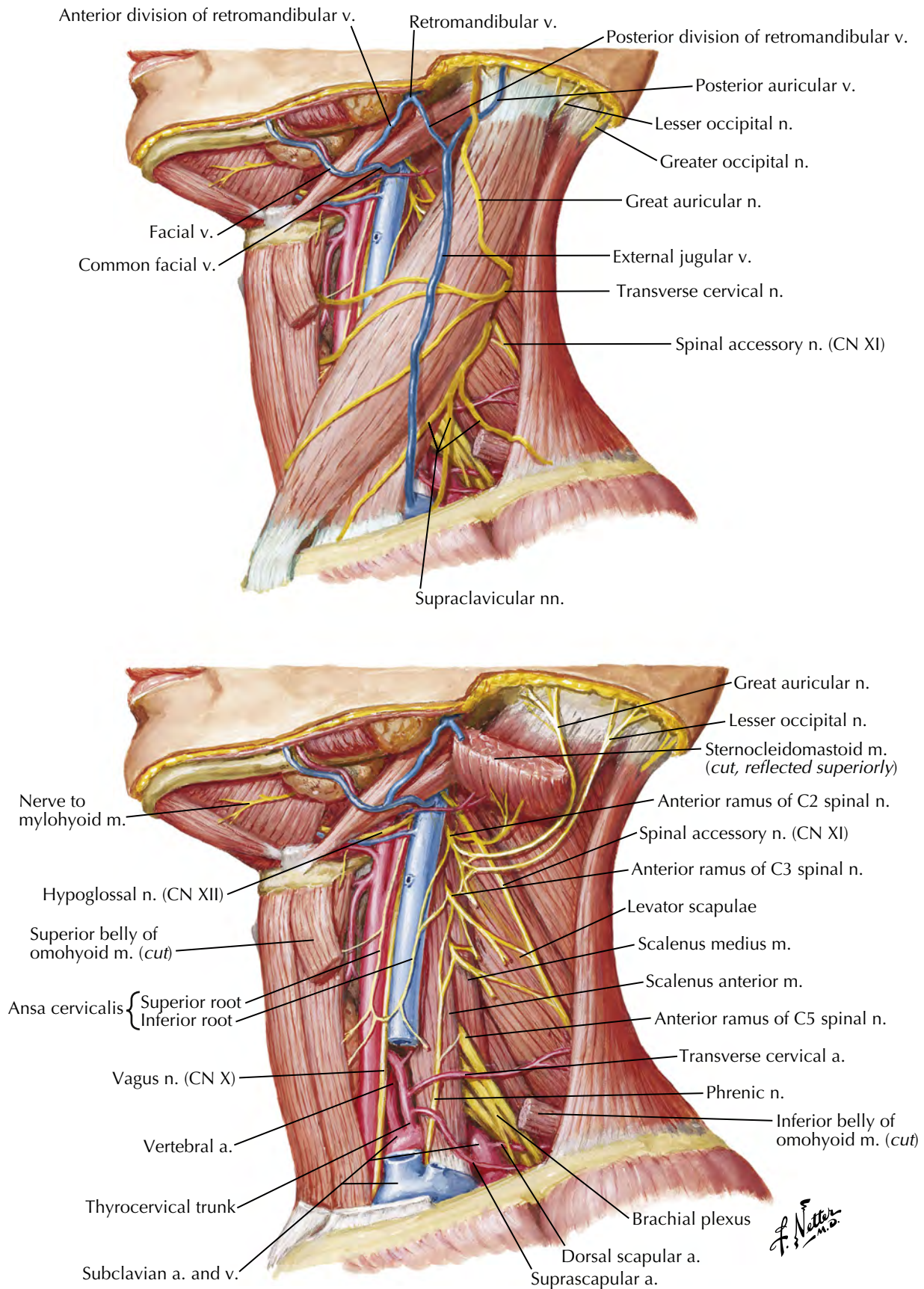




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Anterior and Lateral Cervical Muscles





Cervical plexus: schema

(S = gray ramus communicans from superior cervical ganglion)

Geniohyoid branch of hypoglossal n.

Thyrohyoid branch of hypoglossal n.

Transverse cervical n.

Nerve to superior belly of omohyoid m.

Ansa cervicalis { Superior root
Inferior root

Nerve to sternothyroid m.

Nerve to sternohyoid m.

Nerve to inferior belly of omohyoid m.

Supraclavicular nn.

Hypoglossal n. (CN XII)

Spinal accessory n. (CN XI)

Great auricular n.

Lesser occipital n.

Nerves to rectus lateralis capitis, longus capitis, and rectus anterior capitis mm.

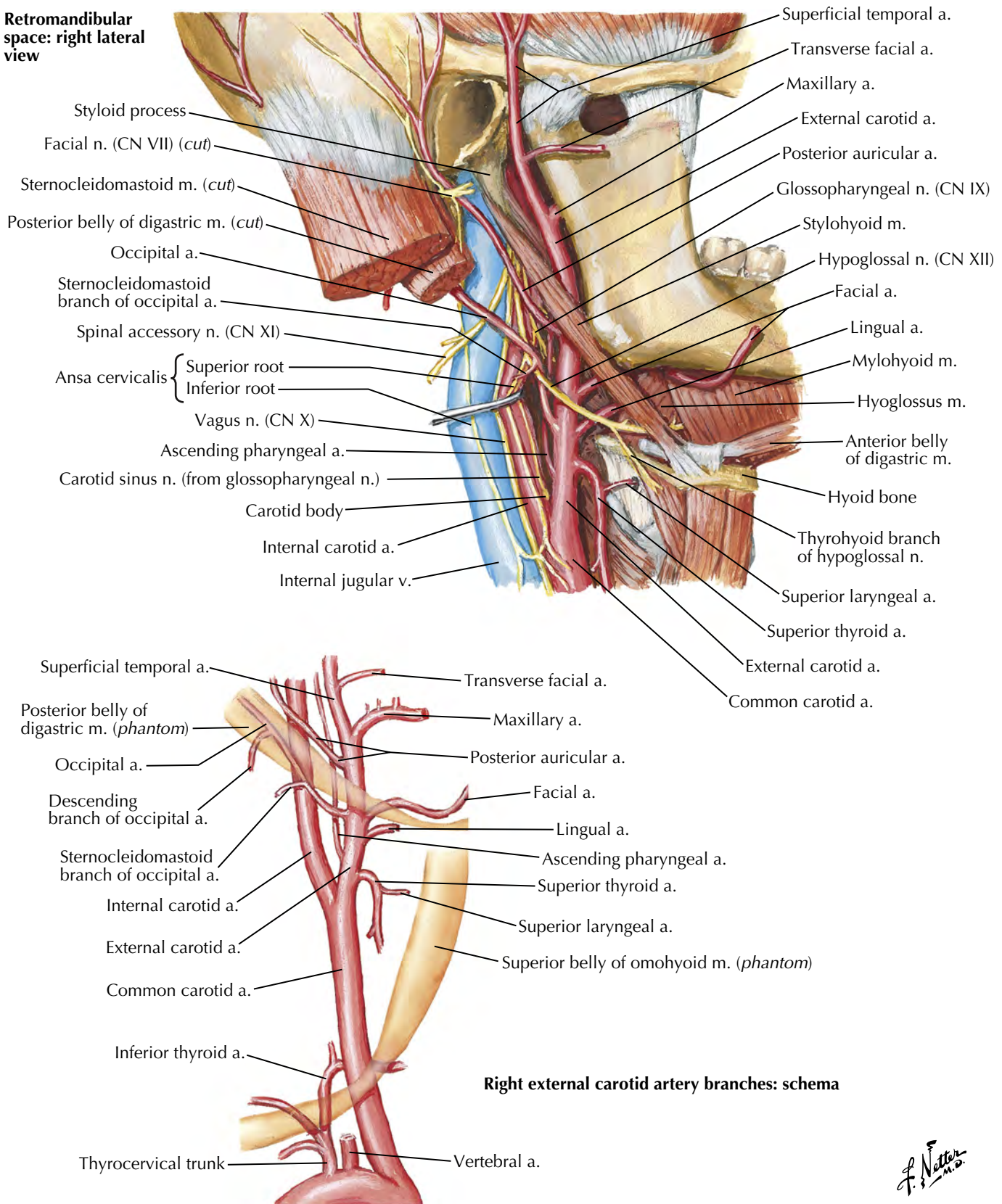
Nerves to longus capitis and longus colli mm.

Nerves to scalenus mm. and levator scapulae

Phrenic n.

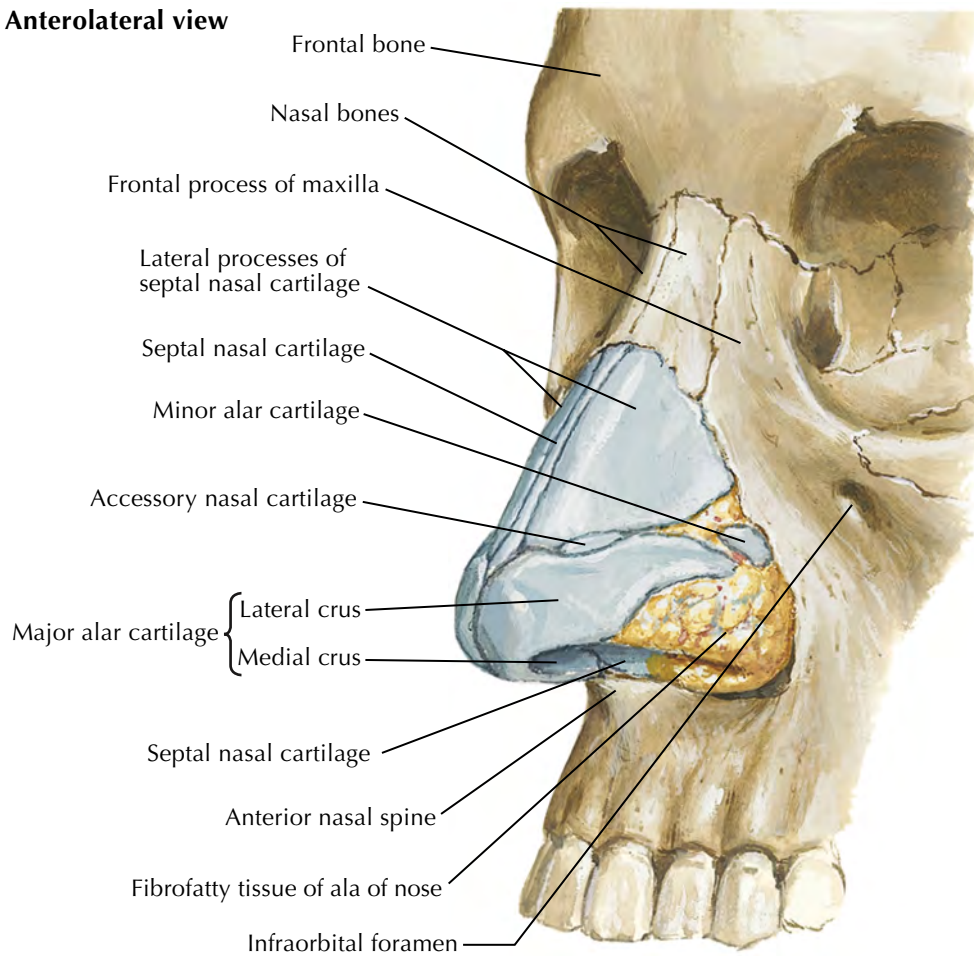
Right anterior view

Internal jugular v.
Common carotid a.
Ascending cervical a.
Phrenic n.
Scalenus anterior m.
Inferior thyroid a.
Transverse cervical a.
Brachial plexus
Suprascapular a.
Dorsal scapular a.
Costocervical trunk
Thyrocervical trunk
Subclavian a. and v.
Thyroid gland (retracted)
Middle cervical ganglion
Vagus n. (CN X)
Vertebral a.
Common carotid a.
Recurrent laryngeal n.
Brachiocephalic trunk
Internal jugular v. (cut)

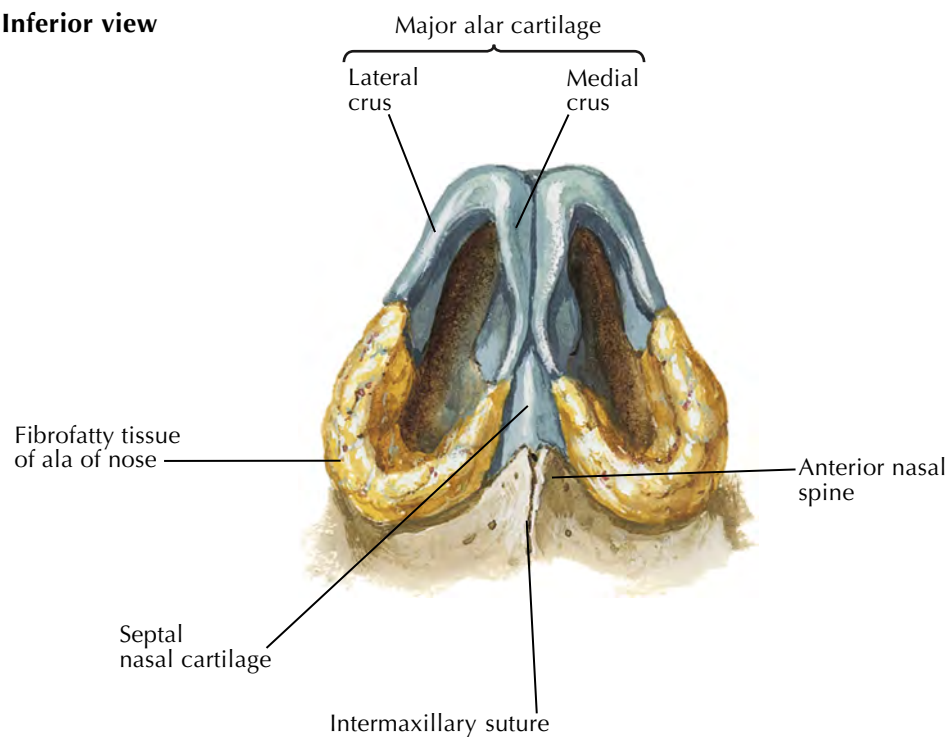
**Retromandibular
space: right lateral
view**

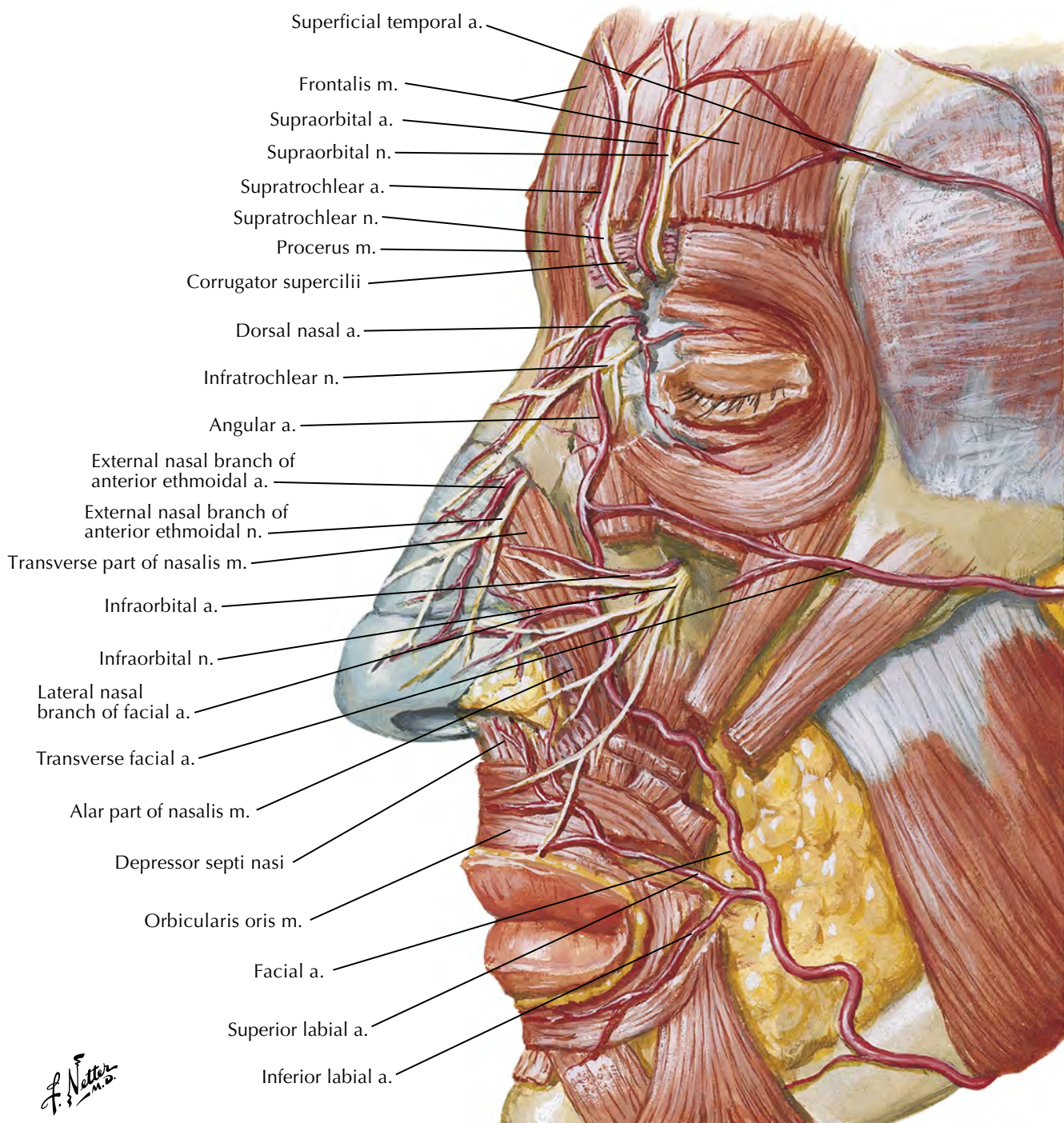
F. Netter M.D.

Anterolateral view

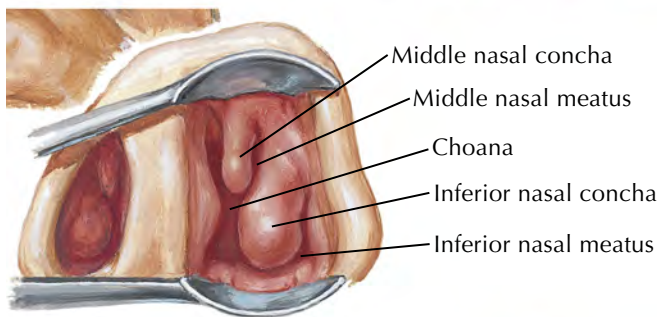
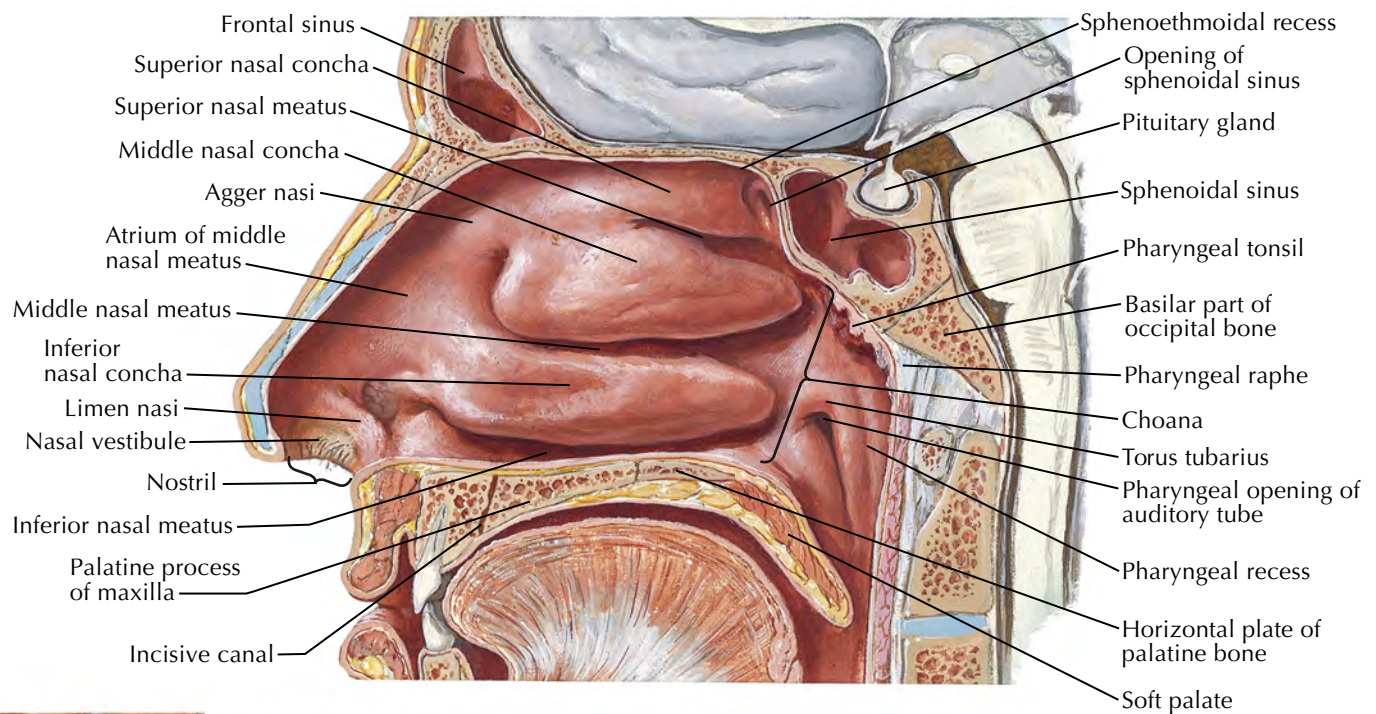


Inferior view

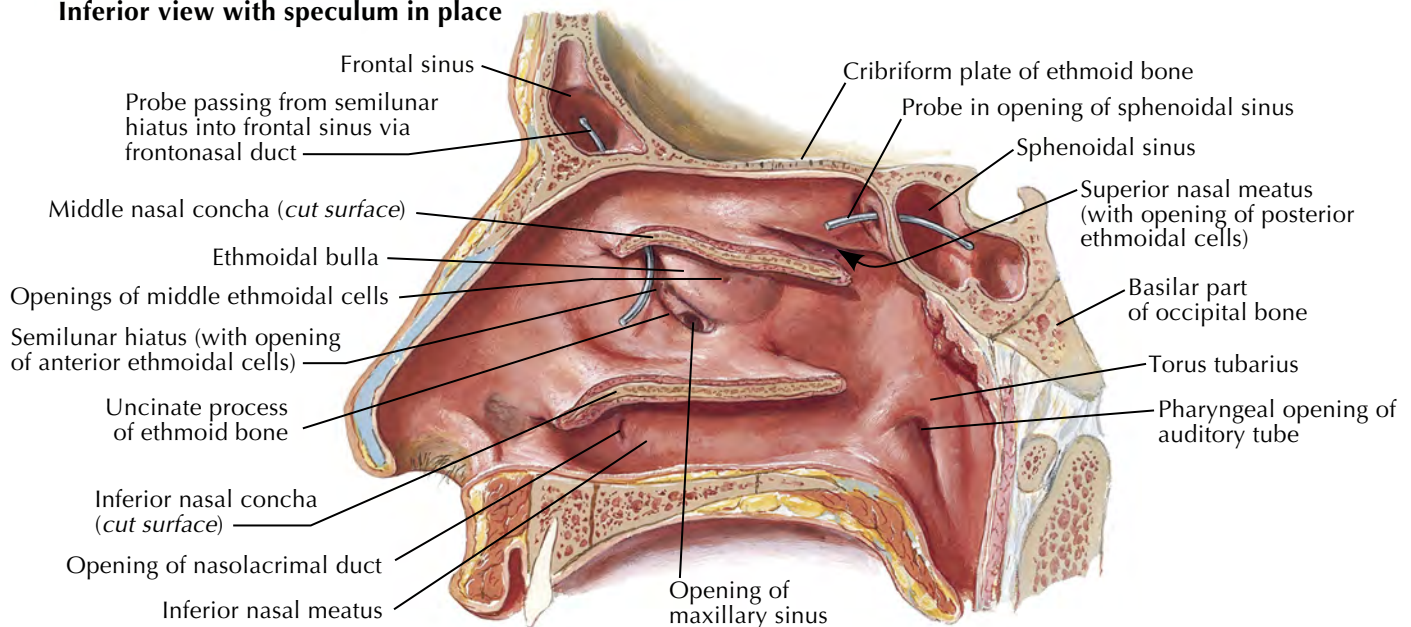


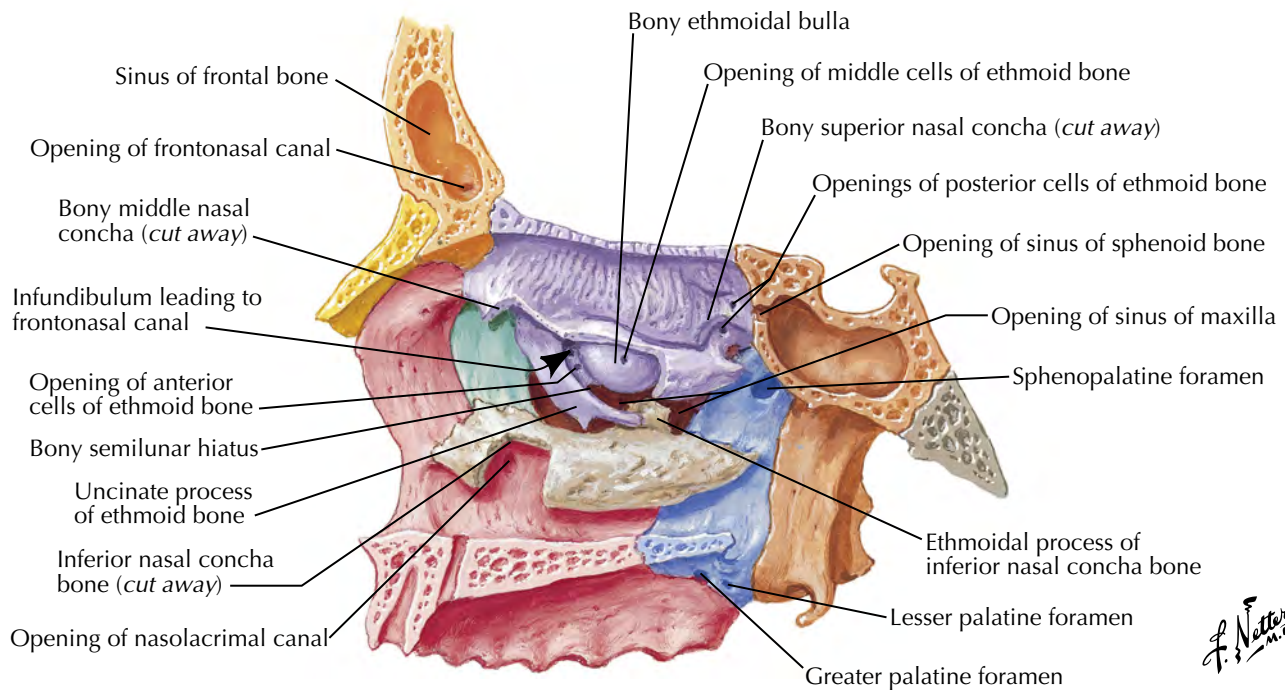
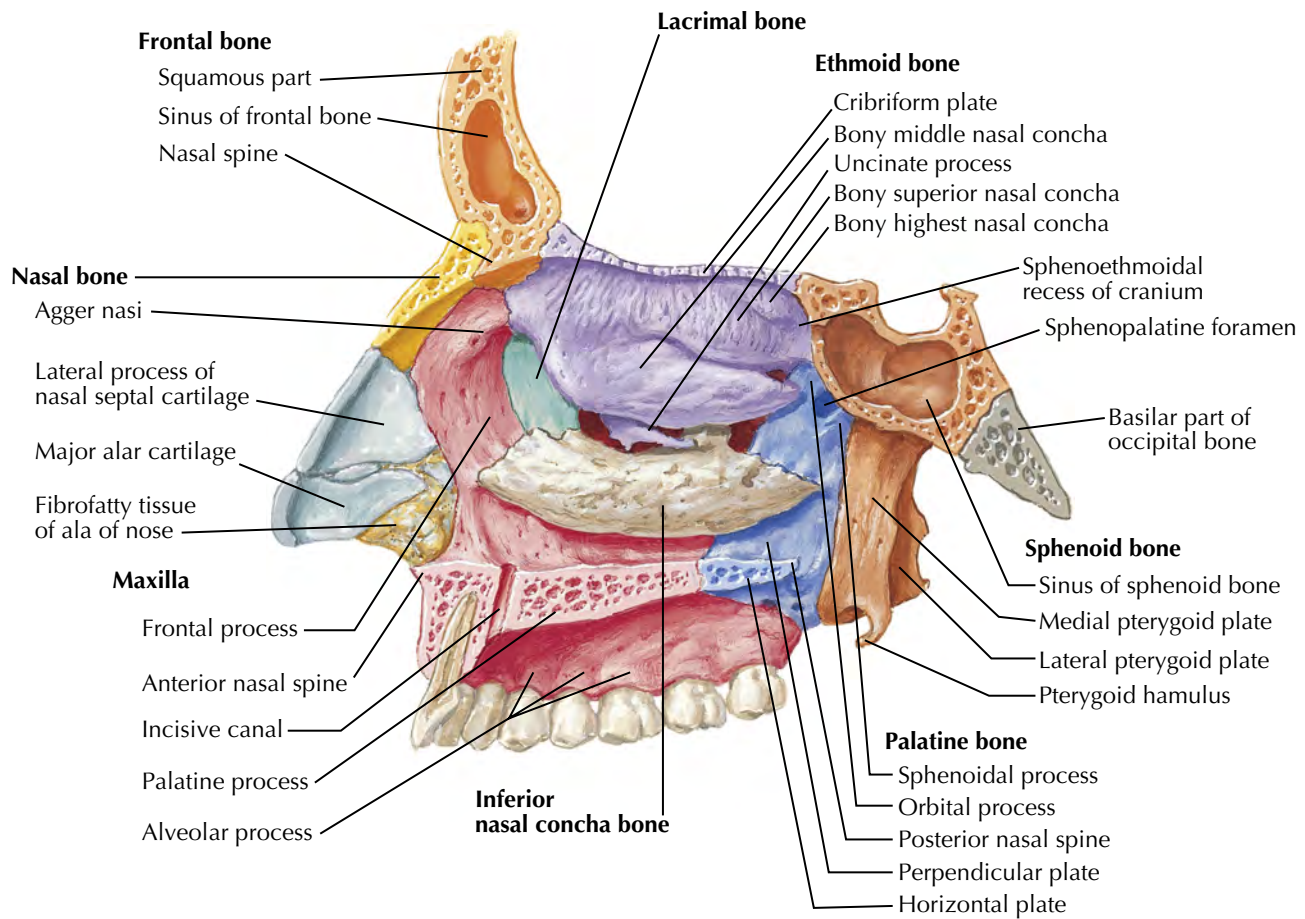


Lateral Wall of Nasal Cavity



Inferior view with speculum in place

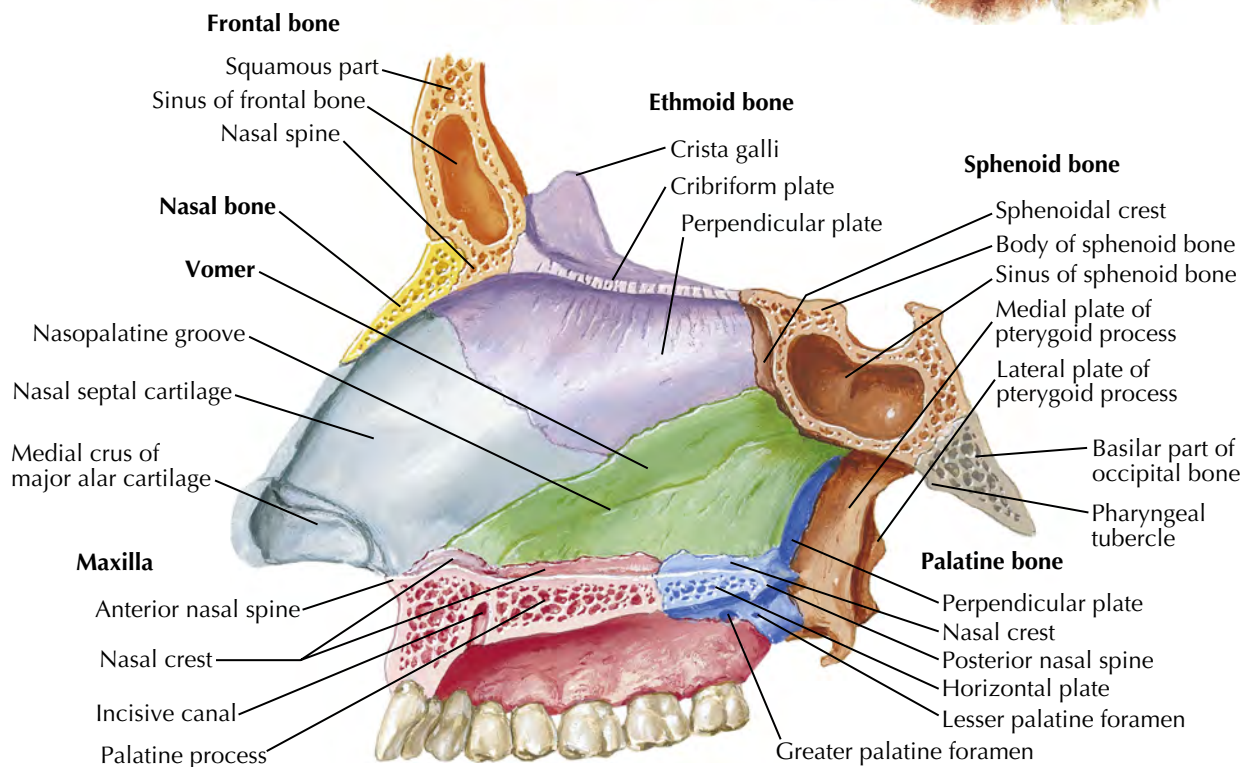
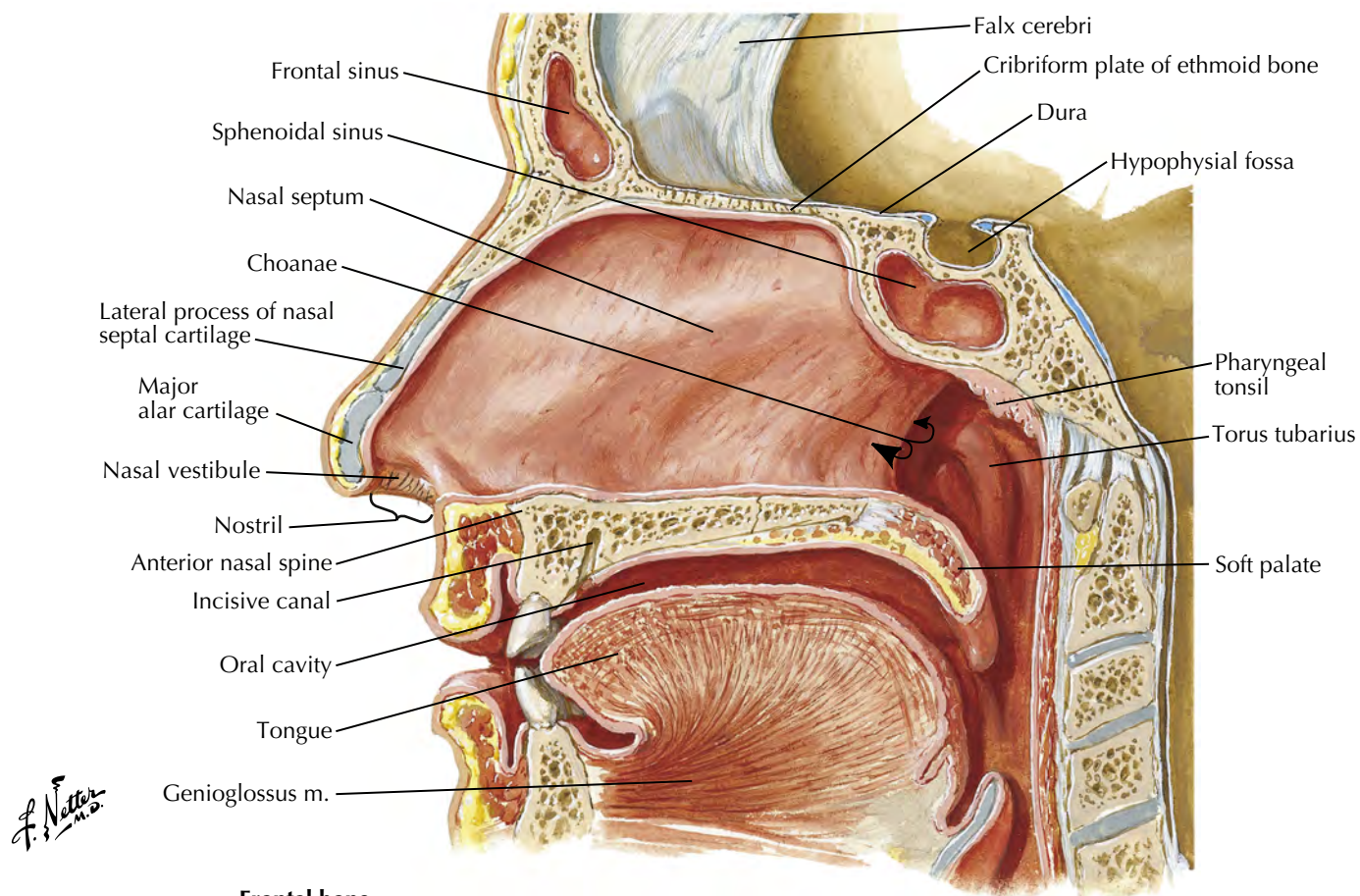




F. Netter M.D.

Medial Wall of Nasal Cavity (Nasal Septum)

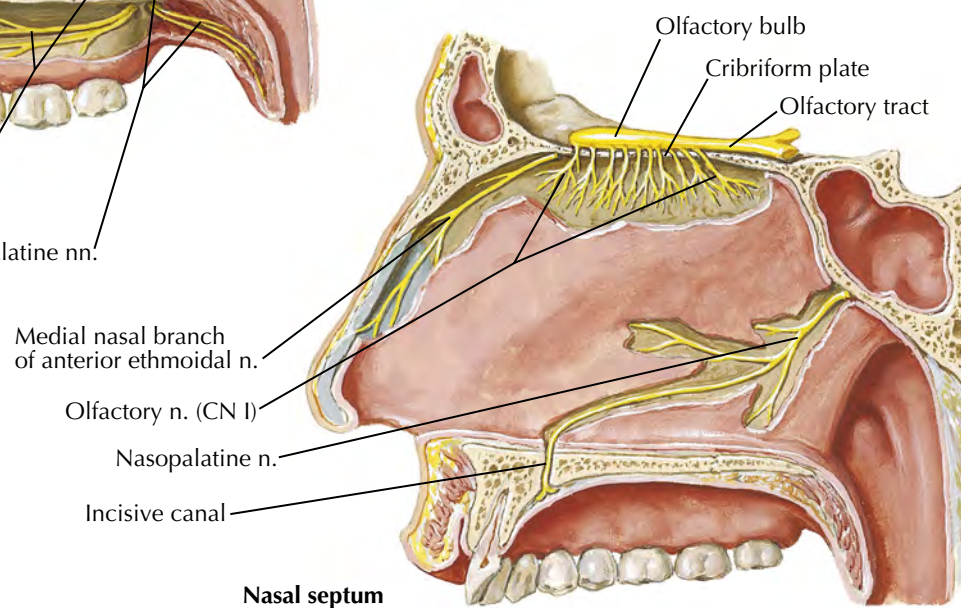
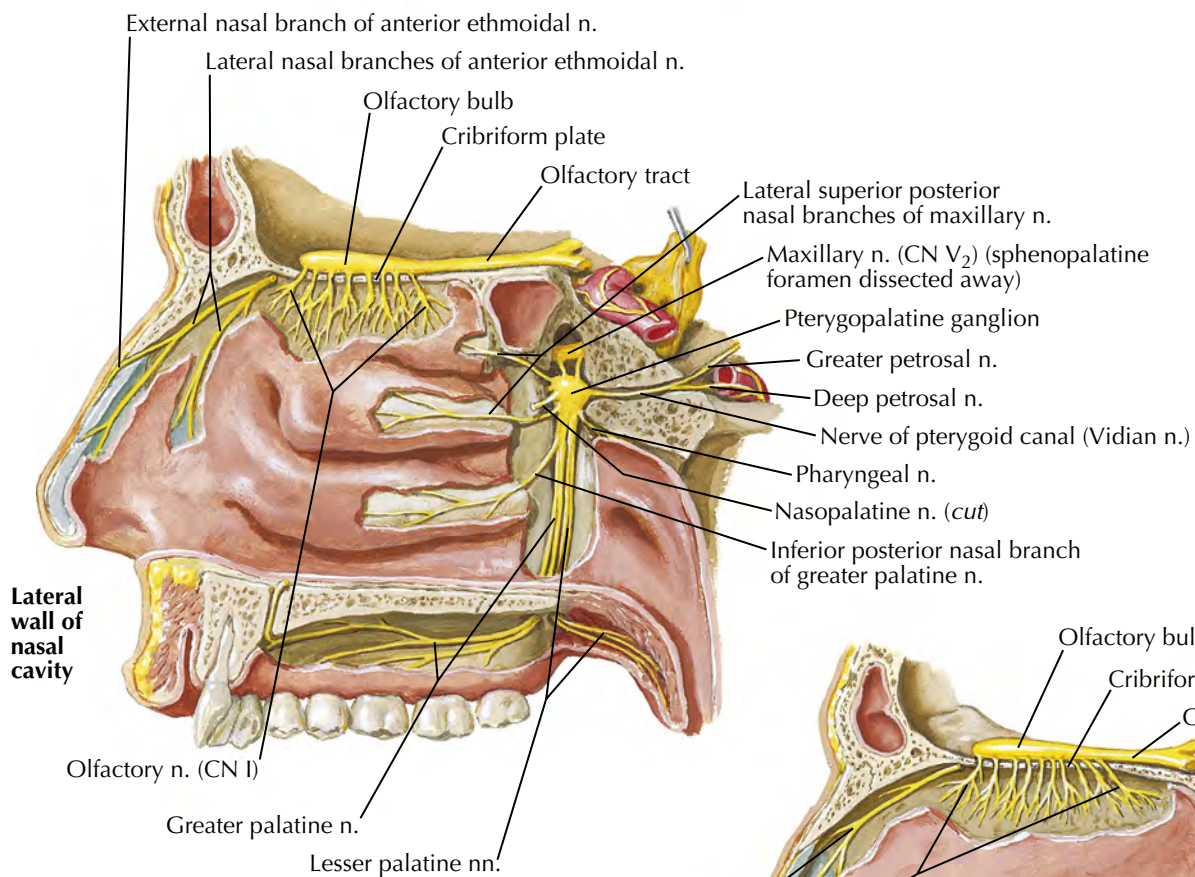
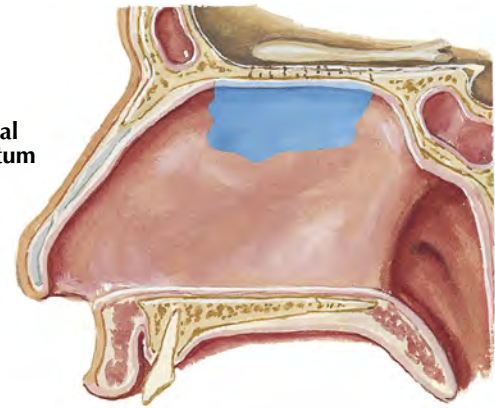
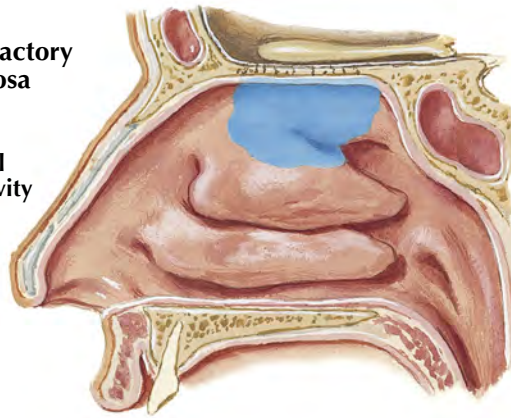
See also Plate 29



Distribution of olfactory part of nasal mucosa (shaded blue)

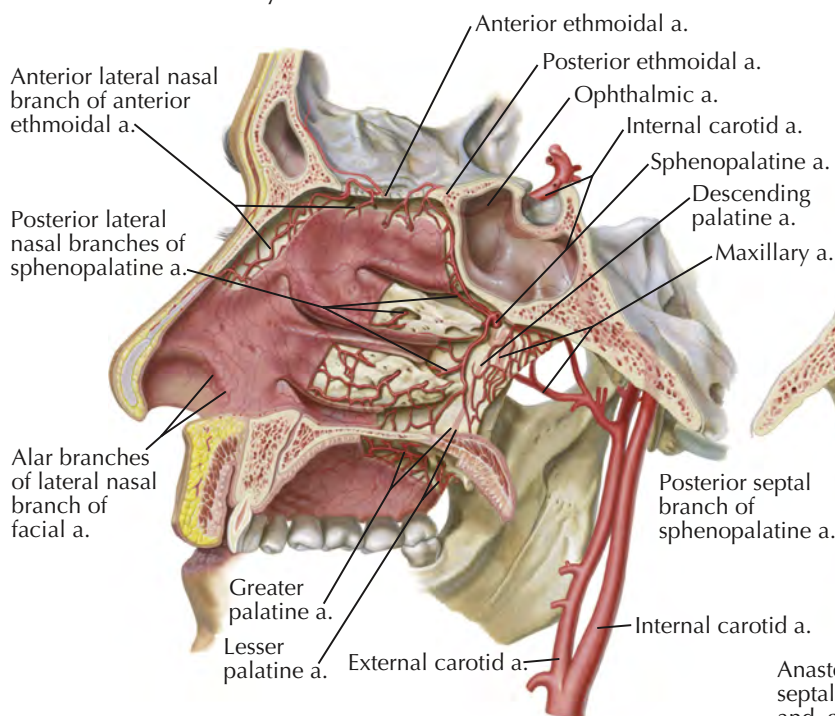
Lateral wall of nasal cavity

Nasal septum

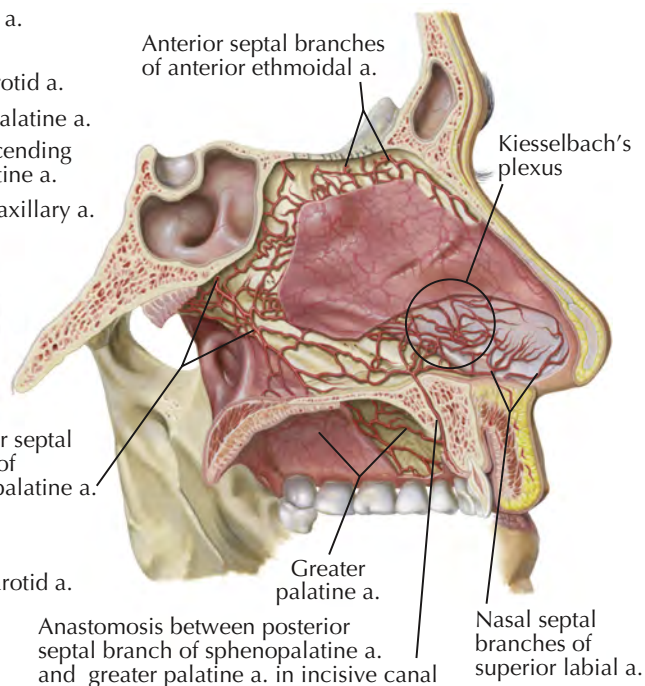


F. Netter M.D.

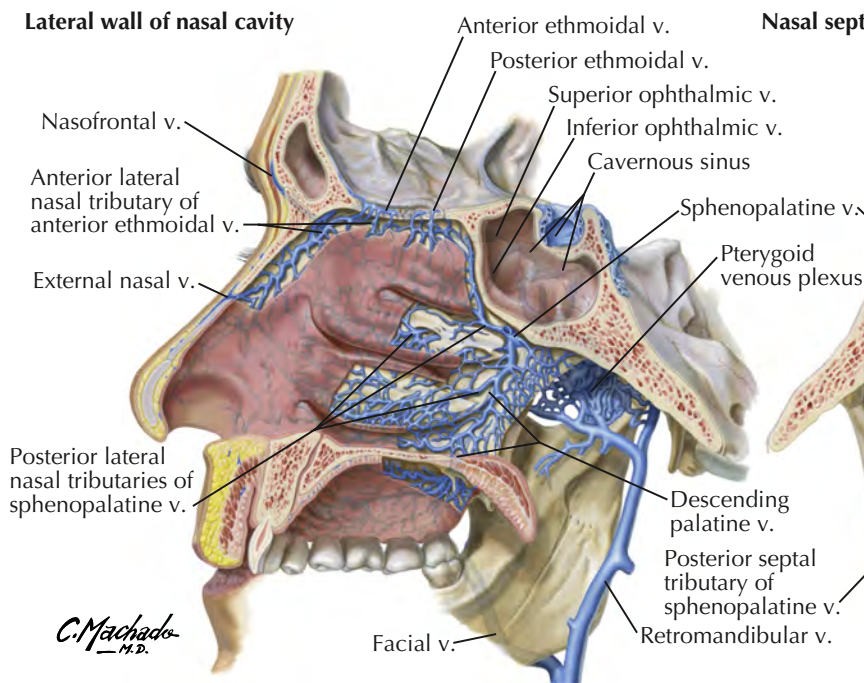
Lateral wall of nasal cavity



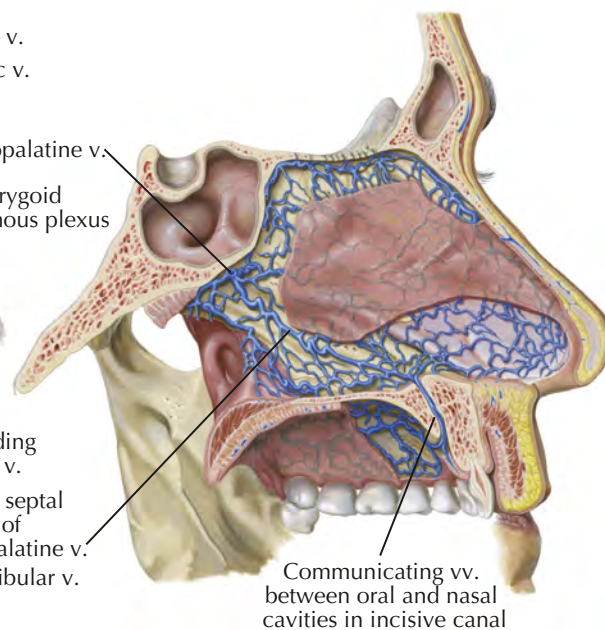
Nasal septum

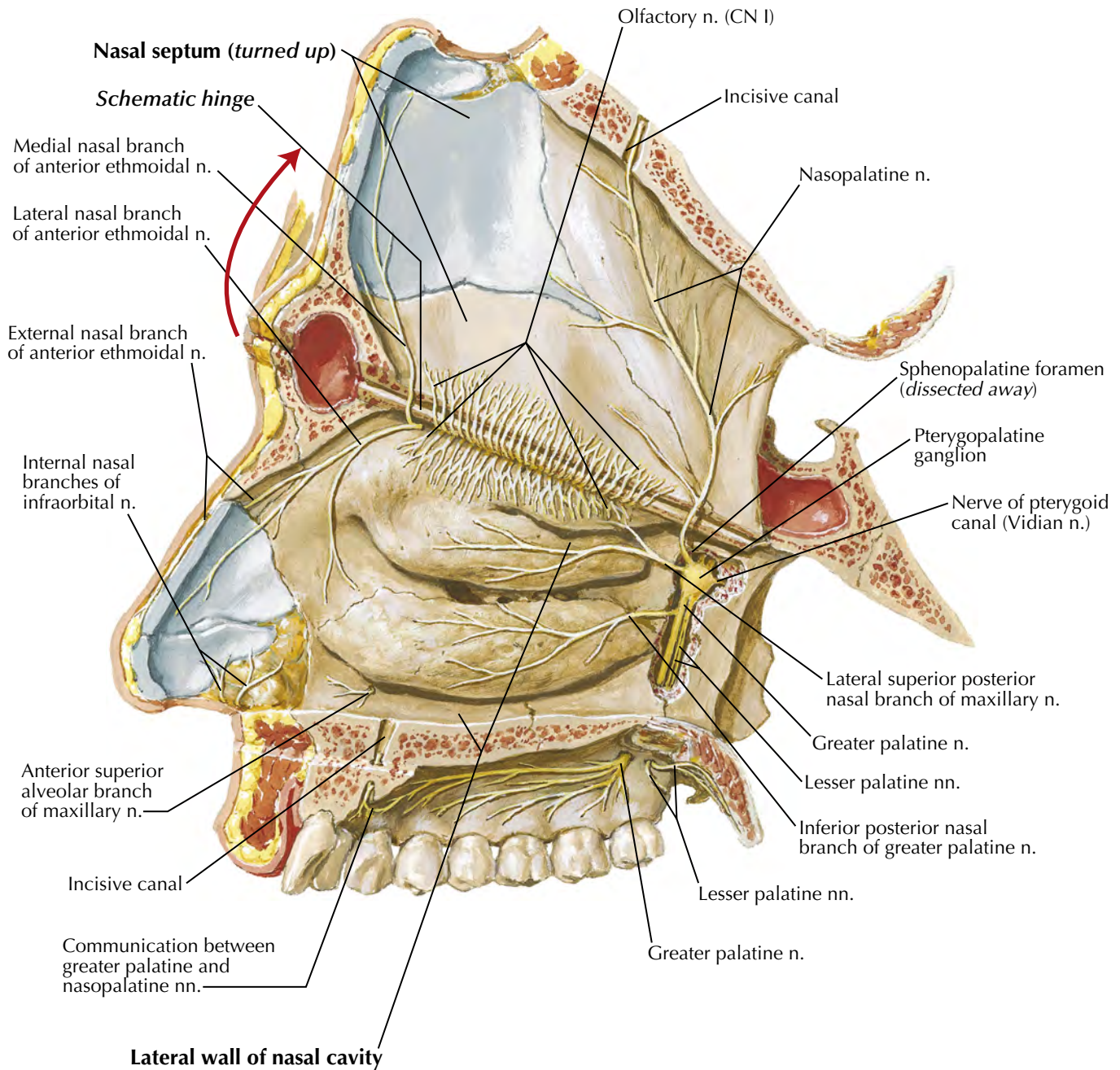


Lateral wall of nasal cavity



Nasal septum

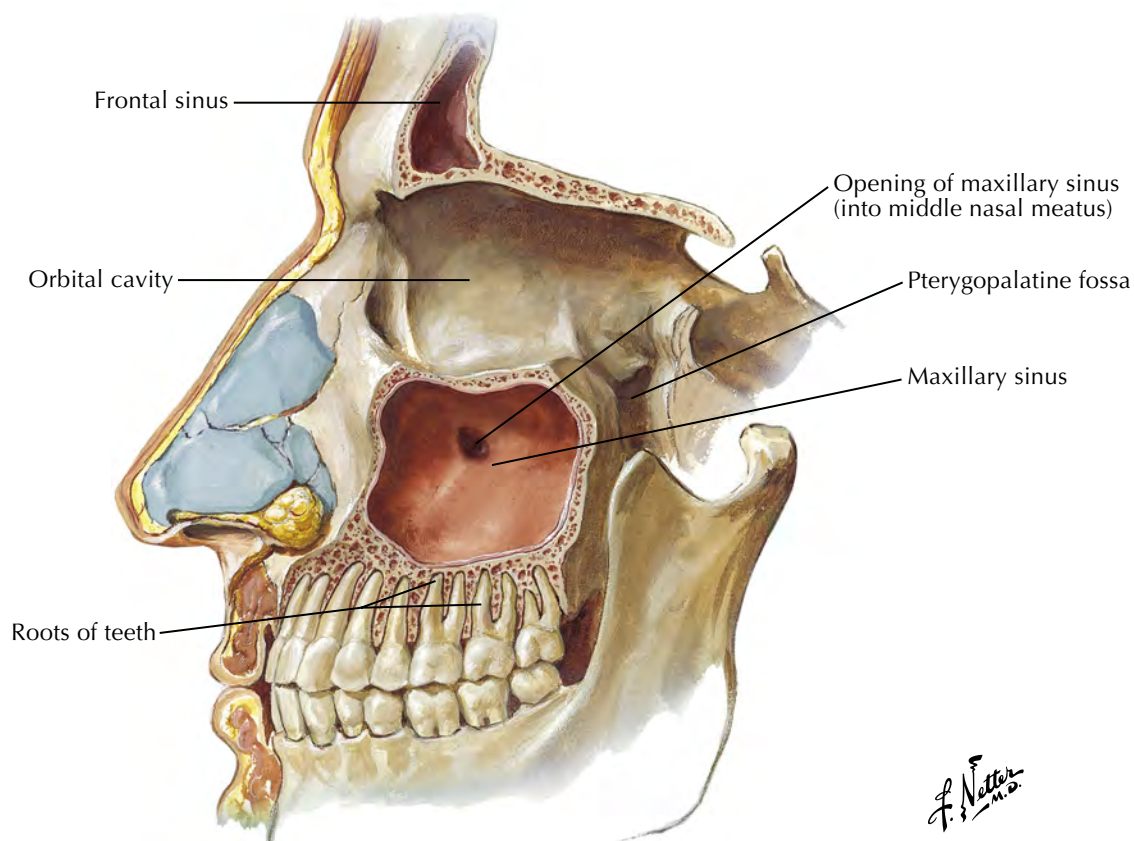
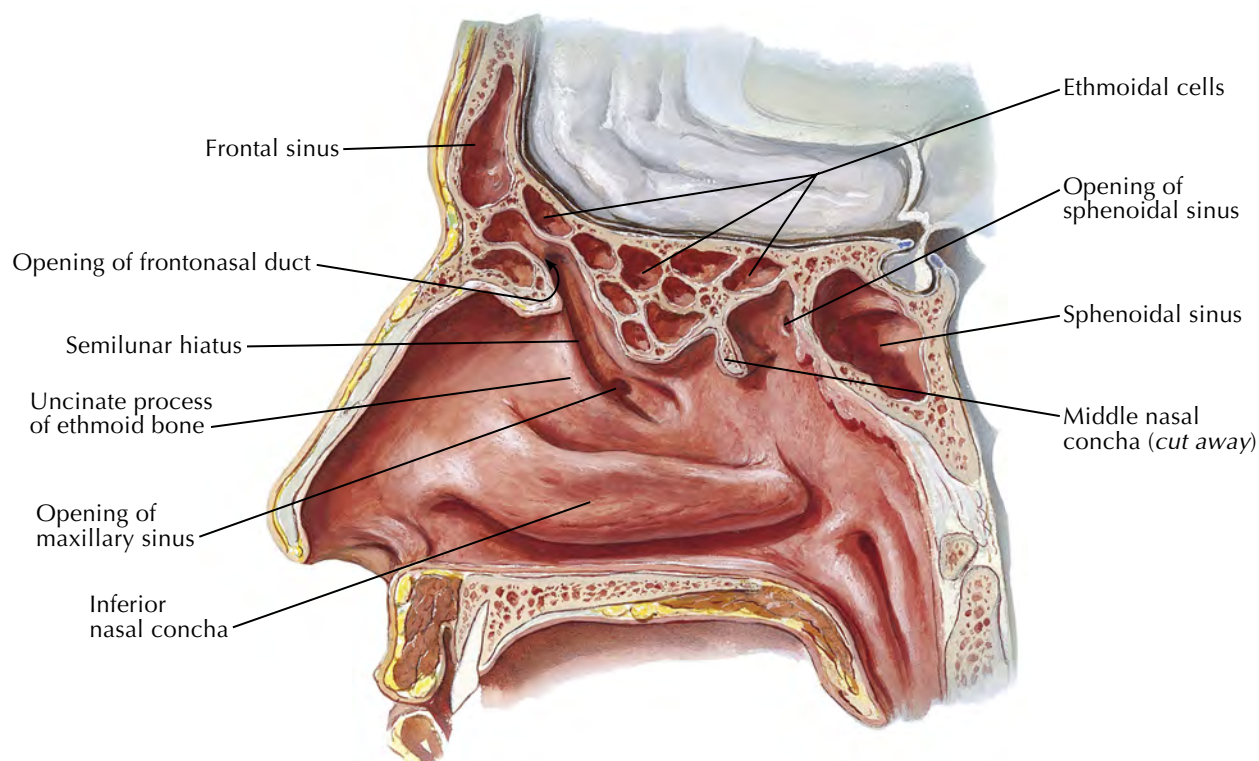




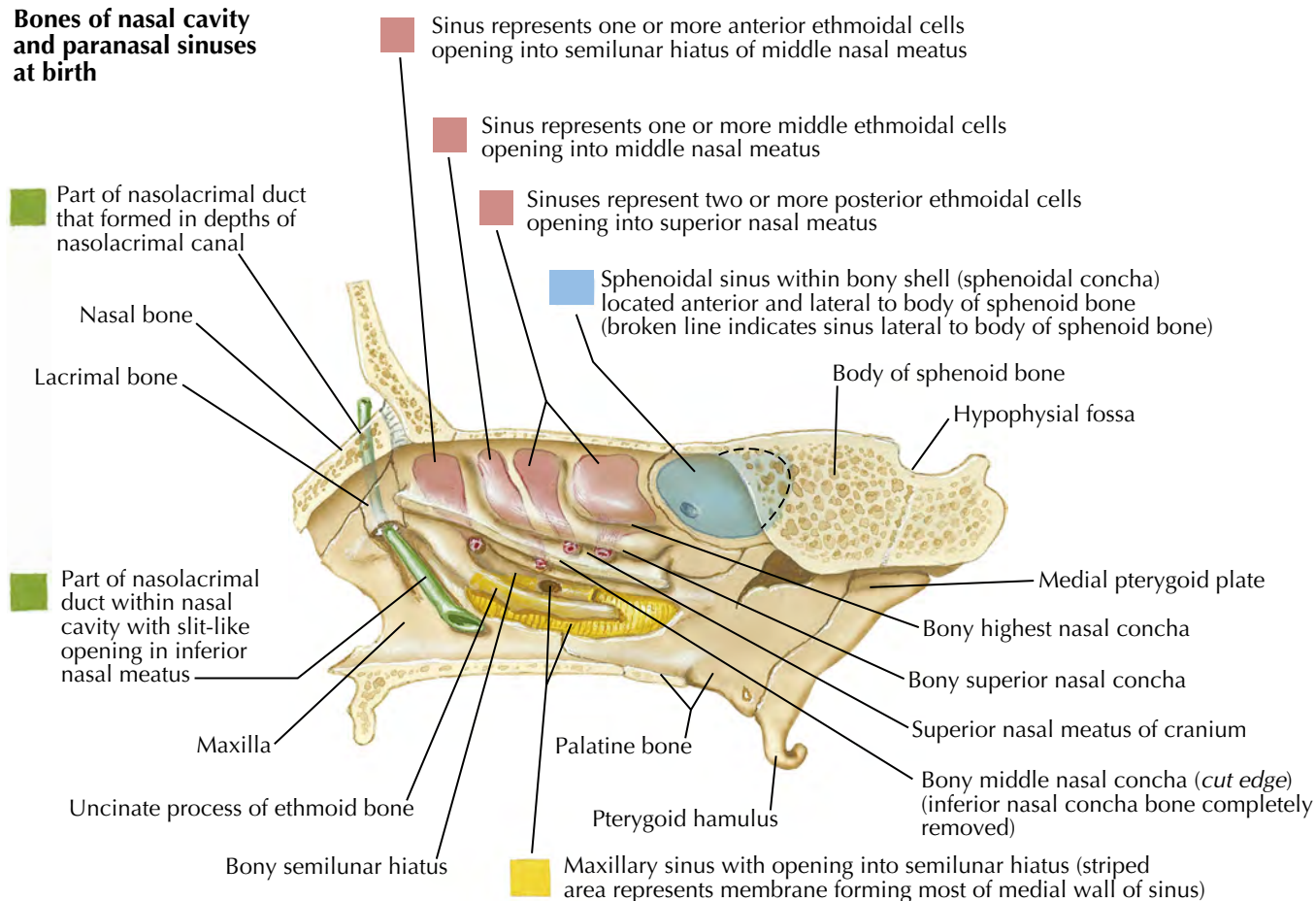
F. Netter M.D.

Paranasal Sinuses: Paramedian Views

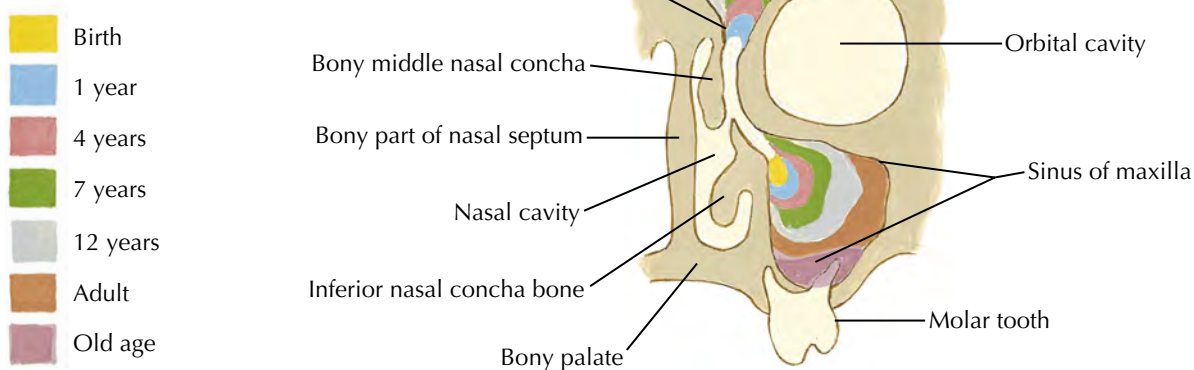
See also Plate 61



Bones of nasal cavity and paranasal sinuses at birth



Growth of sinuses in frontal bone and maxilla throughout life

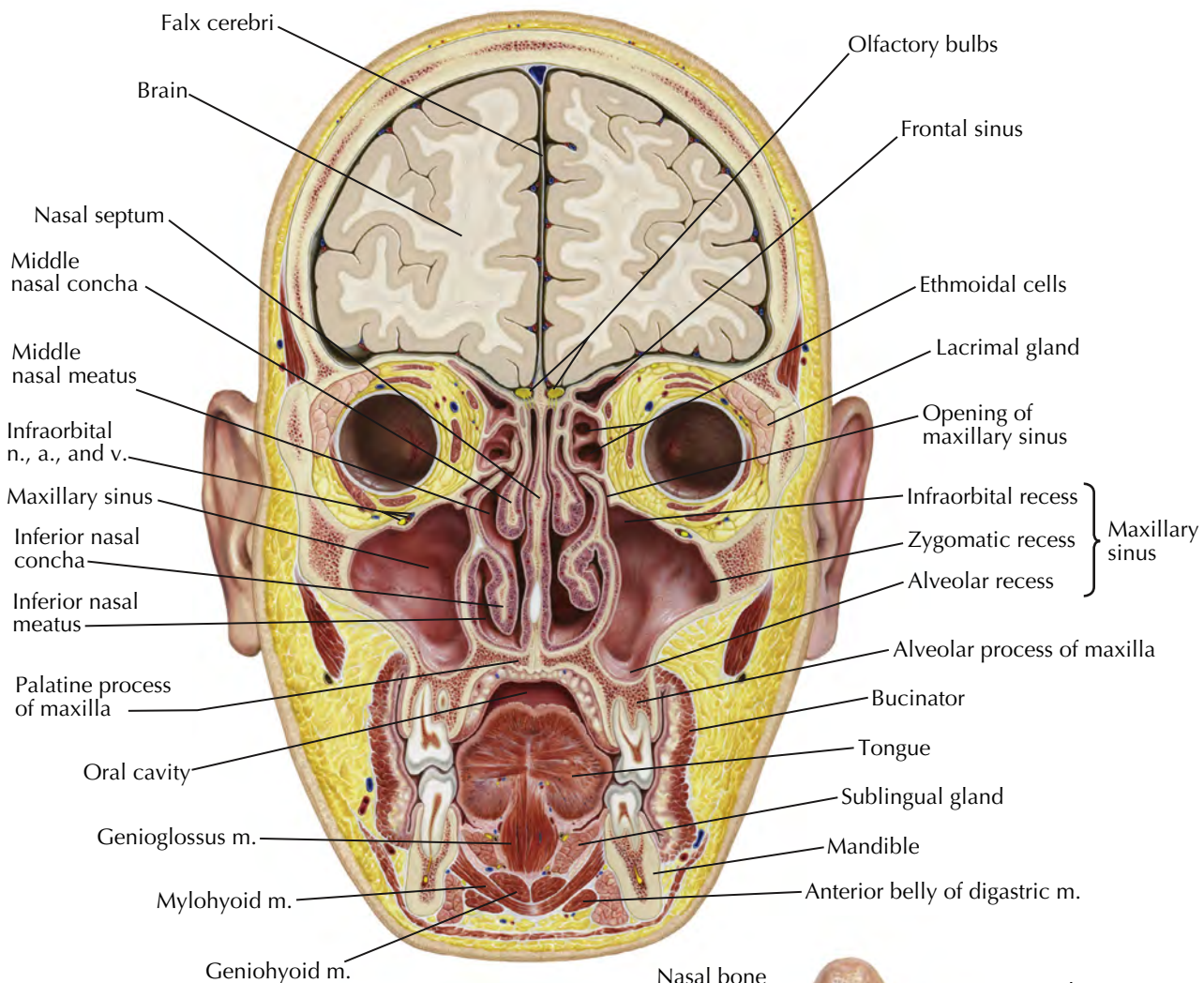


F. Netter M.D.

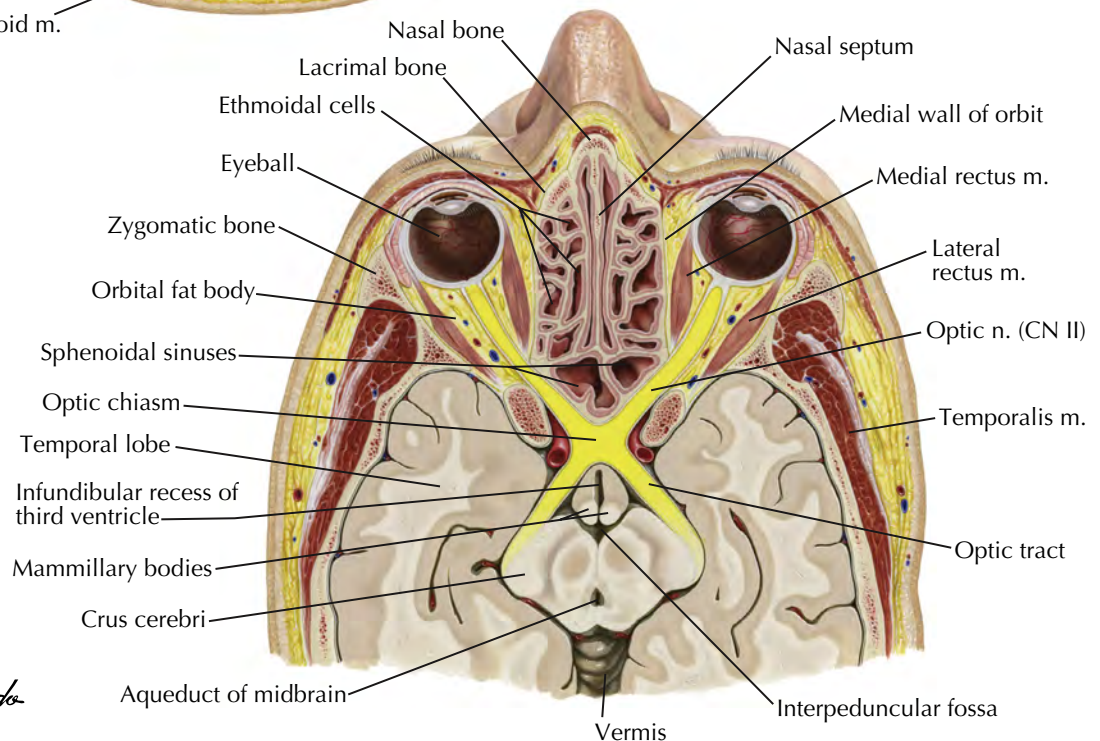
Paranasal Sinuses: Coronal and Transverse Sections

See also Plates BP 22, BP 23

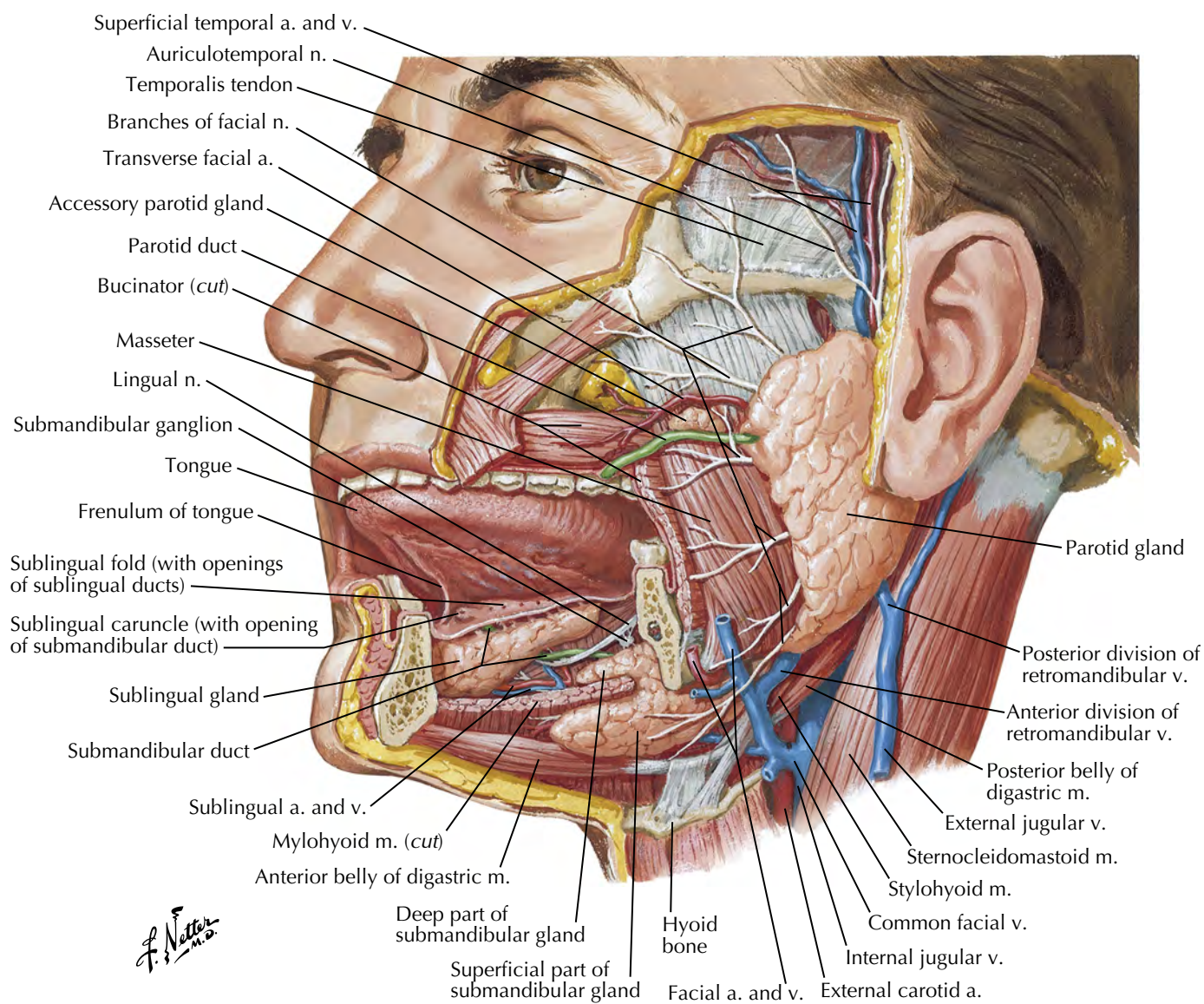
Coronal section

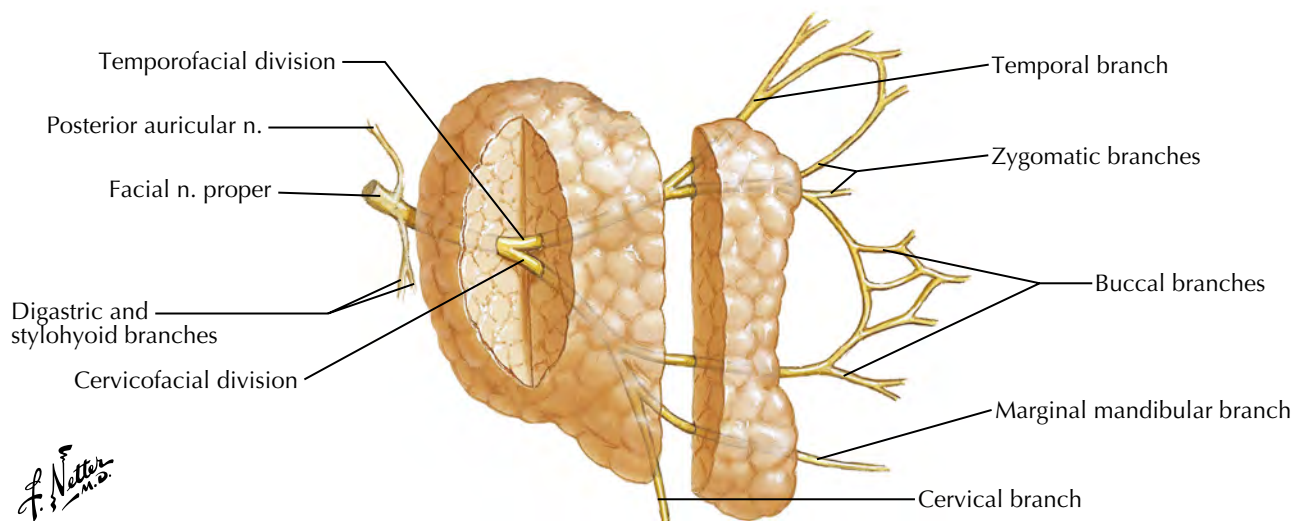
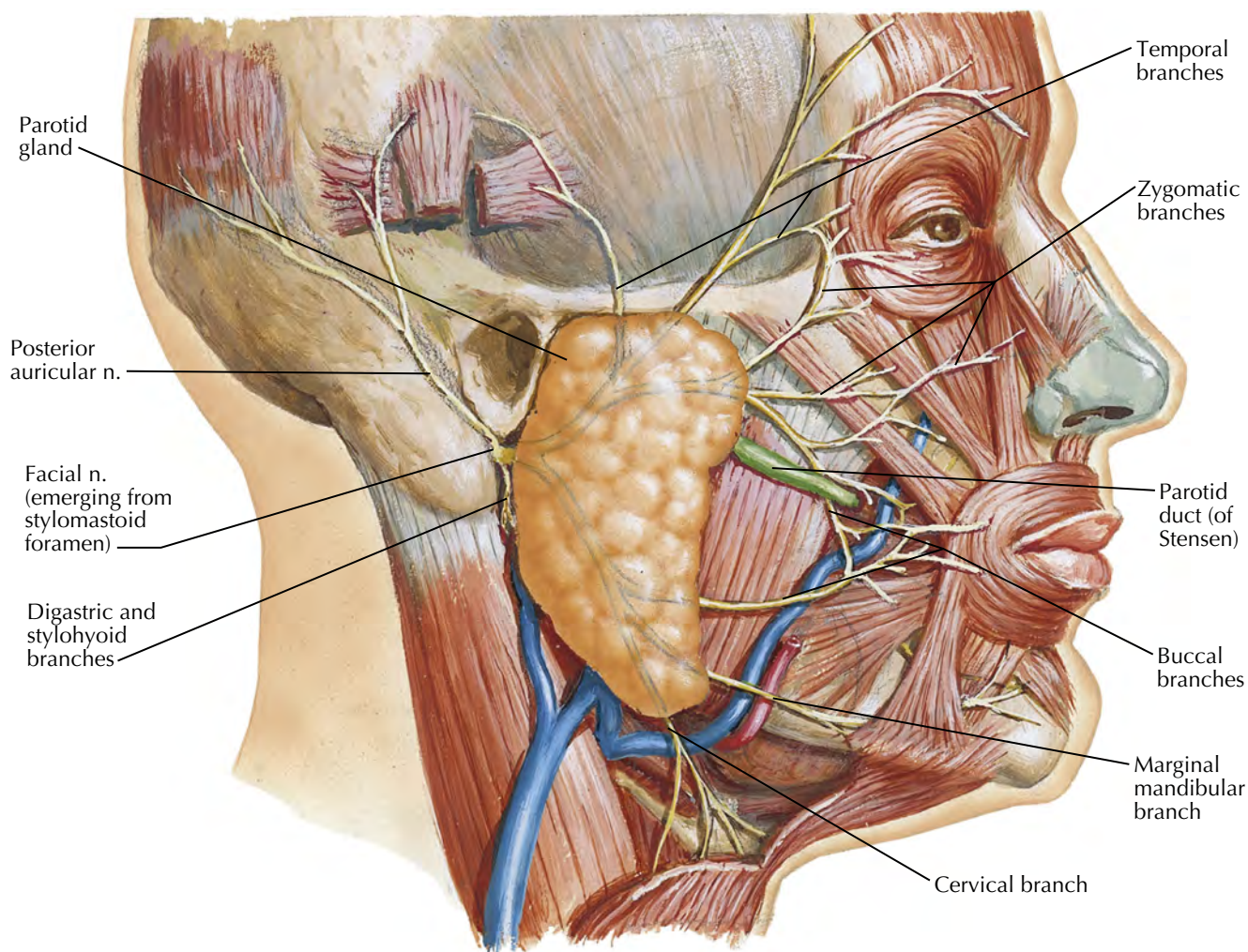


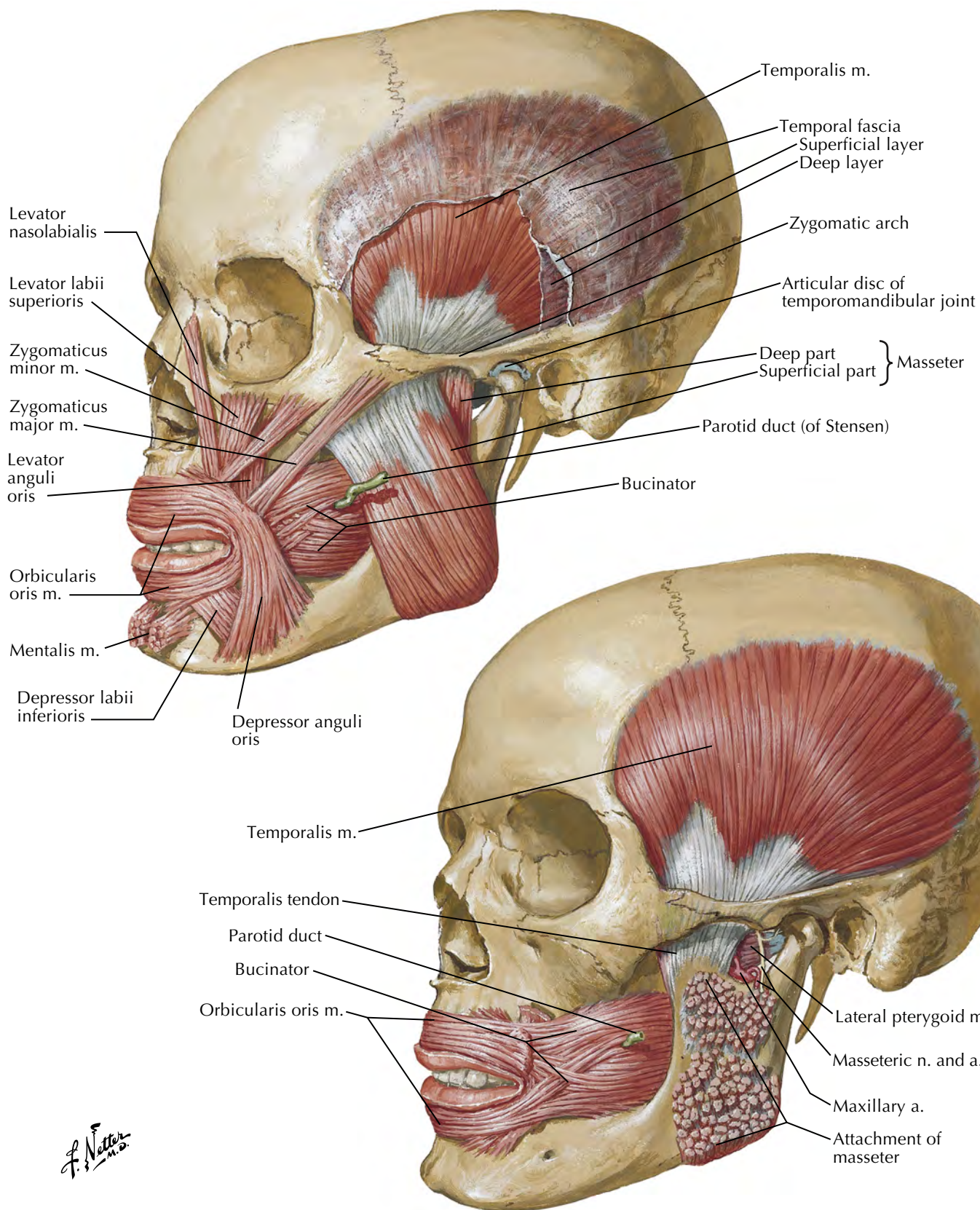
Transverse section



C. Machado M.D.



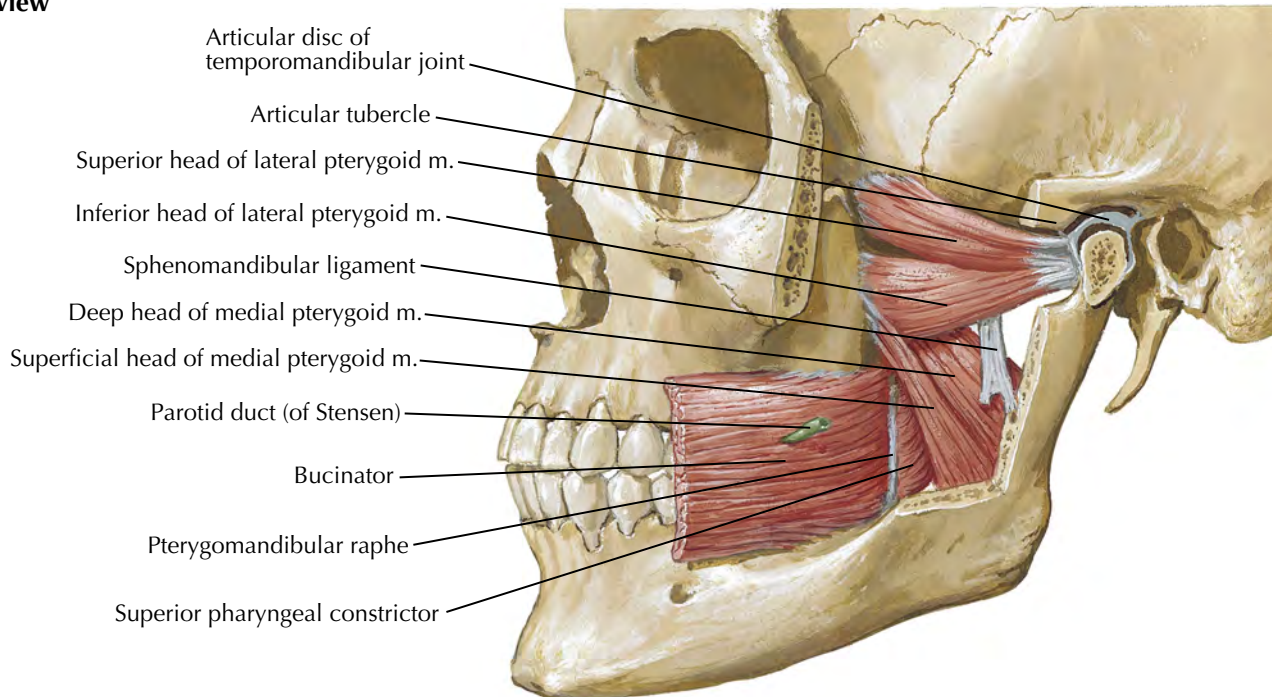




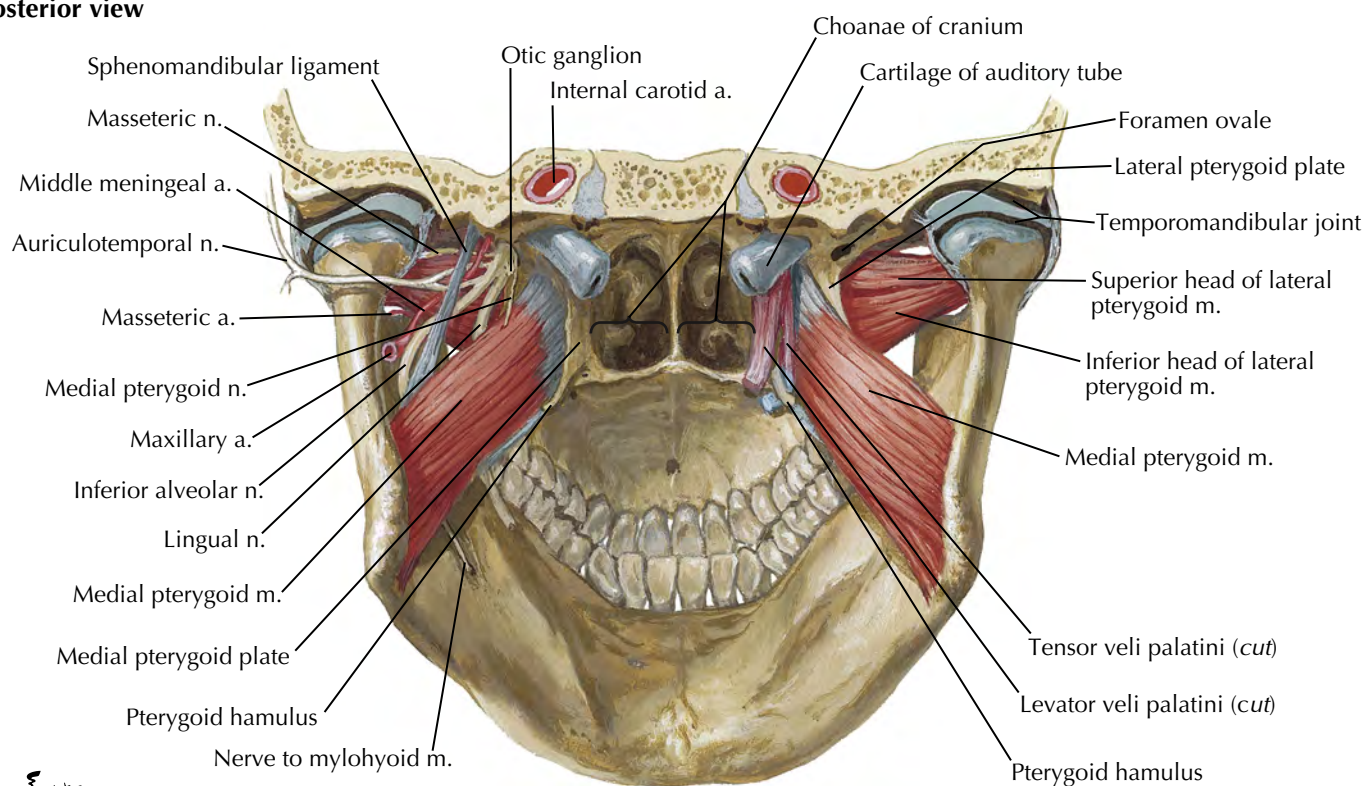
Muscles of Mastication: Pterygoid Muscles

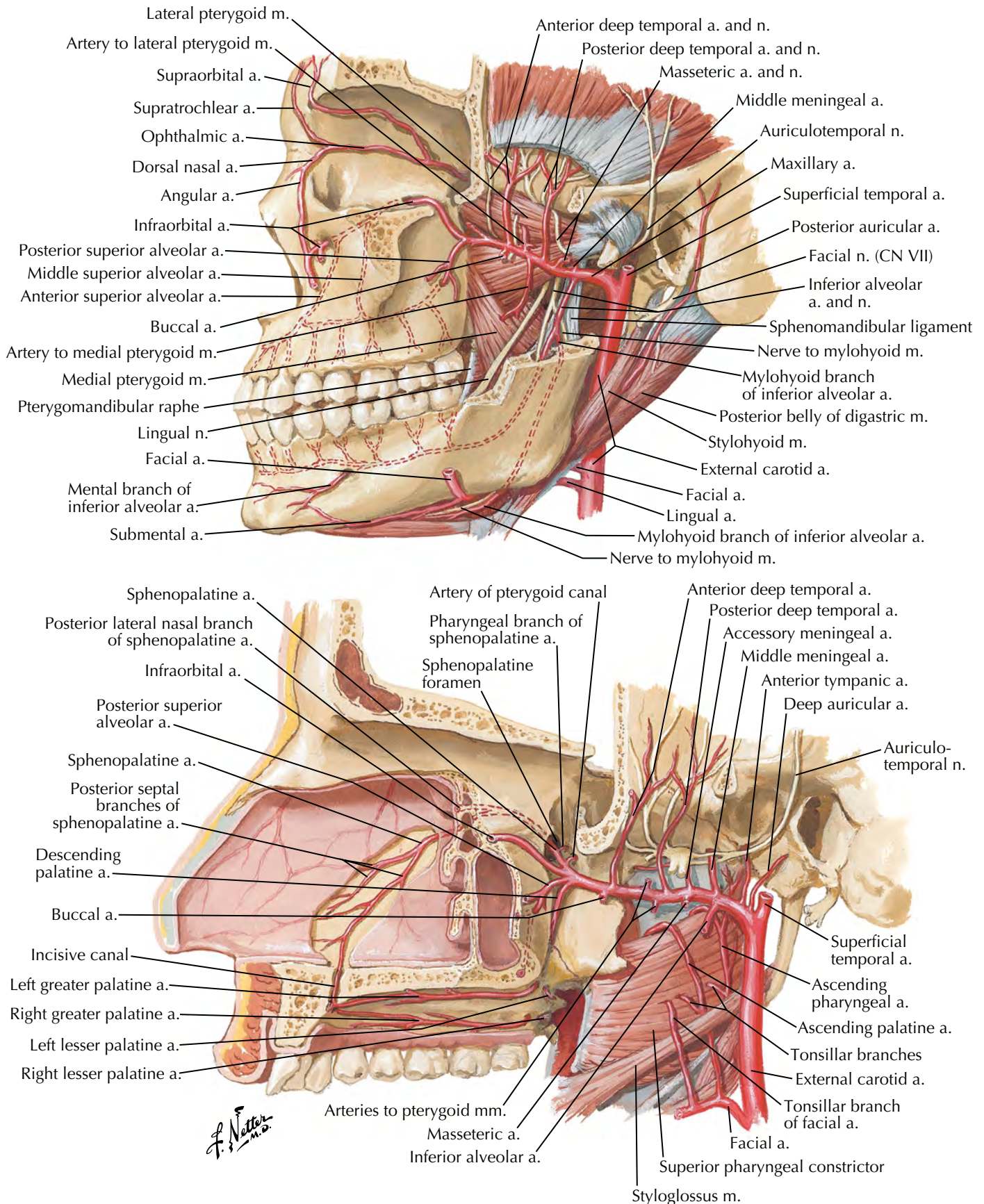
See also Plates 72, 77, BP 25

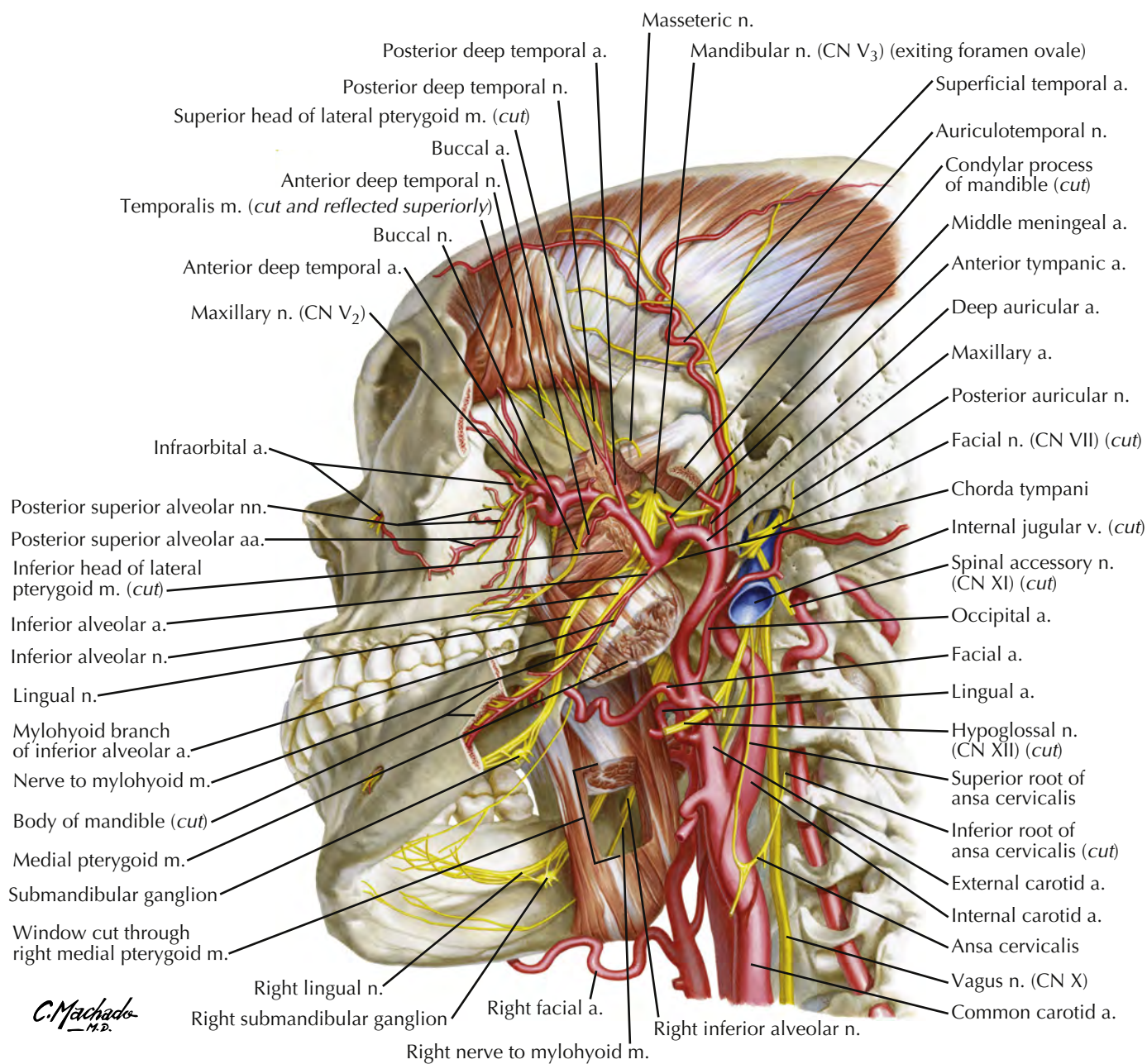
Lateral view



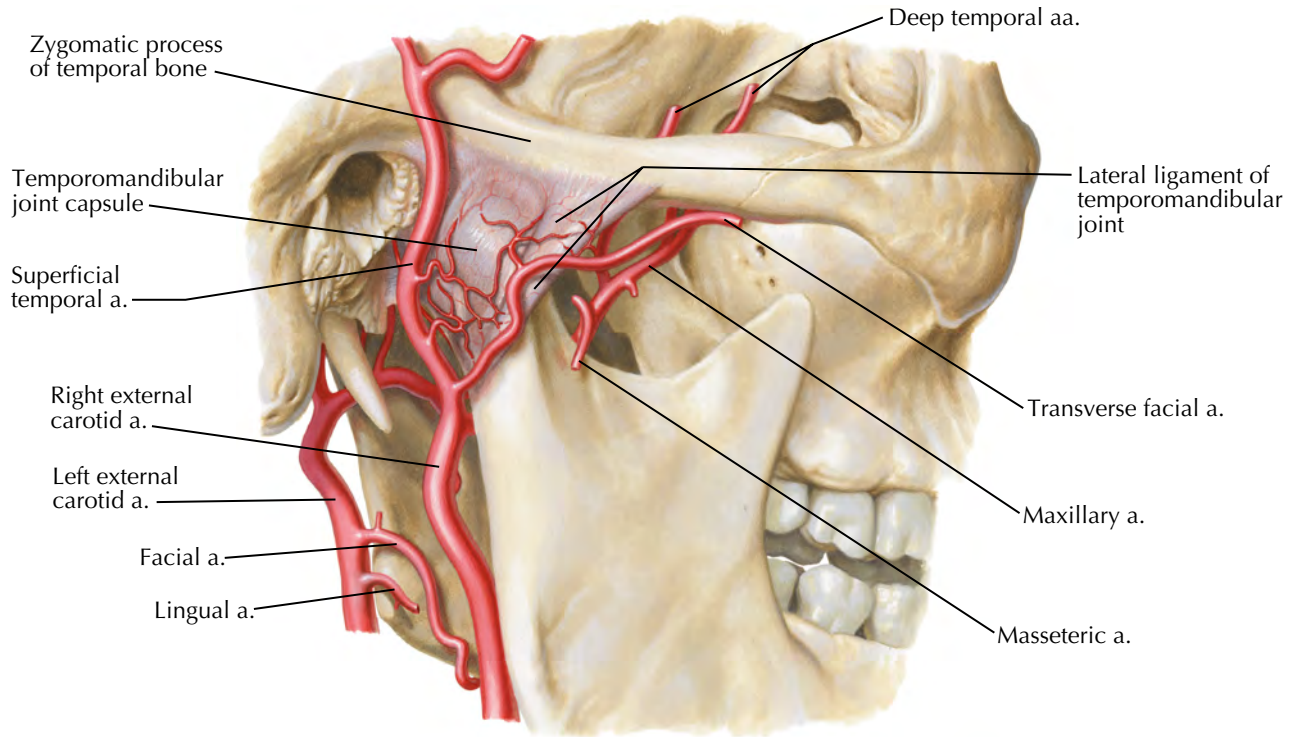
Posterior view



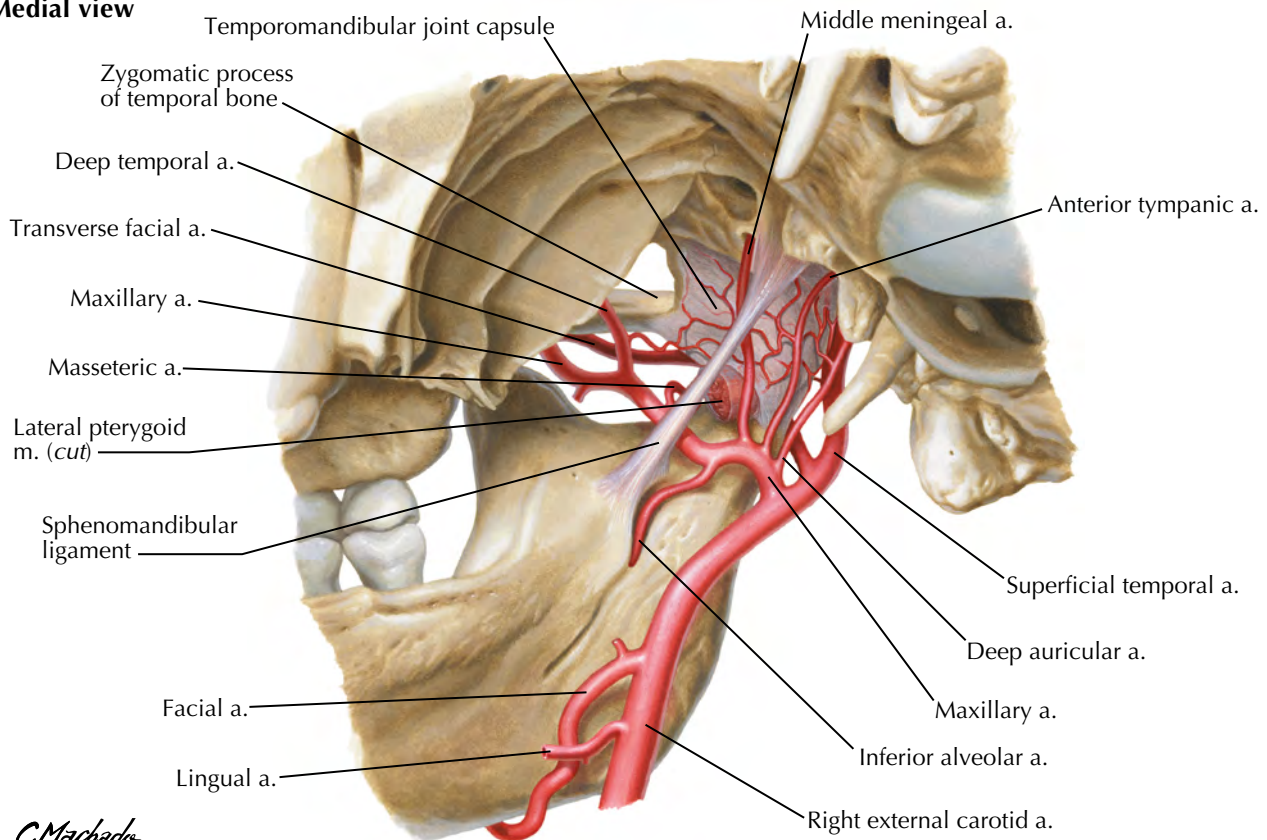




Lateral view



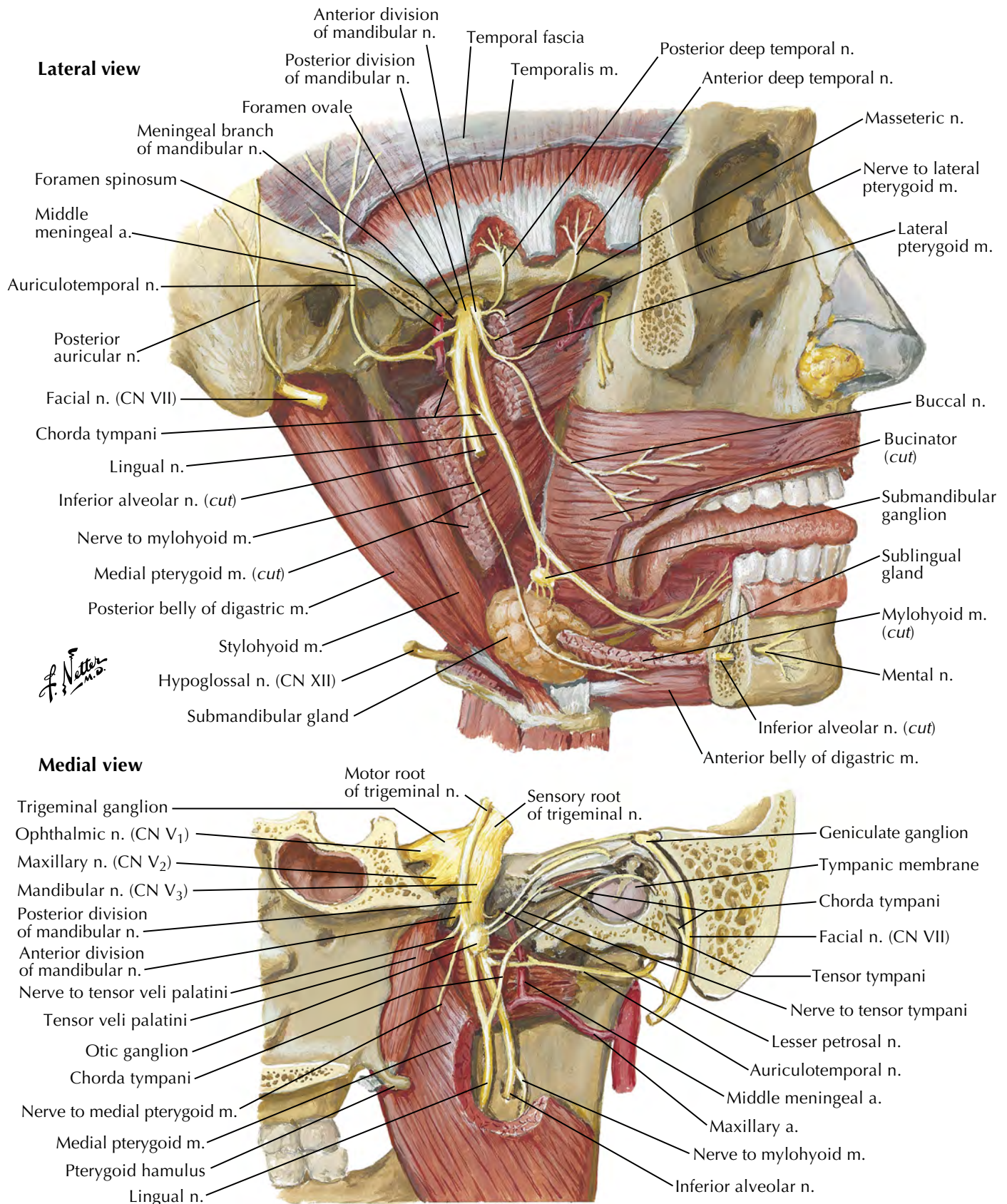
Medial view

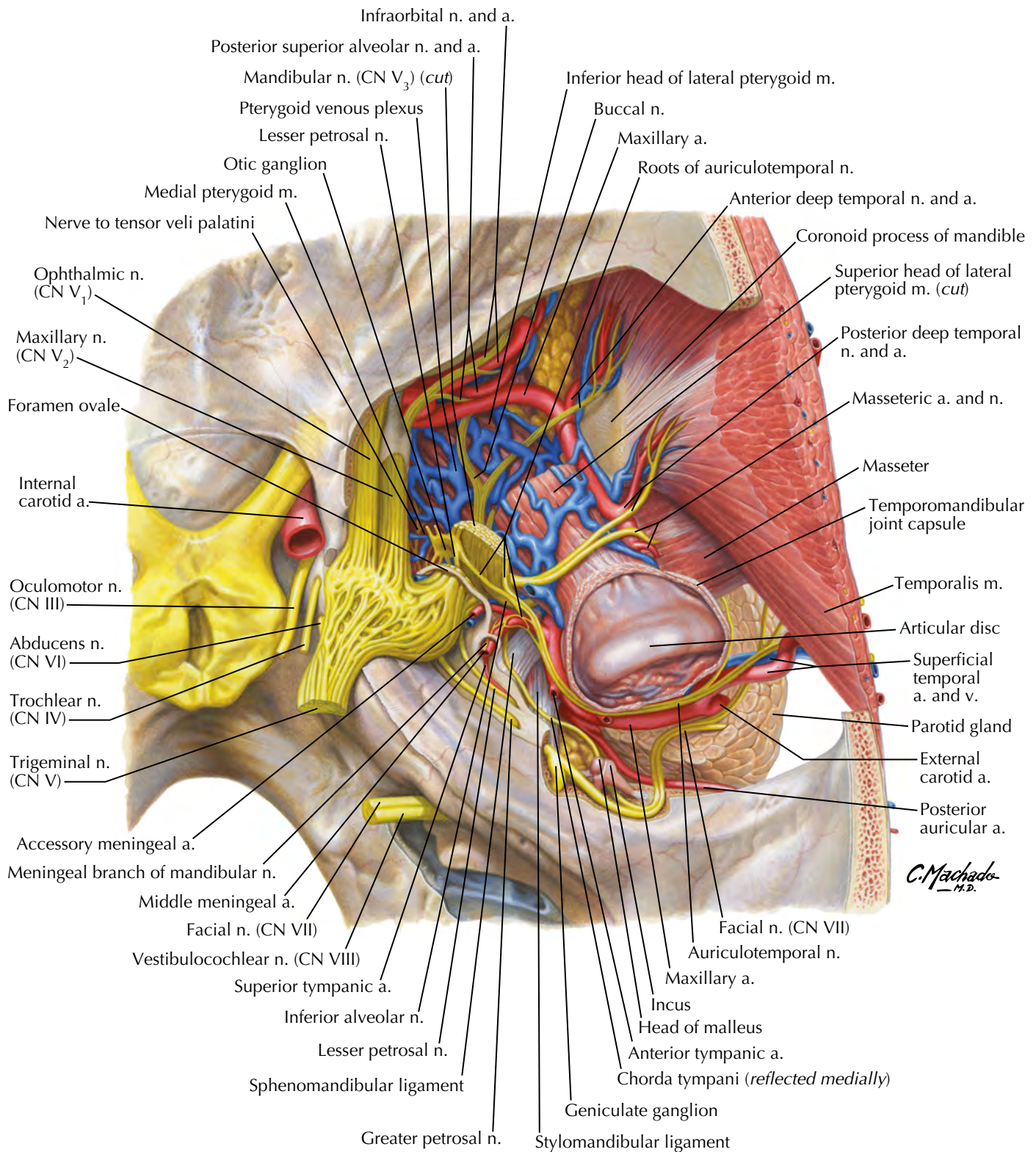


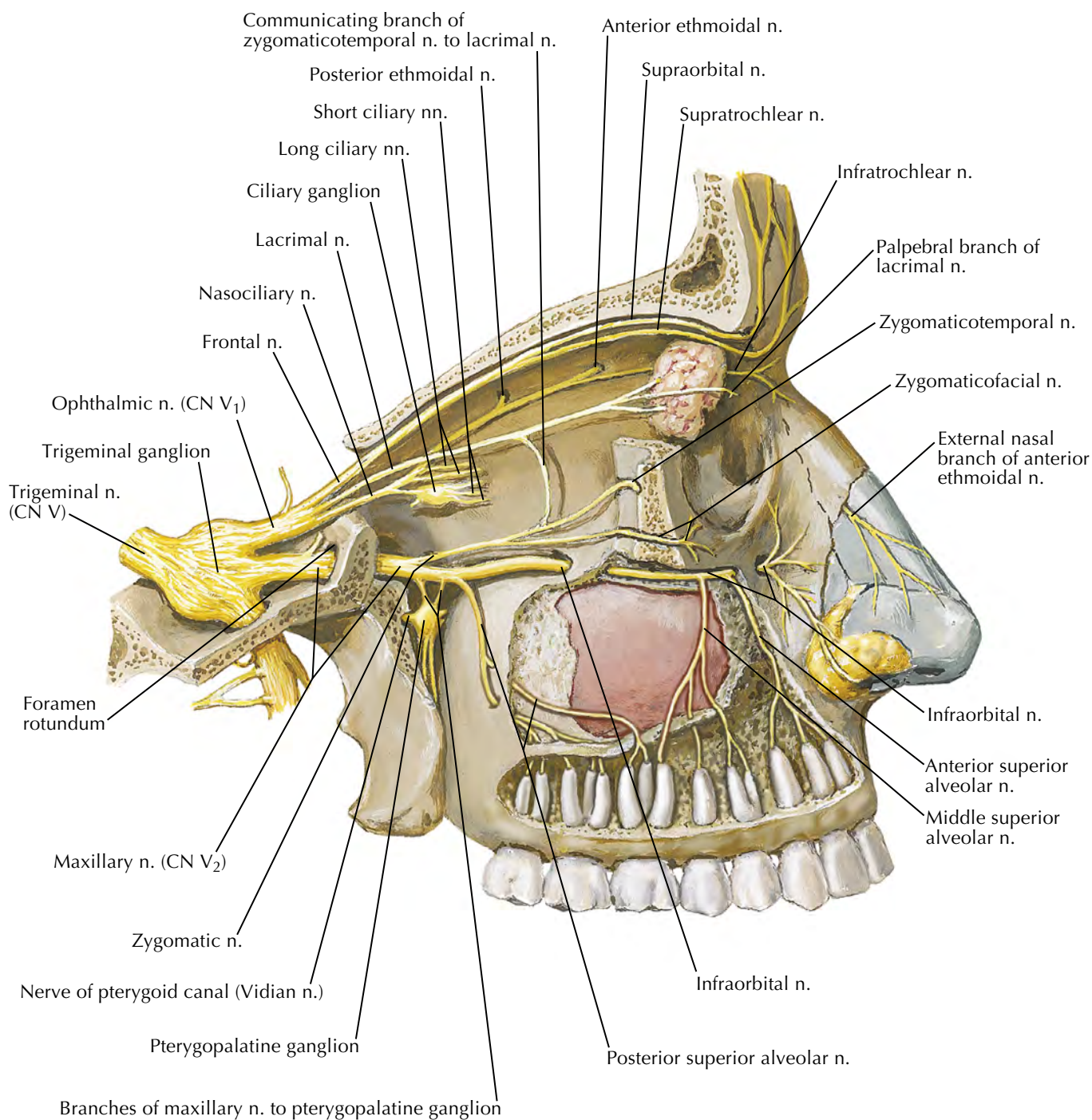
C. Machado
— M.D.

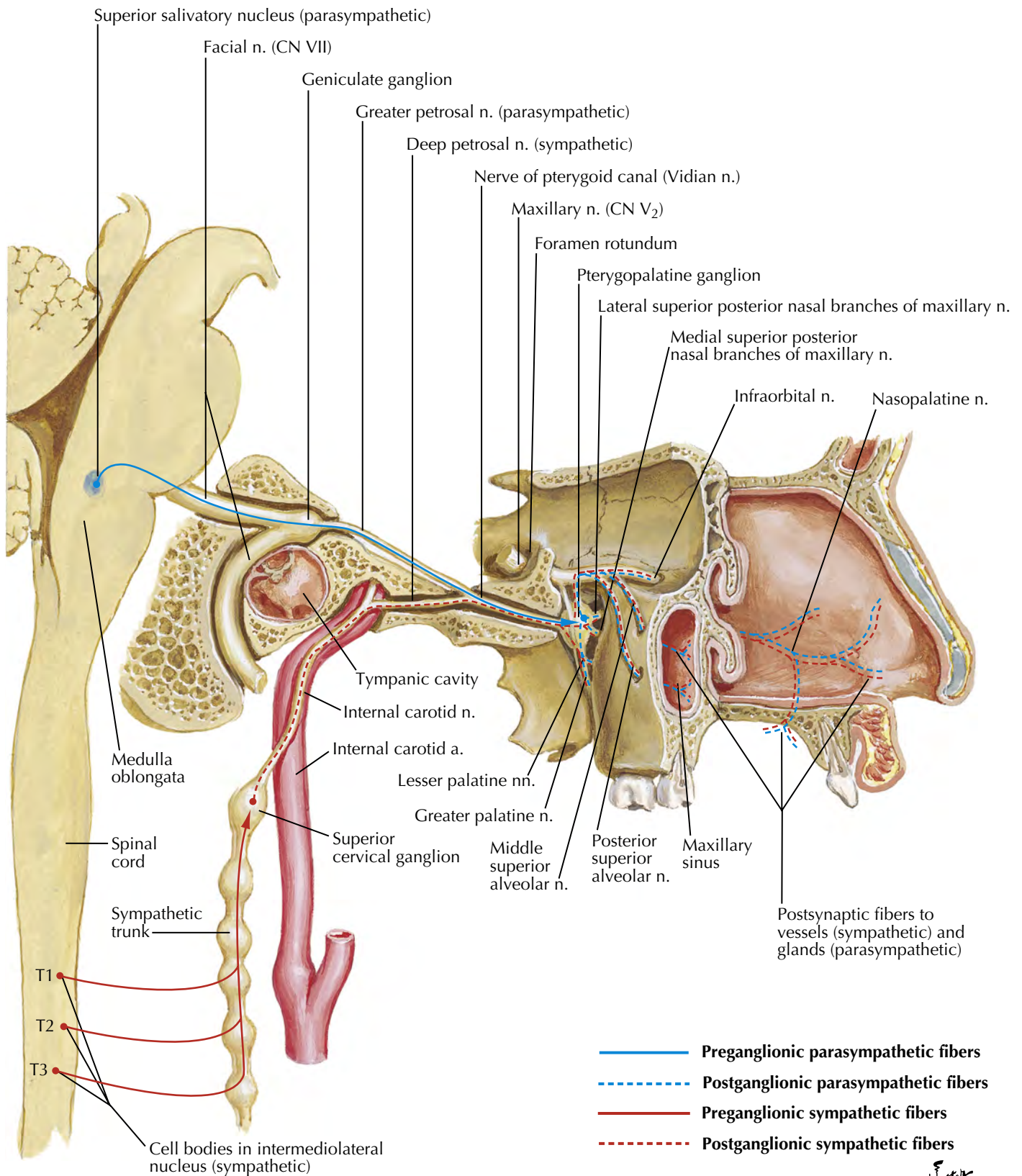
Mandibular Nerve (CN V₃)

See also Plates 98, 149

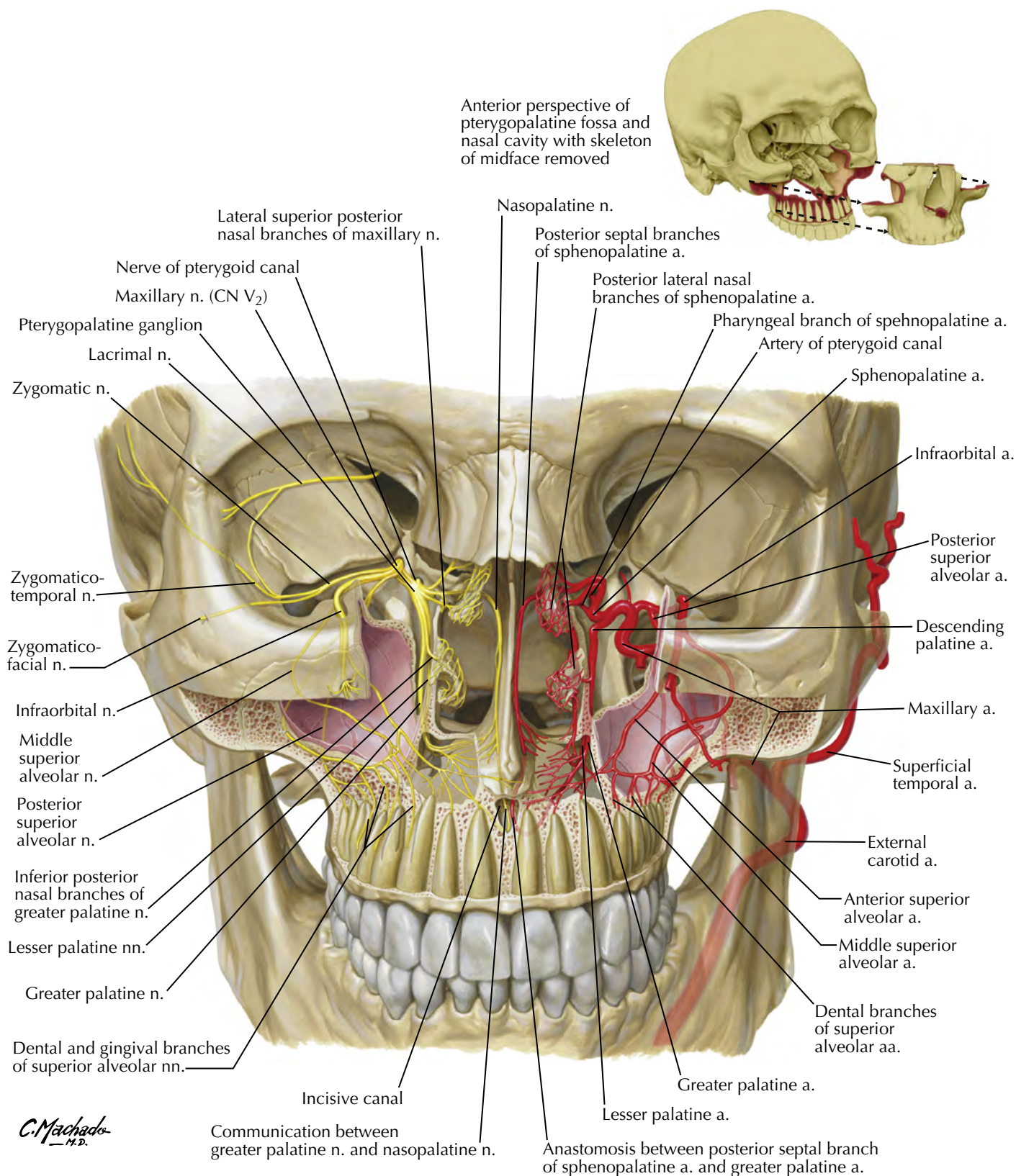




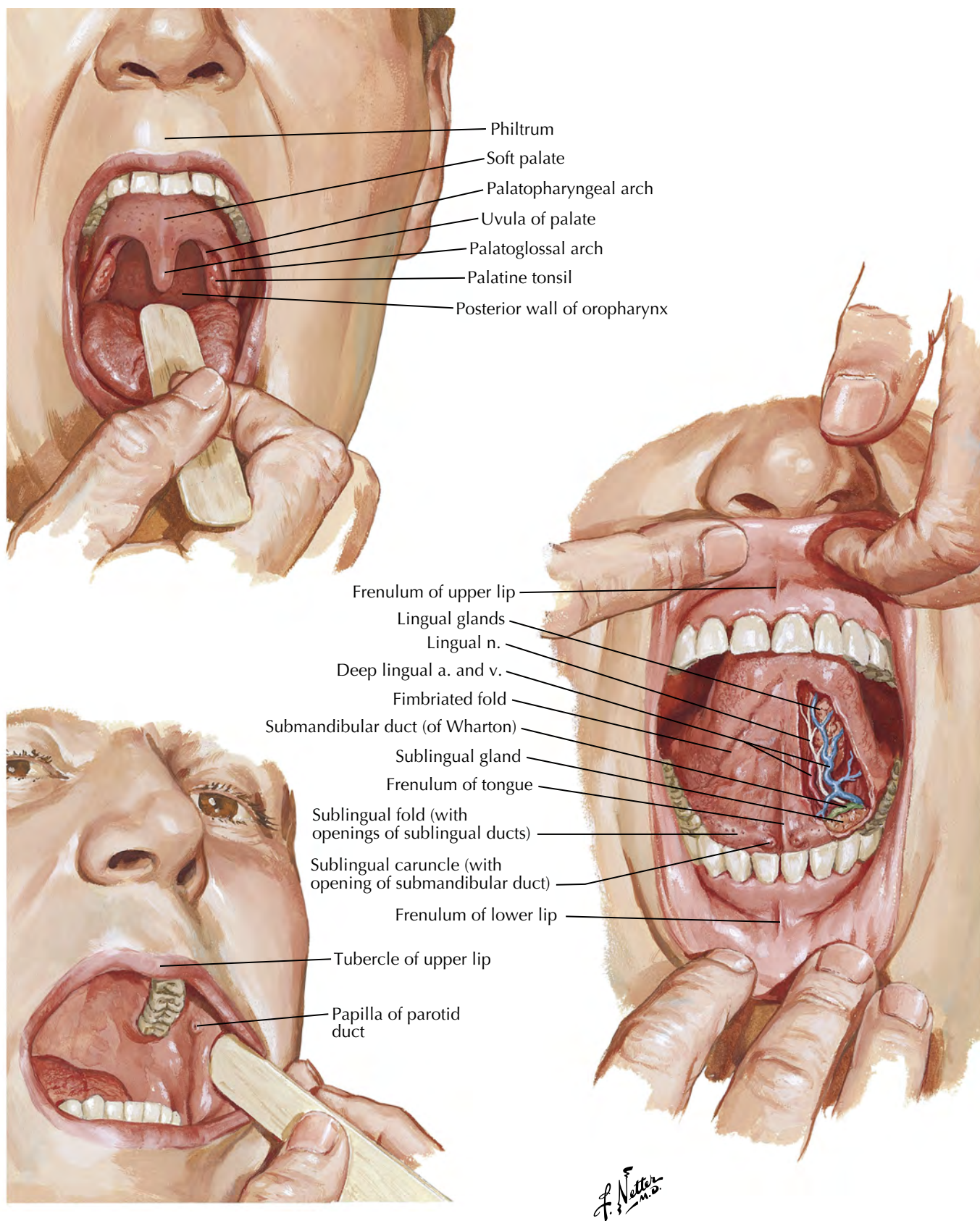


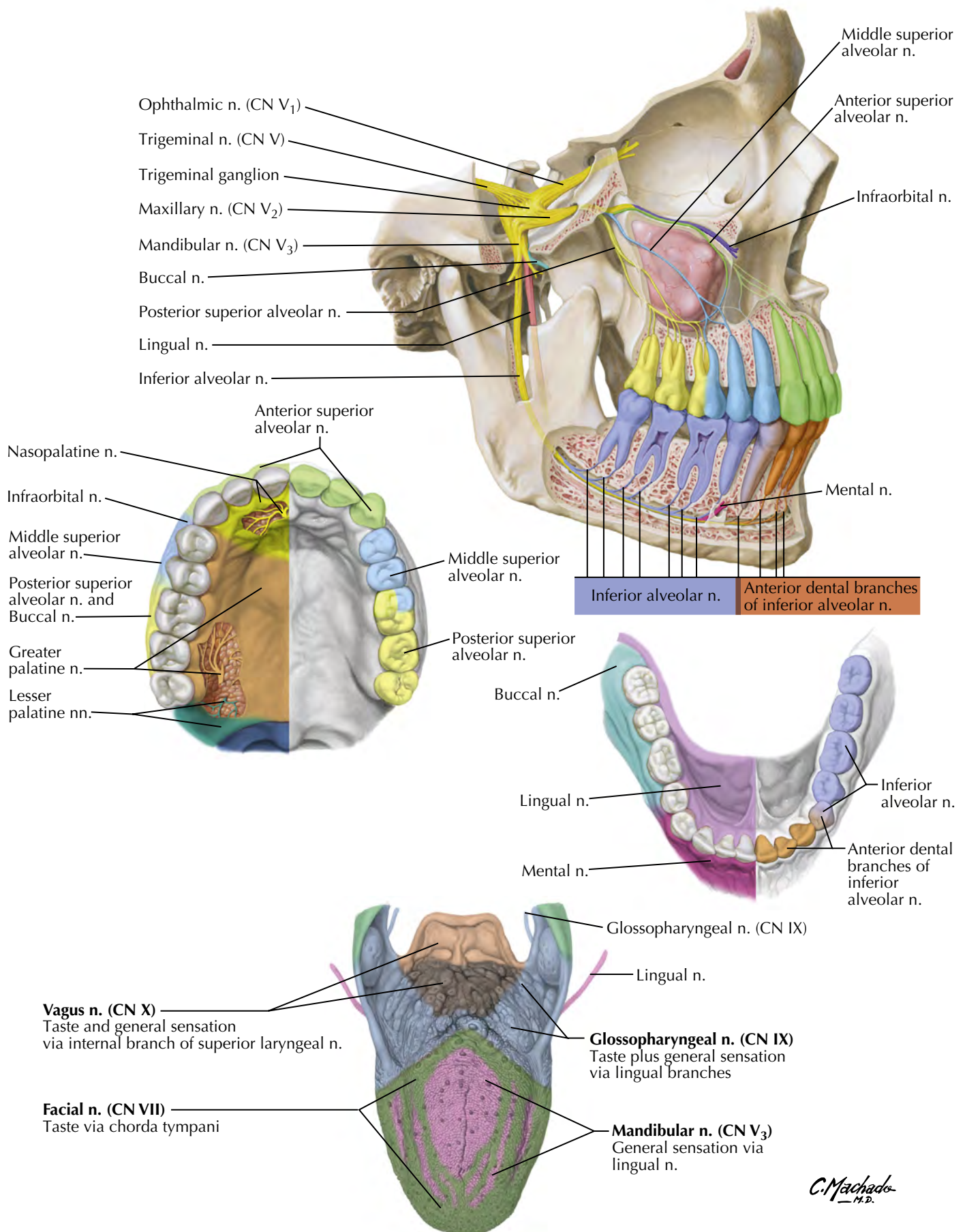


F. Netter M.D.

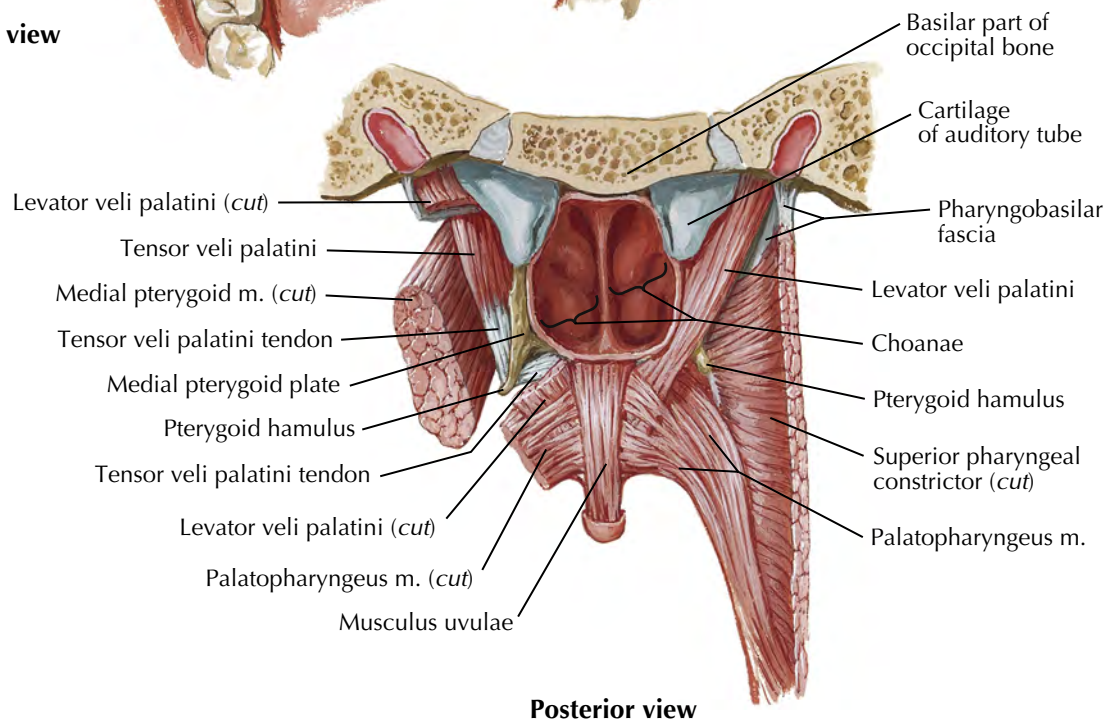
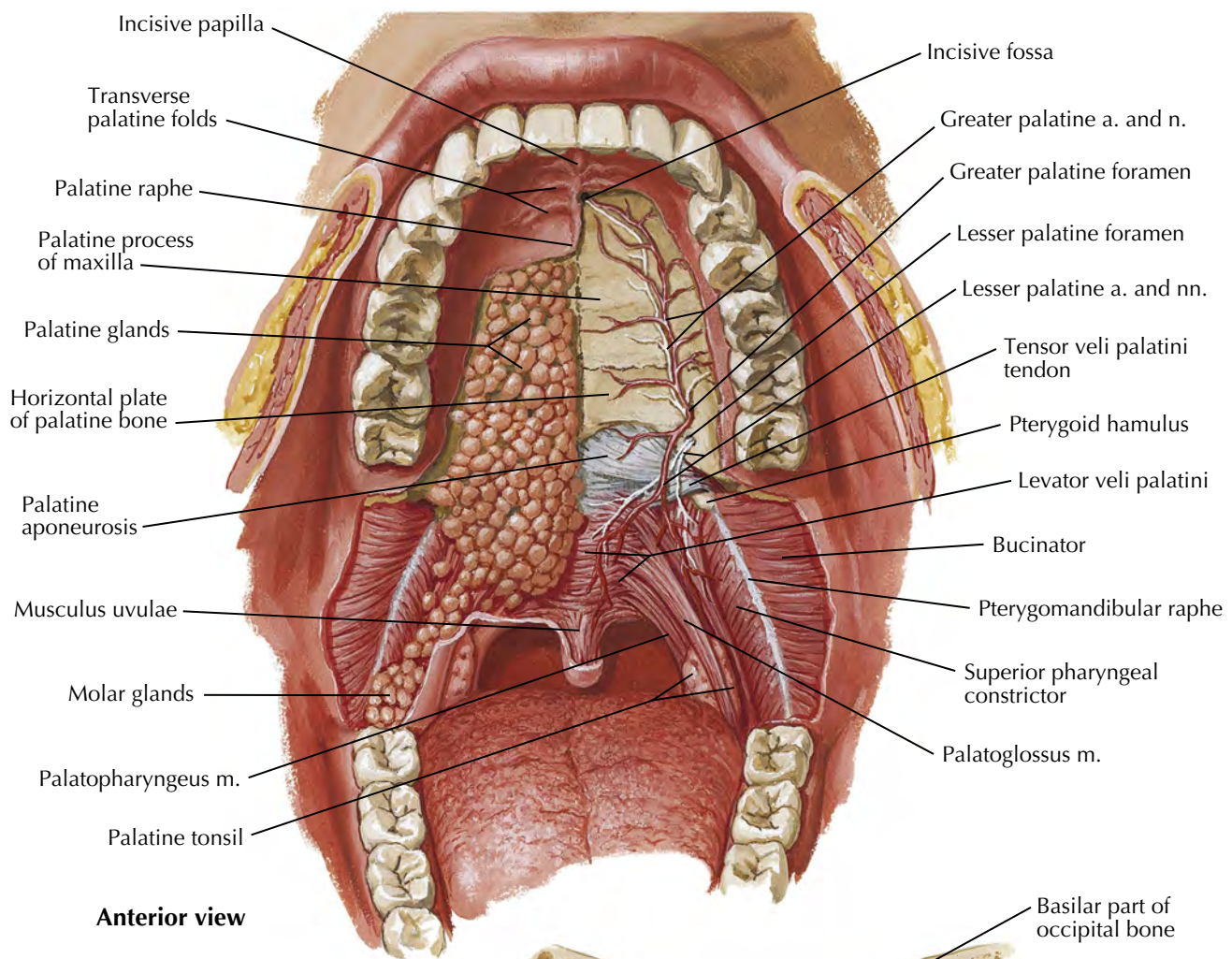






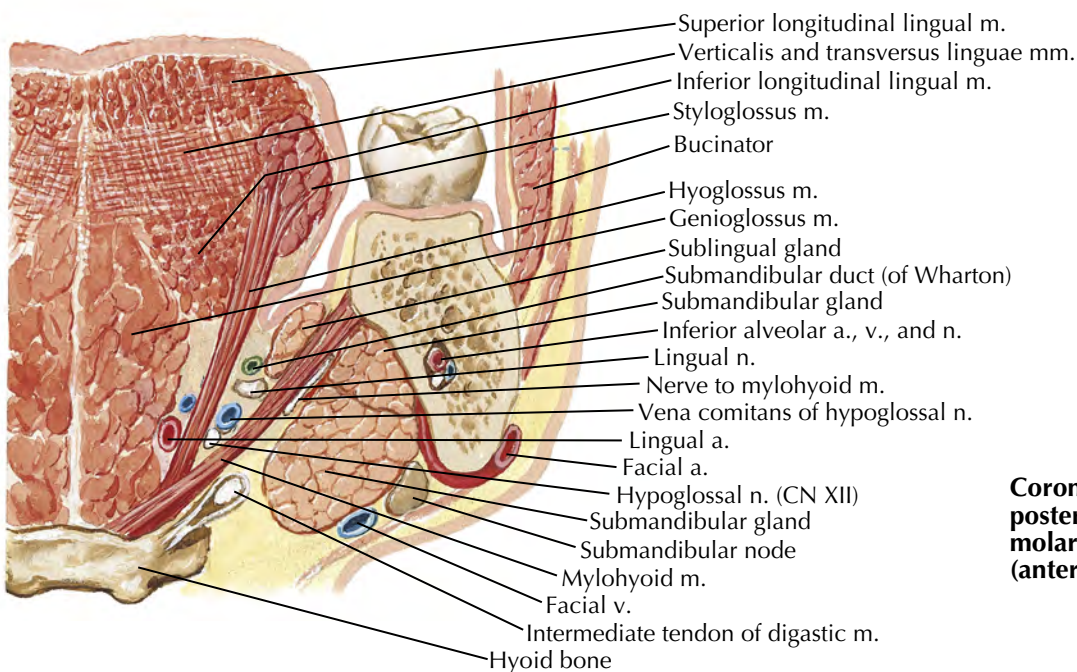
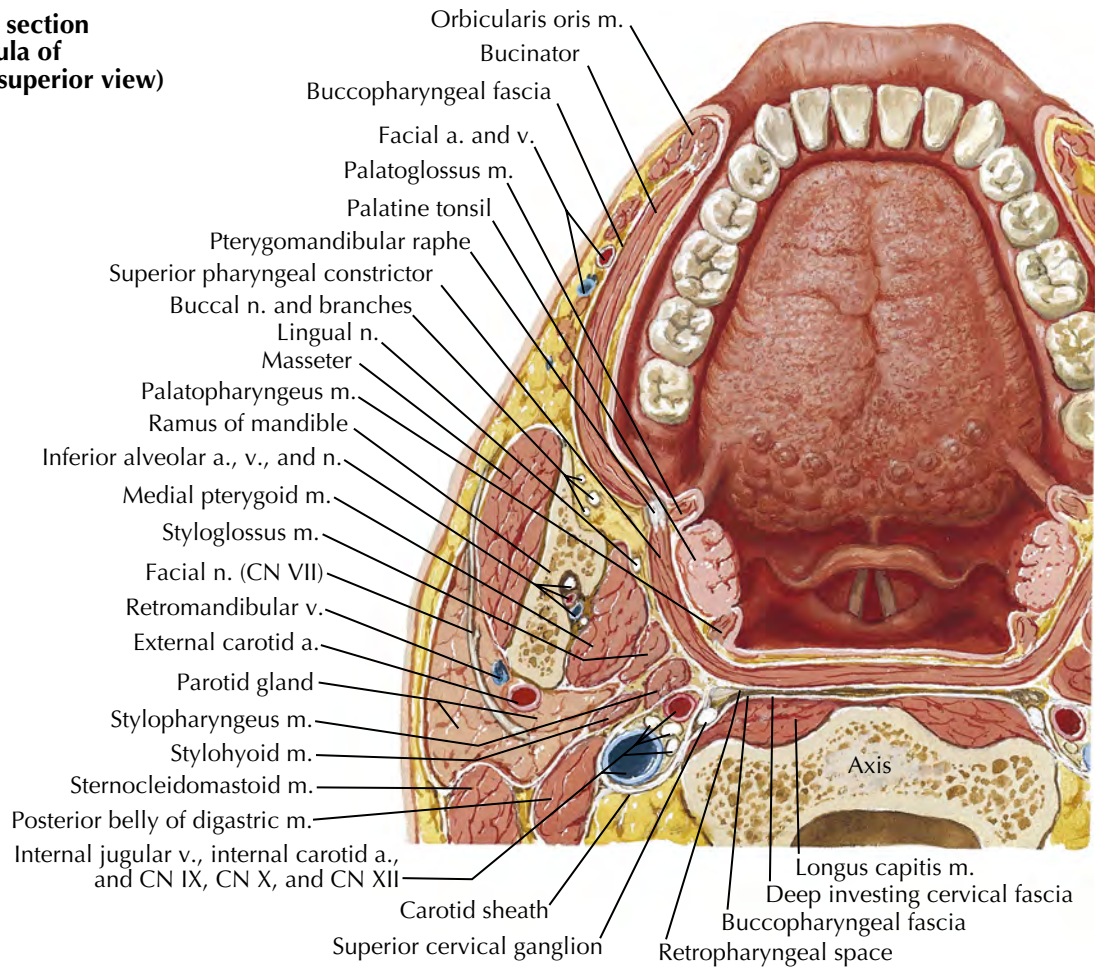


C. Machado
—M.D.

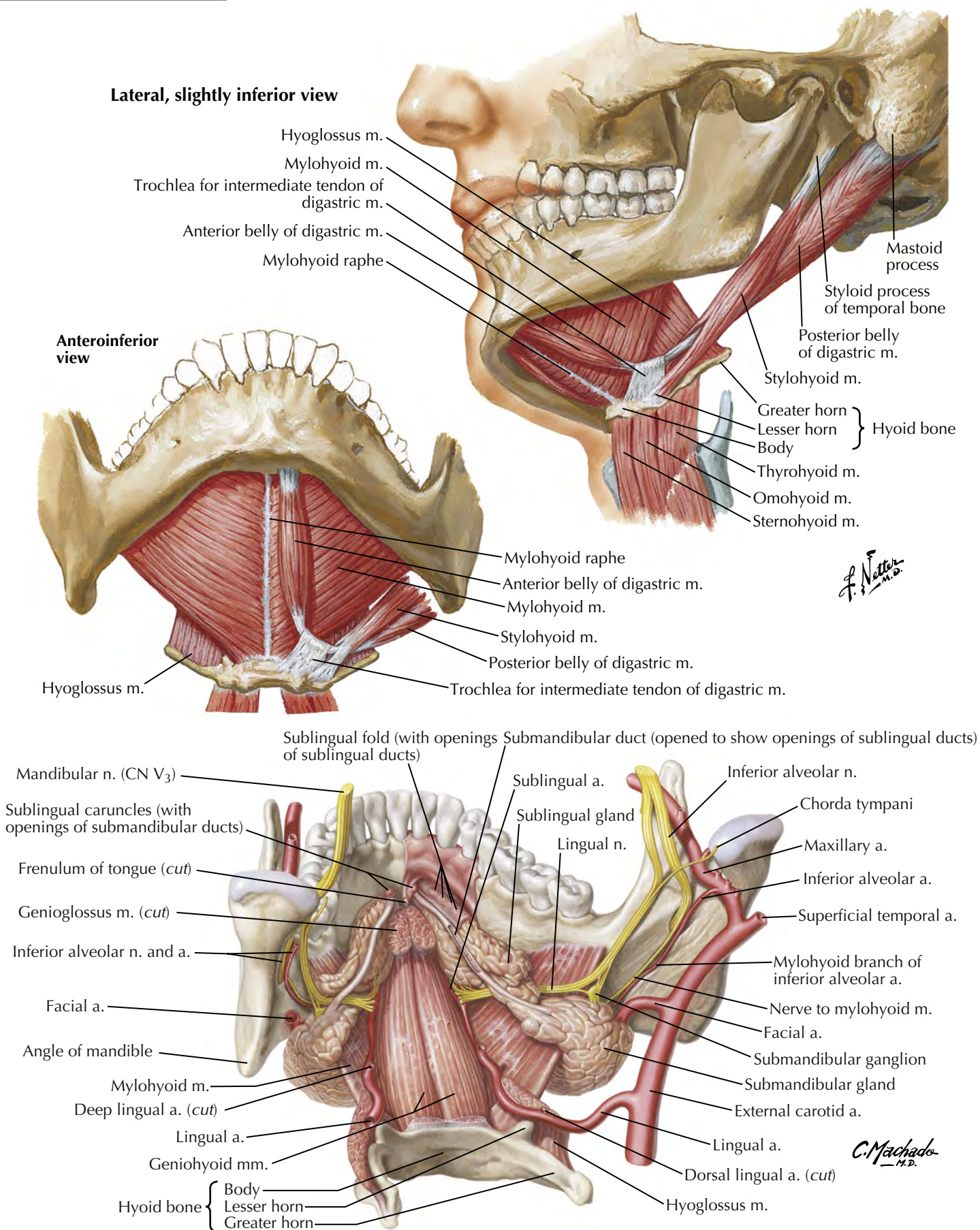


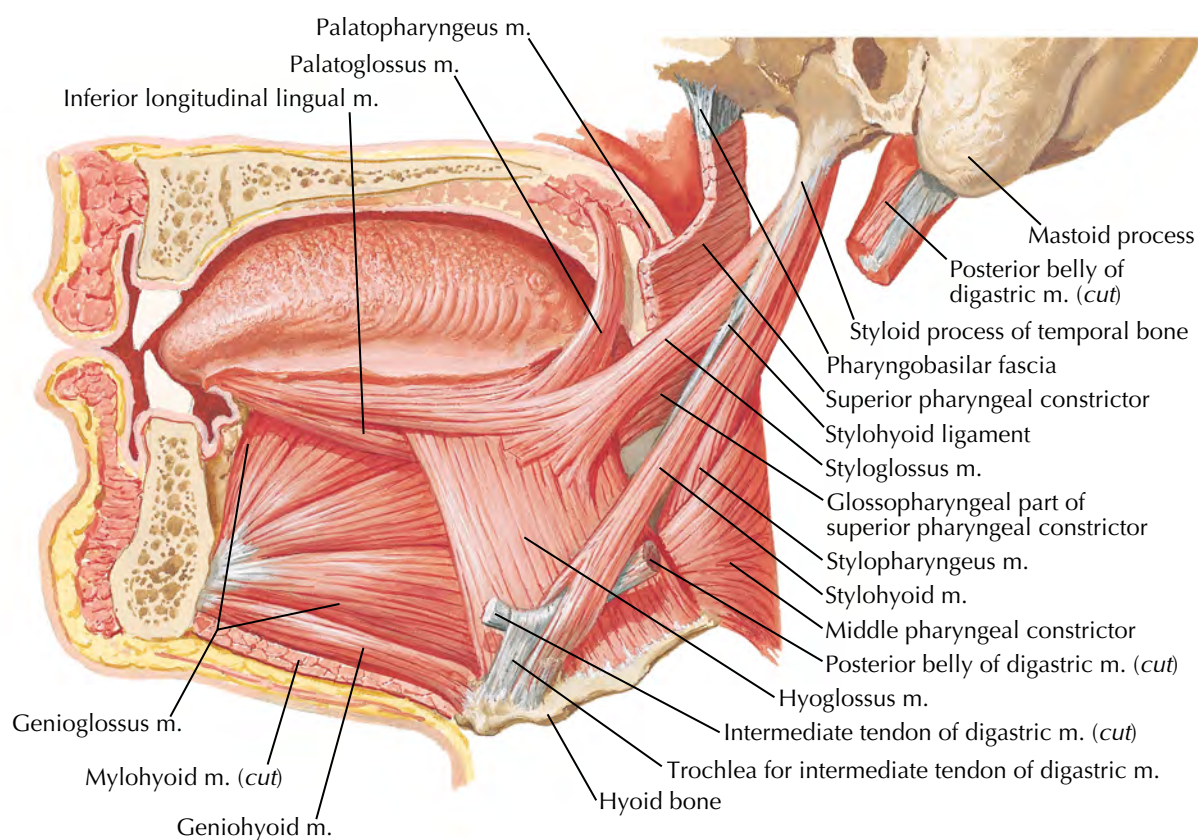
**Horizontal section
below lingula of
mandible (superior view)**

*F. Netter
m.d.*

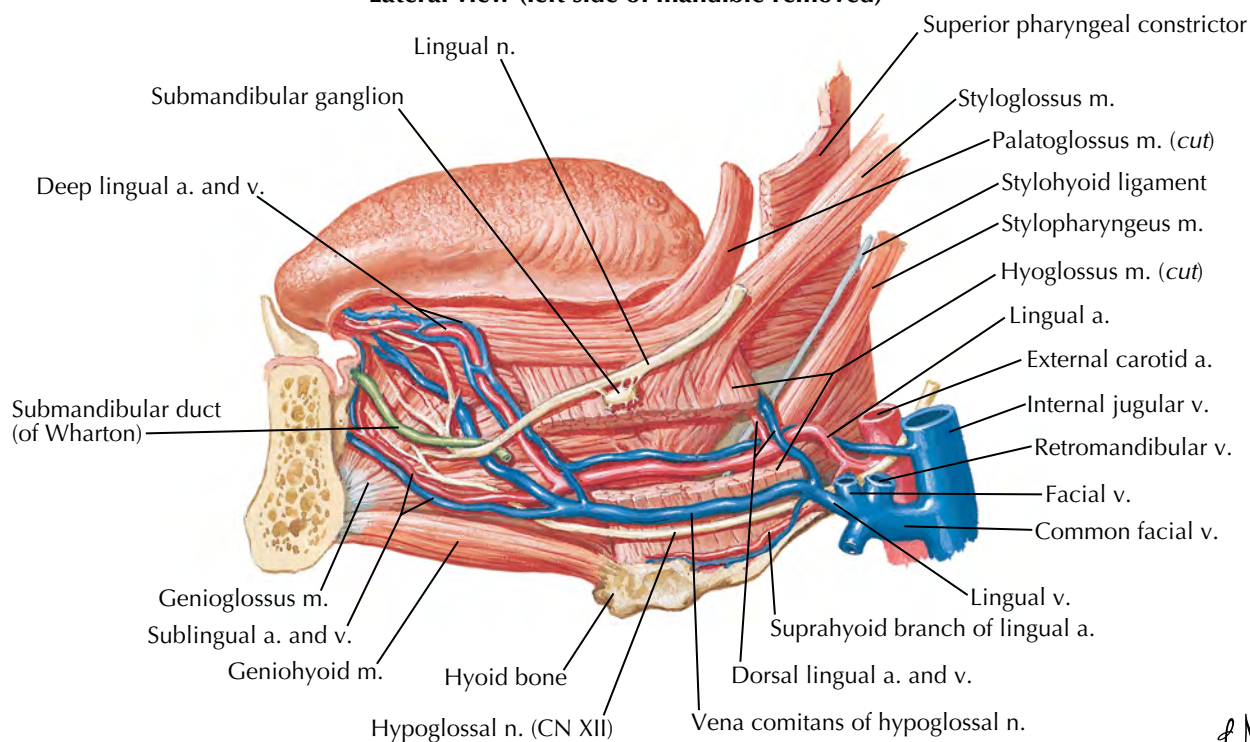


**Coronal section
posterior to 1st
molar tooth
(anterior view)**



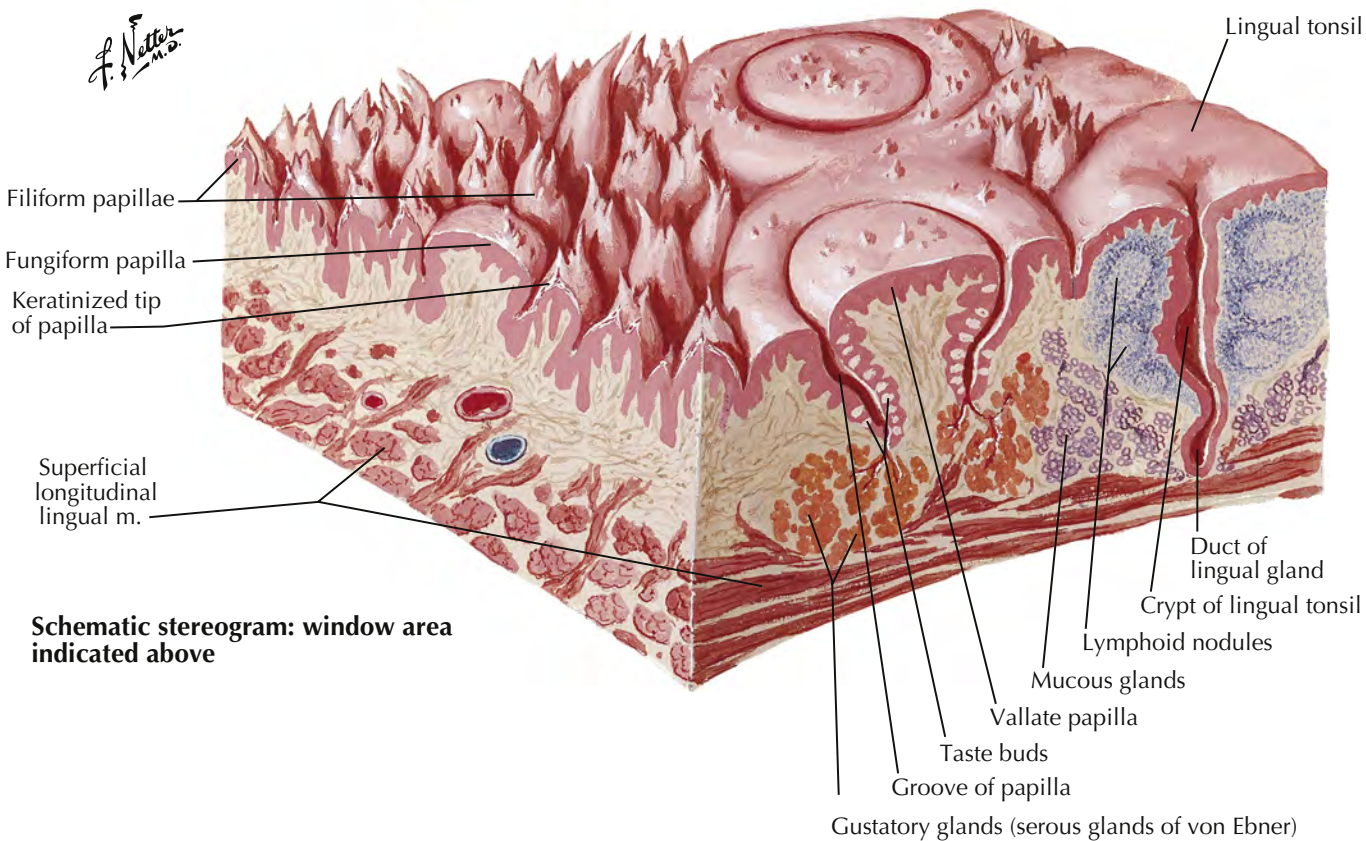
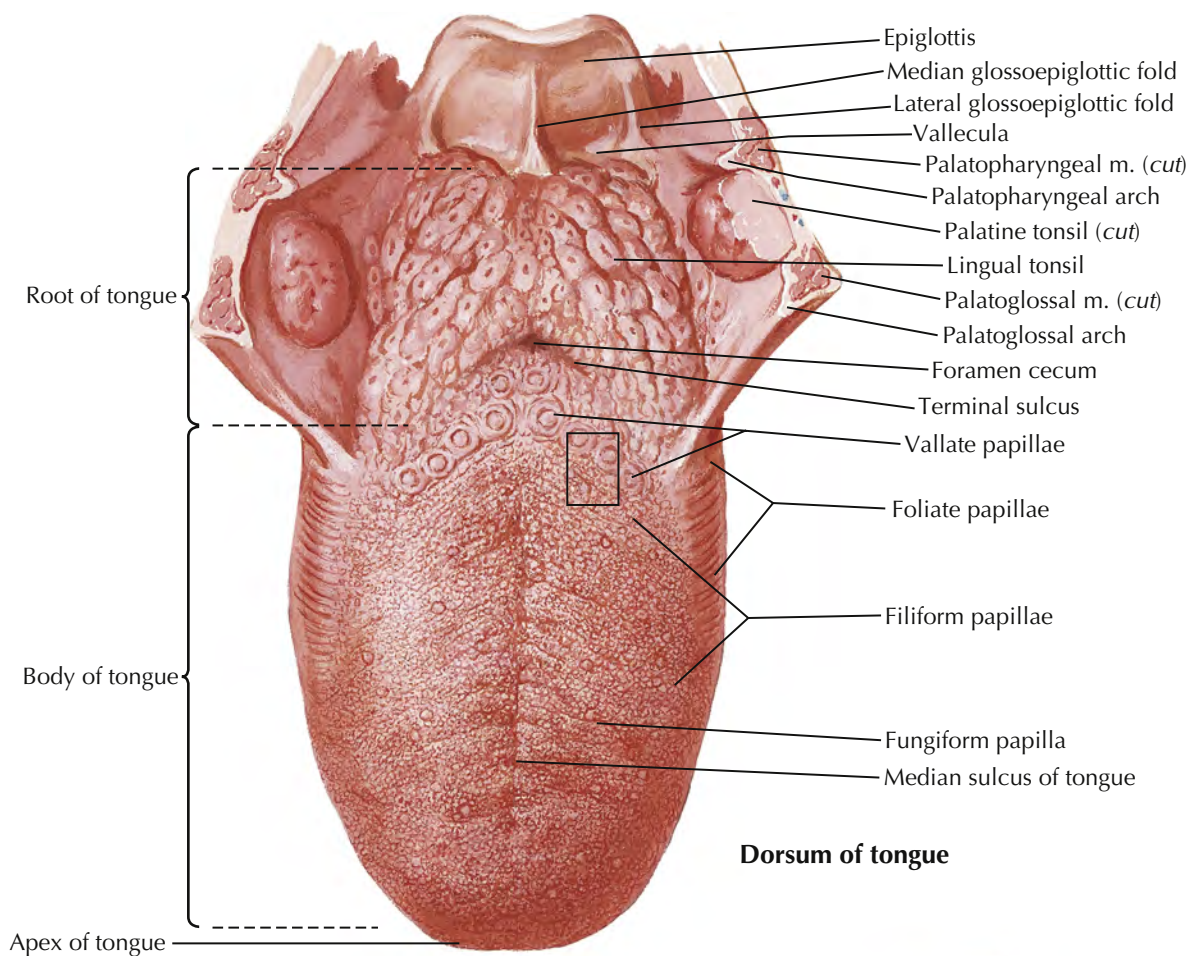


Lateral view (left side of mandible removed)

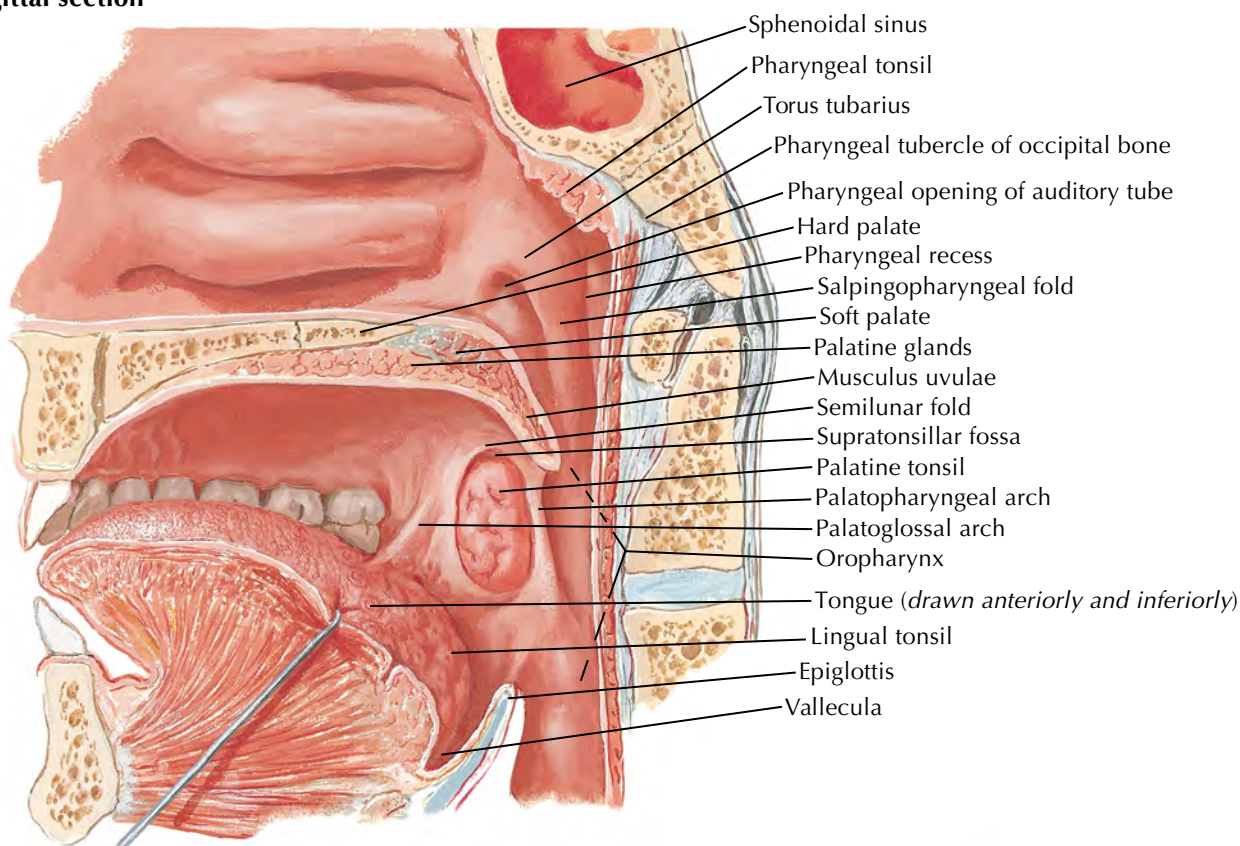


F. Netter M.D.

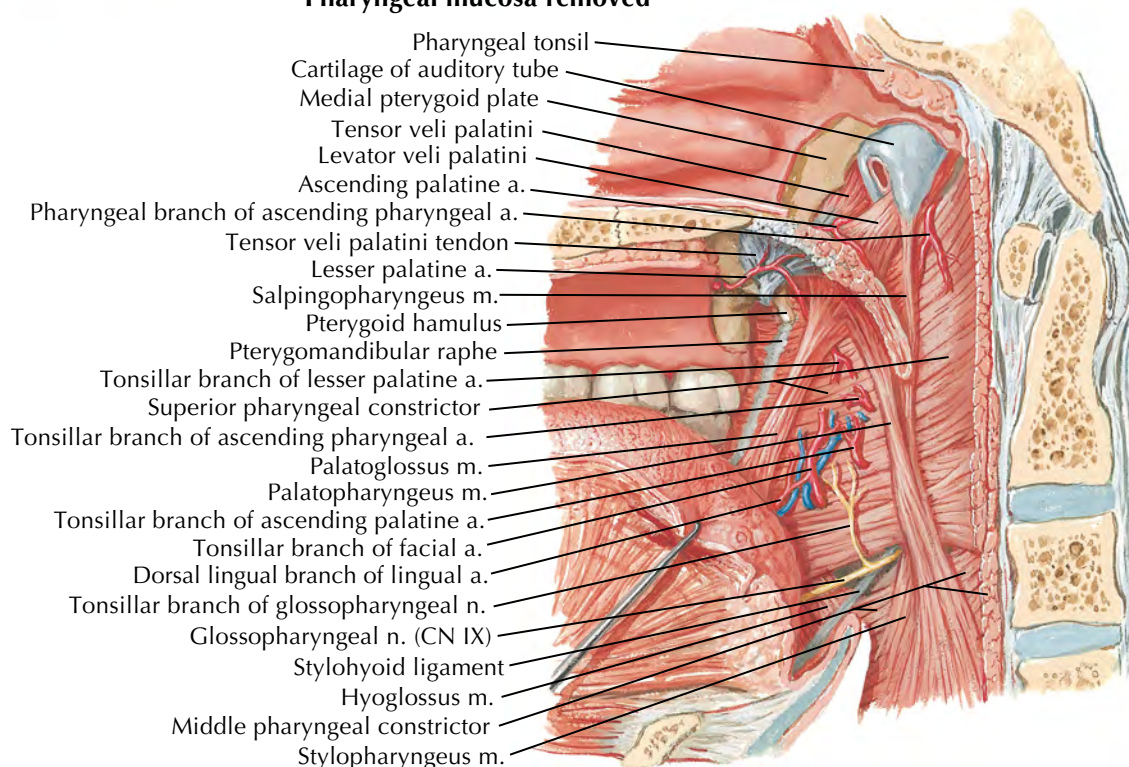
Dorsum of Tongue



Medial view sagittal section



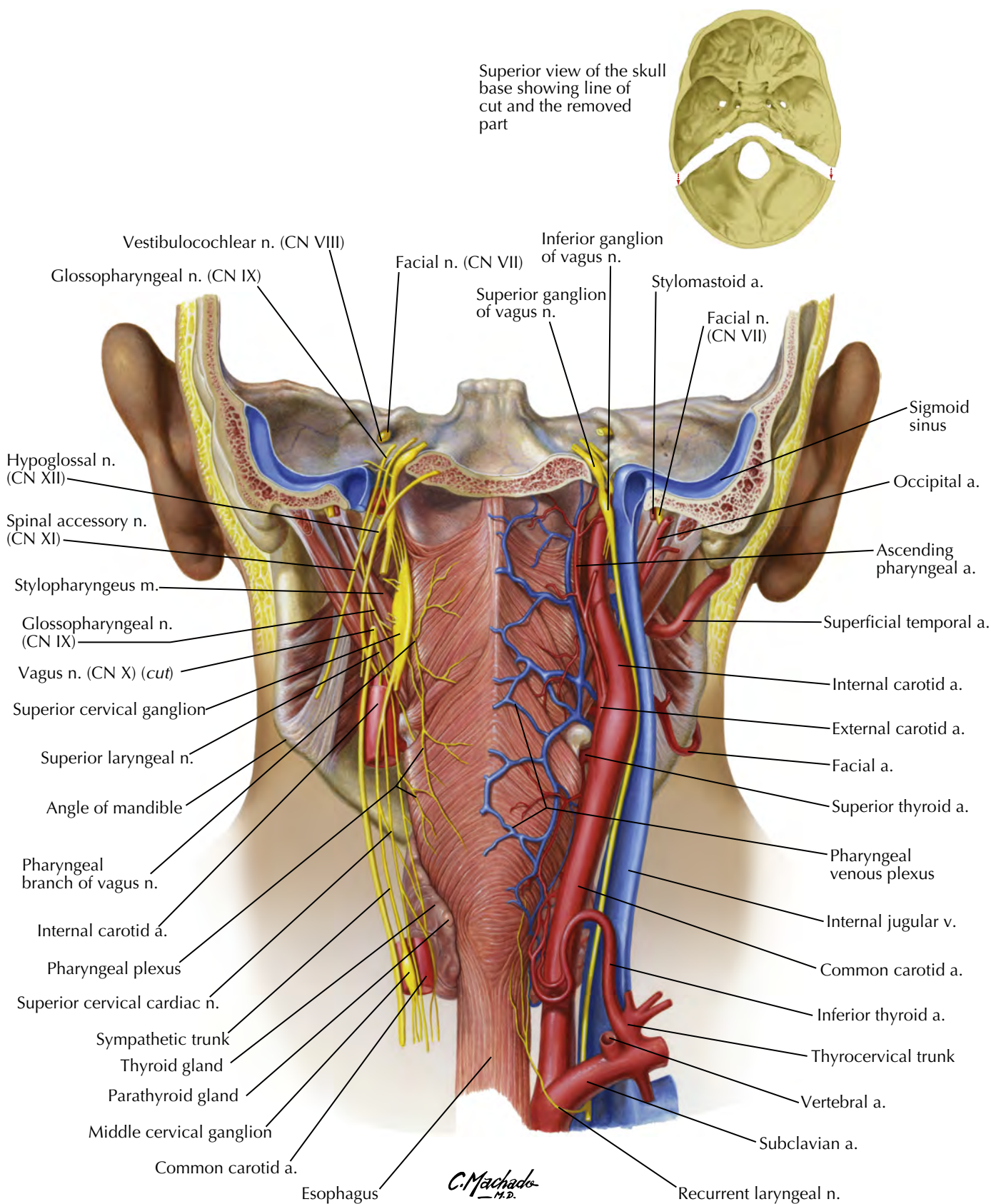
Pharyngeal mucosa removed

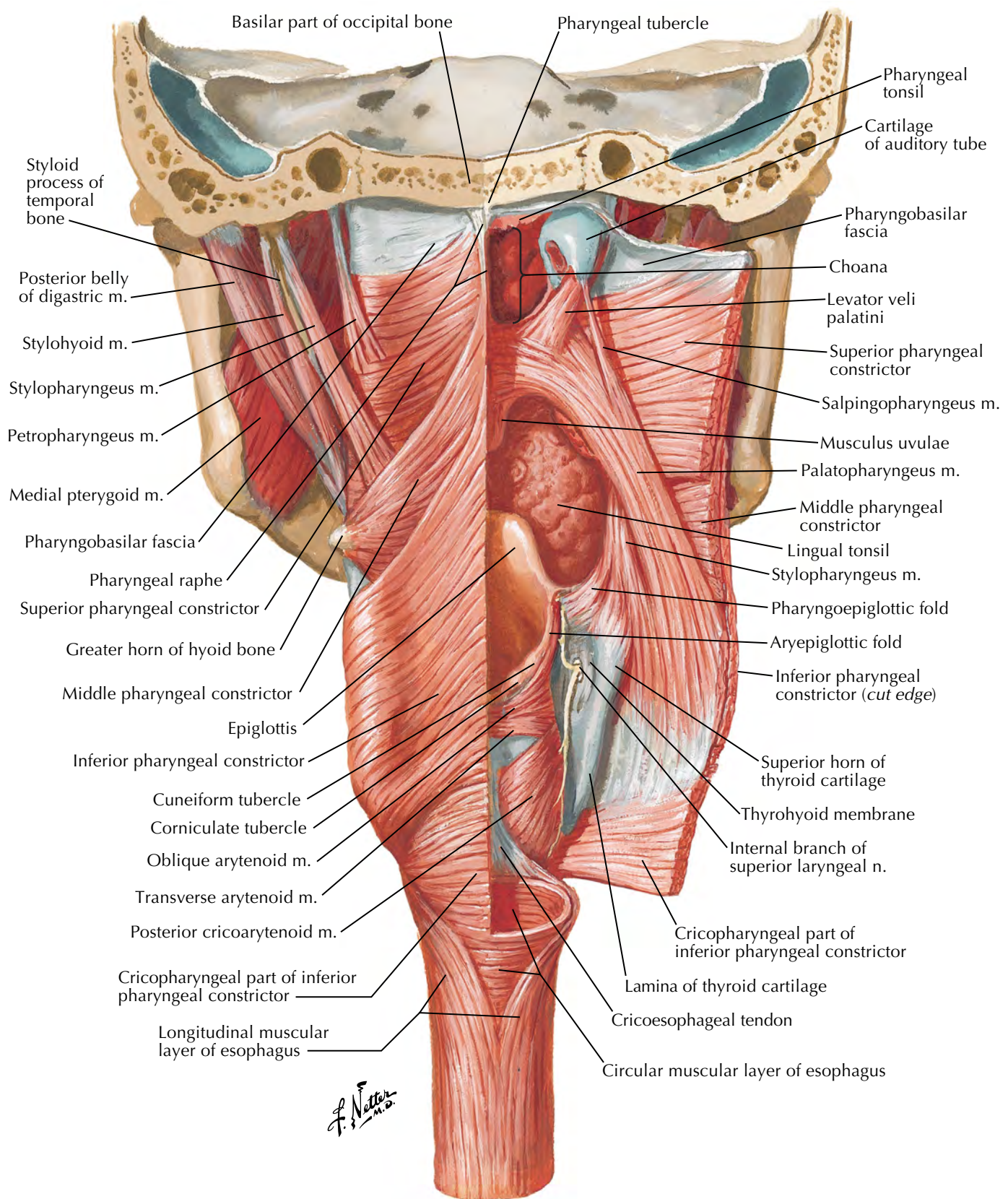


F. Netter M.D.

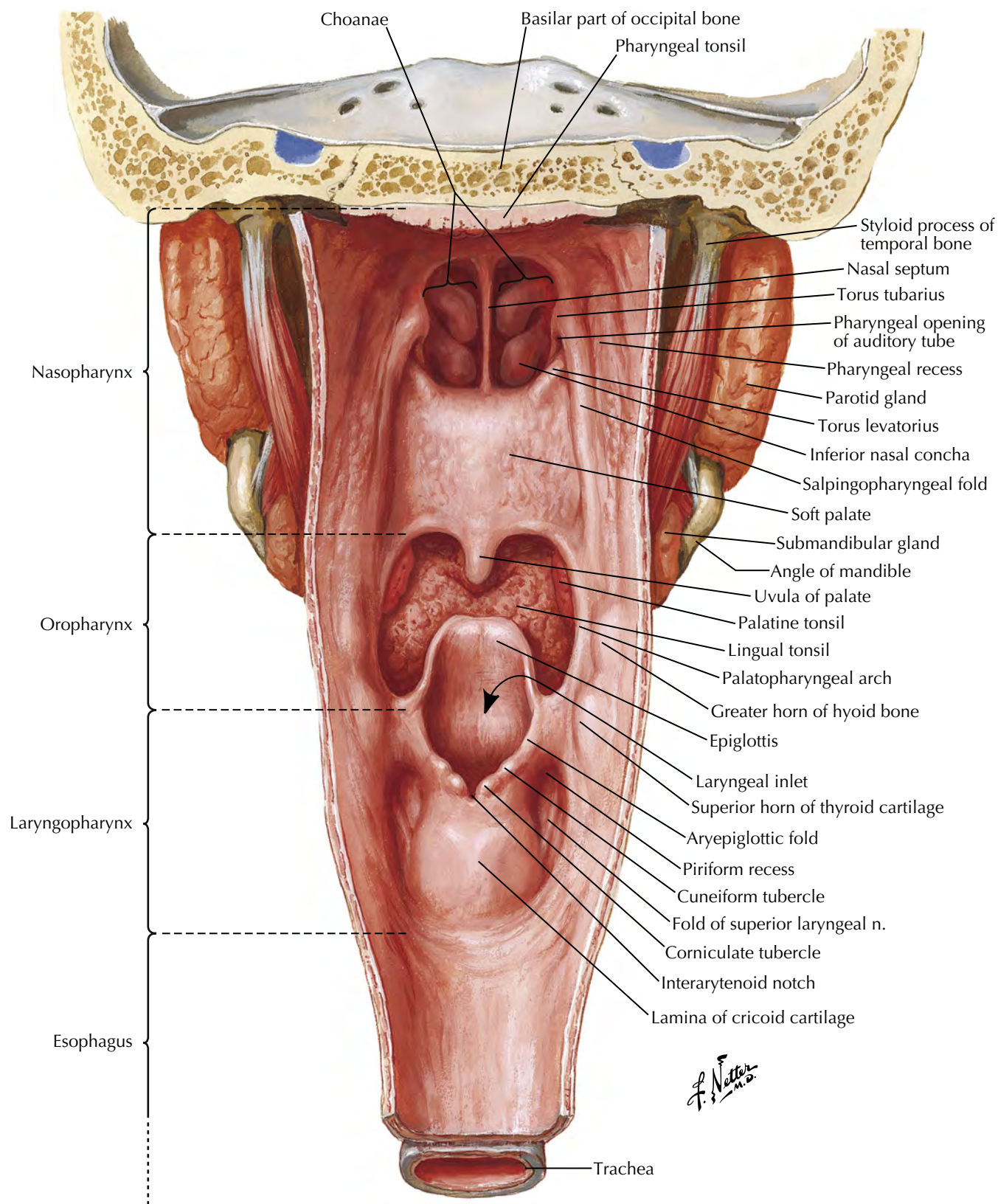
Posterior View of Pharynx: Nerves and Vessels

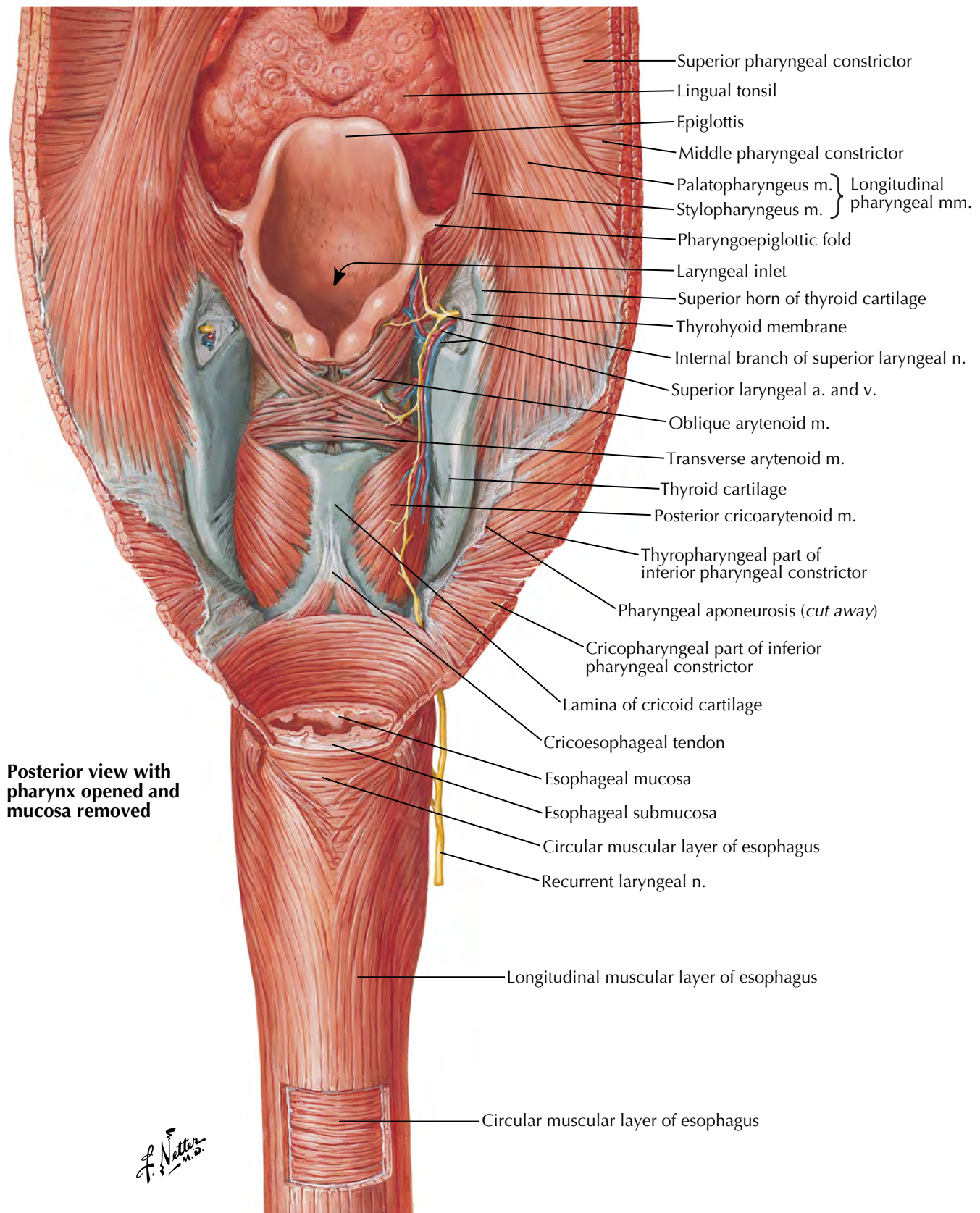
See also Plates 92, 93, 98, 99





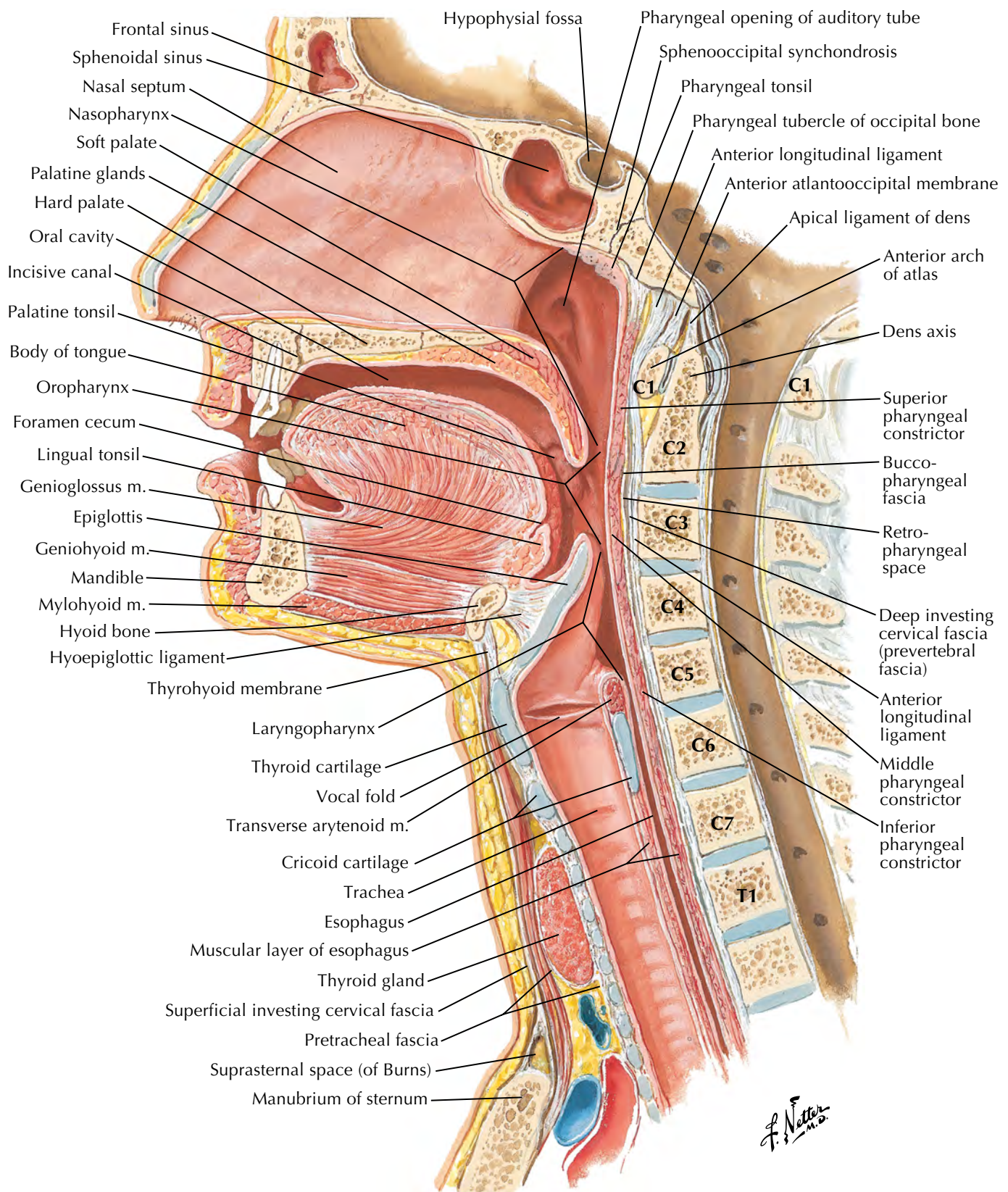
Pharynx: Opened Posterior View

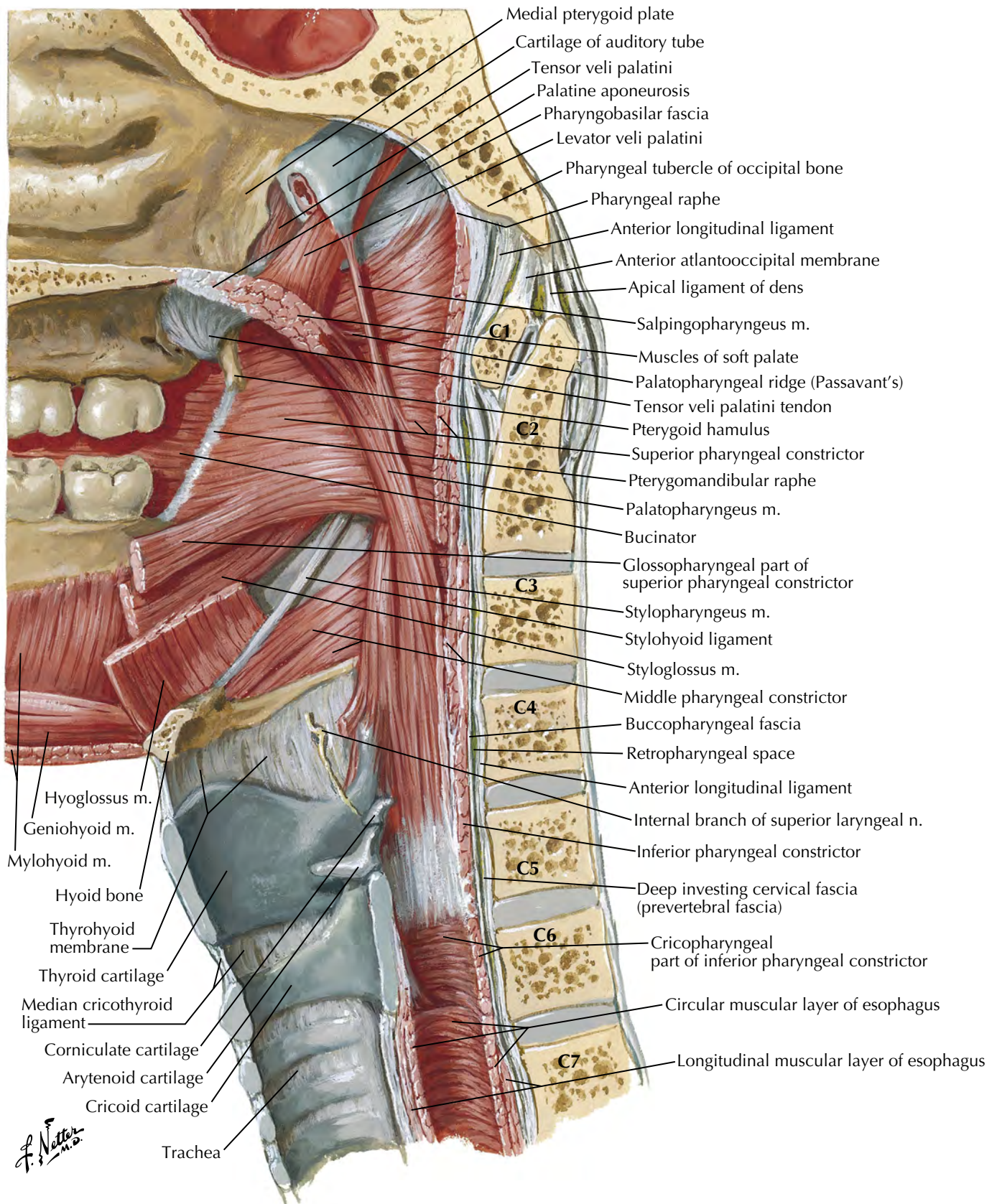




Pharynx: Medial View

See also Plate BP 26





Muscles of Pharynx: Lateral View

See also Plate 92

